Equating for t: $\frac{x+3}{4} = \frac{y}{4}$

$4(x + 5) = 3y$

$\therefore \quad y = \frac{4x}{3} + \frac{20}{3}$

73. Consider a line passing through the point having position vector $-i$. and parallel to the line $i + 2j$. Then, find

a. The vector equation

b. The parametric equation

c. The standard equation

Solution

a. $r = r_o + tv$

$r = (-1, -1) + t(1 + 2)$

or $r = (-1 + t)i + (-1 + 2t)j$

b. $x = x_o + tp = -1 + t$

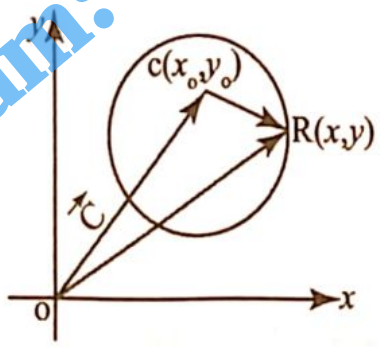
$y = y_o + tq = -1 + 2t$

c. $\frac{x - x_o}{p} = \frac{y - y_o}{q}$

$\frac{x+1}{1} = \frac{y+1}{2}$

5.12 Vectors and Circles

Consider a circle with center c (x_o, y_o) and radius $r > 0$; any point R (x, y) on the circle with position vector $\bar{C}$ and respectively such that:



$\overline{CR} = R - C = r$

$= (x - x_0, y - y_0)$

$|\overline{CR}| = \sqrt{\overline{CR} \cdot \overline{CR}}$

$r = \sqrt{(x - x_0, y - y_0) \cdot (x - x_0, y -}$

$r = \sqrt{(x - x_0)^2 + (y - y_0)^2}$

squaring both side

$(x - x_0)^2 + (y - y_0)^2 = r^2$ ⟵ Standared equation of a circle

center at (x_o, y_o)

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Illustrative Example

74. Write equation of a circle with center (4, -3) and radius 5

Solution: $(x - x_0)^2 + (y - y_0)^2 = r^2$

$(x - 4)^2 + (y + 3)^2 = 25$

75. Find the equation of a circle with diameter at end points (2, 3) and (6, 9)

Solution : Center of the circle is at

$$c\left(\frac{2+6}{6}, \frac{3+9}{2}\right) = c(4,6)$$

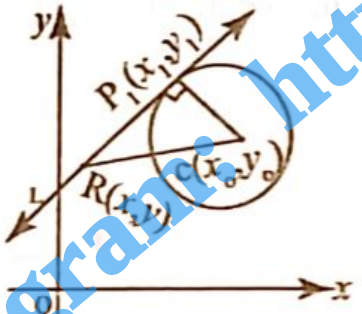
radius, $r = \sqrt{(6-4)^2 + (9-6)^2} = \sqrt{13}$

Then, $(x - 4)^2 + (y - 6)^2 = r^2$

$\therefore \quad (x - 4)^2 + (y - 6)^2 = 13$

5.13 Equation of Tangent Line to a Circle

Consider a circle with center C (x_0, y_0) and radius r and let P (x_1, y_1) be a point of tangency to the circle so that; $(x - x_0)^2 + (y - y_0)^2 = r^2$



$\overline{CP}.\overline{PR} = 0$

The radius of the circle r is perpendicular to the tangent L at the point contact P (x_1, y_1). If R(x, y) is an arbitrary point on L, then

$\overline{PR} \cdot \overline{CP} = 0$, they are perpendicular

$(x - x_1, y - y_1) \cdot (x_1 - x_0, y_1 - y_0) = 0$

$(x - x_1)(x_1 - x_0) + (y - y_1)(y_1 - y_0) = 0$

Add, $(x_1 - x_0)^2 + (y_1 - y_0)^2 = r^2$ into the above equation:

$(x - x_0)(x_1 - x_0) + (y - y_1)(y_1 - y_0) + (x_1 - x_0)^2 + (y_1 - y_0)^2 = r^2$

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Factorizing $(x_1 - x_0)^2$ and $(y_1 - y_0)^2$, we have

$(x - x_0)(x_1 - x_0) + (y - y_1)(y_1 - y_0) + (x_1 - x_0)(x_1 + x_0) + (y_1 - y_0)(y_1 + y_0) = r^2$

$(x - x + x_1 + x_0)(x_1 - x_0) + (y - y_1 + y_1 + y_0)(y_1 - y_0) = r^2$

$$(x + x_0)(x_1 - x_0) + (y + y_0)(y_1 - y_0) = r^2 \longleftarrow \text{Equation of Tangent}$$

And, if the circle is centred at the origin, $(x_0, y_0) = (0, 0)$

$(x + x_0)(x_1 - x_0) + (y + y_0)(y_1 - y_0) = r^2$

$(x + 0)(x_1 - 0) + (y + 0)(y_1 - 0) = r^2$

$xx_1 + yy_1 = r^2 \leftarrow$ Equation of Tangent

Illustrative Examples

76. Find the equation of the tangent line at the given point

a. $x^2 + y^2 = 10$ at point P(3, 1)

b. $(x - 3)^2 + (y + 1)^2 = 8$ at point P(1, 1)

c. $x^2 + y^2 - 8x + 10y + 21 = 0$ at point P(2, -1)

Solution

a. Let R(x,y) is point on the tagent line center C $(x_0, y_0) = (0, 0)$
$P(x_1, y_1) = (3, 1)$

$\overrightarrow{RP} = (3 - x, 1 - y)$ and

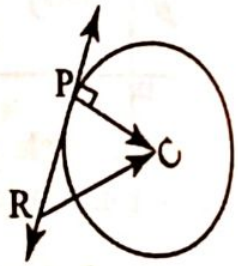
$\overrightarrow{PC} = (0 - 3, 0 - 1) = (-3, -1)$

$\overrightarrow{RP} \cdot \overrightarrow{PC} = 0$

$(3 - x, 1 - y) \cdot (-3, -1) = 0$

$-9 + 3x - 1 + y = 0$

$3x + y = 10$, is equation of tangent line



b. R (x, y), C (3, -1), P(1, 1)

$\overrightarrow{RP} = (1,1) - (x,y) = (1 - x, 1 - y)$

$\overrightarrow{PC} = (3,-1) - (1,1) = (2, -2)$

$\overrightarrow{RP} \cdot \overrightarrow{PC} = 0$ this implies $(1 - x, 1 - y) \cdot (2,-2)$

$2 - 2x - 2 + 2y = 0$ impies that $-2x + 2y = 0$

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$\therefore$ $y = x$ is equation of tangent line

c. R (x, y), C$(4, -5)$, P$(2, -1)$

$\overrightarrow{RP} = (2,-1) - (x, y) = (2 - x, -1 - y)$

$\overrightarrow{PC} = (4,-5) - (2,-1) = (2, -4)$

$\overrightarrow{RP} \cdot \overrightarrow{PC} = 0$ implies $(2 - x, -1 - y) \cdot (2,-4)$

$4 - 2x + 4 + 4y = 0$

$-x + 2y = -4$ ⟵ Equation of tangent line

77. Find the equation of the tangent line to the circle $x^2 + y^2 + 4x - 8y = 0$ at point P$(-4, 8)$

Solution:- $x^2 + y^2 + 4x - 8y = 0$

$x^2 + 4x + y^2 - 8y = 0$

$(x + 2)^2 - 4 + (y - 4)^2 - 16 = 0$. By completing the square

$(x + 2)^2 + (y - 4)^2 - 4 - 16 = 0$

$(x + 2)^2 + (y - 4)^2 = 20$ ← Equation of circle

Since, P$(-4, 8)$, C $(-2, 4)$, R(x, y):

$\overrightarrow{RP} \cdot \overrightarrow{PC} = 0$

$(-4 - x, 8 - y) \cdot (2, -4) = -8 - 2x + -32 + 4y = 0$

$= 2x + 2y = 40$

Addition Examples

78. If $\vec{a} = 2i - 3j$, $\vec{b} = 3i - j$ and $\vec{c} = i - j$ then $\vec{a} \cdot (2\vec{b} - \vec{c})$ is equal to?

A. -10 B. 7 C. 10 D. 13

Solution : $2\vec{b} - \vec{c} = 2(3i - j) - (i - j) = 5i - j$

$\vec{a} \cdot (2\vec{b} - \vec{c}) = (2i - 3j)(5i - j) = (2)(5) + (-3)(-1) = 13$

Answer: D

79. If A $= (2,3)$, B $= (3, 4)$ and c $= (-1, -2)$ then $\overrightarrow{AB} \cdot \overrightarrow{AC}$ is ?

A. -10 B. -2 C. 2 D. -8

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Solution

$\overrightarrow{AB} = (3,4) - (2,3) = (1,1)$ and $\overrightarrow{AC} = (-1, -2) - (2,3) = (-3, -5)$

$\overrightarrow{AB} \cdot \overrightarrow{AC} = (1,1) \cdot (-3,-5) = (1)(-3) + (1)(-5) = -8$

Answer: D

80. Let $\vec{u} = (-1,10)$ and $\vec{V} = (1, 3)$ the unit vector in the direction of $\vec{u} - 2\vec{v}$ is?

A. $\left(\frac{-3}{5}, \frac{-4}{5}\right)$ B. $\left(\frac{-3}{5}, \frac{4}{5}\right)$ C. $\left(\frac{3}{5}, \frac{-4}{5}\right)$ D. $\left(\frac{3}{5}, \frac{4}{5}\right)$

Solution : $\vec{u} - 2\vec{v} = (-1, 10) - 2(1, 3)$

The unit vector in $(-3, 4)$ direction $= \frac{(-3,4)}{\sqrt{(-3)^2 + 4^2}} = \left(\frac{3}{5}, \frac{4}{5}\right)$

Answer: B

81. If $(i + 5j) - \vec{x} = (3i + 2j) - 2(i - 3j)$, then the norm of vector $\vec{x}$ is?

A. 11 B. $\sqrt{10}$ C. 3 D. $\sqrt{3}$

Solution : $\vec{x} = i + 5j - 3i + 2j + 2i - 6j = -3j$

$|\vec{x}| = \sqrt{(-3)^2} = 3$

Answer: C

82. If $|\vec{a}| = 4$, $|\vec{b}| = \sqrt{3}$ and $\vec{a} \cdot \vec{b} = -3$, then $|\vec{a} - \vec{b}|$ is?

A. 5 B. 7 C. $\sqrt{13}$ D. $\sqrt{22}$

Solution : $|\vec{a} - \vec{b}| = \sqrt{(\vec{a} - \vec{b})(\vec{a} - \vec{b})} = \sqrt{|\vec{a}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2}$

$= \sqrt{4^2 - 2(-3) + \left(\sqrt{3}\right)^2} = \sqrt{25} = 5$

Answer: A

83. If $\vec{u} = -i + 2j$, $\vec{V} = -3i + j$ and $\vec{W} = -3j$, then the cosine of the angle between $\vec{u} + \vec{W}$ and $\vec{V}$ is?

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A. $\frac{1}{\sqrt{5}}$ B. $\frac{1}{2}$ C. $\frac{2}{\sqrt{5}}$ D. 0

Solution :- $(\vec{u} + \vec{W}) \cdot \vec{V} = |\vec{u} + \vec{W}|\,|\vec{V}| \cos\theta$

$$\cos\theta = \frac{(\vec{u} + \vec{W}) \cdot \vec{V}}{|\vec{u} + \vec{W}|\vec{V}|} = \frac{(-i - j)\,(-3 + j)}{(\sqrt{2})\,(\sqrt{10})} = \frac{2}{\sqrt{20}} = \frac{1}{\sqrt{5}}$$

Answer: A

84. Let $\vec{b}$ is unit vector and $|\vec{a}| = 3$. The vectors $\vec{a}$ and $\vec{b}$ make angle $\theta = \frac{\pi}{3}$.

What is the value of $|a - 2b|$

A. $\sqrt{5}$ B. $\sqrt{7}$ C. $\sqrt{8}$ D. $\sqrt{19}$

Solution :- $|\vec{a} - 2\vec{b}| = \sqrt{|a|^2 - 4|\vec{a}||\vec{b}| \cos\theta + 4|\vec{b}|^2}\;|\vec{b}|^2 = 1$

$$= \sqrt{9 - 4(3)(1)(0.5) + 4(1)} = \sqrt{7}$$

Answer: B

85. Suppose a triangle has vertices A = (1, 2) B = (3, 4) and C = (2, 5). What is the interior angle of Δ ABC at vertex B?

A. 30^0 B. 45^0 C. 60^0 D. 90^0

Solution : $\overrightarrow{BA} = (1, 2) - (3, 4) = (-2, -2)$

$\overrightarrow{BC} = (2, 5) - (3, 4) = (-1, 1)$

$$\cos\theta = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{|\overrightarrow{BA}|\,|\overrightarrow{BC}|} = \frac{(-2, -2) - (-1, 1)}{4} = \frac{0}{4} = 0$$

$\theta = 90^0$

Answer: D

86. What is vector equation of the line passing through (-1, 4) and parallel to the line $y - 3x - 10 = 0$

A. $(x, y) = (0, -7) + t(1, 3)$ C. $(x, y) = (0, 7) + t(1, -3)$

B. $(x, y) = (1, 10) + t(1, 3)$ D. $(x, y) = (-1, 4) + t(1, 3)$

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Solution:- $y - 3x - 10 = 0 \Rightarrow y = 3x + 10$

Direction vector $\overline{V}$ is obtained by setting $y = 3x$

$\overline{V} = (a, b) = (a, 3a) = a(1, 3) = t(1, 3)$

$(x, y) = (-1, 4) + t(1, 3)$

Answer: D

87. If the equation of a line ℓ: $2x - 3y - 7 = 0$. What is vector equation of the line?

A. $(x, y) = (0, -7) + t(2, -3)$ C. $(x, y) = (-3, 4) + t(1, -6)$

B. $(x, y) = (2, -1) + t(3, 2)$ D. $(x, y) = (-4, 2) + t(1, 5)$

Solution: $2x - 3y - 7 = 0 \Rightarrow 3y = 2x - 7$

$$y = \frac{2}{3}x - \frac{7}{3}$$

Direction vector $\overline{V}$ parallel to ℓ is $y = \frac{2}{3}x$

$$\overline{V} = (a, b) = \left(a, \frac{2}{3}a\right) = t(3,2)$$

If you take any point on $y = \frac{2}{3}x - \frac{7}{3}$, let $(2, -1)$:

$(x, y) = (2, -1) + t(3, 2)$

Answer: B

88. Let $(x, y) = (2, 5) + t(1, -2)$ is vector equation of ℓ. What is standard equation of ℓ?

A. $y + 2x = 9$ B. $3y - 4x = 3$ C. $y = 3x - 5$ D. $-2y + 1 = 3$

Solution:- $(x, y) = (2, 5) + t(1, -2)$

$x = 2 + t$, $y = 5 - 2t$

Solving for t, $t = \frac{x - 2}{1} = \frac{y - 5}{-2} \Rightarrow y + 2x = 9$

Answer: A

89. A circle that passes through A = (2, 3) and center at = (1, 1). The equation of the line tangent to the circle at A is?

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A. $3x + y = 9$ B. $x + 2y = 8$ C. $3x - 2y = 0$ D. $x + y = 5$

Solution:- Let R(x, y) be a point on ℓ. $\overline{CA} \cdot \overline{AR} = 0$

$[(2, 3) - (1, 1)] \cdot [(x - 2, y - 3)] = (1, 2) \cdot (x - 2, y - 3)$

$x + 2y = 8$

Answer: B

90. Which of the following is a vector that lie on the line through $(0,0)$ and $(2,4)$

A. $\vec{u} = (2, 1)$ B. $\vec{u} = (-1, -2)$ C. $\vec{u} = \left(\frac{1}{2}, 2\right)$ D. $\vec{u} = (-2, -6)$

Solution:- Let A $= (0, 0)$ and B $= (2, 4)$, $\overline{AB} = (2, 4) - (0, 0) = (2, 4)$

Let $V = t\overline{AB} = 2t\ (1,2) = 2t\ (-1, -2)$

Answer: B

91. Which of the following is a vector equation of the line tangent to the circle $x^2 + y^2 + 2x - 7 = 0$ at $(1, 2)$

A. $(x, y) = (0, 3) + \lambda\ (-1, 2)$ C. $(x, y) = (0, 3) + \lambda\ (1, -1)$

B. $(x, y) = (1, 2) + \lambda\ (2, -1)$ D. $(x, y) = (1, 2) + \lambda\ (-1, 2)$

Solution:- $x^2 + y^2 + 2x - 7 = 0$

$(x + 1)^2 + (y - 0)^2 = 8$

C $(-1, 0)$, P $(1, 2)$ and R(x, y)

$\overline{CP} = (1, 2) - (-1, 0) = (2, 2)$

$\overline{CP} \cdot \overline{CR}\ (2, 2) \Rightarrow 2x + 2y - 6 = 0\ \ (x - 1, y - 2) = 0$

Direction vector $\vec{V} = (a, b)$ parallel to the line $x + y = 3$ is,

$x + y = 0 \Rightarrow y = -x$

$\vec{V} = (a, b) = (a, -a) = \lambda\ (1, -1)$

Take any point on $x + y = 3$, say, $(0, 3)$

vector equation of ℓ : $(x, y) = (0, 3) + \lambda\ (1, -1)$

Answer: C

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92. Suppose A = 2i - j + 2k and B is a vector such that $|\vec{B}| = \vec{A}\cdot\vec{B}$. If $\vec{U}$ is the unit vector in the direction of $\vec{B}$ then the value of $|\vec{A} + \vec{U}|^2$ is equal to

A. 16 B. 12 C. 10 D. 14

Solution : $|A| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{9} = 3$

since $|\vec{B}| = \vec{A} \cdot \vec{B}$, then $|\vec{B}| = |\vec{A}||\vec{B}| \cos\theta$,

$$\cos\theta = \frac{1}{|\vec{A}|} = \frac{1}{3}$$

$$|\vec{A} + \vec{U}|^2 = |\vec{A}|^2 + 2|A||U|\cos\theta + |\vec{U}|^2 = 3^2 + 2(3)(1)\frac{1}{3} + 1 = 12$$

Answer: B

93. If A = 4i - 3j and $\vec{U}$ is a unit vector such that $|\vec{A} + \vec{U}|^2 = 27$, then the cosine of the angle between $\vec{A}$ and $\vec{U}$ is equal to ___

A. 0.1 B. 0.2 C. 0.3 D. 0.4

Solution

$$|\vec{A} + \vec{U}| = |A|^2 + 2|\vec{A}||\vec{U}|\cos\theta + |\vec{U}|^2$$

$$27 = 25 + 2(5)(1)\cos\theta + 1$$

$$\therefore \quad \cos\theta = \frac{27 - 26}{10} = \frac{1}{10} = 0.1$$

Answer: A

94. What is the value of k, for which the two vector

$$\vec{U} = \begin{pmatrix} 1 \\ k \\ -3 \end{pmatrix} \text{ and } \vec{V} = \begin{pmatrix} 2k \\ -5 \\ 4 \end{pmatrix} \text{ are perpendicular?}$$

A. 4 B. -4 C. 3 D. -3

Solution

$$\vec{U} \cdot \vec{V} = 1 \times 2k + -5k - 12 = 0$$

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$-3k = 12, k = -4$

Answer: B

End of Unit Problem

1. Given the vectors: $\vec{U} = (3, -2)$, $\vec{V} = (-2, -1)$ and $\vec{W} = (5, 1)$. Find
   a. $3\vec{U} - 2\vec{V}$
   b. $\vec{U} + 4\vec{V} - \vec{W}$
   c. $\vec{x}$ if $\vec{x} + \vec{V} = 2\vec{u} - \vec{W}$
   d. $|3\vec{u} - 2\vec{v} + \vec{W}|$
   e. Unit vector in the direction of $2\vec{U} - 3\vec{V}$
2. If $\vec{U} = i + 2j$ $\vec{V} = 2i + 4j$ and $\vec{W} = 5i - j$, then find
   a. $(\vec{u} \cdot \vec{v})\vec{W}$ b. $\vec{u} \cdot (3\vec{W})$ c. $|\vec{v}|^2$
3. If $\vec{U} = 3i + 2j, + 5k$ and $\vec{V} = i + 3j + 6k$, then find
   a. $\vec{u} \times \vec{v}$ b. $\vec{v} \times \vec{u}$
4. If vector $\vec{U} = 2i + 3j$, then find the angle formed by;
   a. $\vec{U}$ and the x - axis b. $\vec{U}$ and the y - axis
5. For what of m will $\vec{U} = i + mk$ and $\vec{V} = 10i + 3j - 2k$ became perpendicular?
6. Let $\vec{U} = (-5, 3, 7)$ and $\vec{V} = (6, -8, 2)$. Determine if these vectors are parallel, perpendicular or neither
7. Given the vectors: $\vec{U} = 6i + j - 3k$ $\vec{U} = 2i + 3j + k$, then find
   a. Area of the parallelogram formed by $\vec{U}$ and $\vec{V}$
   b. Area of a triangle formed by $\vec{U}$ and $\vec{V}$
8. If a line passes through a point (-2, 1) and is parallel to the vector $2i + 3j$, then find
   a. Vector equation b. Parametric equation
   c. Standard equation of the line
9. Find the equation of the line tangent to the circle $x^2 + y^2 = 5$ at point P(-1, 2)
10. Find a unit vector in the direction of

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$\bar{r}_1 - \bar{r}_2$ where, $\bar{r}_1 = i + 2j - k$ and $\bar{r}_2 = 3i + j - 5k$

11. Find the angle between the vectors

a. $\bar{U} = -3i + 2j$ and $\bar{V} = 2i + 3j$

b. $\bar{U} = i + j$ and $\bar{V} = 3i$

12. Given the vectors $\bar{U} = i$, $\bar{V} = i + j$, find scalar k such that

a. $\bar{U} + k\bar{V}$ is perpendicular to $\bar{U}$

b. $\bar{U} + k\bar{V}$ is perpendicular to $\bar{V}$

13. Vector $\bar{u}$ and $\bar{v}$ make angle $\theta = \frac{\pi}{3}$. If $|\bar{u}| = 3$ $|\bar{v}| = 2$, then find

a. $|2\bar{u} + \bar{v}|$ b. Angle betwen $\bar{v}$ and $(\bar{u} + \bar{v})$

14. Let $\bar{u} = (1, 3)$, $\bar{v} = (2,1)$ and $\bar{W} = (3, 4)$, find vector $\bar{x}$ that satisfies $2\bar{u} - \bar{v} + \bar{x} = 7\bar{x} + \bar{W}$

15. Find parametric equation for the line passing through P_1 and P_2

a. $P_1(3, -2), P_2(5, 1)$ b. $P_1(5, -2), P_2(2, 4, 2)$

16. Find equation of a circle with

a. Center (3, 4), radius b. Center (-1, 1) radius $\sqrt{3}$

17. Find equation of a line that is tangent to the circle $x^2 + y^2 = 25$ at point (3,-4)

18. Find symmetric equation of a line that passes through the point (3,2) and is

a. Parallel to vector 2i - 4j b. Perpendicular to vector -i + j

19. What is the power developed by the engine of a car if it exerts (2i + 100j)N force to speed up the car to (20i + 5j) m/s from rest?

20. What is the volume of a parallelepiped whose side are the vectors $\bar{U} = 2i - j + 3k$, $\bar{V} = i + 4j$ and $\bar{W} = 3i + 4j + 5k$

Solution to the end of unit problems

1. a. $3\bar{u} - 2\bar{v} = 3(3,-2) - 2(-2, -1) = (13, -4)$

b. $\bar{u} + 4\bar{v} - \bar{W} = (3,-2) + 4(-2,-1) - (5, 1) = (-10, -7)$

c. $\bar{x} + \bar{v} = 2\bar{u} - \bar{W} \Rightarrow x = (2\bar{u} - \bar{W}) - \bar{v} = (3, -4)$

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d. $13\vec{u} - 2\vec{v} + \vec{W} = \sqrt{(18)^2 + (-3)^2} = \sqrt{333}$

e. $2\vec{u} - 3\vec{v} = (12, -1)$, The unit vector is $\left(\frac{8}{565}, \frac{-1}{\sqrt{65}}\right)$

2. a. $\left(\vec{u} \cdot \vec{v}\right) \vec{W} = (i + 2j).(2i + 4j)\ (5i - j) = 50i - 10j$

b. $\vec{u}\left(3\vec{W}\right) = (i + 2j).3\ (5i - j) = (1)(15) + (2)\ (-3) = 9$

c. $\left|\vec{v}\right|^2 = (2i + 4j) \cdot (2i + 4j) = (2)(2) + (4)\ (4) = 20$

3. a. $\vec{u} \times \vec{v} = -3i - 13j - 7k$ b. $\vec{v} \times \vec{u} = 3i + 13j + 7K$

4. a. $\theta = \cos^{-1}\left(\frac{2\sqrt{13}}{13}\right)$ b. $\theta = \cos^{-1}\left(\frac{3\sqrt{13}}{13}\right)$

5. $\vec{U} \cdot \vec{V} = 0 \Rightarrow (1)\ (10) + (0)\ (3) + (m)\ (-2) = 0 \Rightarrow m = 5$

6. $\vec{U} \cdot \vec{V} = (-5, 3, 7) \cdot (6, -8, 2) = -40 \neq 0$

7. a. $\left|\vec{u} \times \vec{v}\right| = 10\sqrt{5}$ units b. $\frac{1}{2}\left|\vec{u} \times \vec{v}\right| = 5\sqrt{5}$ units

8. a. $(x, y) = (-2, 1) + t(2, 3)$ c. $\frac{x + 2}{2} = \frac{y - 1}{3}$

b. $x = -2 + 2t$, $y = 1 + 3t$

9. C (0, 0), P (-1, 2) and R (x, y).

RP. PC $= (-1 - x,\ 2 - y) \cdot (1, -2) = 0$

$-1 - x\ 4 + 2y = 0$

$\therefore 2y - x = 5$

10. $\vec{r_1} - \vec{r_2} = -2i + j + 4k$, $\hat{u} = \frac{1}{\sqrt{21}}(-2i + j + 4k)$

11. a. $\theta = 90^0$ b. $\theta = 45^0$

12. a. $\left(\vec{u} + k\vec{v}\right) \cdot \left(\vec{u}\right) = 0 \Rightarrow [(1 + k)i + kj] \cdot i = 0,\ k = -1$

b. $\left(\vec{u} + k\vec{v}\right) \cdot \left(\vec{v}\right) = 0 \Rightarrow [(1 + k)i + kj] \cdot (i + j) = 0,\ k = \frac{-1}{2}$

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13. a. $\left|2\bar{u}+\bar{v}\right| = \sqrt{4u^2 + 4uv\cos\theta + v^2} = 5\sqrt{2}$

b. $\theta = \cos^{-1}\left(\frac{1}{57}\right)$

14. $2\bar{u} - \bar{v} + x = 7\bar{x} + \overline{W} \Rightarrow 6\bar{x} = 2\bar{u} - \bar{v} - \overline{W}, x = \left(\frac{-2}{3}, 1\right)$

15.a. $x = 3 + 2t, y = -2 + 3t$

b. $x = 5 - 3t, y = -2 + 6t, z = 1 + t$

16.a. $(x - 3)^2 + (y - 4)^2 = 4$ b. $(x + 1)^2 + (y - 1)^2 = 3$

17. $x = 3 + 4t, y = -4 + 3t$

18.a. $y = -2x + 8$ b. $y = x - 1$

19. $P = \bar{F} \cdot \bar{V} = (2i + 100j) \cdot (20i + 5j) = (2)(20) + (100)(5)$

$P = 540\,W$

20. $V = \bar{u} \cdot \left(\bar{v} \times \overline{W}\right) = (2i - j + 3k)(20i - 5j - 8k)$

$V = (2)(20) + (-1)(-5) + (3)(-8)$

$V = 21$ units cube

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UNIT 6

6. Transformation of the Plane

6.1 What is Transformations?

Transformations are movements of geometric figure

There are two kinds of transformation (motion): rigid and non rigid

(i) Non rigid Motion

- A non rigid transformation that does not preserve distance.
- The size of the figure is changed.

In non - rigid motion transform a figure, by changing the shape and size of figure either enlarged or reduced.

(ii) Rigid Motion

A rigid transformation that preserve distance. The shape, size of the figure is not changed. In rigid motion carries any:

- Segment moved into congruent segment.
- Circle moved into congruent circle.
- Triangle moved into congruent triangle.
- Plane figures into congruent pane figures.

Translation, reflection and rotation are **rigid motion.**

6.2 What is a Translation

A translation is a rigid transformation that slides a figure with out changing the size or shape of the figure. In a translation every point of a figure moved from one location to another location along the same direction in the same distance.

Note: The points on the translated figure are indicated by the prime symbol

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(') to identify them from the original points.

That is P → P '. Point P to change it to point P'

If point P is translated to point P', then the vector $\overrightarrow{PP'}$ is called the translation vector.

How to translate a point or figure?

To translate a point P(x, y) by translating vector. T (a, b):

- add **a** to the x - coordinate and add **b** to the y - coordinate

Symbols : $P(x, y) \xrightarrow{T(a, b)} P'(x + a, y + b)$

Illustrative Example

1. If translation T takes the point P(7, 2) to the point P'(5, 3). What is the image of (3, 4) under the image of (3, 4) under the same translation T?

Solution

First, find translation vector: $T = \overrightarrow{PP'} = (5, 3) - (7, 2) = (-2, 1)$

$\therefore$ The image of $(3, 4) \xrightarrow{T(-2, 1)} (3 - 2, 4 + 1) = (1, 5)$

2. If translation T takes the origin to (3, -2), then find the image of the point A(1, 4) and B(-2, 5).

Solution: Translating vector: T = (3 - 0, -2 - 0) = (3, -2)

- The image of point A (1, 4) is A': (1 + 3, 4 - 2) = (4, 2)
- The image of point B (-2, 5) is B' : (-2 + 3, 5 - 2) = (1, 3)

So that A (1, 4) $\xrightarrow{T(3, -2)}$ A' (4, 2)

B (-2, 5) $\xrightarrow{T(3, -2)}$ B' (1, 3)

Method II you can use matrix addition to translate the point in one step.

Given point			**Translation Vector (matrix)**			**image**	
A	B					A'	B'
$\begin{bmatrix} 1 & -2 \\ 4 & 5 \end{bmatrix}$		+	$\begin{bmatrix} 3 & 3 \\ -2 & -2 \end{bmatrix}$		=	$\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$	

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Thus, the images are A'(4, 2) and B'(1, 3)

3. Triangle ABC has vertices A (-3, 4), B (3, 1), and C(2, -5).

Find the coordinate of the image of vertices of Δ ABC under the translation vector T = (-4, -2)

Solution:- The image of (x, y) under T = (-4, -2) is $(x, y) \rightarrow (x - 4, y - 2)$.

A. $(-3, 4) \xrightarrow{T(-4,-2)} (-3 - 4, 4 - 2) = (-7, 2)$

B. $(3, 1) \xrightarrow{T(-4,-2)} (3 - 4, 1 - 2) = (-1, -1)$

C. $(2, -5) \xrightarrow{T(-4,-2)} (2 - 4, -5 - 2) = (-2, -7)$

∴ The coordinates of the vertices of the image are:

A'(-7, 2), B'(-1, -1) and C(-2, -7)

Method II, we can use matrix addition to translate in one step.

Vetrices				Translating matrix			image		
A	B	C					A'	B'	C'

$$\begin{bmatrix} -3 & 3 & 2 \\ 4 & 1 & -5 \end{bmatrix} + \begin{bmatrix} -4 & -4 & -4 \\ -2 & -2 & -2 \end{bmatrix} = \begin{bmatrix} -7 & -1 & -2 \\ 2 & -1 & -7 \end{bmatrix}$$

Thus the image A' (-7, 2), B'(-1, -1) and C'(-2, -7)

4. Quadrilateral ABCD has vertices A (1, 1), B(-1, 4), C (-4, 2) and D(-4, 0). Find the coordinates of the image of quadrilateral ABCD under the translation T = (5, -3)

Solution:- We can use matrix addition to translate in one step.

Vetrices in matrix form				**Translating vector matrix form**				**image**			
A	B	C	D	T	T	T	T	A'	B'	C'	D'

$$\begin{bmatrix} 1 & -1 & -4 & -4 \\ 1 & 4 & 2 & 0 \end{bmatrix} + \begin{bmatrix} 5 & 5 & 5 & 5 \\ -3 & -3 & -3 & -3 \end{bmatrix} = \begin{bmatrix} 6 & 4 & 1 & 1 \\ -2 & 1 & -1 & -3 \end{bmatrix}$$

∴ The coordinates of the image are A'(6-2), B'(4, 1), C' (1,-1) and D'(1,-3)

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6.2.1 Line Translation

Translation is a parallel movement, so that under translation line moves into parallel lines.

Note that if the translating vector T = (-3, 2), geometrically interpreted as the line (figure) moves 3 units left in the x - direction and 2 units up in the y - direction. So that we replace x by $x + 3$ we replace y by $y - 2$

- The translating vector T = (4, -1), geometrically interpreted as, the line (figure)moves 4 unite right in x - direction and 1 unit down in y - direction. As a result we replace x by $x - 4$ and replace y by $y + 1$.

Illustrative Example

5. Translation T takes the point (7, 2) to the point (5, 3). Find the image of the line

   a. L: $3x - 4y = 7$ b. L: $2x + 3y = 1$

   Solution: Translation vector: T = (5, 3) - (7, 2) = (-2, 1)

   a. Step - 1 Take any point on L: $3x - 4y = 7$, let $x = 1$

   $3 - 4y = 7, y = -1$

   Thus the point (1, -1) on line L

   Step - 2: Translate (1, -1) under translation vector T = (-2, 1)

   $\therefore \quad (1, -1) \xrightarrow{T = (-2, 1)} (1 - 2, -1 + 1) = (-1, 0)$

   Step - 3 Find slope of L: $3x - 4y = 7$, thus slope of L $= \frac{3}{4}$

   Since translation is parallel movement, then the slope of the image of L is L' $\frac{3}{4}$

   Step - 4: we find the equation of L' passing through (-1, 0) in step 2 and step3, with slop $\frac{3}{4}$

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$\therefore$ The image of L is L': $\frac{y-0}{x+1} = \frac{3}{4}$

$L' : 3x - 4y = -3$

Method II We have translation Vector T = (-2, 1)

The line horizontally moves 2 unit left, in this case we replace x by $x + 2$. And the line moves vertically 1 unit up, in this case we replace y by $y - 1$

Thus $L' : 3(x + 2) - 4(y - 1) = 7$

$L' : 3x + 6 - 4y + 4 = 7$

$L' : 3x - 4y = 7 - 10 = -3$

b. We replace x by $x + 2$ and y by $y - 1$

Thus, $L' : 2(x + 2) + 3(y - 1) = 1$

$L' : 2x + 4 + 3y - 3 = 1$

$\therefore$ $L' : 2x + 3y = 0$

6. If a translation T takes (1, -4) to (-2, 3), then find the image of the line $\ell: 4x + 2y + 3 = 0$

Solution: Translation vector: T = (-2, 3) - (1, -4) = (-3, 7) The line ℓ moves 3 units left and 7 units up so that x replaced by $x + 3$ and y replace by $y - 7$

$\ell' : 4(x + 3) + 2(y - 7) + 3 = 0$

$\ell' : 4x + 2y + 12 - 14 + 3 = 0$

$\therefore$ $\ell : 4x - 2y + 1 = 0$

7. If the image of the line $2x - 3y = 7$ under a translation is $2x - 3y = 0$, which one of the following is a translation vector of the translation line?

A. $\vec{u} = (2, -1)$ C. $\vec{u} = (-2, 1)$

B. $\vec{u} = (-1, 2)$ D. $\vec{u} = (1, -2)$

Solution:- Take any point on L: $2x - 3y = 7$: Let $x = 2, y = -1$

(2, -1) on L: $2x - 3y - 7$, Since the image of L is

L : $2x - 3y = 0$, and (0, 0) on L'

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Thus, $(2, -1) \xrightarrow{T = (a, b)} (0, 0)$

$T(a, b) = (0, 0) - (2, -1) = (-2, 1)$

Similarly, let $x = 5$, and $y = 1$ on L: $2x - 3y = 7$

$(5, 1) \xrightarrow{T(-2, 1)} (3, 2)$ on L' $2x - 3y = 0$

Check: Replace x by $x + 2$ and y by $y - 1$, on L : $2x - 3y = 7$

image: L' : $2(x + 2) - 3(y - 1) = 7$

L' : $2x - 3y + 4 + 3 = 7$

$\therefore$ L' : $2x - 3y = 0$

$\therefore$ **Answer: C**

8. In a translation T the image of P(1, -1) is P'(3, 2). If a line L, its parametric equation is given by $(x, y) = (2, -1) + t(1, 1)$ for any $t \in R$ is translated by T. What is the image of L?

A. $y = x + 3$ B. $2x + y = 1$ C. $x - y = 2$ D. $3y - x = 3$

Solution

Translation vector T : $\overrightarrow{PP'} = (3, 2) - (1, -1) = (2, 3)$

L : $x = 2 + t$ and $y = -1 + t$

L : $x - 2 = y + 1$ implies L : $x - y = 3$

The image of L is L' : $(x - 2) - (y - 3) = 3$

L' : $x - y = 2$

Answer: C

6.2.2 Translating a Circle

A translation carries circles into congruent circles. Therefore in order to find the image of a circle, we need to translate a centre of a circle by translation vector.

Illustrative Example

9. A translation T takes the point (2, 1) to (-1, 3), then find the equation of the image of the circle.

a. $x^2 + y^2 = 9$ b. $x^2 + y^2 + 8x - 6y + 24 = 0$

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Solution

The translation vector: $T = (-1, 3) - (2, 1) = (-3, 2)$

so the circle moves 3 units left and 2 units up.

a. In this case, we replace x by $x + 3$ and replace y by $y - 2$

$\therefore \quad (x + 3)^2 + (y - 2)^2 = 9$

b. $x^2 + y^2 + 8x - 6y + 24 = 0$

$x^2 + 8x + y^2 - 6y + 24 = 0$ • completing square

$$x^2 + 8x + \left(\frac{8}{2}\right)^2 + y^2 - 6y + \left(\frac{6}{2}\right)^2 = -24 + 16 + 9$$

$(x + 4)^2 + (y - 3)^2 = 1$, center $(-4, 3)$

Now we translate the center:

$(-4, 3) \xrightarrow{T(-3, 2)} (-4 + (-3), 3 + 2) = (-7, 5)$

$\therefore$ The image of a circle is $C' : (x + 7)^2 + (y - 5)^2 = 1$

10. If a translation T takes the point (7, 2) to the point (5, 3), then find the image of the circle whose equation is $(x - 2)^2 + y^2 = 4$

Solution

Translation vector: $T = (5, 3) - (7, 2) = (-2, 1)$ center of a circle is $(2, 0)$

$(2, 0) \xrightarrow{T(-2, 1)} (-2 + 2, 0 + 1) = (0, 1)$

$\therefore$ The image of a circle is $x^2 + (y - 1)^2 = 4$

11. If a translation T takes the circle $x^2 + y^2 - 2x + 6y + 3 = 0$, into the circle whose equation is $(x + 2)^2 + (y - 4)^2 = 7$ then what is the image of the origin under T ?

A. $(-3, 7)$ B. $(3, -7)$ C. $(1, 2)$ D. $(-2, 4)$

Solution

$C_1 : x^2 - 2x + y^2 + 6y + 3 = 0$

$$x^2 - 2x + \left(\frac{-2}{2}\right)^2 + y^2 + 6y + \left(\frac{6}{2}\right)^2 = -3 + 1 + 9$$

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$(x - 1)^2 + (y + 3)^2 = 7$, center $(1, -3)$

C_2: $(x + 2)^2 + (y - 4)^2 = 7$, center $(-2, 4)$

Translation vector: $T = (-2, 4) - (1, -3) = (-3, 7)$

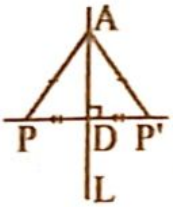
Answer: A

6.3 What is a Reflection

A reflection is a rigid transformation that creates symmetry on the coordinate plane.

A reflection maps a point in the plane to its mirror image, using a specific line as the mirror.

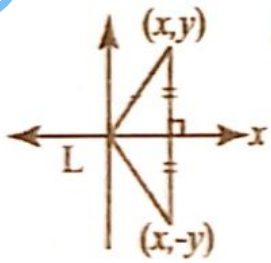
DEFINITION: Let L be a fixed line in a plane.



If P is a point not on a line L, then its image in L is the point P' denoted by M (P), if L is the perpendicular bisector of $\overline{PP'}$

- The line L is called the **line of reflection** or the **axis of reflection**
- If P is on line L, then its image P' is on L is P itself
- $\overline{PP'} \perp L$, where L is a line of reflation
- $\overline{PD} \cong \overline{P'D}$

6.3.1 Reflection in the axes, line y = x and y = -x



i) Reflection over the x - axis (y = 0)

(Angle of line of reflection is $\theta = 0^0$)

To reflect a point over the x - axis, multiply the y - coordinate by - 1.

Symbols: Reflection over x - axis: $(x, y) \longrightarrow (x, -y)$

Example: $P(2, 3) \longrightarrow P'(2, -3)$

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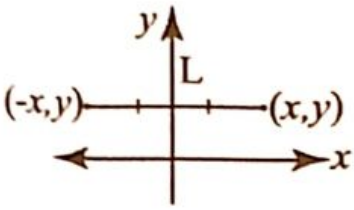
$P(-3, 4) \longrightarrow P'(-3, -4)$

ii) Reflection over the y - axis (x = 0)



$\left(\text{Angle of line of reflection is } \theta = \frac{\pi}{2}\right)$

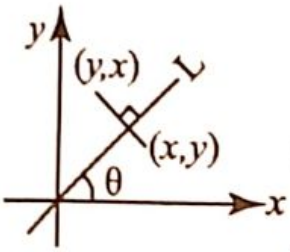
To reflect a point over the y - axis, multiply the x - coordinate by - 1

Symbols: reflection over y - axis $(x = 0)$: $(x, y) \longrightarrow (x, -y)$

$(2, 3) \longrightarrow (-2, 3)$

$(-3, 4) \longrightarrow (3, 4)$

iii) Reflection in the line y = x



Angle of line or reflection is $\theta = \frac{\pi}{4}$

To reflect a point (x, y) in the line of reflection $y = x$, switch the roles of x and y, (interchange x and y).

Symbols: Reflection in the line $y = x$:

$(x, y) \longrightarrow (y, x)$

$(a, b) \longrightarrow (b, a)$

$(2, -3) \longrightarrow (-3, 4)$

iv. Reflection in the line y = -x

- The angle between line of a reflection and x axis is $\theta = \frac{3\pi}{4}$

To reflect a point (x, y) in the line of reflection $y = -x$, interchange the x and y coordinate then multiply each by (-1).

Symbols : Reflection in the line $y = -x$

$(x, y) \longrightarrow (-y, -x)$

$(a, b) \longrightarrow (-b, -a)$

Example: $(2, 3) \xrightarrow{y = -x} (-3, -2)$

$(1, -5) \longrightarrow (5, -1)$

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V. Reflection in the line x = k, where k ∈ ℝ

To reflect a point (x, y) in the line of reflection $x = k$, $k \in \mathbb{R}$ multiply k by 2, then subtract x.

Symbols : Reflection in the line $x = K$, $k \in \mathbb{R}$

$$(x, y) \xrightarrow{x = k} (2k - x, y)$$

$$(a, b) \xrightarrow{x = k} (2k - a, b)$$

As example: $(3, 1) \xrightarrow{x = 0} (2(0) - 3, 1) = (-3, 1)$

$$(3, 1) \xrightarrow{x = 4} (2(4) - 3, 1) = (5, 1)$$

V. Reflection in the line y = t, where t ∈ ℝ

$$(x, y) \xrightarrow{y = t} (x, 2t - y)$$

$$(a, b) \xrightarrow{y = t} (a, 2t - b)$$

Example $(3, 1) \xrightarrow{y = 0} (3, 0 - 1) = (3, -1)$.

$$(3, 1) \xrightarrow{y = 2} (3, 2(2) - 1) = (3, 3)$$

$$(3, 1) \xrightarrow{y = 5} (3, 2(5) - 1) = (3, 9)$$

Illustrative Example

12. Find the image of a point P(3, 5) over the following line of reflection:

a. x - axis
b. y - axis
c. $y = x$
d. $y = -x$
e. $x = -1$
f. $y = -1$

Solution

a. $P(3, 5) \xrightarrow{x\text{-axis}} P'(3, -5)$

b. $P(3, 5) \xrightarrow{y\text{-axis}} P'(-3, 5)$

c. $P(3, 5) \xrightarrow{y = x} P'(5, 3)$

d. $P(3, 5) \xrightarrow{x = -x} P'(-5, -3)$

e. $P(3, 5) \xrightarrow{x = -1} P'(2(-1) - 3, 5) = P'(-5, 5)$

f. $P(3, 5) \xrightarrow{y = -1} P'(3, 2(-1) - 5) = P'(3, -7)$

13. Find the coordinates of A (2, 3), B (1, -4) and C (3, 6) reflected over:

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a. x - axis b. y - axis c. $y = x$ d. $y = -x$

Solution

The image of a coordinate point reflected over:

a. x - axis is A' (2, -3), B' (1, 4) and C' (3, -6)

b. y - axis is A' (-2, 3), (-1, -4), C' (-3, 6)

c. the line $y = x$ is A' (3, 2), B' (-4, 1) and C' (6, 3)

d. the line $y = -x$ is A' (-3, -2), B' (4, -1) and C' (-6, -3)

14. A triangle ABC with A (-3, 2), B (2, 5) and C (6, 3) reflected about the line $x = 3$. Find the image of coordinates of the vertices of a triangle.

Solution

Reflection in the line $x = k$: $(x, y) \xrightarrow{x = k} (2k - x, y)$

A. $(-3, 2) \xrightarrow{x = 3} A'(2(3) - (-3), 2) = A'(9, 2)$

B. $(2, 5) \xrightarrow{x = 3} B'(2(3) - 2, 5) = B'(4, 5)$

C. $(6, 3) \xrightarrow{x = 3} C'(2(3) - 6, 3) = C'(0, 3)$

15. Find the image of a line $L : y = 3x + 4$ over the following line of refection:

a. x - axis c. $y = x$ e. $x = 4$

b. y - axis d. $y = -x$ f. $y = 4$

Solution

a. Reflection over x -axis: $(x, y) \longrightarrow (x, -y)$

$L: y = 3x + 4$, then

$L' : -y = 3x + 4$

$\therefore$ The image of $L : y = 3x + 4$ is $L' : y = -3x + 4$

b. Reflection over y - axis: $(x, y) \longrightarrow (-x, y)$

$\therefore$ The image of $L' : y = 3x - 4$ is $L: y = -3x + 4$.

c. Reflection over the line $y = x$:

$L : y = 3x + 4$ *Interchange x to y and y to x*

$L' : x = 3y + 4 \longleftarrow$ switch x and y.

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$L' : y = \frac{1}{3}x - \frac{4}{3}$ ⟵ solving for y.

$\therefore$ The image of L: $y = 3x + 4$ is $L' : y = \frac{1}{3}x - \frac{4}{3}$

d. Reflection over the line $y = -x$:

$(x, y) \longrightarrow (-y, -x)$. Replace x by $-y$ and y by $-x$

$\therefore$ The image of L: $y = 3x + 4$ is $L' : -x = -3y + 4$

$\therefore$ $L' : y = \frac{1}{3}x + \frac{4}{3}$ ⟵ solving for y

e. Reflection over the line $x = 4$

$(x, y) \xrightarrow{x = 4} (2(4) - x, y) = (8 - x, y)$

Replace x by $8 - x$

$\therefore$ The image of $L : y = 3x + 4$ is $L' : y = 3(8 - x) + 4$

$\therefore$ $L' : y = -3x + 28$

f. Reflection over the line $y = 4$

$(x, y) \xrightarrow{y = 4} (x, 2(4) - y) = (x, 8 - y)$ replace y by $8 - y$:

$L' : 8 - y = 3x + 4$ • solving for y

$L' : -y = 3x - 4$

$\therefore$ $L' : y = -3x + 4$

16. Describe the transformation of parallelogram ABCD with A (-5, 3), B (-2, 5), C (0, 3), and D (-3, 1) if the coordinates of its image are A' (5, 3), B' (2, 5), C' (0, 3) and D' (3, 1)

Solution

This transformation is reflection over y - axis ($x = 0$)

17. What is the image of equation of circle $(x - 2)^2 + (y + 1)^2 = 9$ reflected:

a. Over x - axis
b. Over y - axis
c. In the line $y = x$
d. In the line $y = -x$

Solution:- The center of a circle is (2, -1)

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a. $(2, -1) \xrightarrow{x\text{-axis}} (2, 1)$, image: $(x - 2)^2 + (y - 1)^2 = 9$

b. $(2, -1) \xrightarrow{y\text{-axis}} (-2, -1)$, image: $(x + 2)^2 + (y + 1)^2 = 9$

c. $(2, -1) \xrightarrow{y = x} (-1, 2)$, image: $(x + 1)^2 + (y - 2)^2 = 9$

d. $(2, -1) \xrightarrow{y = x} (1, -2)$, image: $(x - 1)^2 + (y + 2)^2 = 9$

18. Describe the line of reflection of a point A (x, y), if its image is:

a. A' $(-x, y)$ c. A' (y, x) e. $(2k - x, y), k \in R$

b. A' $(-y, -x)$ d. $(x, -y)$ f. $(x, 2k - y), k \in R$

Solution

a. The line of reflection on is: $x = 0$, with angle of inclination $\theta = \frac{\pi}{2}$

b. The line of reflection is: $y = -x$, with angle of inclination $\theta = \frac{3\pi}{4}$

c. The line of reflection is: $y = x$, with $\theta = \frac{\pi}{4}$

d. The line of reflection is: $y = 0$, with $\theta = 0$

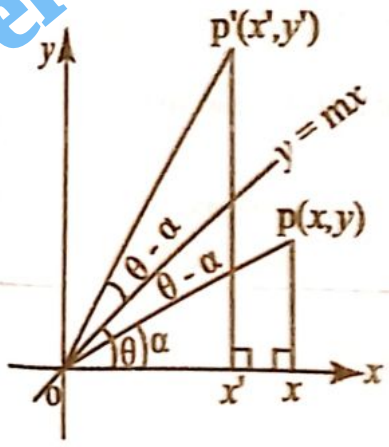
e. The line of reflection is: $x = k$

f. The line of reflection is: $y = k$

6.3.2 Reflection of a Point in the Line y = mx, Where m = tanθ

Let L be a line passing through origin and making an angle θ with the positive x - axis. To reflect a point P(x, y) in the line of reflection

L: $y = mx$, consider the following figure.



Let $\angle POx = \alpha$ and $\overline{OP} = r$

Since the image of P(x, y) is P' (x', y'), then OP' = OP = r, because reflection maps line into congruent line.

Then the coordinate of P(x, y) are:

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$$\cos\alpha = \frac{x}{r}, \; x = r\cos\alpha$$

$$\sin\alpha = \frac{y}{r}, \; y = r\sin\alpha$$

- The angle measured from positive x - axis to $\overline{OP'}$ is $\theta + (\theta - \alpha) = 2\theta - \alpha$
- The coordinate of $P'(x', y')$ are:

$$\cos(2\theta - \alpha) = \frac{x'}{r}, \; x' = r\cos(2\theta - \alpha)$$

$$\sin(2\theta - \alpha) = \frac{y'}{r}, \; y' = r\sin(2\theta - \alpha)$$

Note: $\cos(A + B) = \cos A \cos B - \sin A \sin B$ $\sin(A - B) = \sin A \cos B - \cos A \sin B$ Using these identity:

1. $x' = r\cos(2\theta - \alpha) = r(\cos 2\theta \cos\alpha + \sin 2\theta \sin\alpha)$

   $x' = (r\cos\alpha)\cos 2\theta + r\sin\alpha \sin 2\theta$

   $\therefore \quad \boxed{x' = x\cos 2\theta + y\sin 2\theta}$

2. $y' = r\sin(2\theta - \alpha) = r(\sin 2\theta \cos\alpha - \cos 2\theta \sin\alpha)$

   $y' = (r\cos\alpha)\sin 2\theta - r\sin\alpha \cos 2\theta$

   $\boxed{y' = x\sin 2\theta - y\cos 2\theta}$

Theorem: The image of a point $P(x, y)$ in the line of reflection $L: y = mx$, is given by $P'(x, y)$ such that

$x' = x\cos 2\theta + y\sin 2\theta$ and $y' = x\sin 2\theta - y\cos 2\theta$

where angle θ is the angle of inclination of the line $y = mx$

19. Find the image of the point (4, 2) when reflected about the line

a. $y = \sqrt{3}x$ b. $y = \frac{-1}{\sqrt{3}}x$ c. $y = 3x$

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Solution

a. Because $\tan\theta = \sqrt{3}$, implies $\theta = \frac{\pi}{3}$.

The image of P(4, 2) is P'(x', y'):

$\therefore \quad x' = x\cos 2\theta + y\sin 2\theta$

$= 4\cos\frac{2\pi}{3} + 2\sin\frac{2\pi}{3} = 4\left(\frac{-1}{2}\right) + 2\frac{\sqrt{3}}{2} = -2 + \sqrt{3}$

$y' = x\sin 2\theta - y\cos 2\theta = 4\sin\frac{2\pi}{3} - 2\cos\frac{2\pi}{3}$

$\therefore \quad y' = 4\left(\frac{\sqrt{3}}{2}\right) - 2\left(\frac{-1}{2}\right) = 2\sqrt{3} + 1$

$\therefore \quad P(4, 2) \xrightarrow[\theta = \frac{\pi}{3}]{y = \sqrt{3}x} P\left(-2 + \sqrt{3}, 2\sqrt{3} + 1\right)$

b. Because $m = \tan\theta = \frac{-1}{\sqrt{3},}$ implies $\theta = \frac{-\pi}{6}$

The image of P(4, 2) is P'(x', y'):

$x' = x\cos 2\theta + y\sin 2\theta = 4\cos\left(\frac{-2\pi}{6}\right) + 2\sin\left(\frac{-2\pi}{6}\right)$

$\therefore \quad x' = 4\left(\frac{1}{2}\right) + 2\left(\frac{-\sqrt{3}}{2}\right) = 2 - \sqrt{3}$

$y' = x\sin 2\theta - y\cos 2\theta = 4\sin\left(\frac{-2\pi}{6}\right) + 2\cos\left(\frac{-2\pi}{6}\right)$

$y' = 4\left(\frac{-\sqrt{3}}{2}\right) + 2\left(\frac{1}{2}\right) = -2\sqrt{3} + 1$

$\therefore \quad P(4, 2) \xrightarrow[\theta = \frac{-\pi}{6}]{y = \frac{-1}{\sqrt{3}}x} P'\left(2 - \sqrt{3}, 1 - 2\sqrt{3}\right)$

c. Since $m = \tan\theta = 3$, $\sin\theta = \frac{3}{\sqrt{10}}$ = and $x' = x\cos 2\theta + y\sin 2\theta$

$= x(\cos^2\theta - \sin^2\theta) + y\,2\sin\theta\cos\theta$

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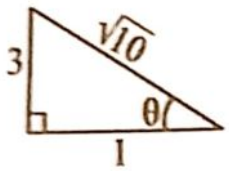
$$= 4\left(\frac{1}{10} - \frac{9}{10}\right) + 2\,(2)\left(\frac{3}{\sqrt{10}} \cdot \frac{1}{\sqrt{10}}\right)$$



$$\therefore \quad x' = \frac{-32}{10} + \frac{12}{10} = \frac{-20}{10} = -2$$

Also, $y' = x\sin 2\theta - y\cos 2\theta$

$$= x\,(2\sin\theta\cos\theta) - y\,(\cos^2\theta - \sin^2\theta)$$

$$= 4 \cdot 2 \cdot \frac{3}{\sqrt{10}} \cdot \frac{1}{\sqrt{10}} - 2\left(\frac{1}{10} - \frac{9}{10}\right)$$

$$y' = \frac{24}{10} + \frac{16}{10} = \frac{40}{10} = 4$$

$\therefore$ The image of P(4, 2) is P'(-2, 4)

20. What is the image of the point (2, 2) reflected in the line passing through the origin and making 30^0 with t x - axis?

A. (2, -2) C. $\left(\sqrt{3} - 1, \sqrt{3} + 1\right)$

B. $(1 + \sqrt{3}, \sqrt{3} - 1)$ D. $\left(\sqrt{3}, -\sqrt{3}\right)$

Solution

$$x' = x\cos 2\theta + y\sin 2\theta = 2\cos 60^0 + 2\sin 60^0$$

$$= 2\left(\frac{1}{2}\right) + 2\left(\frac{\sqrt{3}}{2}\right) = 1 + \sqrt{3}$$

$$y' = x\sin 2\theta - y\cos 2\theta = 2\sin 60^0 - 2\cos 60^0$$

$$= 2\left(\frac{\sqrt{3}}{2}\right) - 2\left(\frac{1}{2}\right) = \sqrt{3} - 1$$

Answer: B

21. Reflection in which line takes to figure with vertices A(0, 0), B(-2, 4) C(-4, 2) and D (-3, 0) to vertices A'(0, 0) B'(-4, 2) C'(-2, 4) and D' (0, 3)?

A. x - axis B. y - axis C. $y = x$ D. $y = -x$

Solution

It is reflection in the line $y = -x$: $(x, y) \xrightarrow{y=-x} (-y, -x)$

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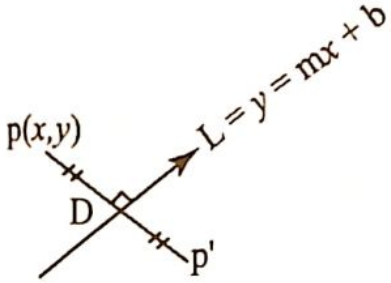
Answer: D

22. Find the image of the point P(2, 6) after it has been reflected a long a line passing through the origin and making an angle of 15^0 with the x - axis.

Solution: Exercise left for you.

6.3.3 Reflection of a Point in the Line L: $y = mx + b$, $m \neq 0$

To find the image of a point P(x, y) when reflected in the line L : $y = mx + b$:



- $\overline{PD} \cong \overline{P'D}$
- D is mid - point

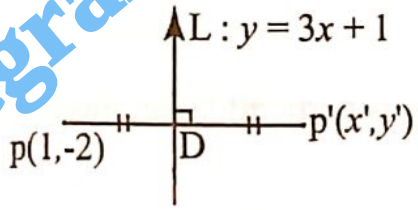
1. Find the slope of the line L, say m.
2. Find the slope of the line segment $\overline{PP'}$ which is $\frac{-1}{m}$
3. Find the equation of the line $\overline{PP'}$, passing through P(x, y) with slope $\frac{-1}{m}$.
4. Find the intersection of line L: $y = mx + b$ and the line $\overline{PP'}$.
5. Using the mid - point D, find the coordinate of P'

Illustrative Example

23. Find the image of the point P(1, -2) after reflection in the line L : $y = 3x + 1$

Solution

Slope of L : $y = 3x + 1$ is 3.



- Because L is perpendicular to $\overline{PP'}$ so that the of $\overline{PP'}$ is $\frac{-1}{3}$
- Equation of $\overline{PP'}$ passing through P (1, -2) with slope $\frac{-1}{3}$ is:

$$\frac{y + 2}{x - 1} = \frac{-1}{3}, \text{ this implies } 3y + x = -5$$

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The intersection of $L: y = 3x + 1$ and $\overline{PP'}: 3y + x = -5$ is the mid-point D:

$$\begin{cases} y = 3x + 1 \\ 3y + x = -5 \end{cases} \text{ implies that } 3(3x + 1) + x = -5$$

$9x + 3 + x = -5$, implies that $10x = -8$, $x = \frac{-4}{5}$

To find y: $y = 3\left(\frac{-4}{5}\right) + 1 = \frac{-12}{5} + \frac{5}{5} = \frac{-7}{5}$

Since $D\left(\frac{-4}{5}, \frac{-7}{5}\right)$ is mid - point of $\overline{PP'}$

$$\Rightarrow \frac{1 + x'}{2} = \frac{-4}{5} \text{ and } \frac{-2 + y'}{2} = \frac{-7}{5}$$

$$\therefore \quad x' = \frac{-13}{5} \text{ and } y' = \frac{-4}{5}$$

$\therefore$ The image of the point $P(1, -2)$ is $P'\left(\frac{-13}{5}, \frac{-4}{5}\right)$.

24. The image of the point $P(-2, 3)$ in reflection is $P'(6, 7)$. Find the equation of line of reflection.

Solution

Mid-Point of $\overline{PP'}$ is $\left(\frac{-2 + 6}{2}, \frac{3 + 7}{2}\right) = (2, 5)$.

Slope of PP' is $\frac{3 - 7}{-2 - 6} = \frac{-4}{-8} = \frac{1}{2}$

- Since the line of reflection and PP' is perpendicular, then slope of line reflection is -2.
- Because the line of reflection passes through the mid - point of PP' : $(2, 5)$

$\therefore$ Equation of line of reflection is $\frac{y - 5}{x - 2} = -2$, this implied $y = -2x + 9$

25. The image of the point $P(-4, 3)$ in reflection is $P'(-3, 4)$

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Find the equation of line of reflection.

Solution:- $(x, y) \rightarrow (-y, -x)$

This is a reflection of a point (-4, 3) to (-3, 4) over the line of reflection $y = -x$,

Reflecting a circle in a line

A reflection carries circle into congruent circle.

To reflect a circle in a line we reflect the center of a circle with the same radius .

26. Find the image of a circle $x^2 + y^2 - 4x + 3y + 9 = 0$ when it is reflected in the line $y = 2x + 3$.

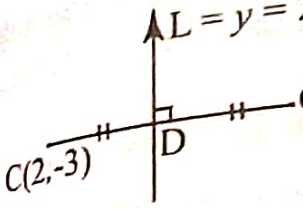
Solution

By using complying the square method

$$x^2 - 4x + \left(\frac{-4}{2}\right)^2 + y^2 + 6y + \left(\frac{6}{2}\right)^2 = -9 + 4 + 9$$

$$(x - 2)^2 + (y + 3)^2 = 4$$

Center of a circle is C(2, -3)



- Slope of $\overline{CC'}$ is $\frac{-1}{2}$
- Equation of $\overline{CC'}$ passing through C(2, -3) is $\frac{y + 3}{x - 2} = \frac{-1}{2}$ implies $2y + x = -4$

- The intersection of L and line segment $\overline{CC'}$ is

$$\begin{cases} 2y + x = -4 \\ y = 2x + 3 \end{cases}$$ solving together

$2(2x + 3) + x = -4$, implies $x = -2$. And for y : $y = 2(-2) + 3 = -1$

The intersection of line L and $\overline{CC'}$ is the mid - point D (-2, -1).

In this case:

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$\frac{x' + 2}{2} = -2$ and $\frac{y' + (-3)}{2} = -1$

$\therefore \quad x' = -6$ and $y' = -2 + 3 = 1$

$\therefore$ The image of center of a circle C (2, -3) is C (-6, 1).

$\therefore$ The image of a circle is $(x + 6)^2 + (y - 1)^2 = 4$

27. What is the image of the circle $x^2 + y^2 - 4x - 6y + 12 = 0$ when it reflected with respect to the line $y = -x$?

A. $(x + 3)^2 + (y + 2)^2 = 1$

B. $(x - 2)^2 + ((y - 3)^2 = 1$

C. $(x + 2)^2 + (y + 3)^2 = 1$

D. $(x - 3)^2 + (y - 2)^2 = 1$

Solution

$x^2 - 4x + y^2 - 6y = -12$

$x^2 - 4x + 4 + y^2 - 6y + 9 = -12 + 4 + 9$

$(x - 2)^2 + (y - 3)^2 = 1$, center (2, 3).

$(2, 3) \xrightarrow{y = -x} (-3, -2)$

$(x + 3)^2 + (y + 2)^2 = 1$

Answer: A

28. The image of the circle $x^2 + y^2 + 2x - 6y = 0$. When it is reflected about the line L is $x^2 + y^2 + 6x + 2y = 0$. Find the equation of the line of reflection.

Solution

A reflection carries circle into congruent circle:

- Center of $x^2 + y^2 + 2x - 6y = 0$ is (-1, 3)
- Center of $x^2 + y^2 + 6x + 2y = 0$ is (-3, -1)
  - Mid - point between the two center of a circle:

$$\left(\frac{-1 + (-3)}{2}, \frac{3 + (-1)}{2}\right) = (-2, 1)$$

  - Solpe of CC' is $\frac{3 - (-1)}{-1 - (-3)} = \frac{4}{2} = 2$

- Slope of equation of the line of reflection is $-\frac{1}{2}$
- The equation of the line of reflection passing through (-2, 1) is

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$L: \frac{y-1}{x+2} = \frac{-1}{2}$, implies $y = \frac{-1}{2}x$

6.3.4 Reflection of a Line in the Line

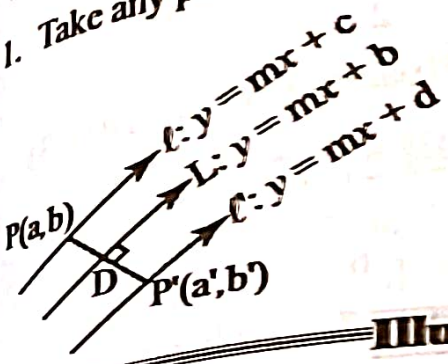
To find the image of the line ℓ after it has been reflected along the line L.

We have two cases:

Case I: If the line of reflection L is parallel to ℓ

To find the image of ℓ under line of reflection L:

1. Take any point P(a, b) on $\ell: y = mx + c$



2. Find the image of point P (a, b) on line of reflection $L: y = mx + b$. (In this case reflection of a point in a line L)
3. Find the equation of ℓ' thought the point P' (a', b')

Illustrative Example

29. Find the image of the line $\ell: y = -x + 5$ after reflection in the line $L: y = -x$

Solution

Take point (3, 2) on $\ell: y = -x + 5$

- The image of P(3, 2) is P ' (-2, -3)
- Slope of ℓ ' is -1, because $L \parallel \ell$
- The equation of the image line ℓ ', passing through P ' (-2, -3) is

$\frac{y+3}{x+2} = -1$, this implies $y = -x - 5$

∴ The image of the line $\ell: y = -x + 5$ after reflection along the line $L: y = -x$ is $\ell': y = -x - 5$.

30. Find the image of the line $\ell: y = x + 3$ after reflection in the line $L: y = x$.

Solution: Take point (1, 4) on $\ell: y = x + 3$.

The image of P(1, 4) is P '(4, 1) Since $\ell \parallel L$, then slope of ℓ ' is 1.

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Equation of the image line ℓ passing through (4, 1).

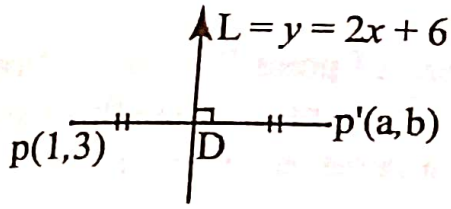
$\frac{y-1}{x-4} = 1$, this implies $\ell' : y = x - 3$

31. Find the image of the line $l : y = 2x + 1$ after reflection along the line $L : y = 2x + 6$

Solution

Take a point P (1, 3) on $\ell : y = 2x + 1$

To find the image of P (1, 3) on $L : y = 2x + 6$:



- Slope of $\overline{PP'}$ is $\frac{-1}{2}$
- Equation of $\overline{PP'}$ passing throught P (1,3):

$$\frac{y-3}{x-1} = \frac{1}{-2} \Rightarrow 2y + x = 7$$

- Find the intersection of L and $\overline{PP'}$: $\begin{cases} y = 2x + 6 \\ 2y + x = 7, \end{cases}$ solving together

$2(2x + 6) + x = 7$, this implies $5x = 7 - 12$, $x = -1$ and $y = 2(-1) + 6 = 4$.

- The intersection of L and $\overline{PP'}$ is the mid - point D (-1, 4).

$$\therefore \quad \frac{a+1}{2} = -1 \text{ and } \frac{b+3}{2} = 4$$

$a = -3$ and $= 5$

$\therefore$ The image of P(1, 3) is P ' (-3, 5). It follows that P (-3, 5) lies on ℓ. since $\ell \parallel L$ implies $\ell \parallel \ell'$, then slope of ℓ' is 2.

Equation of the image line ℓ' passing through P (-3, 5) with slope 2 is:

$$\frac{y-5}{x+3} = 2 \text{ implies } y = 2x + 11$$

$\therefore$ The image of the $\ell : y = 2x + 1$ after reflecting along the line

$L : y = 2x + 6$ is $\ell' : y = 2x + 11$.

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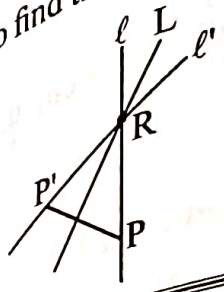
In general, If the line of reflection L is parallel to ℓ :

The image of the line $\ell : y = mx + c$ after reflecting about the line $L : y = mx + b$ is $\ell' : y = mx + 2b - c$

In case II: If the line of reflection L is not parallel to ℓ (intersecting at some point R).

To find the image of ℓ under line of reflection L:



1. Choose any point P on ℓ
2. Find the image of P on L is P '
3. Find the intersection of ℓ and L as R.
4. Find the equation of ℓ' passing through P ' and R

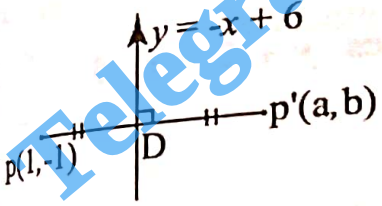
Illustrative Example

32. Find the image of the line $\ell : y = 2x - 3$ after it has been reflected about the line L: $y = -x + 6$

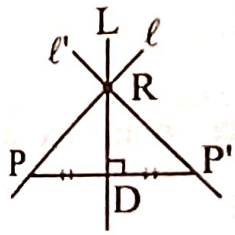
Solution

To find the image of $\ell : y = 2x - 3$ under the line of reflection $L : y = -x + 6$:

1. Choose any point P, on $\ell : y = 2x - 3$ let $x = 1, y = -1$, say P (1, -1)
2. Find the image of P (1, -1) on line of reflection $L : y = -x + 6$



- Slope of $\overline{PP'}$ is 1.
- Equation of the line $\overline{PP'}$ passing through P is $\frac{y + 1}{x - 1} = 1$ implies $y = x - 2$
- The intersection of $y = -x + 6$ and $y = x - 2$ is D (4, 2), mid - point of $\overline{PP'}$



$\frac{a + 1}{2} = 4$, $a = 7$ and $\frac{b + (-1)}{2} = 2$, $b = 5$

$\therefore$ The image of P (1, -1) is P ' (7, 5)

3. Find the intersection of $\ell : y = 2x - 3$ and $L : y = -x + 6$

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$$\begin{cases} y = 2x - 3 \\ y = -x + 6, \end{cases} \Rightarrow 2x - 3 = -x + 6,\ x = 3,\ y = 3$$

$\therefore$ The point R (3, 3)belongs to both line ℓ and L.

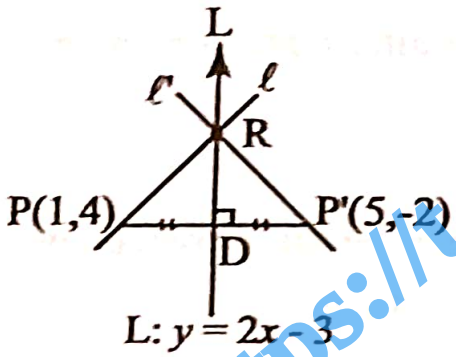
4. The equation of the image line (ℓ') passed through P'(7,5) and R(3,3) is:

$\frac{y - 3}{x - 3} = \frac{3 - 5}{3 - 7}$ implies that $\ell' : y = \frac{1}{2}x + \frac{3}{2}$

33. Find the image of the line $\ell : y = 3x + 1$ when it is reflected along the line L: $y = 2x - 3$

Solution

Choose P(1, 4) is a point on $\ell : y = 3x + 1$ image of P (1, 4) in L: $y = 2x - 3$ is P' (5, -2) on ℓ'



The intersection of L and ℓ

$$\begin{cases} y = 3x + 1 \\ y = 2x - 3 \end{cases} \Rightarrow x = -4,\ y = -11$$

It follows that R (-4, -11), is the point of intersection.

Equation of image line is ℓ' :

$\frac{y + 2}{x - 5} = \frac{-2 - (-11)}{5 - (-4)} = \frac{9}{9} = 1$ this implies $y + 2 = x - 5$

$\therefore \quad \ell' : y = x - 7$

6.4 Rotation

What is a rotation ?

A rotation is a rigid transformation that turns a figure about a fixed point called the center of rotation

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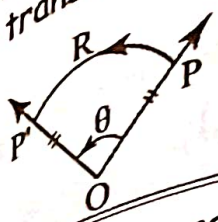
DEFINITION : A rotation R about a point O through an angle θ is a transformation of the plane onto itself which carries every point P of plane into the point P' of the plan such that OP = OP ' and m (∠ POP ') = θ, O is called the center of **rotation** and θ is called **angle of rotation.**



Note: If R is a rotation, through angle θ, denoted by R_θ (x, y).

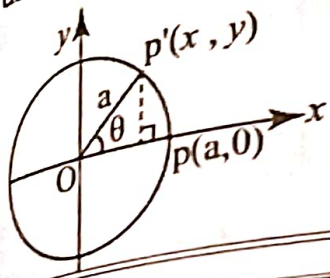
- The rotation is in the counter clock wise, if the angle θ is **positive** and in the clock wise direction if the angle θ is **negative.**

6.4.1 Rotation of a Point P (a, 0) about Origin

The image of a point P(a, 0) when rotated through angle θ about the origin is given by P' (x, y) such that:

$$x = a \cos \theta$$

$$y = a \sin \theta$$



In this case: a = r, so that $x = r\cos\theta$ and $y = r \sin \theta$

Illustrative Example

34. Find the image of point P (4, 0) when it is rotated through the following angle θ about the origin.

a. $\theta = 60^0$
b. $\theta = 150^0$
c. $\theta = 135^0$
d. $\theta = -150^0$
e. $\theta = -60^0$
f. $\theta = -135^0$

Solution

Let the image of point P (a, 0) be P (x, y), then $x = a \cos \theta$ and $y = a \sin \theta$

a. $P(x, y) = P'(a \cos \theta, a \sin \theta) = P'(4 \cos 60^0, 4 \sin 60^0)$

$$= P'\left(4\left(\frac{1}{2}\right), 4\left(\frac{\sqrt{3}}{2}\right)\right) = P'\left(2, 2\sqrt{3}\right)$$

b. $\theta = 150^0$: $P'(x, y) = P'(4 \cos 150^0, 4 \sin 150^0)$

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$$= P'\left(4\left(\frac{-\sqrt{3}}{2}\right), 4\left(\frac{1}{2}\right)\right) = P'\left(-2, \sqrt{3}\right)$$

c. $\theta = 135°$: $P'(x, y) = P'(4\cos 135°, 4\sin 135°)$

$$= P'\left(4\left(\frac{-\sqrt{2}}{2}\right), 4\left(\frac{\sqrt{2}}{2}\right)\right) = P'\left(-2\sqrt{2}, 2\sqrt{2}\right)$$

d. $\theta = -150°$: $P'(x, y) = P'(4\cos(-150°), 4\sin(-150°))$
Recall that $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$
$= P(4\cos 150°, -4\sin 150°)$

$$= P'\left(4\left(\frac{-\sqrt{3}}{2}\right), -4\left(\frac{1}{2}\right)\right) = P'\left(-2\sqrt{3}, -2\right)$$

e. $\theta = -60°$: $P'(x, y) = P(4\cos(-60°), 4\sin(-60°))$

$$= P'\left(4\left(\frac{1}{2}\right), 4\left(\frac{-\sqrt{3}}{2}\right)\right) = P'\left(-2, -2\sqrt{3}\right)$$

f. $\theta = -135°$: $P'(x, y) = P'(4\cos(-135°), 4\sin(-135°))$

$$= P'\left(4\left(\frac{-\sqrt{2}}{2}\right), 4\left(\frac{-\sqrt{2}}{2}\right)\right) = P'\left(-2\sqrt{2}, -2\sqrt{2}\right)$$

Rotation Through Angle: $\frac{\pi}{2}$, π and $\frac{3\pi}{2}$ about the origin

(i) To rotate a figure by $\theta = \frac{\pi}{2}$ counter clockwise about the origin, switch the coordinates of each point and then multiply the new first coordinate by -1.

Symbols: Rotation through angle $\theta = \frac{\pi}{2}$, about (0, 0):

$(x, y) \longrightarrow (-y, x)$ counter clockwise.

$(x, y) \longrightarrow (y, -x)$ clock wise direction.

(ii) To rotate a figure through angle, $\theta = \pi$: about the origin multiply both

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coordinates of each point by - 1

Symbols: Rotation through angle $\theta = \pi$, about (0, 0):

$(x, y) \longrightarrow (-x, -y)$

(iii) Rotation through angle $\theta = \frac{3\pi}{2}$ about (0, 0):

counter clock wise

$(x, y) \longrightarrow (y, -x)$.

clock wise

$(x, y) \longrightarrow (-y, x)$.

Note: 1. Rotation through angle $\theta = 2\pi k$, $k \in R$, about the origin is an identity transformation. That is $(x, y) \longrightarrow (x, y)$

2. i. $R_{90^\circ} = R_{-270^\circ}$

ii. $R_{-90^\circ} = R_{270^\circ}$

iii. $R_{180^\circ} = R_{-180^\circ}$

vi. $R_{90^\circ} = R_{450^\circ}$

Illustrative Example

35. A triangle ABC with vertices A (-3, 2), B (2, 5) and C (4,3).

Find the image of the triangle where rotated about the origin through an angle:

a. $\theta = \frac{\pi}{2}$, counter clock wise diection

b. $\theta = \frac{\pi}{2}$, clock wise diection

c. $\theta = \pi$

d. $\theta = \frac{3\pi}{2}$, counter clock wise

e. $\theta = \frac{3\pi}{2}$, clock wise

Solution: $(x, y) \longrightarrow (-y, x)$

a. A (-3, 2) $\longrightarrow$ A' (-2, -3), B (2, 5) $\longrightarrow$ B' (-5, 2)

C (4, 3) $\longrightarrow$ C' (-3, 4)

$\therefore$ The image of the vertices are A' (-2, -3), B' (-5, 2) and (-3, 4)

b. A' (2, 3) B' (5, -2), and C' (3, -4)

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c. $(a,b) \longrightarrow (-a,-b)$, for both direction A'(3,-2), B'(-2,-5) and C'(-4,3)

d. $(a, b) \xrightarrow{\theta = \frac{3\pi}{2}} (b, -a)$. (counter clock wise)

A' (2,3), B' (5, -2) and C' (3, -4)

e. $(a, b) \xrightarrow{\theta = \frac{-3\pi}{2}} (-b, a)$. clock wise

A (-2, -3), B (-5, 2) and C (-3, 4)

36. Rotation through in which angle θ takes the triangle ABC with vertices A (2, -1), B (-3, -5), and C (4, 0) to A' (1, 2) B' (-3, -5), and C' (0, 4).

A. $\theta = \frac{-3\pi}{2}$ B. $\theta = \frac{-\pi}{2}$ C. $\theta = \frac{-3\pi}{2}$ D. $\theta = \frac{\pi}{2}$

Solution

In this case: the image of (a, b) is (-b, a) when rotated $\theta = \frac{\pi}{2}$ counter clock wise about the origin.

Answer: D

6.4.2 Rotation of a point P(x, y) about the origin

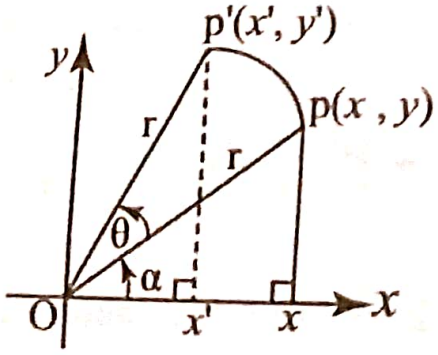
Let R be rotation through an angle θ about the origin.

If $R_\theta (x, y) = (x', y')$ then

$$x' = x \cos\theta - y \sin\theta$$

$$y' = x \sin\theta + y \cos\theta$$

To proof consider the following figure



$$\cos\alpha = \frac{x}{r}, \quad x = r\cos\alpha$$

$$\sin\alpha = \frac{y}{r}, \quad y = r\sin\alpha$$

And $\cos(\theta + \alpha) = \frac{x'}{r}$

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$$x' = r\cos(\theta + \alpha)$$
$$= r(\cos\theta\cos\alpha - \sin\theta\sin\alpha)$$
$$= r\cos\alpha\cos\theta - r\sin\alpha\sin\theta$$

$$\boxed{x' = x\cos\theta - y\sin\theta}$$

$$\sin(\theta + \alpha) = \frac{y'}{r}$$
$$= r\sin(\alpha + \theta)$$
$$y' = r\sin\alpha\cos\theta + r\sin\alpha\sin\theta$$

$$\boxed{y' = y\cos\theta + x\sin\theta}$$

Illustrative Example

37. Find the image of the following points in rotation through the indicated angle about the origin:

a. $(4, 2)$, $\theta = \frac{\pi}{3}$

b. $(2, -3)$, $\theta = \frac{-\pi}{6}$

c. $(1, 4)$, $\theta = \frac{\pi}{4}$

d. $(3, 2)$, $\theta = 930^\circ$

Solution

$x' = x\cos\theta - y\sin\theta$ and $y' = x\sin\theta + y\cos\theta$

a. $x' = 4\cos\frac{\pi}{3} - 2\sin\frac{\pi}{3} = 4\left(\frac{1}{2}\right) - 2\left(\frac{\sqrt{3}}{2}\right) = 2 - \sqrt{3}$

$y' = 4\sin\frac{\pi}{3} + 2\cos\frac{\pi}{3} = 4\left(\frac{\sqrt{3}}{2}\right) + 2\left(\frac{1}{2}\right) = 2\sqrt{3} + 1$

$\therefore \quad R_{\frac{\pi}{3}}(4, 2) = \left(2 - \sqrt{3}, 2\sqrt{3} + 1\right)$

b. $x = 2, y = -3, \theta = \frac{-\pi}{6}$

$x' = 2\cos\left(\frac{-\pi}{6}\right) + 3\sin\left(\frac{-\pi}{6}\right) = 2\left(\frac{\sqrt{3}}{2}\right) + 3\left(\frac{-1}{2}\right) = \frac{2\sqrt{3} - 3}{2}$

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$$y' = 2\sin\left(\frac{-\pi}{6}\right) - 3\cos\left(\frac{-\pi}{6}\right) = 2\left(\frac{-1}{2}\right) - 3\left(\frac{\sqrt{3}}{2}\right) = \frac{-2-3\sqrt{3}}{2}$$

$$\therefore \quad R_{\frac{-\pi}{6}}(2, -3) = \left(\frac{2\sqrt{3}-3}{2}, \frac{-2-3\sqrt{3}}{2}\right)$$

c. $x = 1,\ y = 4,\ \theta = \frac{\pi}{4}$

$$x' = \cos\frac{\pi}{4} - 4\sin\frac{\pi}{4} = \frac{\sqrt{2}}{2} - 4\left(\frac{\sqrt{2}}{2}\right) = \frac{-3\sqrt{2}}{2}$$

$$y' = \sin\frac{\pi}{4} + 4\cos\frac{\pi}{4} = \frac{\sqrt{2}}{2} + 4\left(\frac{\sqrt{2}}{2}\right) = \frac{5\sqrt{2}}{2}$$

$$\therefore \quad R_{\frac{\pi}{4}}(-1, 4) = \left(\frac{-3\sqrt{2}}{2}, \frac{5\sqrt{2}}{2}\right)$$

d. $x = 3,\ y = 2$ and $\theta = 930^0 = 2 \times 360^0 + 210^0$

$$x' = 3\cos 930^0 - 2\sin 930^0 = 3\cos 210^0 - 2\sin 210^0$$

$$= 3\left(\frac{-\sqrt{3}}{2}\right) - 2\left(\frac{-1}{2}\right) = \frac{-3\sqrt{3}+2}{2}$$

$$y' = 3\sin 210^0 + 2\cos 210^0 = 3\left(\frac{-1}{2}\right) + 2\left(\frac{-\sqrt{3}}{2}\right) = \frac{-3-2\sqrt{3}}{2}$$

$$R_{930^0}(3, 2) = R_{210^0}(3, 2) = \left(\frac{-3\sqrt{3}+2}{2}, \frac{-3-2\sqrt{3}}{2}\right).$$

38. What is the image of the point P(2, 3) after being rotated through angle $\theta = -540^0$ about the origin ?

A. (-3, -2) B. (-2, -3) C. (-2, 3) D. (2, -3)

Solution

We use coterminal angle definition.

$\alpha = \theta + 360^0K$, K is integer and $\alpha = -540^0 + 720^0 = 180^0$

Now, $x' = 2\cos 180^0 - 3\sin 180^0 = 2(-1) - 3(0) = -2$

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$y' = 2 \sin 180^0 + 3 \cos 180^0 = 0 + 3(-1) = -3$

$\therefore \quad R_{-540^0}(2, 3) = R_{180^0}(2, 3) = (-2, -3)$

Answer: B

39. Find the image of each line in rotation through the indicated angle about the origin.

a. $L: 3x + 2y = 8, \ \theta = \frac{3\pi}{2}$ b. $L: y = 2x + 1, \ \theta = \frac{-\pi}{2}$

Solution: use $x' = x \cos\theta - y \sin\theta$ and $y' = x \sin\theta + y \cos\theta$

a. Take any two points on $L: 3x + 2y = 8$

Let $x = 2$, then $y = 1$, $(2, 1)$ on L

Let $x = 0$, then $y = 4$, $(0, 4)$ on L

Image of $(2, 1)$ when rotated by $\theta = \frac{3\pi}{2}$

$x' = 2\cos\frac{3\pi}{2} - 1\sin\frac{3\pi}{2} = 0 - (1)(-1) = 1$

$y' = 2\sin\frac{3\pi}{2} - 1\cos\frac{3\pi}{2} = -2 - 0 = -2$

$\therefore \quad R_{\frac{3\pi}{2}}(2, 1) = (1, -2)$

Image of $(0, 4)$ when rotated by $\theta = \frac{3\pi}{2}$:

$R_{\frac{3\pi}{2}}(0, 4) \longrightarrow (4, 0)$

$\therefore$ The equation of the line passing through $(1, -2)$ and $(4, 0)$ is given by:

$L': y + 2 = \frac{0 + 2}{4 - 1}(x - 1) = \frac{2}{3}(x - 1)$

$L': 3y + 6 = 2x - 2$

$\therefore$ The image of the line is $L': 3y - 2x = -8$

b. Take any two points on $L: y = 2x + 1$

Let $x = 1, y = 3$, $(1, 3)$ on L

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Let $x = -1, y = -1, (-1, -1)$ on L

Image of $(1, 3)$ when rotated by $\theta = \frac{-\pi}{2}$:

$$x' = 1\cos\left(\frac{-\pi}{2}\right) - 3\sin\left(\frac{-\pi}{2}\right) = 0 + 3 = 3$$

$$y' = 1\sin\left(\frac{-\pi}{2}\right) + 3\cos\left(\frac{-\pi}{2}\right) = -1 - 0 = -1$$

Image of $(1, 3)$ is $(3, -1)$

Image of $(-1, -1)$ when rotated by $\theta = \frac{-\pi}{2}$:

$$x' = -1\cos\left(\frac{-\pi}{2}\right) - (-1)\sin\left(\frac{-\pi}{2}\right) = -1$$

$$y' = -1\sin\left(\frac{-\pi}{2}\right) + (-1)\cos\left(\frac{-\pi}{2}\right) = 1$$

$\therefore$ Image of $(-1, -1)$ is $(-1, 1)$

Equation of the line passing through $(3, -1)$ and $(-1, 1)$ is:

$$y - 1 = \frac{1 - (-1)}{-1 - 3}(x + 1) = \frac{-1}{2}(x + 1) = 2y - 2 = -x - 1$$

$\therefore$ The image of the line is $L' : y = \frac{-1}{2}x + \frac{1}{2}$

40. What is the image of the circle $x^2 + y^2 - 4x + 2y = 0$ after being rotated through angle $\theta = \frac{-3\pi}{2}$ about the origin?

A. $x^2 + y^2 - 2y + 4x = 0$

B. $x^2 + y^2 + 2y + 4x = 0$

C. $x^2 + y^2 - 2x + 4y = 0$

D. $x^2 + y^2 - 2x - 4y = 0$

Solution

The center of a circle is $(2, -1)$ and radiu $= \sqrt{5}$

$$x' = 2\cos\left(\frac{-3\pi}{2}\right) - (-1)\sin\left(\frac{-3\pi}{2}\right) = 0 + 1 = 1$$

$$y' = 2\sin\left(\frac{-3\pi}{2}\right) + (-1)\cos\left(\frac{-3\pi}{2}\right) = 2$$

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$R(2, -1) \xrightarrow{\theta = -3\pi/2} (1, 2)$

$\therefore$ Image of the circle $C' : (x - 1)^2 + (y - 2)^2 = 5$ **Answer: D**

6.4.3 Rotation about any Point (a, b)

If $P'(x', y')$ is the point image of a point $P(x, y)$ when it is rotated through an angle θ about the (a, b), then:

$x' = a + (x - a)\cos\theta - (y - b)\sin\theta$,

$y' = b + (x - a)\sin\theta + (y - b)\cos\theta$

To rotate $P(x, y)$ in angle θ about point (a, b), we have the following steps:

Step 1: Translate $(a, b) \longrightarrow (0, 0)$

Translation vector: $T = (-a, -b)$

Step 2: Translate $(x, y) \xrightarrow{T = (-a, -b)} (x - a, y - b)$

Step 3: Translate $(x', y') \xrightarrow{T = (-a, -b)} (x' - a, y' - b)$

Step 4: Replace step 2 and step 3 in rotation of a point $P(x, y)$ about the origin:

$x' \longrightarrow x' - a$ and $y' \longrightarrow y' - b$,

$x \longrightarrow x - a$ and $y \longrightarrow y - b$

$x' = a + (x - a)\cos\theta - (y - b)\sin\theta$

$y' = b + (x - a)\sin\theta + (y - b)\cos\theta$

Illustrative Example

41. Find the image of the point P (3, 1) when it is rotated through $\theta = \frac{\pi}{2}$ about (-2, 3).

Solution: $(a, b) = (-2, 3)$, P (3, 1), and $\theta = \frac{\pi}{2}$

$x' = a + (x - a)\cos\theta - (y - b)\sin\theta$

$= -2 + (3 - (-2))\cos\frac{\pi}{2} - (1 - 3)\sin\frac{\pi}{2}$

$= -2 + 0 + 2 = 0$

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$y' = b + (x - a)\sin\theta + (y - b)\cos\theta$

$= 3 + (3 - (-2))\sin\frac{\pi}{2} + (1 - 3)\cos\frac{\pi}{2}$

$= 3 + 5 + 0 = 8$

$\therefore$ The image of the point P(3, 1) after rotating $\frac{\pi}{2}$ about (-2, 3) is (0, 8)

42. Find the image of the point P (0, -3) when it is rotated through $\theta = \frac{17\pi}{6}$ about (0, 1).

Solution

First, find the measure of angle between 0^0 and 2π coterminal with $\theta = \frac{17\pi}{6}$. To find a coterminal angle, subtract 2π.

$\frac{17\pi}{6} - 2\pi = \frac{5\pi}{6}$, thus $\frac{17\pi}{6}$ and $\frac{5\pi}{6}$ are coterminal

To find the image, point P ' (x', y'), about (a, b) is:

$x' = a + (x - a)\cos\theta - (y - b)\sin\theta$

$y' = b + (x - a)\sin\theta + (y - b)\cos\theta$

Then the image of P_1 (0, -3) about (0, 1)is

$x' = 0 + (0 - 0)\cos\frac{5\pi}{6} - (-3 - 1)\sin\frac{5\pi}{6} = 4\left(\frac{1}{2}\right) = 2$

$y' = 1 + (0 - 0)\sin\frac{5\pi}{6} - (-3 - 1)\cos\frac{5\pi}{6} = 1 - 2\sqrt{3}$

$\therefore$ The image of P_1 (0, -3) is $\left(2, 1 - 2\sqrt{3}\right)$

43. When the plane is rotated 45^0 about the point (1, -2), then what would be image of the point P(2, 4)?

A. $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

B. $\left(\frac{5\sqrt{2}}{2}, \frac{7\sqrt{2}}{2}\right)$

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C. $\left(\frac{2-5\sqrt{2}}{2}, \frac{-4+7\sqrt{2}}{2}\right)$ D. $\left(\frac{2-\sqrt{2}}{2}, \frac{-4+\sqrt{2}}{2}\right)$

Solution: $(a, b) = (1, -2)$ and $P(x, y) = P(2, 4)$ and $\theta = 450^0$

$x' = a + (x - a)\cos\theta - (y - b)\sin\theta$

$= 1 + (2 - 1)\cos 45^0 - (4 - (-2))\sin 45^0$

$= 1 + \frac{\sqrt{2}}{2} - 6\frac{\sqrt{2}}{2} = 1 - \frac{5\sqrt{2}}{2} = \frac{2 - 5\sqrt{2}}{2}$

$y' = b + (x - a)\sin\theta + (y - b)\sin\theta$

$= -2 + (2 - 1)\sin 45^0 + (4 - (-2))\sin 45^0$

$= -2 + \frac{\sqrt{2}}{2} + \frac{6\sqrt{2}}{2} = -2 + \frac{7\sqrt{2}}{2} = \frac{-4 + 7\sqrt{2}}{2}$

Answer: C

44. Find the image of the line $L : y = 2x - 3$ after a rotation of $\theta = \frac{3\pi}{2}$ about the point (4, 3).

Solution

Step 1: choose any two points on $L : y = 2x - 3$

Let P_1 (2, 1) and P_2 (-1, -5) are points on $L : y = 2x - 3$

Step 2: Find the images of the two points P_1 (2, 1) and P_2 (-1, -5):

$x' = a + (x - a)\cos\theta - (y - b)\sin\theta$

$(a, b) = (4, 3)$ and $\theta = \frac{3\pi}{2}$

For $P_1(2, 1)$: $x' = 4 + (2 - 4)\cos\frac{3\pi}{2} - (1 - 4)\sin\frac{3\pi}{2}$

$= 4 + (-2)(0) + 3(-1) = 1$

$y' = b + (x - a)\sin\theta + (y - b)\cos\theta$

$= 3 + (2 - 4)\sin\frac{3\pi}{2} + (1 - 3)\cos\frac{3\pi}{2}$

$= 3 + -2\sin\frac{3\pi}{2} - 2\cos\frac{3\pi}{2}$

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$= 3 - 2(-1) - 2(0) = 5$

$\therefore$ Image of P_1 (2, 1) is (1, 5)

For $P_2(-1, -5)$: $x' = 4 + (-1 - 4)\cos\frac{3\pi}{2} - (-5 - 3)\sin\frac{3\pi}{2}$

$= 4 - 5(0) + 8(-1) = -4$

$y' = 3 + (-1 - 4)\sin\frac{3\pi}{2} + (-5 - 3)\sin\frac{3\pi}{2}$

$= 3 + (-5)(-1) + -8(-1) + -8(-1) = 16$

$\therefore$ Image of P_2 (-1, -5) is (-4, 16)

Step 3: we find the equation of the image of the line passing through (1, 5) and (-4, 16).

$$L' : y - 5 = \frac{16 - 5}{-4 - 1}(x - 1) = \frac{11}{-5}(x - 1)$$

$$L' : y = \frac{-11}{5}x - \frac{36}{5}$$

45. Find the image of the line $L: 3x - 2y = 6$. When it is rotated through $\theta = \frac{7\pi}{2}$ about (-1, 1)

Solution

First, find the measure of angle between 0 and 2π coterminal with $\theta = \frac{7\pi}{2}$. Thus $\frac{7\pi}{2} - 2\pi = \frac{3\pi}{2}$ is coterminal angle with $\frac{7\pi}{2}$.

Choose two points P_1 (2, 0) and P_2 (0, - 3) on $L : 3x - 2y = 6$

To find the image, of $P_1(x_1, y)$ is $P_1'(x', y')$, about (a, b) given by:

$x' = a + (x_1 - a)\cos\theta - (y_1 - b)\sin\theta$

$y' = b + (x_1 - a)\sin\theta + (y_1 - b)\cos\theta$

Where $(x_1, y_1) = (2, 0)$, $(a, b) = (-1, 1)$

$x' = -1 + (2 - (-1))\cos\frac{3\pi}{2} - (0 - 1)\sin\frac{3\pi}{2}$

$= -1 + 0 - 1 = -2$

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$y' = 1 + (2 - (-1)) \sin \frac{3\pi}{2} + (0 - 1) \cos \frac{3\pi}{2}$

$= 1 + 3(-1) + 0 = -2$

To find the image of P_2 (0, -3) about (-1, 1) is

$x' = -1 + (0 - (-1)) \cos \frac{3\pi}{2} - (-3 - 1) \sin \frac{3\pi}{2}$

$= -1 + 0 + (-1) = -2$

$y' = 1 + (0 - (-1)) \sin \frac{3\pi}{2} + (-3 - 1) \cos \frac{3\pi}{2}$

$= 1 + (1)(-1) + 0 = 0$

$\therefore$ The image of P_2 (0, -3) is P' (-2, 0) thus imaged line L' containing the point (-2, -2) and (-2, 0) is the vertical line L' : $x = -2$.

46. Find the image of the circle $(x - 2)^2 + (y + 3)^2 = 16$ when it is rotated through $\theta = \pi$ about (5, -6)

Solution

To find the image of the circle we rotate the center of a circle (2, -3) we use the formula:

$x' = a + (x - a) \cos \theta - (y - b) \sin \theta$

$y' = b + (x - a) \sin \theta + (y - b) \cos \theta$

Where $(x, y) = (2, -3)$ and $(a, b) = (5, -6)$

$x' = 5 + (2 - (5)) \cos \pi - (-3 + 6) \sin \pi$

$= 5 + -3(-1) + 0 = 8$

$y' = -6 + (2 - (5)) \sin \pi + (-3 + 6) \cos \pi = -6 - 3 = -9$

$\therefore$ The equation of the image of the circle is $(x - 8)^2 + (y + 9)^2 = 16$

47. What is the vertices of the image of quadrilateral ABCD with A (-4, 2), B (-2, 4), C (0, 2), and D(-2, -4) reflected over the line $y = -x$, then translated by the vector T = (-2, -1) ?

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Solution

$(x, y) \xrightarrow[\text{by } y=-x]{\text{reflected}} (-y, -x)$

A. $(-4, 2) \xrightarrow{y=-x} A'(-2, 4) \xrightarrow{T(2,-1)} (0, 3)$

B. $(-2, 4) \xrightarrow{y=-x} B'(-4, 2) \xrightarrow{T(2,-1)} (-2, 1)$

C. $(0, 2) \xrightarrow{y=-x} C'(-2, 0) \xrightarrow{T(2,-1)} (0, -1)$

D. $(-2, -4) \xrightarrow{y=-x} D'(4, 2) \xrightarrow{T(2,-1)} (6, 1)$

$\therefore$ The image of the quadrilateral are

A" (0, 3), B" (-2, 1) C" (0, -1) and D" (6,1)

48. The vector equation of the line L is given by $(x, y) = (-2, 1) + \lambda(1, 1), \lambda \in R$. What is the equation of the image of L after being translated by the vertical $\bar{u} = (2, -1)$ followed by a rotated through 45^0 about the origin?

A. $y = \sqrt{2}\, x$ B. $y = -2\sqrt{x}$ C. $y = \sqrt{2}$ D. $x = 0$

Solution: $(x, y) = (-2, 1) + \lambda(1, 1)$

$x = -2 + \lambda$ and $y = 1 + \lambda$

$\lambda = x + 2$ then $y = 1 + x + 2 = x + 3$

The image of $y = x + 3$ when translated by $\bar{u} = (2, -1)$ is:

$y + 1 = (x - 2) + 3$ implies $y = x$

The image of $y = x$ when rotated by angle 45^0: Let $(1, 1)$ and $(2, 2)$ pt. on L.

$x' = x\cos\theta - y\sin\theta = x\cos 45^0 - x\sin 45^0 = 0$

$y' = x\sin\theta + y\cos\theta = y\sin 45^0 + y\sin 45^0$

$$= \frac{\sqrt{2}}{2}y + \frac{\sqrt{2}}{2}y = \frac{2\sqrt{2}\,y}{2} = \sqrt{2}\,y$$

$\therefore$ Image of (1, 1) is $\left(0, \sqrt{2}\right)$

Image of (2, 2) is $\left(0, 2\sqrt{2}\right)$

$\therefore$ The image of the equation of the line is $x = 0$.

Answer: D

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49. If R is rotation through $\theta = \frac{\pi}{2}$ and is followed by reflection in the x - axis. What is the image of circle given by $(x - 1)^2 + (y + 2)^2 = 3^2$ under the two transformation?

A. $(x + 1)^2 + (y - 2)^2 = 3^2$

B. $(x + 2)^2 + (y - 1)^2 = 3^2$

C. $(x - 2)^2 + (y - 1)^2 = 3^2$

D. $(x - 2)^2 + (y + 1)^2 = 3^2$

Solution: Center of a circle is (1, -2) with radius 3.

$(1, -2) \xrightarrow{\text{Rotation of } \theta = \pi/2} (2, 1)$

To reflect a point (2, 1) over the x - axis

$(2, 1) \xrightarrow{\text{over } x \text{ - axis}} (2, -1)$

$\therefore$ Equation of image of a circle is $(x - 2)^2 + (y + 1)^2 = 3^2$

Answer: D

50. Triangle ABC with A (4, 8), B (7, 5), and C (2, -1) reflected over the line $y = -x$, then rotated 90^0 about the origin. Find the image of the vertices.

Solution

To reflect a point over the line $y = -x$ switch the coordinated of each point and then multiply both coordinates of each point by -1.

$(x, y) \xrightarrow{y = -x} (-y, -x)$ then $(a, b) \xrightarrow{90^0 \text{ rotation}} (-b, a)$

A. $(4, 8) \xrightarrow{y = -x} (-8, -4) \xrightarrow{90 \text{ - rotation}} A'(4, -8)$

B. $(7, 5) \xrightarrow{y = -x} (-5, -7) \xrightarrow{90 \text{ - rotation}} B'(7, -5)$

C. $(2, -1) \xrightarrow{y = -x} (1, -2) \xrightarrow{90 \text{ - rotation}} C'(2, 1)$

$\therefore$ The image of the vertices are A'(4, -8), B'(7,-5) and C'(2,1)

51. If the point M (3, 2) is reflected in the line $y = 2x$, then rotated clock wise through $\theta = \frac{3\pi}{2}$ about the origin. What is the image of the point under the two transformation.

Solution: The angle of inclination of $y = 2x$ m $= \tan\theta = 2$.

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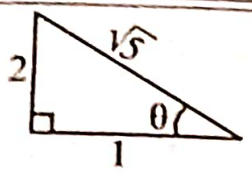

Then: $\sin\theta = \frac{2}{\sqrt{5}}$, and $\cos\theta = \frac{1}{\sqrt{5}}$

From the definition reflection, through angle θ:

$x' = x\cos 2\theta + y\sin 2\theta$ and

$y' = x\sin 2\theta - y\cos 2\theta$, where $x = 3$, and $y = 2$

$\cos 2\theta = \cos(\theta + \theta) = \cos^2\theta - \sin^2\theta$

$$= \left(\frac{1}{\sqrt{5}}\right)^2 - \left(\frac{2}{\sqrt{5}}\right)^2 = \frac{-3}{5}$$

$$\sin 2\theta = \sin(\theta + \theta) = \left(\frac{2}{\sqrt{5}}\right)\left(\frac{1}{\sqrt{5}}\right) = \frac{4}{5}$$

As result of this

$$x' = x\cos 2\theta + y\sin 2\theta = 3\left(\frac{-3}{5}\right) + 2\left(\frac{4}{5}\right) = \frac{-1}{5}$$

$$y' = x\sin 2\theta - y\cos 2\theta = 3\left(\frac{4}{5}\right) - 2\left(\frac{-3}{5}\right) = \frac{18}{5}$$

Clock wise rotation of $\theta = \frac{3\pi}{2}$ is means $\theta = -\frac{3\pi}{2}$.

Rotation through angle θ about the origin is:

$x' = x\cos\theta - y\sin\theta$ and $y' = x\sin\theta + y\cos\theta$ where $(x, y) = \left(\frac{-1}{5}, \frac{18}{5}\right)$

$$x' = \frac{-1}{5}\cos\left(\frac{-3\pi}{2}\right) - \frac{18}{5}\sin\left(\frac{-3\pi}{2}\right) = 0 - \frac{18}{5} = \frac{-18}{5}$$

$$y' = \frac{-1}{5}\sin\left(\frac{-3\pi}{2}\right) - \frac{18}{5}\cos\left(\frac{-3\pi}{2}\right) = \frac{-1}{5}$$

Answer: C

52. Let ℓ be a line given $2x - y = 10$. What is the equation of the image of ℓ after a reflection in the line $y = 2x - 5$ followed by rotation through the angle $\theta = \frac{\pi}{2}$ about the origin ?

A. $x + 2y = 0$ B. $2x + y = 0$ C. $x + 2y = 5$ D. $x - 2y = 5$

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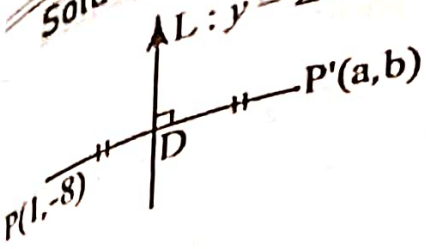
Solution



Take a point P(1, -8) on ℓ: $2x - y = 10$

To find the image of P (1, -8) on L : $y = 2x - 5$

- Equation of $\overline{PP'}$ passing through P(1, - 8):

$$\frac{y + 8}{x - 1} = \frac{-1}{2}, \text{ imples } 2y + x = -15$$

- Find the intersection of $L : y = 2x - 5$ and $\overline{PP'} : 2y + x = -15$

$\Rightarrow 2(2x - 5) + x = -15$, $x = -1$ and $y = -7$.

Since (-1, -7) is the mid point of $\overline{PP'}$

$$\frac{a + 1}{2} = -1, \text{ and } \frac{b - 8}{2} = -7$$

$a = -3$ and $b = -14 + 8 = -6$

- Equation of the imaged line passing through (-3, -6) with slope 2 is

$$\frac{y + 6}{x + 3} = 2, \text{ implies } y = 2x$$

- Image of $\ell' : y = 2x$ when rotated by $\theta = \frac{\pi}{2}$ about the origin.

Take two point , say (1, 2) and (-4, -8) on $\ell' : y = 2x$

$$(x, y) \xrightarrow{R = \pi/2} (-y, x)$$

$$(1, 2) \longrightarrow (-2, 1)$$

$$(-4, -8) \longrightarrow (8, -4)$$

$\therefore$ Equation of the line passing through (-2, 1) and (8, -4) is:

$$\frac{y - 1}{x + 2} = \frac{-4 - 1}{8 + 2} = \frac{-5}{10} = \frac{-1}{2}$$

$\Rightarrow 2y - 2 = -x - 2$ implies $2y + x = 0$

Answer: A

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53. What is the image of the line given by $(x, y) = (-1, 0) + t(3, 6)$, under the translation that makes $(1, 0)$ to $(0, 1)$ followed by the reflection about the line $y = 2x$?

A. $y = 2x + 3$ B. $y = 2x - 3$ C. $y = 2x + 6$ D. $y = 2x - 5$

Solution

$(x, y) = (-1, 0) + t(3, 6)$

$x = -1 + 3t$ and $y = 0 + 6t = 6t$, solving and equating for t:

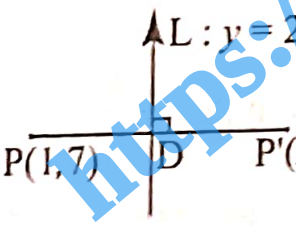
$$\frac{x+1}{3} = \frac{y}{6} \Rightarrow y = 2x + 2$$

(i) Under translation: $(1, 0) \xrightarrow{T = (-1, 1)} (0, 1)$

In this case the line is translated by horizontally $x + 1$ and vertically: $y - 1$. As a result, the new line is:

$$y - 1 = 2(x + 1) + 2 \text{ implies } y = 2x + 5$$

(ii) Under reflection, to find the image of $y = 2x + 5$ on axis of reflection $y = 2x$, take point P $(1, 7)$ on L : $y = 2x + 5$.



Slop of $\overline{PP'}$ is $-\frac{1}{2}$

Equation of the line passing through P and P' is:

$$\frac{y-1}{x-7} = \frac{-1}{2} \text{ implies } 2y + x = 9$$

The intersection of $y = 2x$ and $2y + x = 9$ $\left\{\frac{9}{5}, \frac{18}{5}\right\}$

The intersection point is the mid - point of $\overline{PP'}$, so that

$$\frac{x+1}{2} = \frac{9}{5} \text{ implies } x = \frac{13}{5} \text{ and } \frac{y+7}{2} = \frac{18}{5}, \text{ implies } y = \frac{1}{5}$$

The imaged line passing through with slope 2 is:

$\left(\frac{13}{5}, \frac{1}{5}\right)$ with slope 2 is: $y - \frac{1}{5} = 2\left(x - \frac{13}{5}\right)$ implies $y = 2x - 5$

Answer: D

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Concept check

54. What is transformations?

Answer: Transformations are movements of a geometric figure

55. What is non - rigid transformation? Given example of non - rigid transformation.

Answer: A transformation that does not preserve distance.

scaling a figure is an example of non - rigid transformation.

56. What is rigid transformation?

Answer: A mapping of a figure onto itself is called a rigid motion if of preserves distances.

57. What is the image of a triangle under a rigid motion

Answer: A rigid motion carry triangles into congruent triangle.

58. List some types of rigid motion

Answer: Translation, reflection and rotation are some examples of rigid motion.

59. A translation takes the point A (1, 2) into A' (-3, 5).

a. What is the value of translation vector?

b. Into what point does it take Q (7, 4)?

Solution

a. Translation vector: $T = \overline{AA'} = (-3, 5) - (1, 2) = (-4, 3)$.

b. $Q(7, 4) \xrightarrow{T = (-4, 3)} (7 - 4, 4 + 3) = (3, 7)$.

60. What is a reflection?

Answer: A reflection in a line L is one - to one mapping of a figure onto itself

61. If the x - axis is the axis of reflection, what is the image of ?

a. M (-2, -5) b. M(2, 4) c. M (a, b)

Solution

a. M (-2, -5) = (-2, 5) b. M(2, 4) = (2, -4) c. M (a, b) = (a, -b)

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62. If (-3, -5) is the image of (5, 3), what is the equation of the mirror?

Solution:- Reflection over the line: $y = -x$

$$M(x, y) = (-y, -x)$$

63. Find the image of the point P(x, y) when reflected about the line $y = mx$ where $m = \tan\theta$

Solution: Let P (x, y) is its image:

$$x' = x\cos 2\theta + y\sin 2\theta \quad \text{and} \quad y' = x\sin 2\theta - y\cos 2\theta$$

64. If the plane is rotated 90^0 counter clock wise about the origin. What are the images of:

a. (2, 5) b. (1, -3) c. (a, b)

Solution

a. $R_{90^0}(2, 5) = (-5, 2)$

b. $R_{30^0}(1, -3) = (3, 1)$

c. $R_{90^0}(a, b) = (-b, a)$

65. If the plane rotated 180^0 about the origin, what are the images of:

a. (2, 5) b. (1, -3) c. (a, b)

Solution

a. If $\theta = 180^0$, then $R_\theta(2, 5) = (-2, -5)$

b. If $\theta = 180^0$, then $R_\theta(1, -3) = (-1, 3)$

c. If $\theta = 180^0$, then $R_\theta(a, b) = (-a, -b)$.

66. If the plane is rotated through $\theta = \frac{3\pi}{2}$ about the origin, what are the image of:

a. (2, 5) b. (1, -3) c. (a, b)

Solution

a. If $\theta = \frac{3\pi}{2}$, then $R_\theta(2, 5) = (5, -2)$

b. If $\theta = \frac{3\pi}{2}$, then $R_\theta(1, -3) = (-3, -1)$

c. If $\theta = \frac{3\pi}{2}$, then $R_\theta(a, b) = (b, -a)$

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67. If the plane is rotated through angle θ about (h,k), what are the image of (a, b)?

Solution

R_θ (a, b) = (a cos θ - b sinθ, a sinθ + b cosθ)

68. If the plane is rotated through angle θ about (h, k), what are the image of (a, b)?

Solution

$$R_\theta\ (a, b) = \begin{bmatrix} h + (a - h)\cos\theta - (b - k)\sin\theta) \\ k + (a - h)\sin\theta + (b - k)\cos\theta \end{bmatrix}$$

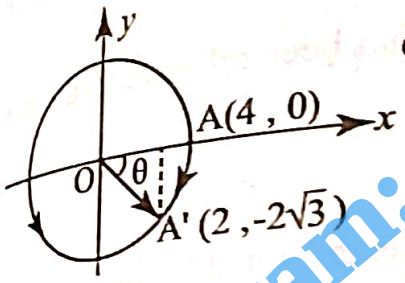
69. In rotation R, the image of A (4, 0) is $A'\ \left(2, 2\sqrt{3}\right)$ about the center of ration O (0, 0). Find the a mount of rotation of angle θ

Solution

The angle between $\overline{OA}$ and $\overline{OA'}$ is the amount of rotation.

$$\overline{OA} \cdot \overline{OA'} = \left|\overline{OA}\right| \left|\overline{OA'}\right| \cos\theta$$

$$\cos\theta = \frac{\overline{OA} \cdot \overline{OA'}}{\left|\overline{OA}\right| \left|\overline{OA'}\right|} = \frac{(4, 0) \cdot \left(2, -2\sqrt{3}\right)}{\sqrt{4^2 + 0^2} \cdot \sqrt{4 + 13}}$$



$$= \frac{8 + 0}{(4)\ (4)} = \frac{1}{2}$$

$$\therefore \quad \theta = 300^0 \text{ or } \theta = -60^0$$

70. In rotation R, the image of (3, 5) is A (-2, 4) about center of rotation C (1, 2). Find the angle θ.

Solution

To find the angle, we use vector dot product between $\overline{CA}$ and $\overline{CA'}$.

$\overline{CA}$ = (3, 5) - (1, 2) = (2, 3) $\overline{CA'}$ = (-2, 4) - (1, 2) = (-3, 2)

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$$\cos\theta = \frac{\overrightarrow{CA} \cdot \overrightarrow{CA'}}{|\overrightarrow{CA}|\,|\overrightarrow{CA'}|} = \frac{(2, 3) \cdot (-3, 2)}{\sqrt{4+9} \cdot \sqrt{9+4}} = \frac{-6+6}{13} = 0$$

$$\therefore \quad \theta = \frac{\pi}{2}$$

Review Exercise 6

1. A translation T takes the point (2, 3) to the point (0, 4). In to what po does it take (-1, 6)?
2. If translation T takes the point (4, 2) to the point (2, 5), then find t image of :
   a. P(3, 1)
   b. the line $2x - 3y = 6$
   c. the circle $x^2 + y^2 = 5$
   d. the circle $x^2 + y^2 - 2x + 4y - 5 = 0$
   e. the line $x + 2y = 5$
3. A translation T takes the origin to the point (2, 5). Find the image of th line whose equation is $3x + 2y = 1$
4. Find the image of the point P(3, 4), after it has been reflected in the line
   a. x - axis
   b. y - axis
   c. $y = x$
   d. $y = -x$
   e. $x = -2$
   f. $y = -2$
5. Find the image of the point P(1, 3) when reflected about the line $y = 2x +$
6. The image of the point A (2, 5) in a reflection is A' (-3, 1). Find th equation of the line of reflection.
7. If (2, -3) the image of (-6, 1), under the line of reflection. What is the equation of the line of reflection.
8. Find the image of (2, 4) after it has been reflected along a line passing through the origin and making an angle of 60^0 with the x - axis
9. Find the image of the circle $x^2 + y^2 - 4x + 6y + 12 = 0$ after it has been reflected about the line $x = -4$.

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10. Find the image of the point (0, 3) when is rotated through $\frac{\pi}{3}$ about (-1, 5).
11. Find the image of P(3, 2) after a clockwise rotation through 90^0 about the point (-1, 4).
12. Rotate the point (2, 1), 300^0 clock wise about the origin.
13. Find the image of the line L : $2x + 3y = 1$ when it is rotated clock wise through $\theta = 90^0$:
    a. about the origin
    b. about the point (-1, 3)
14. Find the image of the point (-2, -3) when it is rotated through $\theta = 180^0$ about (-4, 1).
15. Find the image of the circle $x^2 + y^2 + 6x - 4y + 9 = 0$ when it is rotated through $\theta = 180^0$ about (-1, 4).

Answer to Review Exercise 6

1. $(2, 3) \xrightarrow{T = (-2, 1)} (0, 4)$, then $(-1, 6) \xrightarrow{T = (-2, 1)} (-3, 7)$.
2. Translation vector: $T = (-2, 3)$
   a. $P(3, 1) \xrightarrow{T = (-2, 3)} (1, 4)$
   b. Replace x by $x + 2$ and y by $y - 3$:
      $L' : 2(x + 2) - 3(y - 3) = 6$ implies $L' : 2x - 3y = -7$
   c. $(x + 2)^2 + (y - 3)^2 = 5$
   d. $(x + 2)^2 + (y - 3)^2 - 2(x + 2) + 4(y - 3) - 5 = 0$
      $x^2 + y^2 + 2x - 2y - 8 = 0$
   e. $x + 2y = 9$
3. $3x + 2y = 17$
4. a. P' (3, -4)
   b. P' (-3, 4)
   c. P' (4, 3)
   d. P' (-4, -3)
   e. P' (-7, 4)
   f. P' (3, -8)
5. $P'\left(\frac{-11}{5}, \frac{23}{5}\right)$
6. $8y - 10x = 29$
7. $y = 2x + 3$
8. $x' = x \cos 2\theta + y \sin 2\theta$ and $y' = x \sin 2\theta - y \cos 2\theta$

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$x' = 2\cos 120^0 + 4 \sin 120^0$ and $y' = 2 \sin 120^0 - 4 \cos 120^0$.

$x' = 2\left(\frac{-1}{2}\right) + 4\left(\frac{\sqrt{3}}{2}\right) = -1 + 2\sqrt{3}$ and $y' = \left(\frac{\sqrt{3}}{2}\right) - 4\left(\frac{-1}{2}\right) = 2 + \sqrt{3}$

$\therefore$ The image of the point is $\left(-1 + 2\sqrt{3}, 2 + \sqrt{3}\right)$

9. Center of a circle (2, -3) with radius: r = 1

$\therefore$ Image of circle is $(x + 10)^2 + (y + 3)^2 = 1$

10. $x' = a + (x - a) \cos\theta - (y - b) \sin\theta$

$x' = -1 + (0 - (-1)) \cos\frac{\pi}{3} - (3 \quad 5) \sin\frac{\pi}{3} = \frac{-1 + 2\sqrt{3}}{2}$

$y' = b + (x - a) \sin\theta + (y - b) \cos\theta$

$= 5 + (0 - (-1)) \sin\frac{\pi}{3} + (3 - 5) \cos\frac{\pi}{3} = 4 + \frac{\sqrt{3}}{2}$

$\therefore$ The image of the point (0, 3) is $\left(\frac{-1 + 2\sqrt{3}}{2}, \frac{4 + \sqrt{3}}{2}\right)$

11. Since the rotation is clockwise, then $\theta = -90^0$

$x' = -1 + (-3 - (-1)) \cos(-90^0) - (2 - 4) \sin(-90^0)$

$= -1 + 0 + 2(-1) = -3$

$y' = 4 + (3 - (-1)) \sin(-90^0) + (2 - 4) \cos(-90^0)$

$= 4 + 4(-1) + 0 = 0$

$\therefore$ The image of the point (3, 2) is (-3, 0).

12. $x' = x \cos\theta - y \sin\theta$ and $y' = x \sin\theta + y \cos\theta$

$x' = -2 \cos(-300^0) - 1 \sin(-300^0)$ and

$y' = 2 \sin(-300^0) + 1. \cos(-300^0)$

$x' = 2\left(\frac{1}{2}\right) - 1\left(\frac{\sqrt{3}}{2}\right)$ and $y' = 2\left(\frac{\sqrt{3}}{2}\right) + \frac{1}{2}$

$x' = \frac{2 - \sqrt{3}}{2}$ and $y' = \frac{1 + 2\sqrt{3}}{2}$

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13. Take (-1, 1) and (2, -1) point on L : $2x + 3y$

a. image of (-1, 1) is (1, 1) and image of (2, -1) is (-1, -2)

Equation L ' : $y = \frac{3}{2}x - \frac{1}{2}$

b. Image of (-1, 1) is: $x' = a + (x - a)\cos\theta - (y - b)\sin\theta$

$= -1 + (1 - (-1))\cos(-90^0) - (1 - 3)\sin(-90^0) = -1 + -2 = -3$

$y' = b + (x - a)\sin\theta + (y - b)\cos\theta$

$= 3 + (-1 - (-1)\sin(-90^0) + (1 - 3)\cos(-90^0)$

$= 3 + 0 = 3$

$\therefore$ Image of (-1, 1) is (-3, 3)

Image of (2, -1) is: $x' = -1 + (2 - 1)\cos(-90^0) - (-1 - 3)\sin(-90^0)$

$= -1 - 4 = -5$

$y' = 3 + (2 - (-1))\sin(-90^0) + (-1 - 3)\cos(-90^0)$

$= 3 - 3 + 0 = 0$

$\therefore$ Image of (2, -1) is (-5, 0)

$\therefore$ The equation of imaged line is L' : $2y - 3x = 15$

14. $x' = a + (x - a)\cos\theta - (y - b)\sin\theta$

$= -4 + (-2 - (-4))\cos 180^0 - (-3-1)\sin 180^0 = -4 - 2 + 0 = -6$

$y' = b + (x - a)\sin\theta + (y - b)\cos\theta$

$= 1 + (-2 - (-4))\sin 180^0 + (-3 - 1)\cos 180^0 = 1 + 0 + 4 = 5$

$\therefore$ The image of the point (-2, -3) is (-6, 5)

15. Center (-3, 2), radius r = 2

$x' = -1 + (-3 - (-1))\cos 180^0 - (2 - 4)\sin 180^0$

$= -1 + 2 + 0 = 1$

$y' = 4 + (-3 - (-1))\sin 180^0 - (2 - 4)\cos 180^0 = 4 + 0 - 2 = 2$

$\therefore$ The image of a circle is $x^2 + y^2 + 6x - 4y + 9 = 0$ is

$(x - 1)^2 + (y - 2)3^2 = 2^2$

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UNIT 7

7. Statistic

What is Statistics?

Statistics is the science which deals with method of collecting, classifying, presenting, comparing and interpreting numerical date collected.

7.1 Types of Statistical Data

The word **data** means information or a set of given facts.

A **variable** is a characteristics or attribute that can assume different value. These date can be classified into **quantitative** or **qualitative** date.

(i) **Qualitative Data** is not numerical data, we can expressed in number.

Example: Colour, gender, religion, interest, beauty, honesty, educational qualification ..... etc.

(ii) **Quantitative Data** is the data whose value is measured in number

Example: Age, height, weight, time, temperature, distance income, size, score in test, etc.

Quantitative data classified into **Discrete** and **continuous data**

- **Discrete Data** Can be assigned value such as 0, 1, 2, 3, ,,,, and assume values that can be counted.

  Example: Number of student, number of house, population size, number of car, number of yellow, marble, rank, etc.

- **Continuous Data** is a quantitative data whose value is expressed in fraction or decimals. They are obtained by measuring.

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Example: Height, weight, age, size, temperature, distance, income, length of time.

Conceptual Example

1. Classify each variable as **qualitative or quantitative**
   a. Number of desks in class room.
   b. Weight of desks in class room.
   c. Classification of student learning achievement in a class room (slow, average, fast)
   d. Colors of automobiles in the Addis Ababa city.
   e. Weight of fish caught in Hidas Dam (Abbay Gidib).
   f. Volume of water in Gibee Dam.
   g. State of birth of new born infants at certain health centre.
   h. Educational qualification back ground of Ethiopian Student parents.

Solution

a. quantitative
b. quantitative
c. qualitative
d. qualitative
e. quantitative
f. quantitative
g. qualitative
h. qualitative

2. Classify each Variable as **discrete or continuous.**
   a. Number of white colors automobiles in Addis Ababa.
   b. Number of desks in a class room
   c. Weight of desks in a class room
   d. Volume of water hold in Hidas Dam
   e. Water temperature of the saunas at given health spa.
   f. Life span of dogs.
   g. Income of single parents
   h. The time it takes to drive to work
   i. The number of car accident in Dradaw city.

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Solution

a. Discrete
b. Discrete
c. Continuous
d. Continuous
e. Continuous
f. Continuous
g. Continuous
h. Continuous
i. Discrete

How to Organize Data ?

The most convenient method of organizing data is **frequency distribution**

(i) **A frequency Distribution** is the organization of raw data in table form, using classes and frequencies.

(ii) **Grouped Data** is data organized and summarized as in the frequency distribution.

7.2 Grouped Frequency Distribution

When the range of the data is large the data must be grouped into class that are more than one unit in width.

Formation of Classes

When forming classes from given data in raw form:

1. Classes should be non - overlapping
2. There should be no gaps between the class.
3. The class must be equal in width.
4. As far as possible, open ended class should be avoided
5. The number of class should be between 5 and 20.

Class Intervals and Class Limit

(i) **Class intervals** is the numerical width of class in a particular distribution. As an example: a data defining such as: 60 - 62, 69 - 71 are called **class intervals.**

(ii) **Class Limit:** is the minimum value and the maximum value of the class interval.

- The end numbers 60 and 62 are called **class limit.**
- The smaller number 60 is the **lower class limit** and the larger

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number is the upper class limit.

- **Class boundaries:** these numbers are used to separate the classes so that there are no gaps in the frequency distribution.
- The boundaries are obtained by:
- Subtracting 0.5 from the lower class limit.
- adding 0.5 to the upper class limit.

Lower limit - 0.5 = 60 - 0.5 = 59.5 = lower boundary.

Upper limit + 0.5 = 62 + 0.5 = 62.5 = upper boundary.

Hence 59.5 - 62.5, 62.5 - 65.5, 65.5 - 68.5 are class boundaries.

- **Class width** is the difference between the lower and upper class boundaries of the class interval. It is equal to the difference between two successive lower class limit or two successive upper class limit.
- **Class mark** is the midpoint of the class interval and is obtained by adding the lower and upper class limit and dividing by 2.

$$\text{Class mark} = \frac{(\text{lower class limit}) + (\text{upper class limit})}{2}$$

The Procedure For Constructing Frequency Distribution

STEP 1: Determine the number of classes

STEP 2: Find the largest and the smalls numbers in the raw data

STEP 3: Find the class with : $\dfrac{(\text{largest value}) - (\text{smalles value})}{\text{number of class}}$

Cumulative Frequency is the sum of the frequency of a particular class and frequency of all classes preceding that class that is, cumulative frequency of a particular class

= (frequency of this class) + (frequencies) of preceding classes.

Illustrative Example

1. Consider the data table, for the marks obtained by 40 students in mathematics test is given below.

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38	11	40	6	26	15	5	45	7	32		
2	18	42	8	31	27	4	12	35	15		
28	46	29	10	7	5	1	17	22	23		
35	7	17	42	45	48	15	3	8	18		

a. Construct grouped frequency distribution table with 8 classes
b. Find the class width, class mark, class boundaries
c. Find the lower limit of the 4th class
d. Find the upper class limit of the 7th class
e. How many students scoring less than 30.
f. How many students scoring more than 30.

Solution

$$\text{Class width} = \frac{(\text{Maimum mark}) - (\text{minimum mark})}{8}$$

$$= \frac{48 - 1}{8} \approx 6$$

The class intervals are: 1 - 6, 7 - 12, 13 - 18, 19 - 24, 25 - 30, 31 - 36, 37 - 42 and 43 - 48.

The class boundaries are: 0.5 - 6.5, 6.5 - 12.5, 12.5 - 18.5, 18.5 - 24.5, 24.5 - 30.5, 30.5 - 65.5, 36.5 - 42.5 and 42.5 - 48.5

The class marks are 3.5, 9.5, 15.5, 21.5, 27.5 33.5, 9.5, and 45.5

Frequency Distribution Table

Class Interval	Class Boundaries	Class Mark	Frequency	Cumulative Frequency
1 - 6	0.5 - 6.5	3.5	7	7
7 - 12	6.5 - 12.5	9.5	8	15
13 - 18	12.5 - 18.5	15.5	7	22
19 - 24	18.5 - 24.5	21.5	2	24
25 - 30	24.5 - 30.5	27.5	4	28
31 - 36	30.5 - 36.5	33.5	4	32

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37 - 42	36.5 - 42.5	39.5	4	36
43 - 48	42.5 - 48.5	45.5	4	40

c. The 4th class is 19 - 24, so that the lower class limit is 19.

d. The 7th class is 37 - 42, so that the upper class limit is 42.

e. 28 students score less than 30.

f. 12 students score less than 30.

2. The class marks of distribution are: 8, 12, 16, 20, 24, 28 and 32. Find

a. The class width b. Class interval c. Class boundaries

Solution

a. Class width equal to the difference the successive class marks of the data.

$\therefore$ Class width $= 12 - 8 = 4$

b. Lower limit is the first class interval $= 8 - \frac{4}{2} = 6$

c. Upper limit of the first class interval $= 8 + \frac{4}{2} = 10$

Hence the class interval are: $6 \leq x \leq 10$, $10 \leq x \leq 14$, $14 \leq 18$, $18 \leq x <$ 22, $22 \leq x < 26$, $26 \leq x < 30$, ans $30 \leq x < 34$.

The class boundaries 5.5 - 10.5, 10.5 - 14.5, 14.5 - 18.5, 18.5 - 22.5, 22.5 - 26.5, 26.5 - 3.5, and 30.5 - 37.5

Note: The quantity measured in an experimenter is called a variable or variate, where as the number of times it appears is called its frequency.

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7.3 Graphic Representation of Grouped Data

After the data have been organized into a frequency distribution, they can be presented in graphical form. It is easier for most people to comprehend the meaning of data presented graphically than data presented numerically in tables to represent data in frequency distribution graphically using **histograms and frequency polygons.**

DEFINITION

(i) The **histogram** is a graph that displays the data by using contiguous vertical bars of various heights to represent the frequencies of the classes. Another way to represent the same data set is by using frequency polygon.

(ii) The **frequency polygon** is a graph that displays the data by using lines that connect points plotted for the frequencies at the midpoints of the classes. The frequencies are represented by the heights of the points.

Illustrative Example

3. The following scores were obtained in a mathematics exam:

84	74	80	65	85	98	72	76	72	93
70	75	75	60	74	75	63	78	87	90

a. Construct a grouped frequency distribution into four classes:

b. Display the results in a histogram and frequency

c. How many students are in the class 70 - 80.

d. How many students are below 80.

e. How many students are above 70.

Solution

a. The highest mark is 98 and the smallest mark is 60, so that the range is 98 - 60 = 38. Since four class intervals are used, the class width is $\frac{38}{4} \approx 10$ class interval: 60 - 70, 80, 80 - 90, and 90 - 100

Mark	Class Mark	Number of Student Frequency	Cumulative Frequency
$60 \leq x < 70$	65	3	3
$70 \leq x < 80$	75	10	13
$80 \leq x < 90$	85	4	17
$90 \leq x < 100$	95	3	20

b. The histogram and frequency polygon are shown below.

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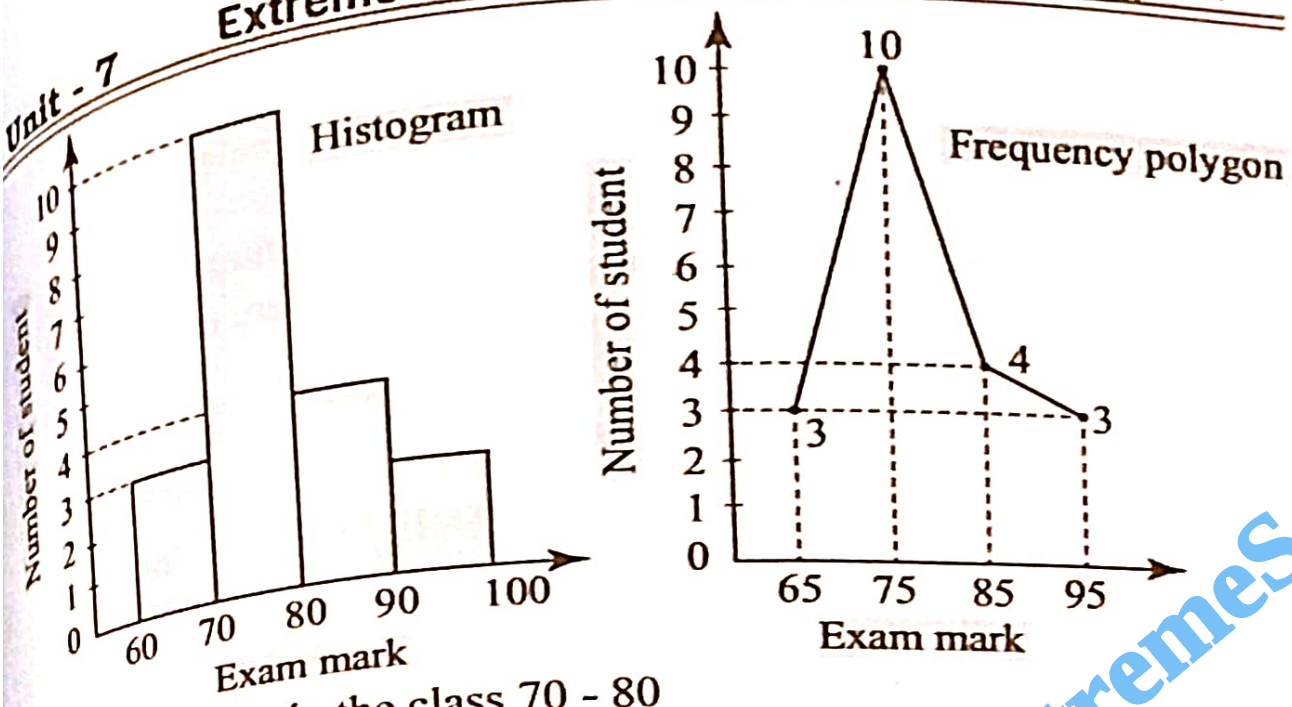

c. 10 students are in the class 70 - 80

d. 13 students are scored below 80.

e. 17 student are seconder above 70.

7.4 Measures of Central Tendency

Statics is the student of data analysis and interpretation. The mean the **median**, and the **mode** are single, centeral values that helps to describe a set of data. They are called **measures of centre tendency.**

- Mean is sum of data value divided by the total number of data values
- Median is middle value or mean of the two middle values that has been
- arranged in order of magnitude.
- Mode is most frequent data value.

7.4.1 Calculation of Mean of Data

(i) The mean of set of N data: $x_1, x_2, x_3, \ldots\ldots x_N$ is denoted by $\bar{x}$ and is defined as

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \ldots\ldots + x_N}{N}$$

(ii) If the data set $x_1, x_2, x_3, \ldots. x_n$ occur $f_1, f_2, f_3, \ldots. f_n$ times respectively, then the means is

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$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + ... + f_n x_n}{f_1 + f_2 + ... + f_k}$$

(iii) If f_1 numbers have mean m_1, f_2 numbers have mean m_2, ... f_k numbers have mean m_k, then mean of all the numbers is

$$\bar{x} = \frac{f_1 m_1 + f_2 m_2 + ... + f_k m_k}{f_1 + f_2 + ... + f_k}$$

(iv) Calculation of mean for Grouped frequency Distribution.

In frequency distribution we use the midpoint as representatives of the class intervals. $\frac{\sum_{i=1}^{n} f_i m_i}{\sum fi}$ where, is class mark.

$\bar{x} = \frac{\sum_{i=1}^{n} f_i m_i}{\sum_{i=1}^{n} f_i}$, where m_i is the class mark of class interval.

Illustrative Example

4. Marks obtained by 9 students in maths are given as follows: 50, 75, 40, 70, 43, 65, 35, 48, 40. Find their mean mark.

Solution

$$\bar{x} = \frac{50 + 75 + 40 + 70 + 43 + 65 + 35 + 48 + 40}{9} = \frac{468}{9} = 5$$

5. Calculate the mean monthly income (in thousand Birr) from the following table.

Monthly income (x)	2.5	3.0	3.5	4.0	4.25	5.0
Number of worker (f)	6	8	4	5	6	4

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Solution

$$\text{Mean: } \bar{x} = \frac{6 \times 2.5 + 8 \times 3 + 4 \times 3.5 + 5 \times 4 + 6 \times 4.25 + 5 \times 4}{6 + 8 + 4 + 5 + 6 + 4}$$

$$= \frac{15 + 24 + 14 + 20 + 27 + 20}{30} = 4.0$$

Hence, the mean monthly income of worked is Birr 4000.

6. Find the mean for grouped data is given below.

Class	6 - 10	11 - 15	16 - 20	21 - 25	26 - 30	31- 35	36 - 40
frequency (f)	1	2	3	5	4	3	2

Solution: The process for finding the mean for grouped data is

Step 1: Find the class mark (midpoints) of each class:

m = 8, 13, 18, 23, 28, 33 and 38

Step 2: multiply the frequency by the class mark

Step 3: Find the sum of step 2.

Class	6 - 10	11 - 15	16 - 20	21 - 25	26 - 30	31 - 35	36 - 40
x_m = class mark	8	13	18	23	28	33	38
f	1	2	3	5	4	3	2
$f \cdot m$	8	26	54	11 5	11 2	99	76

$$\therefore \text{Mean: } \bar{x} = \frac{\sum f \cdot m}{\sum f} = \frac{8 + 26 + 54 + 115 + 112 + 99 + 76}{20} = 245$$

7. Calculate the mean marks for each student in class from the following frequency distribution.

Marks (x)	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
No. of student (f)	12	18	27	20	17	6

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Solution

The mid - points of the class interval are m = 5, 15, 25, 35, 45 and 55.

x	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
x_m	5	15	25	35	45	55
f	12	18	27	20	17	6
$f_i m_i$	60	270	675	700	765	330

$$\bar{x} = \frac{\sum_{i=1}^{6} f_i m_i}{\sum f_i} = \frac{60 + 270 + 675 + 700 + 765 + 330}{12 + 18 + 27 + 20 + 17 + 6} = \frac{2800}{100} = 28$$

Hence, the mean is 28.

8. **Combined mean:** If the mean marks of one class of 500 students are 30 and the mean marks of another class. What is the mean of all the 150 students?

Solution

If $\bar{x}_1$, $\bar{x}_2$ are means of n_1, n_2 observations respectively, then the combined mean is

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{50 \times 30 + 100 \times 40}{50 + 100} = 36.4$$

9. The mean of 5 numbers is 27. If one is exclude their mean is 25. Find the excluded number.

Solution

- Total of 5 numbers = $27 \times 5 = 135$. Let the excluded number be = x
  Total of remaining 4 numbers = $4 \times 25 = 100$

$$100 + x = 135$$

$$x = 135 - 100 = 35$$

10. The mean of 11 number is 35. If the mean of first 6 number is 32 and that of last 6 number is 37, find the 6th number.

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Solution

$$6^{th} \text{ number} = \begin{pmatrix}\text{total of first}\\ 6\text{ - numbers}\end{pmatrix} + \begin{pmatrix}\text{total of last}\\ 6\text{ - numbers}\end{pmatrix} - \begin{pmatrix}\text{total of 11-}\\ \text{number}\end{pmatrix}$$

$$= (6 \times 32) + (6 \times 37) - (11 \times 35)$$

$$= 192 + 222 - 385 = 29$$

11. In a class, the mean score of girls in an examination is 73 and that of the boys is 71. The mean score for the whole class is 71.8. Find the percentage of girls and boys in the class.

Solution

Let n = the number of boys and m = the number of girls, then

$$\text{Combined mean} = \frac{71n + 73m}{n + m} = 71.8$$

$$71n + 73m = 71.8n + 71.8m$$

$$71.8n - 71n = 73m - 71.8m$$

$$0.8n = 1.2m \text{ implies } n = \frac{1.2}{0.8}m$$

$$n = \frac{3}{2}m, \text{ then } \frac{n}{n+m} \times 100\% = \frac{\frac{3}{2}m}{\frac{3}{2}m + m} \times 100 = 40\%$$

That is percentage of girls 40% and boys are 60%.

12. The mean of 31 results 60. If the mean of first 16 results is 58 and that of last 16 results is 62, find the 16th result.

Solution

Total of 31 results = $n\bar{x} = 31 \times 60 = 1860$

Total of first 16 result = $n\bar{x} = 16 \times 58 = 928$

Total of last 16 results = $n\bar{x} = 16 \times 62 = 992$

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$$\text{Value of } 16^{th} \text{ result} = \begin{pmatrix}\text{total of}\\ \text{first 16}\end{pmatrix} + \begin{pmatrix}\text{total of}\\ \text{last 16}\end{pmatrix} - \begin{pmatrix}\text{total of}\\ \text{31 results}\end{pmatrix}$$

$$= (928 + 992) - (1860) = 60$$

13. The mean weight of 150 students in a class is 60kg. The mean weigh of the boys is 70kg while that of the girls is 55 kg. Find the number boys and girls in the class.

Solution

Let the number of boys be n

Then the number of girls = 150 - n

$$\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2} = \frac{n.70 + (150 - n)55}{150} = 60$$

$$70n + 8250 - 55n = 9000$$

$$15n = 9000 - 8250 = 750$$

$$n = \frac{750}{15} = 50$$

Hence number of boys is 50 and that of girls is 100.

14. The average height of 30 students is 150cm. It was detected later that one values of 165cm was wrongly copied as 135cm for the computation of mean. find the correct mean

Solution

$$\text{correct mean} = \begin{pmatrix}\text{In correct}\\ \sum X\end{pmatrix} - \begin{pmatrix}\text{in correct}\\ \text{items}\end{pmatrix} + \begin{pmatrix}\text{correct}\\ \text{items}\end{pmatrix}$$

$$\sum X = n\bar{X} = 30 \times 150 = 4500$$

But this $\sum X$ is incorrect as 165 was misread as 135

correct value of $\sum X = 4500 - 135 + 165 = 4530.$

$$\text{correct mean} = \frac{\text{correct } \sum X}{n} = \frac{4530}{30} = -151cm$$

15. The mean marks second by 100 students was found to be 40. Later on it

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was observed that a score of 53 was mused as 83. Find the correct mean

Solution

$\sum X = n\bar{X} = (100)(40) = 4000$

Incorrect value of $\sum \bar{X} = 4000$

correct value of $\sum X = \begin{pmatrix}\text{In correct}\\ \text{valur of } \sum \bar{X}\end{pmatrix} - \begin{pmatrix}\text{in correct}\\ \text{item}\end{pmatrix} + \begin{pmatrix}\text{correct}\\ \text{item}\end{pmatrix}$

$= 4000 - 83 + 53 = 3970$

$$\text{correct mean} = \frac{\text{correct } \sum X}{n} = \frac{3970}{100} = 39.70$$

7.4.2 Calculation of Median

If all the values of a variable in the series are arranged in increasing or decreasing order of magnitude, then the middle . most or central value of the series gives the measure of the median

Let n = total number of items in the observation.

Case I: If n is odd, then median $= \left(\frac{n+1}{2}\right)^{th}$ item of the series.

Case II: If n is even, then median $= \frac{\left(\frac{n}{2}\right)^{th} \text{ item} + \left(\frac{n}{2}+1\right)^{th} \text{ item}}{2}$

Illustrative Example

16. Find the median of the following data: 31, 15, 47, 91, 12, 56, 29, 61, 66

Solution

Step 1: Arrange the data in order:

Step 2: select the middle value. 12, 15, 29, 31, 47, 56, 61, 65, 66, 91

Since there is an even number of n = 10 item

The middle value is 5^{th} and 6^{th} item MD $= \frac{47+56}{2} = \frac{103}{2} = 51.5$

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17. Find the median for the age of preschool childem. The ages are 1, 3, 4, 2, 3, 5 and 1

Solution

There are 7 observations : 1, 1, 2, 3, 3, 4, 5

$$\text{Madian} = \left(\frac{n}{2} + 1\right)^{th} \text{ item} = \left(\frac{7+1}{2}\right)^{th} \text{ item} = 4^{th} \text{ item} = 3$$

18. Weights of 10students (in kg) are 57, 53, 62, 40, 54, 44, 51, 63, 49 and 37.

a. Find the median weight

b. If the weight 63kg is replaced by 36kg, find the new median weight

Solution

a. Arrange the data in order 37, 40, 44, 48, 51, 53, 54, 57, 62, 63
Here, n = 10, which is even

$$\text{Madian} = \frac{\left(\frac{n}{2}\right)^{th} \text{ item} + \left(\frac{n}{2}+1\right)^{th} \text{ item}}{2} = \frac{(5\text{th} + 6\text{th}) \text{item}}{2}$$

$$= \frac{51+53}{2} = \frac{104}{2} = 52$$

b. If the weigh 63kg is replaced by 36 kg: The new increasing order: 36, 67, 40, 44, 49, 51, 54, 57, 62

$$\text{New median} = \frac{5^{th} \text{ item} + 6^{th} \text{ item}}{2} = \frac{49+51}{2} = 50$$

19. Consider the following data set 19, 25, 59, 48, 35, 31, 30, 32, 56.

a. Find the median of the 52,

b. If 25 is replaced by 52, what will be the new median.

Solution

a. Arrange the data in order 19, 25, 30, 31, 32, 35, 48, 56, 59
Here, n = 9, which is odd

$$\text{Median} = \left(\frac{n+1}{2}\right)^{th} \text{ item} = \left(\frac{9+1}{2}\right)^{th} \text{ item} = 32$$

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b. Arrange the data in new order: 19, 30, 31, 32, 35, 48, 52, 56, 59

Median = 5^{th} item = 35.

20. Out of total of 16 observations, arranged in descending order, the 8^{th} and 9^{th} observation are 21 and 26 respectively.

What is the median value of the 16 observations?

Solution

Here, n = 16, which is even

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{th} \text{ithem} + \left(\frac{n}{2}+1\right)\text{th item}}{2}$$

$$= \frac{8\text{th item} + 9\text{th item}}{2} = \frac{21+26}{2} = 23.5$$

21. Find the median of the following data:

x	6.7	6.2	8.4	6.9	7.1
f	3	5	7	7	8

A. 7.65 B. 8.4 C. 7 D. 7.3

Solution

First arrange the data in increasing order:

x	6.2	6.7	6.9	7.1	8.4
f	5	3	7	8	7
cum f	5	8	15	23	30

Here n = 30, which is even:

$$\therefore \quad \text{Median} = \frac{\left(\frac{n}{2}\right)^{th} \text{item} + \left(\frac{n}{2}+1\right)^{th} \text{item}}{2}$$

$$= \frac{15^{th} \text{ item} + 16\text{th item}}{2} = \frac{6.9+7.1}{2} = 7$$

Answer: C

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Median For Grouped Frequency Distribution

(i) First calculates the cumulative frequency corresponding to each class interval

(ii) The class which contains the medians is called the medians class

(iii) The median class corresponds to the middle of the total frequency $\left(\frac{n}{2}\right)$.

$$\text{Median} = B_L + \left(\frac{\frac{n}{2} - C f_B}{f_c}\right) \cdot i$$

Where, B_L = Lower limit of the median class.

n = total frequency

Cf_B = Cummulative frequency just before the median class

i = the size of class interval.

Illustrative Example

22. Calculate the median for a frequency distribution is as follows:

Class interval (x)	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80	80-90
Number of student (f)	2	18	24	20	8	4

Solution

First calculate the cummulative frequency

x	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80	80 - 90
f	2	18	24	20	8	4
cum . f	2	20	44	64	72	76

The median class corresponds to $\left(\frac{n}{2}\right)^{th}$ item $= \left(\frac{76}{2}\right)^{th}$ item $= 38^{th}$ item lies in class interval: 50 - 60.

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Hence, $B_L = 50$, $\frac{n}{2} = 38$, $f_c = 24$, $C f_B = 20$ and $i = 10$

$$Md = B_L + \left(\frac{\frac{n}{2} - c f_B}{f_c}\right).i$$

$$\text{Median} = 50 + \left(\frac{38 - 20}{24}\right).10 = 50 + \frac{180}{24} = 57.5$$

23. Calculate the median for frequency distribution of the following table:

x	6 - 10	11 - 15	16 - 20	21 - 25	26 - 30	31 - 35
f	2	3	4	6	8	1

Explanation

First make the data series make continuous by finding class - boundary and then calculate the cummulative frequency class 5.5 - 10.5 10.5 - 15.5 - 20.5 20.5 - 25.5 25.5 - 30.5 boundaries

Class boundries	5.5-10.5	10.5 -15.5	15.5-20.5	20.5-25.5	25.5-30.5	30.5-35.5
frequncy	2	3	4	6	8	1
cum. freq	2	5	9	15	23	24

The median class correaponds $\left(\frac{n}{2}\right)^{th}$ = item $\left(\frac{24}{2}\right)^{th}$ item

$= 12^{th}$ item lies in class interval 20.5 - 25.5

Hence, $B_L = 20.5$, $\frac{n}{2} = 12$, - $c f_B = 9$, $f_C = 6$ and i = 10.5 - 5.5 = 5

$$\therefore \quad \text{Median} = B_L + \left(\frac{\frac{n}{2} - C f_B}{fc}\right).i$$

$$\therefore \quad \text{Median} = 20.5 + \left(\frac{12 - 9}{6}\right)(5) = 23$$

24. Given below is the distribution of 50 students obtaining marks (x)in a

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certain examination

Mark (x)	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
Cum. frequency	5	13	28	44	50

What is the median of the data?

A. 31 B. 28 C. 29 D. 31

Solution

$$\text{Median} = \left(\frac{n}{2}\right)^{th} = \left(\frac{50}{2}\right)^{th} \text{ item} = 25^{th} \text{ item}$$

$\therefore$ Median class is 20 - 30. $B_L = 20$, i = 10

Frequncy of median class is: $f_c = 28 - 13 = 15$

$$\text{Median} = B_L + \frac{i}{f_c}\left(\frac{n}{2} - C f_B\right)$$

$$= 20 + \frac{10}{15}(25 - 13) = 28 \text{ marks}$$

Answer: B

25. The marks of 50 students is given below:

Marks (V)	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
No. of student	5	8	f_1	10	f_2

If the median of the data is 26, what are the values of f_1 and f_2?

A. $f_1 = 7, f_2 = 20$ C. $f_1 = 20, f_2 = 7$

B. $f_1 = 12, f_2 = 5$ D. $f_1 = 15, f_2 = 12$

Solution

V	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
f	5	8	f_1	10	f_2
$C.f$	5	13	$13 + f_1$	$23 + f_1$	$23 + f_1 + f_2$

Since, $23 + f_1 + f_2 = 50$, then $f_1 + f_2 = 50 - 23 = 27$

The median is 20 - 30 and $B_L = 20$, $Cf_B = 13$ and i = 10

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$$Md = B_L + \left(\frac{\frac{n}{2} - cf_m}{fc}\right) i = 20 + \left(\frac{25 - 13}{f_1}\right) 10 = 26$$

$\frac{120}{f_1} = 26 - 20 = 6$, hence $f_1 = \frac{120}{6} = 20$ **Answer: C**

7.4.3 Mode

The third measure of average is called the mode. The mode is the value that occurs most often in the data set is sometimes said to be most typical value.

The value that occurs most often in a data set is called the mode.

A data set can have more than one mode or no mode at all.

Example: the set 3, 3, 6, 8, 10, 10, 10, 11, 11, 12, 13 and 17 has **mode 10.**
Example: The set 4, 7, 10, 15, 17 and 18 has **no mode.**

Example: The set 5, 6, 7, 7, 7, 6, 6, 8, 8, 8, and 10 has **two modes**, 7 and 8, and is called **bimodal.**

In the case of grouped frequency distribution, mode is given by

$$\text{Mode} = B_L + \left(\frac{d_1}{d_1 + d_2}\right) i$$

Where B_L = lower limit of the modal class.

$d_i = |f_{modal} - f_1|$ and $d_2 = |f_{modal} - f_2|$

f_{modal} is frequency of model class

f_1 is frequency of pre modal class

f_2 is frequency of next class

i = size of class interval

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Illustrative Example

26. Calculate the mode of wage from the following data

Wage	0 - 100	100-200	200-300	300-400	400-500	500-600
Number of worker	8	12	25	15	10	6

Solution

- Since the highest frequency is 25, the modal class is 200 - 300
- B_L = lower limit of modal class = 200
- $f_{modal} = 25$, $d_1 = f_{modal} - f_1 = 25 - 12 = 13$, $d_2 = f_{modal} - f_2 = 25 - 15 = 10$
  $i = 100$

$$\text{Model} = B_L + \left(\frac{d_1}{d_1 + d_2}\right) i$$

$$= 200 + \left(\frac{13}{13 + 10}\right) 10 = 200 + \frac{13}{23} \times 100 = 256.32$$

27. Find the mode of the following data:

Class - interval	4 - 7	8 - 11	12 - 15	16 - 19	20 - 23
frequency	10	14	26	18	7

Solution

- Since the highest frequency is 26, the modal class is 11.5 - 15.5
- $B_L = 11.5$, $d_1 = 26 - 14 = 12$ and $d_2 = 26 - 18 = 8$
- $i = 7.5 - 3.5$

$$\text{Model} = B_L + \left(\frac{d_1}{d_1 + d_2}\right) i$$

$$= 11.5 + \left(\frac{12}{12 + 8}\right) 4 = 11.5 + 2.4 = 13.9$$

28. The expenditure of 100 families is given below:

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Expenditure	0 - 9	10 - 19	20 - 29	30 - 39	40 - 49
No. of families	14	23	f_1	21	f_2

If the mode of the data is 23.5, what are values of f_1 and f_2 ?

A. $f_1 = 27, f_2 = 15$

B. $f_1 = 15, f_2 = 27$

C. $f_1 = 25, f_2 = 17$

D. $f_1 = 17, f_2 = 25$

Solution

The mode of the grouped frequency distribution of data is given by:

$$\text{Mode} = B_L + \left(\frac{d_1}{d_1 + d_2}\right) i, \text{ where, } i = 10$$

$B_L = 19.5$, $d_1 = |f_1 - 23|$ and

$d_2 = |f_{mod} - f_{next}| = |f_1 - 21|$

$$23.5 = 19.5 + 10\left(\frac{f_1 - 23}{f_1 - 23 + f_1 - 21}\right)$$

$$10\left(\frac{f_1 - 23}{2f_1 - 44}\right) = \frac{4}{10} = 0.4 \text{ implies } f_1 - 23 = 0.8f_1 - 17.6$$

$$f_1 - 0.8f_1 = -17.6 + 23 = 5.4$$

$$0.2f_1 = 5.4 \text{ implies } f_1 = \frac{5.4}{0.2} = 27$$

Also, $f_2 = 100 - 85 = 15$

Answer: A

7.5 The Measurer of Position: Quartiles, Deciles and Percentiles

Here, we going to identify the position (or location) of a data value in a data set. The three measures of position are **quartiles deciles,** and **percentiles.**

- Median divides the data set into two equal parts.
- The quartile, Q_1, Q_2, and Q_3, which divides the data set into four equal parts.
- The nine values D_1, D_2, ... D_9 which divides the data set into ten equal

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parts are called *deciles*.

- The 99 values, P_1, P_2, P_3, ... P_{99}, which divides the data set into 100 equal part are called **percentile.**

7.5.1 Quartiles and Calculation of Quartile

In a set of data, the **quartiles** are values that divides the data set into four equal groups. The quartiles are denoted by Q_1, Q_2 and Q_3 named by the **lower quartile, middle quartile** and **upper quartile respectively.**

(i) The **lower quartiles** (Q_1) divides the lower half of the data into two equal parts.

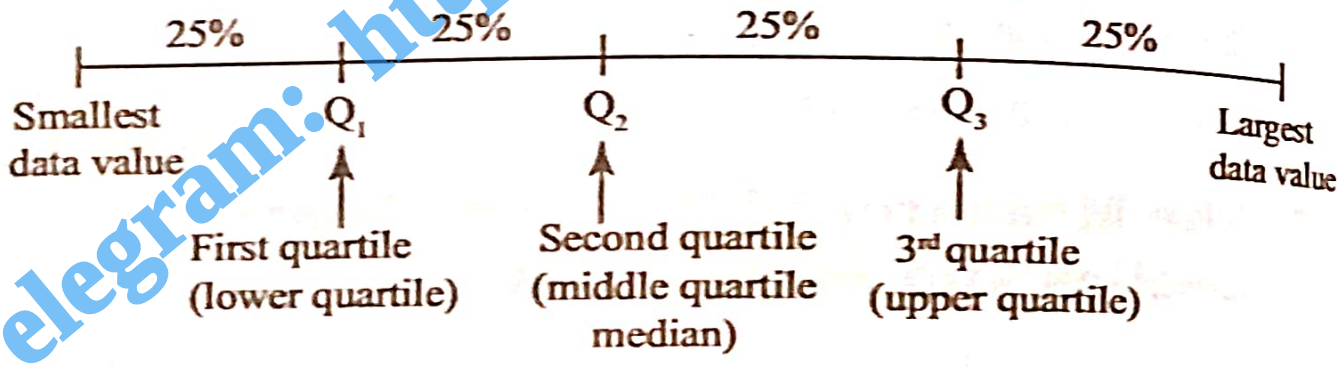
- 25% of the data value will fall less than or equal to Q_1 and 75% of the data value will fall more than or equal to Q_1.

(ii) The **middle quartiles** (Q_1) divides the data set into two equal parts.

- 50% of the data value will fall less than equal to Q_2 and 50% of the data value will fall more than or equal to Q_2.

(iii) The **upper quartile** (Q_2) divides the upper half of set into two equal parts.

- 75% of the data value will fall less than or equal to Q_3 and 25% of the data value will fall more than or equal to Q_3.



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DEFINITION: Interquartile Range

The difference between the upper quartile (Q_3) and the lower quartile (Q_1) is called the interquartile range . IQR = $Q_3 - Q_1$

Calculation of quartile For Ungrounded Data

Arrange the data in increasing order.

Let n = total number of items in the observation

Case I: If n is odd, then the K^{th} quartile: $Q_k = \left(\frac{k(n+1)}{4}\right)^{th}$, item.

Case II: If n is even then the K^{th} quartile:

$$Q_k = \left(\frac{\left(\frac{kn}{4}\right) + \left(\frac{kn}{4} + 1\right)}{2}\right)^{th} \text{ item}$$

Illustrative Example

29. Find the quartiles Q_1, Q_2, Q_3 and interquartile range of the data: 1, 8, 3, 6, 10, 27, 25, 19, 14, 17, 22, 26, 29. Interpret each quartiles.

Solution

STEP 1: Arrange the data in order:

1, 3, 6, 8, 10, 14, [17], 19, 22, 25, 26, 27, 29

STEP 2: Find the median Q_2: $n = 13$ median $= \frac{(n+1)^{th}}{2}$ item $= 7^{th} = 17$

$\therefore$ Q_2 is 17.

STEP 3: Find the median of the data values less than 17:

1, 3, [6, 8] 10, 14

$$Q_1 = \frac{6 + 8}{2} = \frac{14}{2} = 7$$

STEP 4: Find the median of the data values greater than 17:

19, 22, [25, 26], 27, 29.

$$Q_3 = \frac{25 + 26}{2} = 25.5$$

$\therefore$ $Q_1 = 7$, $Q_2 = 17$, $Q_3 = 25.5$ and IQR: $Q_3 - Q_1 = 2.5 - 7 = 18.5$

Interpretation:

- Lower quartile: $Q_1 = 7$ means 25% of the data set are less than or equal to 7 and 75% of the data set are greater than 7.
- Median: $Q_2 = 17$ means 50% of the data set are less than or equal and 50%

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of the data are greater than 17.

- Upper quartiles: $Q_3 = 25.5$ means 75% of the data are less than or equal 25.5 and 25% of the data set are greater than 25.5.

30. Find the qaurtiles Q_1, Q_2, and Q_3 for the data:
18, 22, 26, 34, 46, 20, 44, 42, 40, 28.

Solution

Arrange the data in increasing order,

18, 20, 22, 26, 28, 34, 40, 42, 44, 46. Here, n = 10, which is even: then the quartile: $Q_k = \left(\frac{\left(\frac{kn}{4}\right) + \left(\frac{kn}{4} + 1\right)}{2}\right)^{th}$ item.

$$Q_1 = \left(\frac{\frac{1.10}{4} + \frac{10}{4} + 1}{2}\right)^{th} \text{item} = \left(\frac{6}{2}\right)^{th} \text{item} = 3^{rd} \text{ item}$$

$$\therefore \quad Q_1 = 22$$

$$Q_2 = \left(\frac{\frac{2 \times 10}{4} + \frac{20}{4} + 1}{2}\right)^{th} \text{item} = (5.5)^{th} \text{ item}$$

$$Q_2 = 5^h \text{ item} + \frac{1}{2}\left(6^h - 5^{th}\right) \text{item}$$

$$Q_2 = 28 + \frac{1}{2}(34 - 28) = 28 + 3 = 31$$

$$Q_3 = \left(\frac{\frac{30}{4} + \frac{30}{4} + 1}{2}\right)^{th} \text{item} = 8^h \text{ item} = 42$$

$$\therefore \quad Q_1 = 22,\ Q_2 = 31 \text{ and } Q_3 = 42$$

31. The table shown below is the distribution of math scores in a class of

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33 students.

Mark	34	56	72	90	95	98
No. of Student	2	6	10	9	5	1

a. Calculate the quartile Q_1, Q_2, Q_3 and interquartile range.

b. Interprete each quartile division

Solution

Here, $n = 33$, which is odd, then the k^{th} quartile:

$$Q_k = \left(\frac{k(n+1)}{4}\right)^{th} \text{ item.}$$

a. i) $Q_1 = \left(\frac{1(33+1)}{4}\right)^{th}$ item $= \left(\frac{34}{4}\right)^{th}$ item $= (8.5)^{th}$ item

$$\therefore \quad Q_1 = 8^{th} \text{ item} + \frac{1}{2}\left(9^{th} \text{ item} - 8^{th} \text{ item}\right)$$

$$56 + \frac{1}{2}(72 - 56) = 64$$

ii) $Q_2 = \left(\frac{2(33+1)}{4}\right)^{th}$ item $= 17^{th}$ item $= 72$

iii) $Q_3 = \left(\frac{3(33+1)}{4}\right)^{th}$ item $= (25.5)^{th}$ item

$$\therefore \quad Q_3 = 25^{th} \text{ item} + \frac{1}{2}\left(26^{th} - 25^{th}\right) \text{ item}$$

$$= 90 + \frac{1}{2}(90 - 90) = 90$$

Inter quartile range: $IQR = Q_3 - Q_1 = 90 - 64 = 2$

b. Lower quartile: $Q_1 = 64$, means 25% of students are scoring less than or equal 64 and 75% of students are scoring more than 64.

- The median: $Q_2 = 72$, means 50% of students are scoring less than 72 mark and 50% of students area scoring more than or equal to 72 marks.

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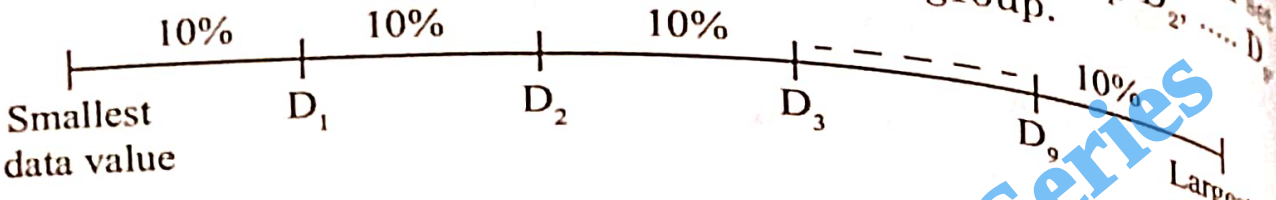
- The upper quartile: $Q_3 = 90$, means 75% of students are scoring mark less than or equal 90 and 25% of students are scoring than or equal to 90.

7.5.2 Deciles and Calculation of Deciles

In a set of data, the deciles are values that divides the data into 10 equal groups. The deciles are denoted by $D_1, D_2, \ldots D_9$ and divides the distribution into ten equal group.



- 10% of the data values will fall less than or equal to D_1
- 20% of the data values will fall less than or equal to D_2.
- 30% of the data values will fall below D_3, and so on.

The j^{th} deciles of the data means j% of the data will fall below the value D_j

Calculation of Deciles for Ungrouped Data Set.

Arrange the data in increasing order.

Let n = total number of items in the observation.

Case I: If n is odd, then the j^{th} deciles: $D_j = \left(\frac{j(n+1)}{10}\right)^{th}$ item

Case II: If n is even, then the j^{th} deciles:

$$D_j = \left(\frac{\left(\frac{jn}{10}\right) + \left(\frac{jn}{10} + 1\right)}{2}\right)^{th} \text{ item}$$

Illustrative Example

32. Find D_2, D_3 and D_8 for each of the following data set:

a. 78, 74, 80, 72, 76, 82, 86, 71, 90, 91, 95, 97, 98

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b. 60, 58, 48, 64, 66, 63, 65, 68, 70, 72, 46, 74.

Solution:

a. Arrange the data set in increasing order:

71, 72, 74, 76, 78, 80, 82, 86, 90, 91, 95, 97, 98

Here, n = 13, which is odd we use the formula, decile:

$$D_j = \left(\frac{j(n+1)}{10}\right)^{th} \text{ item}$$

$$D_2 = \left(\frac{2(13+1)}{10}\right)^{th} \text{ item} = (2.8)^{th} \text{ item}$$

$= 2^{nd}$ item $+ 0.8\ (3^{rd} - 2^{nd})$ item

$= 72 + 0.8\ (74 - 72)$

$D_2 = 73.6 \longleftarrow$ 20% of the data is below 73.6

$$D_3 = \left(\frac{3(13+1)}{10}\right)^{th} \text{ item} = (4.2)^{th} \text{ item}$$

$= 4^{th}$ item $+ 0.2\left(5^{th} - 4^{th}\right)$ item

$\therefore \quad D_3 = 76 + 0.2\ (78 - 76) = 76.4$

30% of the data fall below 76.4

$$D_8 = \left(\frac{8(13+1)}{10}\right) \text{ item} = (11.2)^{th} \text{ item}$$

$= 11^{th} + 0.2(12^{th} - 11^{th})$ item

$= 95 + 0.2\ (97 - 95)$

$D_8 = 95 + 0.4 = 95.4$

$\therefore$ 80% of the data will lie below 95.4 and the rest are above it.

b. Arrange the data in increasing order: 46, 48, 58, 60, 63, 64, 65, 66, 68, 70, 72, 74. Here, n = 12, which is even, we use the formula:

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$$\text{Decile: } D_j = \left(\frac{\left(\frac{jn}{10}\right) + \left(\frac{jn}{10} + 1\right)}{2} \right)^{th} \text{ item}$$

$$D_2 = \left(\frac{\frac{2\times12}{10} + \frac{2\times12}{10}}{2} \right)^{th} \text{ item} = (2.9)^{th} \text{ item}$$

$D_2 = 2^{nd}$ item $+ 0.9\ (3^{rd} - 2^{nd})$ item

$= 48 + 0.9\ (58 - 48) = 57$

$\therefore$ 20% of the data value below 57.

$$D_3 = \left(\frac{\frac{3\times12}{10} + \frac{3\times12}{10} + 1}{2} \right)^{th} \text{ item} = (4.1)^{th} \text{ item}$$

$= 4^{th}$ item $+ 0.1\ (5^{th} - 4^{th})$ item

$\therefore \quad D_3 = 60 + 0.21\ (63 - 60) = 60.3$

$\therefore$ 30% of the data value lies below 60.3 and the rest above it.

$$D_8 = \left(\frac{\frac{8\times12}{10} + \frac{8\times12}{10} + 1}{2} \right)^{th} \text{ item} = (10.2)^{th} \text{ item}$$

$= 10^{th}$ item $+ 0.2\ (11^{th} - 10^{th})$ item

$\therefore \quad D_8 = 70 + 0.2\ (72 - 70) = 70.4$

80% of the data value lies below 70.4.

33. Consider the following frequency distribution table:

Value	5	3	9	6	14
frequency	3	6	6	3	2

a. What is the sixth decile?

A. 7.5 B. 7 C. 6.5 D. 5

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Solution

First arrange the data in increasing order:

V	3	5	6	9	14
f	6	3	3	6	2
$c.f$	6	9	12	18	20

Here, $n = 20$, thus $D_6 = \left[\frac{\frac{6(20)}{10} + \frac{6(20)}{10} + 1}{2}\right]^{th}$ item $= (12.5)^{th}$ item

$= 12^{th}$ item $+ 0.5\,(13^{th} - 12^{th})$ item $= 6 + 0.5\,(9 - 6) = 7.5$

Answer: A

b. What is the ninth decile?

A. 9 B. 9.5 C. 11 D. 11.5

Solution:- Using the formula:

$D_9 = (18.5)^{th}$ item $+ 0.5\,(19^{th} - 18^{th})$ item

$= 9 + 0.5\,(14 - 9) = 9 + 2.5 = 11.5$

Answer: D

7.5.3 Percentiles

Percentiles divide the data set into 100 equal groups.

Percentiles are position measures used in education fields to indicate the position of an individual in a group. For example, in examination such as the National leaving examination certificate are taken by students in Grade - 8. A student's scores are compared with those of other students locally and nationally by using percentile ranks.

Note: Percentiles are not the same as percentage. As example.

- If Hana scored 78 corresponding to to the 84^{th} percentile, means, she did better than 84% of the student in her class.
- The t^{th} Percentiles are denoted by $P_1, P_2, P_3, \ldots\ldots P_{99}$ and divide the data set into 100 groups.

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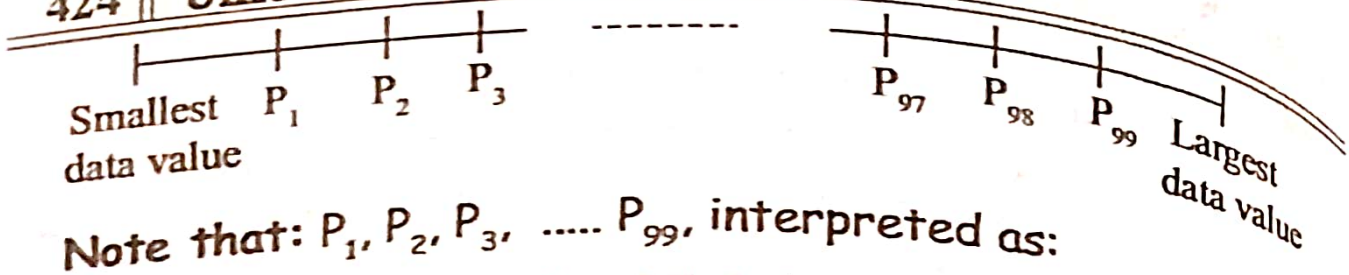

Note that: $P_1, P_2, P_3, \ldots P_{99}$, interpreted as:

- 1% of the data falls below P_1.
- 3% of the data falls below P_3
- 60% of the data falls below P_{60}
- t% of the data falls below the t^{th} percentile P_t.

Calculation of Percentile for Ungrounded Data

Arrange the data in order for smallest to largest. Let n = total number of item.

- **Case I :** If n is odd, then the percentile: $P_t = \left(\frac{t\,(n+1)}{100}\right)^{th}$ item
- **Case II:** If n is even, then the percentile: $P_t = \left[\frac{\left(\frac{tn}{100}\right) + \left(\frac{tn}{100}+1\right)}{2}\right]^{th}$ item

Illustrative Example

34. A teacher gives a 25 - point test to 10 students. The scores are shown here: 23, 20, 17, 11, 13, 7, 10, 25, 15, 8. Find the value corresponding to the

a. 25^{th} percentile b. 87^{th} percentile c. 60^{th} percentile

Solution

Arrange the data in order from smallest to the largest. 7, 8, 10, 11, 13, 15, 17, 20, 23, 25. Then substitute in to the formula. Hence. n = 10 which is even.

a. $P_{25} = \left[\frac{\frac{25\times 10}{100} + \left(\frac{25\times 10}{100}+1\right)}{2}\right]^{th}$ item $= 3^{rd}$ item $= 10$

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Hence, 10 corresponds to the 25th percentile. Thus, a student whose score was 10 did better than 25% of the class.

b. $P_{87} = \left[\frac{\left(\frac{87 \times 10}{100}\right) + \left(\frac{87 \times 10}{100} + 1\right)}{2}\right]^{th}$ item $= (9.2)^{rd}$ item

$P_{87} = 9^{th}$ item $+ 0.2\ (10^{th} - 9^{th})$ item

$\therefore \quad P_{87} = 23 + 0.2\ (25 - 23) = 23.4.$

Hence, 23.4 corresponds to the 87th percentile. Anyone scoring 23.4 would have done better than 87% of the class. In other word: P_{87} = 23.4 means 87% of the student are scoring less than 23.4.

c. $P_{60} = \left[\frac{\frac{60 \times 10}{100} + \frac{60 \times 10}{100} + 1}{2}\right]^{th}$ item $= (6.5)^{rd}$ item

$P_{60} = 6^{th}$ item $+ 0.5\ [7^{th}$ item $- 6^{th}$ item$]$

$P_{60} = 15 + 0.5\ [17 - 15] = 16$

Hence, 16 is 60th percentile. Thus, a student whose score was 16 did better than 60% of the class. In other word, $P_{60} = 16$ means 60% of the students are scoring less than 16.

35. Consider the following frequency distribution table:

V	10	12	14	16	18	20
f	2	7	10	8	7	5

Find: a. 48th percentile and interprete

b. 69th percentile and interprete

c. 87th percentile and interprete

Solution

Substitute in the formula. Here, n = 39, is odd, then

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Percentile: $P_t = \left(\frac{t\,(n+1)}{100}\right)^{th}$ item

a. $P_{48} = \left(\frac{48\,(39+1)}{100}\right)^{th}$ item $= (19.2)^{th}$ item

$= 19^{th}$ item $+ 0.2\,(20^{th} - 19^{th})$ item

48^{th} percentile $= 14 + 0.2\,(18 - 14) = 14.8$

Interpretation: $P_{48} = 14.8$ means 48% of the data value are less than 14

b. $P_{69} = \left(\frac{69\,(39+1)}{100}\right)^{th}$ item $= (27.6)^{th}$ item

$= 27^{th}$ item $+ 0.6\,(28^{th} - 27^{th})$ item

$\therefore$ 69^{th} percentile: $P_{69} = 16 + 0.6\,(20 - 16) = 18.4$

Interpretation: $P_{69} = 18.4$ means 69% of the data are below 18.4 and t rest are above it

c. $P_{87} = \left(\frac{87\,(39+1)}{100}\right)^{th}$ item $= (34.8)^{th}$ item

$= 34^{th}$ item $+ 0.8\,(35^{th} - 34^{th})$ item

$\therefore$ 87^{th} percentile: $P_{87} = 18 + 0.8(20 - 18) = 19.6$

Interpretation: $P_{87} = 19.6$ means 87% of the data are below 19.6.

Finding Percentile Formula For given Data Value

The percentile corresponding to a given value X is calculate by using

$$\text{Percentile} = \frac{(\text{Number of value below X}) + 0.5}{\text{total number of value}} \times 100\%$$

Illustrative Example

36. A teacher gives a 30 - point to 10 students. The scores are shown her 28, 25, 22, 16, 18, 12, 13, 15, 30, 20

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a. Find the percentile rank of a student whose score was 22.

b. Find the percentile rank of a student whose score was 16.

Solution

First arrange the data in order from smallest to largest.

12, 13, 15, 16, 18, 20, 22, 25, 28, 30.

a. Substitute into formula:

$$\text{Percentile} = \frac{(\text{Number of value below 22}) + 0.5}{\text{total number of value}} \times 100\%$$

There are six values blow 22. Thus, percentile =

$$= \frac{6 + 0.5}{10} \times 100\% = 65 \text{ percentile}$$

Thus, a student whose score was 22 did better than 65% of the class

b. $$\text{Percentile} = \frac{(\text{Number of value below 16}) + 0.5}{\text{total number of value}} \times 100\%$$

$$= \frac{3 + 0.5}{10} \times 100\% = 35 \text{ percentile}$$

37. For each test score in the data set 18, 22, 26, 28, 32, 37, 39, 40. What value corresponds to the interquartile range (IQR)?

A. 19.5 B. 10 C. 14 D. 19

Solution: Step 1: First find the median: 18, 22, 26, $\boxed{28, 32}$, 37, 39, 40

$$\text{Median}: \frac{28 + 32}{2} = 30$$

Step 2: Find median of the data values less than 30:

$$18, \boxed{22, 26}, 28, \text{ so that } Q_1: \frac{22 + 26}{2} = 24$$

Step 3: Find the median of the data value greater than 30 are

$$32, \boxed{37, 39}, 40, \text{ so that } Q_3: \frac{37 + 39}{2} = 38$$

$$\therefore \quad IQR = Q_3 - Q_1 = 38 - 24 = 14$$

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38. For each test score in the data set 18, 22, 26, 28, 32, 37. What value corresponds to the 60th percentile?

A. 27.2 B. 26 C. 28.4 D. 28

Solution: n = 6

$$P_{60} = \frac{\left(\frac{60 \times 6}{100} + \frac{60 \times 6}{100} + 1\right)^{th}}{2} \text{ item} = (4.1)^{th} \text{ item}$$

$= 4^{th}$ item + 0.1 (5^{th} - 4^{th}) item

60th percentile: = 28 + 0.1 (32 - 28) = 28.4

39. For each test score in the data set: 42, 35, 12, 47, 28, 49, 50. What value corresponds to the 72nd percentile?

A. 47 B. 48.52 C. 35.5 D. 48

Solution: n = 7: First arrange the data in order: 12, 28, 35, 42, 47, 49, 50

$$P_{72} = \left(\frac{72(7+1)}{100}\right)^{th} \text{ item} = (5.76)^{th} \text{ item}$$

$= x_5 + 0.76 (x_6 - x_5) = 47 + 0.76 (49 - 47) = 48.52$

40. The data shown below represent the scores on 30 - point maths test for the group of 11th grade students

Mark . x	12	16	17	19	21	22	30
No. student	9	12	13	7	6	5	1

Find the value corresponding to the D_4, D_8, P_{24}, P_{64} and P_{98}.

Solution: n = 53

To find the decile: $D_j = \left(\frac{j(n+1)}{10}\right)^{th}$ item

$$D_4 = \left(\frac{4(53+1)}{10}\right)^{th} \text{ item} = (21.6)^{th} \text{ item}$$

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$D_4 = 21^{th}$ item $+ 0.6(22^{th} - 21^{th})$ item

$\therefore D_4 = 16 + 0.6\,(17 - 16) = 16.6$

$\therefore$ To find $D_8 : D_8 = \left(\frac{8(53+1)}{10}\right)^{th}$ item $= (43.2)^{th}$ item.

$D_8 = 43^{rd}$ item $+ 0.2(44^{th} - 43^{rd})$ item

$\therefore D_8 = 21 + 0.2\,(21 - 21) = 21$

To find percentile: $P_t = \frac{(t(n+1))^{th}}{100}$ item

The value corresponding to the

$P_{24} = \left(\frac{24\,(53+1)}{100}\right)^{th}$ item $= (12.96)^{th}$

$= 12^{th}$ item $+ 0.96\,(13^{th} - 12^{th})$ item

$\therefore P_{24} = 16 + 0.96\,(16 - 16) = 16$

The value corresponding to the P_{64} is:

$P_{64} = \left(\frac{64(53+1)}{100}\right)^{th}$ item $= (34.56)^{th}$ item

$P_{64} = 34^{th}$ item $+ 0.56\,(35^{th} - 34^{th})$ item

$\therefore P_{64} = 17 + 0.56\,(19 - 17) = 18.12$

The value corresponding to the P_{98} is:

$P_{98} = \left(\frac{98(53+1)}{100}\right)^{th}$ item $= (52.92)^{th}$ item

$= 52^{th}$ item $+ 0.92\,(53^{th} - 52^{th})$ item

$\therefore P_{98} = 22 + 0.92\,(30 - 22) = 29.36$

41. The following table shows a simple frequency distribution of the mark that students scored in test.

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Score (mark)	4	5	6	7	8	9	10
Frequency	5	8	9	14	7	5	3

42. Which of the following is **not** true about frequency distribution?

A. The inter - quartile range of the mark is 3

B. The 25th percentile of the mark is 5.

C. The top 25% of scores are 8 and above.

D. The 75th percentile of the mark is 7.

Solution $n = 5 + 8 + 9 + 14 + 7 + 5 + 3 = 51$

i. Q_1 is $\left(\frac{1(51+1)}{4}\right)^{th}$ item $= 13^{th}$ item $= 5$

25^{th} percentile $= Q_1 = P_{25} = 5$

ii. Q_3 is $\left(\frac{3(51+1)}{4}\right)^{th}$ item $= 39^{th}$ item $= 8$

$Q_3 = P_{75} = 8$

$Q_3 = P_{75} = 8$ means 75% of scores are 8 or below (or 25% of scores are 8 or above)

$IQR = Q_3 - Q_1 = 8 - 5 = 3$

Answer: D

7.6 Calculating Quartile, Decile and Percentiles of Grouped Frequency Data

Calculating and Interpretation of Quartile For grouped data

For grouped frequency distribution of data, the K^{th} quartile value is obtained by:

$$Q_k = B_L + \frac{i}{f_c}\left(\frac{kn}{4} - c f_B\right), k = 1, 2, 3$$

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Where

- The K^{th} quartile class is $\left(\frac{kn}{4}\right)^{th}$ item
- B_L = Lower limit of the K^{th} quartile class,
- n = total number of observation
- f_c = frequency of the K^{th} quartile class
- i = the size of the class interval.
- Cf_B = Cummulative frequency before the K^{th} quartile.

Calculating Interpretation of Deciles for grouped data

The j^{th} decile of a grouped frequency distribution data is obtained by:

$$D_j = B_L + \frac{i}{f_c}\left(\frac{tn}{10} - Cf_B\right), j = 1, 2, 3, ... 9$$

Where

- The j^{th} decile class is: $\left(\frac{jn}{10}\right)^{th}$ item
- B_L = Lower limit of the j^{th} decile class.
- i = The size (width) of the class interval.
- f_c = The frequency of the j^{th} decile class.
- Cf_B = Cummulative frequency before the j^{th} decile class.
- n = Total number of observation.

Calculating and interpretation of the t^{th} percentile of group data

The t^{th} percentile of a grouped frequency distribution data is obtained by:

$$P_t = B_L + \frac{i}{f_c}\left(\frac{tn}{100} - Cf_B\right), t = 1, 2, 3, ... 9$$

Where

- n = total number of observation
- The t^{th} percentile class is $\left(\frac{tn}{100}\right)^{th}$ item

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- B_L = Lower limit (boundary) of the class
- i = The size of the class
- f_c = The frequency of the t^{th} percentile class
- Cf_B = Commulative frequency before the t^{th} percentiles

Illustrative Example

43. The following distribution represents the data for weights of grade students

Weight	52.5-55.5	55.5-58.5	58.5-61.5	61.5-64.5	64.5-67
frequency	9	12	18	21	15

Find a. the quartiles: Q_1, Q_2, Q_3 and IQR

b. the deciles: D_1, D_3, D_9 and Interprete

c. the percentiles: P_{15}, P_{48}, P_{80} and interprete

Solution

First determine the corresponding cummulative

Weight	52.5-55.5	55.5-58.5	58.5-61.5	61.5-64.5	64.5-67.5
frequency	9	12	18	21	15
Cum. Frequency	9	21	39	60	75

quareile the K^{th} quartile: $Q_k = B_L + \frac{i}{f_c}\left(\frac{kn}{4} - C f_B\right)$

a. The first quartile Q_1 class is obtained by

$$\left(\frac{Kn}{4}\right)^{th} \text{ item} = \left(\frac{1 \times 75}{4}\right)^{th} \text{ item} = (18.75)^{th} \text{ item}$$

It lies in 2nd class: 55.5 - 58.5

$B_L = 55.5$, $i = 3$, $f_c = 12$, and $C f_B = 9$

$$\therefore \quad Q_1 = 55.5 + \frac{3}{12}(18.75 - 9) = 57.94$$

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The 2nd quartile Q_2 class is

$\left(\frac{kn}{4}\right)^{th}$ item $= \left(\frac{2 \times 75}{4}\right)^{th}$ item $= (37.5)^{th}$ item

Q_2 lies in 3rd class :58.5 - 61.5

$B_L = 58.5$, i = 3, $f_c = 17$, and $Cf_B = 21$

$\therefore$ $Q_2 = 58.5 + \frac{3}{18}\left(\frac{2 \times 75}{4} - 21\right) = 61.25$

$\therefore$ The 3rd quartile class is $\left(\frac{3n}{4}\right)^{th}$ item $= \left(\frac{3 \times 75}{4}\right)^{th}$ item $= (56.25)^{th}$ item

Q_3 lies in 4th class: 61.5 - 64.5

$B_L = 61.5$, i = 3, $f_c = 21$ and $Cf_B = 39$

$\therefore$ $Q_3 = 61.5 + \frac{3}{21}(56.25 - 39) = 61.5 + 2.46 = 63.96$

$\therefore$ $Q_1 = 57.94$, Q_2, 61.25 and $Q_3 = 63.94$

Hence, 25% of the student weight less than or equal to 57.94

- 50% of the student weight less than or equal to 61.25
- 75% of the student weight less than or equal to 63.96

b. The jth decile: $D_j = B_L + \frac{i}{f_c}\left(\frac{jn}{10} - c f_B\right)$

The 1st decile class is

$\left(\frac{jn}{10}\right)^{th}$ item $= \left(\frac{1 \times 75}{10}\right)^{th}$ item $= (7.5)^{th}$ item

It lies in 1st class: 52.5 - 55.5, $B_L = 52.5$, i = 3, $f_c = 9$, and $Cf_B = 0$.

$\therefore$ $D_1 = 52.5 + \frac{3}{9}(7.5 - 0) = 25.5 + 2.5 = 55$

The 3rd decile class is

$\left(\frac{jn}{10}\right)^{th}$ item $= \left(\frac{3 \times 75}{10}\right)^{th}$ item $= (22.5)^{th}$ item

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In lies in 3rd class: 58.5 - 61.5

$B_L = 58.5$, $i = 3$, $f_c = 18$ and $Cf_B = 21$

$$\therefore \quad D_3 = 58.5 + \frac{3}{18}(22.5 - 21) = 58.5 + 0.25 = 58.75$$

The 9^{th} decile, class is $\left(\frac{jn}{10}\right)^{th}$ item

$$= \left(\frac{9 \times 75}{10}\right)^{th} \text{ item} = (67.5)^{th} \text{ item}$$

It lies on 5^{th} class: 64.5 - 67.5, $B_L = 64.5$, $i = 3$, $f_c = 15$ and $Cf_B = 60$

$$\therefore \quad D_9 = 64.5 + \frac{3}{15}(67.5 - 60) = 64.5 + 1.5 = 66$$

Hence, 10% students weight is 55 or less, 30% of student weight is 58.75 or less and 90% of student weight 66 less than or equal to 66.

C. Ex

44. The following table shows a frequency distribution of score on final examination in physics

Scores	30-39	4-49	50-59	60-69	70-79	80-89	90-100
No. of Student	1	3	11	21	43	32	9

a. Find the quartile: Q_1, Q_2, Q_3 and interpret

b. Find the decile: D_2, D_6, D_9 and interpret

c. Find the percentile: P_{12}, P_{40}, P_{90} and interpret.

Solution: $n = 1 + 3 + 11 + 21 + 43 + 32 + 9 = 120$

a. i. 1st quartile class : Q_1 is $\left(\frac{1 \times 120}{4}\right)^{th}$ item $= 30^{th}$ item

Q_1 lies in 4^{th} class: 60 - 69

$B_L = 60$, $i = f_c = 21$, and $Cf_B = 1 + 3 + 11 = 15$

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$$\therefore \quad Q_1 = B_L + \left(\frac{n}{4} - c f_B\right)\frac{i}{f_c} = 60 + (30 - 15)\left(\frac{10}{21}\right) = 67$$

ii. The 2nd quartile class: Q_2 is $\left(\frac{2 \times 120}{4}\right)^{th}$ item $= 60^{th}$ item

Q_2 lies on 5th class: 70 - 79

$B_L = 70, i = 10, f_c = 43$, and $C f_B = 1 + 3 + 11 + 21 = 36$

$$\therefore \quad Q_2 = B_L + \left(\frac{2n}{4} - C f_B\right)\frac{i}{f_c} = 70 + (60 - 36)\left(\frac{10}{43}\right) = 70 + 5.6 = 75.6$$

iii. Q_3 is $\left(\frac{3 \times 120}{4}\right)^{th}$ item $= 90^{th}$ item

Q_3 lies n 6th class: 80 - 89

$B_L = 80, i = 10, f_c = 32, Cf_B = 1 + 3 + 11 + 21 + 43 = 79$

$$\therefore \quad Q_3 = B_L + \left(\frac{3n}{4} - C f_B\right)\frac{i}{f_c} = 80 + (90 - 79)\frac{10}{32} = 80 + 3.4 = 83.4$$

Interpretation:

- Lower quartile: $Q_1 = 67$ means 25% of student scored 67 or below (or 75% scored 67 or above)
- Middle quartile: $Q_2 = 75.6$ means 50% scored 75.6 or below.
- Upper quartile: $Q_3 = 83.4$ means 75% scored 83.4 or below (or the rest 25% scored 83.4 or above)

b. Exercise left of you

c. Exercise left for you

45. Measurement is grouped into five class intervals with the following frequency distribution

Class-interval	5 - 15	15 - 25	25 - 35	35 - 45	45 - 55
Frequency	22	40	68	50	20

What are the corresponding values of the first, Q_3 and the 75th percentile

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P_{75} of the measurement

A. $Q_1 = 20, P_{75} = 40$

B. $Q_1 = 22, P_{75} = 40$

C. $Q_1 = 20, P_{75} = 39$

D. $Q_1 = 22, P_{75} = 39$

Solution: $n = 22 + 40 + 68 + 50 + 20 = 200$

The lower quartile:Q_1 is $\left(\frac{1 \times 200}{4}\right)^{th}$ item $= 50^{th}$ item Q_1 is lies on 2^{nd} class: 15 - 25

$B_L = 15, i = 10, f_c = 40, Cf_B = 22$ and $\frac{kn}{4} = 50$

$$Q_1 = B_L + \frac{i}{f_c}\left(\frac{kn}{4} - Cf_B\right) = 15 + \frac{10}{40}(50 - 22)\ 22$$

The 75^{th} percentile: P_{75} is $\left(\frac{75 \times 200}{100}\right)^{th}$ item $= 150^{th}$ item

75^{th} percentile is lies on 4^{th} class: 35 - 45

$B_L = 35, i = 10 f_c = 50, Cf_B = 22 + 40 + 68 = 130$

$$P_{75} = 35 + \frac{10}{50}\left(\frac{75 \times 200}{100} - 130\right) = 35 + 4 = 39$$

Answer: D

46. Find the quartile Q_1, Q_2, and Q_3 of the following frequency distribution table. It is mark obtained in a mathematics exam (out of 60)

Score	30 - 34	35 - 39	40 - 44	45 - 49	50 - 54	55 - 59
No. of student	7	12	3	5	3	6

a. If students in the top 25% are to be awarded certificate of "BEST IN MATH", what is the minimum mark for a certificate

b. If students whose scores are in the bottom 25% of the marks are consider as failures, then what is the maximum failing mark?

c. If student get more than 75 percentile, find the minimum 75 percentile, find the minimum mark he/she get.

d. Find the pass mark if 70% of the students to pass.

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Solution

a. Q_3 is $\left(\frac{3 \times 36}{4}\right)^{th}$ item $= 27^{th}$ item

Q_3 lies on 4^{th} class: 45 - 49

$B_L = 45, i = 5, f_c = 5, CF_B = 7 + 12 + 3 = 22$

$Q_3 = 44.5 + \frac{5}{5}(27 - 22) = 44.5 + 5 = 49.5$

$\therefore$ $Q_3 = 49.5$ is a minimum mark to award certificate

b. Q_1 is $\left(\frac{1 \times 36}{4}\right)^{th}$ item $= 9^{th}$ item

Q_1 lies on 2^{nd} class: 35 - 39

$B_L = 34.5, i = 5, f_c = 12$, and $CF_B = 7$

$Q_1 = 34.5 + \frac{5}{12}(9 - 7) = 34.5 + 0.8 = 35.3$

$Q_1 = 35.3$ is a maximum failing mark.

c. It is the same as $Q_1 = P_{25} = 35.3$.

d. If 70% is to pass then 30% is to fail, thus it is the 30 percentile.

P_{30} is $\left(\frac{30 \times 36}{100}\right)^{th}$ item $= (10.8)$ item

P_{30} is lies on 2^{nd} class: 35 - 39.

$P_{30} = 34.5 + \frac{5}{12}(10.5 - 7) = 36.08$

$\therefore$ The minimum passing mark is 36.08.

47. The following table gives the distribution of marks secured by grade 8 - students in examination.

Marks	0 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80
No. of Student	42	38	120	84	48	36	31

What is the percentage failure if the minimum to pass is 34 mark

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A. 32% B. 30% C. 40% D. 66%

Solution

The minimum mark 34 lies on interval: 30 - 40

Let x = The number of students less than 34 is the cummulative frequency corresponding to 34 marks

$\therefore \quad B_L = 30, i = 10, f_c = 120, C f_B = 42 + 38 = 80$

We use: $B_L + \frac{i}{f_c}(x - C f_B) = 34$

$$30 + \frac{10}{120}(x - 80) = 34$$

$$\therefore \quad x - 80 = (34 - 30)(12) = 48$$

$$x = 48 + 80 = 128$$

The percent of failure $= \frac{128}{399} \times 100 = 32\%$

Answer: A

48. Given below the age-wise distribution of 420 employees in extreme book printing company.

Age (year)	Below 20	20 - 25	25 - 30	30 - 35	35 - 40	40 - 45	45 - 50	Above 50
No. of Employees	13	29	46	60	112	94	45	21

What is the maximum age of 240 employer in Extreme book printing?

A. 35 years B. 39.1 years C. 42.9 years D. 44 years

Solution

Consider the cummulative frequency table

Age (years)	Below 20	20 - 25	25 - 30	30 - 35	35 - 40	40 - 45	45 - 50	> 50
frequency	13	29	46	60	112	94	45	21
Cum. frequency	13	42	88	148	260	354	399	420

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240^{th} item (employees) belongs to the class: 35 - 40.

$B_L = 35, i = 5, f_c = 112, Cf_B = 148$

Then: $35 + \frac{5}{112}(240 - 148) = 39.107$

Answer: B

Conceptual example

Describe which **measure of central tendency:**

A. Mean, B. Median C. Mode

Was probably used in each situation for 50 - 54

49. Most people preferred convertibles over any other colour.

Answer: C - Mode

50. One - half of the students in a school scores more than 58 and one - half scores less than 58 in a semester exam.

Answer : B - median

51. The average number of children per family in Ethiopia is 2.4

Answer: A - Mean

52. The most common fear today is fear of speaking in public.

Answer: C - mode

53. The daily high temperature during a week in December.

Answer: A - Mean.

54. Define percentile

Explanation

Percentiles divide the data set into 100 equal groups.

55. What is the difference between a percentage and a percentile?

Explanation:

A **percentile** is a relative measurement of a position but a percentage is absolute measure of the part to the total.

56. What is the relationship between quartile and percentile?

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Explanation

$Q_1 = P_{25}$, $Q_2 = P_{50}$ and $Q_3 = P_{75}$

57. How are deciles rated to percentile?

Explanation

$D_1 = P_{10}$, $D_2 = P_{20}$, $D_3 = P_{30}$, ... $D_9 = P_{90}$

58. To which percentile, quartile and decile does the median correspond?

Explanation: The median is the same as P_{50}, Q_2 and D_5

i.e Median = $Q_2 = D_5 = P_{50}$.

Review Exercise 7

1. In a school 85 boys and 35 girls appeared in a university entrance examination. The mean marks of boys was found to be 40%, where as the mean marks of girls was 60%?. Determine the average markspercentage of the school.
2. The mean of 7 numbers is 25. If one number is excluded, their mean is 21. Find the excluded number
3. The mean marks scored by 70 students was found to be 60. Later on it observed that a score of 47 was misread as 74. Find the correct mean.
4. The table shown below is a simple frequency distribution of data below.

x	14	16	17	18	20
f	2	5	6	5	2

a. Find The mean, median and mode

b. Find The quartile, Q_1 and Q_3

c. Find The decile: D_4, D_7 and D_9

d. Find The percentile: P_{10}, P_{80}

5. For the data given: 31, 15, 47, 12, 56, 29, 61, 65, 66.
   a. Find the quartile: Q_1, Q_2, and Q_3
   b. Find the deciles: D_2, D_7 and D_9
   c. Find the percentile: P_{24}, P_{65} and P_{85}

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6. For the following frequency distribution of a data set of age of a group of people.

Age	$3 \le t < 12$	$12 \le t < 22$	$22 \le t < 32$	$32 \le t < 42$	$42 \le t < 52$
f	4	12	10	7	2

Find a. The quartiles Q_1 and Q_3

b. The deciles : D_4, D_8

c. The percentile: P_{10}, P_{24}, P_{68} and P_{95}

Answer to Review Exercise 7

1. Average Marks $= \frac{\text{Total score}}{85 + 40} = \frac{85 \times 40 + 35 \times 60}{85 + 40} = 45.8\%$

2. Mean of 6 numbers $= \frac{\text{New total - A}}{6} = \frac{7 \times 125 - A}{6} = 21$

Hence the excluded number: $A = 175 - 126 = 49$

3. Correct mean $= \frac{70 \times 60 - 74 + 47}{70} = \frac{4\,173}{70} = 58.61$

4. a. Mean: $\bar{x} = 17$ Median: $\tilde{x} = 17$, Mode: $\hat{x} = 17$

b. The quartile: $Q_1 = \frac{5^{th} \text{ item} + 6^{th} \text{ item}}{2} = (5.5)^{th} \text{ item} = 16$

$$Q_3 = \left(\frac{15^{th} + 16^{th}}{2}\right) \text{item} = (15.5)^{th} \text{ item} = 18$$

c. The decile: $D_2 = \left(\frac{4^{th} + 5^{th}}{2}\right) \text{ item} = 4.5^{th} \text{ item} = 16$

$$D_7 = (14.5)^{th} \text{ item} = 18$$

$$D_9 = \left(\frac{18^{th} + 19^{th}}{2}\right) \text{ item} = (18.5)^{th} \text{ item}$$

$$= 18^{th} \text{ item} + 0.5\left(19^{th} - 18^{th}\right) \text{item}$$

$$= 18 + 0.5\,(20 - 18) = 19$$

d. The percentile: $P_{10} = (2.5)^{th}$ item

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$= 2^{nd}$ item $+ 0.5\ (3^{rd} - 2^{nd})$ item

$= 14 + 0.5\ (16 - 14) = 15$

$P_{80} = (16.5)^{th}$ item $= 16^{th}$ item $+ 0.5\ (17^{th} - 16^{th})$ item

$= 18 + 0.5\ (18 - 18) = 18$

5. First arrange the data in order 12, 15, 29, 31, 47, 56, 61, 65, 66

a. $Q_1 = \left(\frac{1 \times (9+1)}{4}\right)^{th}$ item $= (2.5)^{th}$ item

$= 2^{nd}$ item $+ 0.5\ (3^{rd}$ item $- 2^{nd}$ item$) = 15 + 0.5\ (29 - 15) = 22$

$Q_3 = \left(\frac{3 \times (9+1)}{4}\right)^{th}$ item $= (7.5)^{th}$ item

$= 7^{th}$ item $+ 0.5\ (8^{th} - 7^{th})$ item $= 61 + 0.5\ (65 - 61) = 63$

b. $D_2 = \left(\frac{2 \times (9+1)}{10}\right)^{th}$ item $= 2^{nd}$ item $= 15$

$D_7 = \left(\frac{7 \times (9+1)}{10}\right)^{th}$ item $= 7^{th}$ item $= 61$

$D_9 = 9^{th}$ item $= 66$

c. $P_{24} = \left(\frac{24\ (9+1)}{100}\right)^{th}$ item $= (2.4)^{th}$ item

$= 2^{nd}$ item $+ 0.4\ (3^{rd} - 2^{nd})$ item $= 15 + 0.4\ (29 - 15) = 20.6$

$P_{65} = \left(\frac{65\ (9+1)}{100}\right)^{th}$ item $= (6.5)^{th}$ item

$= 6^{th}$ item $+ 0.5\ (7^{th} - 6^{th})$ item

$= 56 + 0.5\ (61 - 56) = 58.5$

$P_{85} = \left(\frac{85\ (9+1)}{100}\right)^{th}$ item $= (8.5)^{th}$ item

$= 65 + 0.5\ (66 - 65)\ 65.5$

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6. The corresponding quartile: $Q_k = B_L + \frac{i}{f_c}\left(\frac{k \times n}{4} - Cf_B\right)$

a. $Q_1 = \left(\frac{1 \times 35}{4}\right)^{th}$ item $= (8.75)^{th}$ item

Q_1 lies in 2nd class: 12 - 22, $B_L = 12$

$i = 10$, $f_c = 12$ and $Cf_B = 4$

$\therefore \quad Q_1 = 12 + \frac{10}{12}(8.75 - 4)$

$Q_3 = \left(\frac{3 \times 35}{4}\right)^{th}$ item $= (26.25)^{th}$ item

Q_3 lies in 4rd class: 32 - 42

$B_L = 32$, $i = 10$, $f_c = 7$ and $Cf_B = 4 + 12 + 10 = 26$

$Q_3 = 32 + \frac{10}{7}(26.25 - 26) = 32.36$

b. The decile: $D_j = B_L + \frac{i}{f_c}\left(\frac{j \times n}{10} - f_{cB}\right)$

$D_4 = \left(\frac{4 \times 35}{10}\right)^{th}$ item $= 14^{th}$ item

D_4 lies on 12, $12 \le t < 22$, $t < 22$, $B_L = 12$,

$i = 10$, $f_c = 12$, and $Cf_B = 4$

$\therefore \quad D_4 = 12 + \frac{10}{12}(14 - 4) = 20.33$

$D_8 = \left(\frac{8 \times 35}{10}\right)^{th}$ item $= 28^{th}$ item

D_8 lies on 4th class: $32 \le t < 42$

$B_L = 32$, $i = 10$ $f_c = 7$ and $C f_B + 4 + 12 + 10 = 26$

$D_8 = 32 + \frac{10}{7}(28 - 26) = 34.86$

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c. $B_L + \frac{i}{f_c}\left(\frac{t \times n}{100} - C f_B\right)$

i) $P_{10}\left(\frac{10 \times 35}{100}\right)^{th} item = (3.5)^{th}$ item

So that P_{10} lies on 1st class: $3 \leq t < 12$

$B_L = 3, i = 10, f_c = 4$ and $C f_B = 0$

$P_{10} = 3 + \frac{10}{4}(3.5 - 0) = 11.75$

ii. $P_{24} = \left(\frac{24 \times 35}{100}\right)^{th}$ item $= (8.4)^{th}$ item

So that P_{24} lies on 2nd class: 12 - 22,

$B_L = 12, i = 10, f_c = 12$ and $C f_B = 4$

$\therefore \quad P_{24} = 12 + \frac{10}{12}(8.4 - 4) = 15.7$

iii. $P_{68} = \left(\frac{68 \times 35}{100}\right)^{th}$ item $= (23.8)^{th}$ item

So that P_{68} lies on 3rd class $22 \leq t < 32$,

$B_L = 22, i = 10, \quad f_c = 10$ and $C f_B = 4 + 12 = 16$

$\therefore \quad P_{68} = 22 + \frac{10}{10}(23.8 - 16) = 22 + 7.8 = 29.8$

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UNIT 8

8. Measures of Dispersion and Skewness

The variation, or scattering or deviation of the different values of a data or variable from their central value (average) is known as **Dispersion.**

- A measure of dispersion in any data shows the extent to which the values tend to spread about the center (average).
- A measure of dispersion helps us to know the degree of uniformity and consistency in the series.
- If the difference between items is large the dispersion is large and vice versa.

8.1 Types of Measures of Dispersion

There are two measures of dispersion. These are :

- Absolute measures of dispersion
- Relative measures of dispersion (coefficient of dispersion).

Absolute measure of dispersion are :

- Range
- Interquartile range
- Mean deviation
- Quartile deviation
- Standard deviation

Relative measure of dispersion are :

- Coefficient of range
- Coefficient of quartile deviation

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- Coefficient of mean deviation
- Coefficient of variation.

(i) Absolute Measure of Dispersion

An absolute measure of dispersion measures the variablity of data in terms of the same units of the data.

An absolute measure of dispersion are measured by range, quartile deviation, mean deviation, and standard deviation.

(ii) Relative Measure of Dispersion

The relative measure of dispersion is called **coefficient of dispersion.**

Coefficient of dispersion is used to compare the variability of two or more series which differ widely in their average or which are measured in different units, we calculate the coefficients of dispersion, which being ratios are numbers independent of the units.

The coefficients of dispersion based on range, mean deviation, quartile deviation, and standard deviation :

$$\text{Coefficients of Dispersion} = \frac{\text{Absolute Dispersion}}{\text{Average}}$$

Coefficients of Dispersion is based on :

(i) Range is coefficient of range : $CR = \frac{L-S}{L+s}$

(ii) Quartile deviation is coefficient of quartile deviation :

$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

(iii) Mean deviation is coefficient of mean deviation

$$CMD = \frac{\text{Mean - deviation}}{\text{Average}}$$

The average is may be mean, median or mode

(iv) Standard deviation : $\frac{S.D}{\text{Mean}} = \frac{\sigma}{\bar{X}}$

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• Coefficient of variation . It is the percentage variation in the mean

$$C.V = \frac{S.D}{mean} \times 100$$

8.1.1 Range and Coefficient of Range

• The range (R) is the difference between the largest value (L) and the smallest values (S) of a data set.

• For grouped data, the range is the difference between the upper limit of the last class interval and the lower limit of the first class interval.

• Range $R = L - S$

Relative range is given by : **Coefficient of range** $= \frac{L - S}{L + S}$

Illustrative Example

1. For the following set of data 12, 22, 24, 27, 31

Find a. the range

b. the coefficient of range

Solution :

a. Range : $R = L - S = 31 - 12 = 19$

b. Coefficient of range : $CR = \frac{L - S}{L + S} = \frac{31 - 12}{31 + 12} = \frac{19}{43} = 0.44$

2. For the following set of a data distribution.

x	3 - 12	13 - 22	23 - 32	33 - 42	43 - 52
f	4	12	10	7	12

a. What is the range of the distribution

b. What is the coefficient of range ?

Solution :

a. Range is an absolute measure of dispersion given by :

range = 52.5 - 2.5 = 50.

b. Coefficient of the range is the relative measure is given by

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$$C.R = \frac{52.5 - 2.5}{52.5 + 2.5} = \frac{50}{55} = \frac{19}{43} = 0.9$$

3. What is the merits and uses of the range ?

Solution :

Merits and uses of range :

- It is simple to compute and easy to understand
- It is time saving

4. What is the disadvantage (limitation) of using range as measure of dispersion ?

Solution :

The limitation of range :

- It is not a precise measure
- It depends only on the extreme values. It is not a reliable measure of dispersion.

8.1.2 Mean Deviation and Coefficient of Mean Deviation

Mean deviation is also called average deviation. The mean deviation of a data is defined as the mean of all absolute deviations of each data items from the average (mean, median or mode).

- Mean deviation about the average A is $MD(A) = \frac{\sum_{i=1}^{n} f_i |x_i - A|}{\sum_{i=1}^{n} f_i}$

Where A represents (mean, median or mode) and x_i is the class mark (or mid – value) of class interval when the value of the variable are given in the form of class.

(i) If mean deviation about the mean, then it is written as $MD(\bar{X})$:

$$MD(\bar{X}) = \frac{\Sigma |X_i - \bar{X}|}{n} \longleftarrow \text{For individual series}$$

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$$MD(\bar{X}) = \frac{\sum_{i=1}^{n} f_i |X_i - \bar{X}|}{f_1 + f_2 + .. + f_n} \longleftarrow \text{For frequency distribution}$$

• Coefficient of mean deviation from mean is

$$C.MD = \frac{\text{mean deviation about mean}}{\text{mean}} = \frac{MD(\bar{X})}{\bar{X}}$$

(ii) If mean deviation about the median (M_d), then it is written as MD (M_d)

• $MD(\text{median}) = \frac{\sum |X_i - \text{median}|}{n}$ ....., For individual series

• $MD(\text{median}) = \frac{\sum f_i |X_i - \text{median}|}{\sum f_i}$ ....., For frequency distribution

• **Coefficient of mean deviation about the median is :**

$$CMD = \frac{MD(M_d)}{\text{median}}$$

(iii) If mean deviation about mode (M_o), then it is written as MD (M_o)

$$MD(\text{Mode}) = MD(M_o) = \frac{\sum |X_i - \text{Mode}|}{n} \text{....., Individual series}$$

$$M.D(M_o) = \frac{\sum f_i |X_i - \text{Mode}|}{\sum f_i} \text{ ....., for grouped frequency}$$

• **Coefficient of mean deviation about the mode :**

$$CMD = \frac{MD(M_o)}{\text{Mode}}$$

Note Since the mean deviation is based on all the values of the variate, it is a better measures of dispersion than range or quartile deviation.

Illustrative Example

5. For the following set of data : 6, 8, 11, 13, and 7

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calculate the mean deviation about the mean and median and their coefficients of mean deviation.

Solution :

Arrange the data in order : 6, 7, 8, 11, 13.

$$\text{Mean}:(\bar{X}) = \frac{6+7+8+11+13}{5} = \frac{45}{5} = 9$$

$$\text{Median}:\tilde{X} = \left(\frac{n+1}{2}\right)^{th} \text{item} = \left(\frac{5+1}{2}\right)^{th} \text{item} = 3^{rd} \text{ item} = 8$$

Mean deviation about the mean is :

(i) $$MD(\bar{X}) = \frac{|6-9|+|7-9|+|8-9|+|11-9|+|13-9|}{5}$$

$$= \frac{3+2+1+2+4}{5} = \frac{12}{5} = 2.4$$

Coefficient of mean deviation from mean is :

$$CMD = \frac{MD}{Mean} = \frac{2.4}{8} = 0.3$$

(ii) Mean deviation about the median is

$$MD\,(M_d) = \frac{|6-8|+|7-8|+|8-8|+|11-8|+|11-8|}{5}$$

$$= \frac{2+1+0+3+5}{5} = \frac{11}{5} = 2.2$$

Coefficient of mean deviation from the median is

$$CMD = \frac{MD}{Median} = \frac{2.2}{9} = 0.24$$

6. Consider the following data set 7, 8, 10, 7, 10, 6, 13, 10, 6, 8

   a. Calculate the mean deviation about the mean, median and mode.

   b. Calculate the coefficient of mean deviation from the mean, medi and mode.

   c. Calculate the inter quartile range and coefficient of inter quart

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range

Solution :

First arrange the data in order

x	6	7	8	10	13
f	2	2	2	3	1

- Mean $= \dfrac{2\times6+2\times7+2\times8+3\times10+1\times13}{2+2+2+3+1} = \dfrac{85}{10} = 8.5$

- Median $= \dfrac{\left(\frac{10}{2}\right)^{th}\text{ item}+\left(\frac{10}{2}+1\right)^{th}\text{ item}}{2} = \left(\dfrac{5^{th}+6^{th}}{2}\right)\text{item} = \dfrac{8+8}{2} = 8$

- Mode is the most frequent value is 10.

a. i Mean deviation about the mean : $\bar{X} = 8.5$

$$MD(\bar{X}) = \frac{2|6-8.5|+2|7-8.5|+2|8-8.5|+3|10-8.5|+|13-8.5|}{10}$$

$$= \frac{2(2.5)+2(1.5)+2(0.5)+3(1.5)+4.5}{10} = \frac{18}{10} = 1.8$$

ii. Mean deviation about the median $\bar{X} = 8$ is

$$MD(Med) = \frac{2|6-8|+2|7-8|+2|8-8|+3|10-8|+|13-8|}{10}$$

$$= \frac{4+2+0+6+5}{10} = 1.7$$

iii. Mean deviation about the mode $\left(\hat{x}\right) = 10$ is

$$MD(M_o) = \frac{2|6-10|+2|7-10|+2|8-10|+3|10-10|+|13-10|}{10}$$

$$= \frac{8+6+4+0+3}{10} = 2.1$$

b. i. The coefficient of mean deviation from the mean is :

$$CMD = \frac{\text{Mean deviation}}{\text{Mean}} = \frac{1.8}{8.5} = 0.212$$

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ii. The coefficient of mean deviation from the median is :

$$CMD = \frac{\text{Mean deviation}}{\text{Median}} = \frac{1.7}{8} = 0.213$$

iii. The coefficient of mean deviation from the mode is :

$$CMD = \frac{\text{Mean deviation}}{\text{Mode}} = \frac{2.1}{10} = 0.21$$

c. $Q_1 = \left(\frac{\left(\frac{n}{4}\right)+\left(\frac{n}{4}+1\right)}{2}\right)^{th}$ item $= \left(\frac{\left(\frac{10}{4}\right)+\left(\frac{10}{4}+1\right)}{2}\right)^{th}$ item $= 3^{rd}$ item

$Q_1 = 7$

$Q_3 = \left(\frac{\left(\frac{Kn}{4}\right)+\left(\frac{Kn}{4}+1\right)}{2}\right)^{th}$ item $= \left(\frac{\left(\frac{3\times10}{4}\right)+\left(\frac{3\times10}{4}+1\right)}{2}\right)^{th}$ item

$= 8^{th}$ item $= 10$

$\therefore$ Inter quartile range : $IQR = Q_3 - Q_1 = 10-7 = 3$

Coefficient of inter quartile range is

$$CQR = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{10-7}{10+7} = \frac{3}{17} = 0.18$$

7. Consider the following grouped data distribution

Class interval	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
Frequency	12	15	28	25	20

a. Calculate the mean deviation about the mean, median and mode

b. Calculate the coefficient of mean deviation from the mean, median and mode

c. Calculate inter quartile range and coefficient of inter quartile range

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Solution :

Class - interval	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
Class mark (X)	5	15	25	35	45
Frequency (f)	12	15	28	25	20
Cum frequency	12	27	55	80	100

- $\text{Mean}(\bar{X}) = \frac{fX}{\Sigma f} = \frac{5\times12+15\times15+25\times28+35\times25+45\times20}{12+15+28+25+20}$

$$\bar{X} = \frac{60+225+700+875+900}{100} = \frac{2760}{100} = 27.6$$

- $\text{Median class} = \left(\frac{n}{2}\right)^{th}$ item $= 50^{th}$ item .

Median lies on 3rd class : 20 - 30

$B_L = 20,\ i = 10,\ f_c = 28,\ \frac{n}{2} = 50$, and $Cf_B = 27$

$$Md = B_L + \frac{i}{f_C}\left(\frac{n}{2} - Cf_B\right) = 20 + \frac{10}{28}(50-27) = 28$$

- Mode is given by : mode $= \hat{X} = B_L + \left(\frac{d_1}{d_1+d_2}\right)i$

$B_L = 20,\ i = 10\ f_c = 28$ and $d_1 = f_{modal} - f_{pre\text{-}mod} = 28 - 15 = 13$

$d_2 = f_{modal} - f_{next\text{-}mode} = 28 - 25 = 3$

$$\therefore \text{Mode} = 20 + 10\left(\frac{13}{13+3}\right) = 28$$

3. i. Mean deviation about the mean is

$$MD(\bar{X}) = \frac{12|5-27.6|+15|15-27.6|+28|25-27.6|+25|35-27.6|+20|45-27.6|}{100}$$

$$= \frac{12(22.6)+15(12.6)+28(2.6)+25(7.4)+20(17.4}{100}$$

$$\therefore MD(\bar{X}) = \frac{271.2+189+72.8+185+348}{100} = 10.66$$

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ii. Mean deviation about the median is

$$MD(\tilde{X}) = \frac{12|5-288|+15|15-28|+28|25-28|+25|35-28|+20|45-28|}{100}$$

$$= \frac{12(23)+15(13)+28(3)+25(7)+20(17)}{100}$$

$$MD(M_d) = \frac{276+195+84+175+340}{100} = 10.7$$

iii. Mean deviation about the mode is

$$MD(M_o) = \frac{12|5-28|+15|15-28|+28|25-28|+25|35-28|+20|45-28|}{100}$$

$$= \frac{276+195+84+175+340}{100} = 10.7$$

b. The coefficient of mean deviation from the mean is :

$$CMD = \frac{\text{Mean deviation}}{\text{Mean}} = \frac{10.66}{27.6} = 0.39$$

The coefficient of mean deviation from the median is

$$CMD = \frac{\text{Mean deviation}}{\text{Median}} = \frac{10.7}{28} = 0.382$$

The coefficient of mean deviation from the mode is

$$CMD = \frac{\text{Mean deviation}}{\text{Mode}} = \frac{10.7}{28} = 0.382$$

c. (i) The 1st quartile class $= \left(\frac{1\times100}{4}\right)^{th}$ item $= 25^{th}$ item

Q_1 lies on 2nd class: 10-20

$B_L = 10$, $i = 10$, $f_C = 15$ and $Cf_B = 12$

$$Q_1 = 10+\frac{10}{15}(25-12) = 18.7$$

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ii. The 3rd quartile class $= \left(\frac{3\times100}{4}\right)^{th}$ item $= 75^{th}$ item

Q_3 lies on 4^{th} class: 30 - 40

$B_L = 30$, i = 10, $f_C = 25$ and $Cf_B = 55$

$$Q_3 = 30 + \frac{10}{25}(75-55) = 38$$

$\therefore$ Inter quartile range (IQR) $= Q_3 - Q_1 = 38 - 18.7 = 19.3$

Coefficient of inter - quartile range is

$$CQR = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{38-18.7}{38+18.7} = \frac{19.3}{56.7} = 0.34$$

8. Ten different teams played foot ball over the summer. After the summer the top goal scores from each team scores the following number of goals :

10, 5, 18, 2, X, 4, 10, 15, 11, 4. If the mean number of goals scored is 9, What is the mean deviation from the mean of the data ?

A. 2.4 B. 3.4 C. 4.2 D. 4.3

Solution :

$$\text{Mean} = \frac{10+5+18+2+X+4+10+15+11+4}{10} = 9$$

$X = 90 - 79 = 11$, substitute $X = 11$

Mean deviation about the mean:

$$MD(\bar{X}) = \frac{2|10-9|+|5-9|+|18-9|+|2-9|+2|11-9|+2|4-9|+|15-9|+}{10}$$

$$= \frac{2(1)+4+9+7+4+2(5)+6}{10} = \frac{42}{10} = 4.2$$

Answer : C

What is the coefficient of mean deviation ?

Solution :

$$CMD = \frac{M.D}{\bar{X}} = \frac{4.2}{9} = 0.47$$

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9. The following is data on a mid - examination results of 40 students

Mark	16 - 20	21 - 25	26 - 30	31 - 35	36 - 40
No . of students	2	6	10	12	10

Which one of the following is the mean deviation about the modal mark of the students ?

A. 4.60 B. 4. 75 C. 4. 80 D. 4.82

Solution :

- Class mark for each class is 18, 23, 28, 33, and 38.
- The modal class of the data is 31 - 35, so that,

$B_L = 30.5, i = 5$ $d_1 = 12 - 10 = 2$ and $d_2 = 12 - 10 = 2$.

$$\therefore \text{Mode} = B_L + \left(\frac{d_1}{d_1 + d_2}\right) i = 30.5 + \left(\frac{2}{2+2}\right)(5) = 33$$

Mean deviation about the mode:

$$MD(M_o) = \frac{2|18-33| + 6|23-33| + 10|28-33| + 12|33-33| + 10|38-33|}{2+6+10+12+10}$$

$$= \frac{190}{40} = 4.75$$

Answer : B

10. What is the coefficient of MD of # 9 ?

Solution :

$$(CMD) = \frac{MD}{Mo} = \frac{4.75}{33} = 0.144$$

11. For the following frequency distribution table of student exam :

Mark	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
No. of student	5	8	15	16	6

What is the mean deviation about the median

A. 9.56 B. 28 C. 10.4 D. 9

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Solution :

Mid value	5	15	25	35	45
f	5	8	15	16	6
$C.f$	5	13	28	44	50

The median class is $\left(\frac{n}{2}\right)^{th}$ item $= \left(\frac{50}{2}\right)^{th}$ item $= 25^{th}$ item

The mediain class is 20-30, $B_L = 20$, $i = 10$, $f_C = 15$, $Cf_B = 13$

Median : $M_d = B_L + \frac{i}{f_C}\left(\frac{N}{2} - Cf_B\right) = 20 + \frac{10}{15}(25-13) = 28$

Mean deviation from median : $MD(M_d) = \frac{\Sigma f|X-Md|}{\Sigma f}$

$$M.D(M_d) = \frac{5|5-28|+8|15-28|+15|25-28|+16|35-28|+6|45-28|}{5+8+15+16+6}$$

$$= \frac{478}{50} = 9.56$$

Answer : A

12. A manufacturer has two types of products, A and B. The mean and the mean deviation about the mean of product A and product B are given below.

Product	Mean	Mean deviation
A	1800	810
B	1640	786

a. Calculate the coefficient of mean deviation from mean of each product.

b. Which product has the greater absolute dispersion ?

c. Which product has the greater relative dispersion ?

d. Which product has more consistent ?

Solution :

a. Coefficient of mean deviation from mean :

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For product A is $CMD = \frac{\text{M.D of A}}{\text{Mean of A}} = \frac{810}{1800} = 0.45$

For product B is $CMD = \frac{\text{M.D of B}}{\text{Mean of A}} = \frac{786}{1640} = 0.48$

b. Mean deviation measures an absolute dispersion, so the absolut dispersion of product A, is $MD_A = 810$ and of B is $MD_B = 786$. Thus product A has the greater absolute dispersion.

c. Relative dispersion measures the coefficient of deviation. The coefficient of mean deviation from the mean for :

product A is $CMD_A = \frac{810}{1800} = 0.45$

product B is $CMD_B = \frac{786}{1640} = 0.48$

Thus product B has the greater relative dispersion

d. Product A has more consistent because $CMD_A < CMD_B$.

13. The grouped frequency distribution of data is given in the following table.

Class interval	8 - 12	13 - 17	18 - 22	23 - 27	28 - 32
Frequency	4	8	10	5	3

What is the median (M_d) and mean deviation (MD) about the median ?

A. $M_d = 19$, MD = 4.2

B. $M_d = 20$, MD = 4.7

C. $M_d = 19$, MD = 4.7

D. $M_d = 20$, MD = 4.2

Solution :

The median is calculated by: $M_d = B_L + \frac{i}{f_C}\left(\frac{n}{2} - Cf_B\right)$

Median class (M_d) belongs to $\left(\frac{n}{2}\right)^{th}$ item $= \left(\frac{30}{2}\right)^{th}$ item $= 15^{th}$ item

Median class lies on 3rd class: 18 - 22

$B_L = 17.5,\ i = 5,\ f_C = 10, Cf_B = 4 + 8 = 12,\ \frac{n}{2} = 15$

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$\therefore M_d = 17.5 + \frac{5}{10}(15-12) = 19$

The class marks of the data : 10, 15, 20, 25, 30.

The mean deviation about the median is calculated as :

$$MD = \frac{4|10-19|+8|15-19|+10|20-19|+5|25-19|+3|30-19|}{4+8+10+5+3}$$

$$\therefore MD = \frac{36+32+10+30+33}{30} = \frac{141}{30} = 4.7$$

Answer : C

14. What is the coefficient of mean deviation from the median in the above table of solved problem # 13 ?

A. 0.25 B. 0.21 C. 4.04 D. 4.7

Solution :

Coefficient of mean deviation from the median is

$$CMD = \frac{MD}{Median} = \frac{4.7}{19} = 0.25$$

15. The marks obtained by 50 students in exam are given below :

Mark	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
No of student	5	a	20	b	10

The mean deviation about the median and coefficient of mean deviation about the median are 9 and 0.36 respectively. Calculate the missing frequency a and b.

Solution :

Coefficient of mean deviation about the median :

$$CMD = \frac{MD(M_d)}{M_d} \text{ implies } 0.36 = \frac{9}{M_d}$$

$$\text{The median : } M_d = \frac{9}{0.36} = 25$$

$\therefore$ Median class is 20-30

$B_L = 20, i = 10, f_C = 20, \frac{n}{2} = \frac{50}{2} = 25$ and $Cf_B = 5 + a$

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$\therefore\ M_d = B_L + \frac{i}{f_c}\left(\frac{n}{2} - Cf_n\right)$

$25 = 20 + \frac{10}{20}(25 - (5+a))$ this implies $\frac{1}{2}(25 - 5 - a) = 5$

$20 - a = 10$ implies $a = 20 - 10 = 10$

Also total number of students:

$5 + a + 20 + b + 10 = 50$

$5 + 10 + 20 + b + 10 = 50$

$\therefore\ b = 50 - 45 = 5$

8.1.3 Quartile Deviation and Coefficient of Quartile Deviation

The quartile deviation (Q . D) is a type of range based on the quartile. The quartile deviation is half the difference between the lower (Q_1) and upper (Q_3) quartiles.

- Quartile deviation is an absolute measure of dispersion given by :

$$Q.D = \frac{Q_3 - Q_1}{2}$$

The relative measure is given by the Coefficient of $Q.D = \frac{Q_3 - Q_1}{Q_3 + Q_1}$

Quartile deviation definitely a better measure of dispersion than range as it makes uses of 50 % of the data.

Illustrative Example

16. Calculate the quartile deviation and coefficient of quartile deviation of the following data : 21, 23, 25, 28, 30, 32, 38, 39, 46, 48.

Solution :

$$Q_1 = \left(\frac{\left(\frac{n}{4}\right) + \left(\frac{n}{4} + 1\right)}{2}\right)^{th} \text{item} = \left(\frac{\left(\frac{10}{4}\right) + \left(\frac{10}{4} + 1\right)}{2}\right)^{th} \text{item}$$

$= 3^{rd}$ item $= 25$

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$$Q_3 = \left(\frac{\left(\frac{Kn}{4}\right)+\left(\frac{Kn}{4}+1\right)}{2}\right)^{th} \text{ item} = \left(\frac{\left(\frac{3\times10}{4}\right)+\left(\frac{3\times10}{4}+1\right)}{2}\right)^{th} \text{ item}$$

$$= 8^{th} \text{ item} = 39$$

$\therefore$ Quartile deviation is $Q.D = \frac{Q_3 - Q_1}{2} = \frac{39-25}{2} = 7$

Coefficient of quartile deviation is

$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{39-25}{39+25} = \frac{14}{74} = 0.22$$

17. Calculate the quartile deviation and coefficient of quartile deviation of the following data : 28, 41, 45, 49, 34, 32, 24, 12, 9, 5, 7, 50, 55.

Solution :

First arrange the data in increasing order : 5, 7, 9, 12, 24, 28, 32, 34, 41, 45, 49, 50, 55

$$Q_1 = \left(\frac{\left(\frac{K(n+1)}{4}\right)^{th} \text{ item} = \frac{1(13+1)}{4}}{2}\right)^{th} \text{ item} = (3.5)^{th} \text{ item}$$

$$Q_1 = 3^{rd} \text{ item} + 0.5\left(4^{th} - 3^{rd}\right)\text{item} = 9 + 0.5(12-9) = 10.5$$

$$Q_3 = \left(\frac{3(13+1)}{4}\right)^{th} \text{ item} = (10.5)^{th} \text{ item}$$

$$Q_3 = 10^{th} \text{ item} + 0.5\left(11^{th} - 10^{th}\right)\text{item}$$

$$Q_3 = 45 + 0.5\,(49-45) = 47$$

$\therefore$ Quartile deviation is $Q.D = \frac{Q_3 - Q1}{2} = \frac{47-10.5}{2} = \frac{36.5}{2} = 18.25$

Coefficient of quartile deviation is $CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{47-10.5}{47+10.5} = 0.63$

18. The following table gives the distribution of marks secured by some

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students in an examination.

Marks	35 - 40	40 - 45	45 - 50	50 - 55	55 - 60
No of students	4	8	12	7	2

Calculate the quartile deviation and the coefficient of quartile deviati of the mark.

Solution :

The K^{th} quartile for grouped frequency distribution is

$$Q_K = B_L + \left(\frac{Kn}{4} - Cf_B\right)\frac{i}{f_C}$$

Q_1 belongs to $\left(\frac{1\times 33}{4}\right)^{th}$ item $= (8.25)^{th}$ item

Which lies on the 2^{nd} class: 40-45

$$B_L = 40,\ i = 5,\ \frac{Kn}{4} = 8.25,\ Cf_B = 4,\ f_C = 8$$

$$Q_1 = 40 + (8.25-4)\left(\frac{5}{8}\right) = 42.67$$

Q_3 belongs to $\left(\frac{3\times 33}{4}\right)^{th}$ item $= (24.75)^{th}$ item

Which lies in the class: 50 -55

$$B_L = 50,\ i = 5,\ f_C = 7,\ \frac{Kn}{4} = 24.75,\ Cf_B = 4+8+12 = 24$$

$$\text{Thus, } Q_3 = B_L + \frac{i}{f_C}\left(\frac{Kn}{4} - Cf_B\right) = 50 + \frac{5}{7}(24.75-24) = 50.$$

There fore, Quartile deviation of the data is

$$Q.D = \frac{Q_3 - Q_1}{2} = \frac{50.54 - 42.67}{2} = 3.94$$

Coefficient of quartile deviation of the data is

$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{50.54 - 42.67}{50.4 + 42.67} = \frac{7.87}{93.21} = 0.084$$

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8.4 Standard Deviation, Variance and Co-efficient of Variation

Another measure of dispersion or variation is the standard deviation a measure of how each value in a data set varies or deviates from the mean. The standard deviation is denoted by the symbol σ (sigma).

It is the most important measure of dispersion.

Standard deviation is the square root of the average of the squares of all the deviation from the mean.

In short, it may be defined as the root - mean - square deviation from the mean.

In symbol :

If $X_1, X_2, \ldots\ldots, X_n$ be a data of values of a variable, and $(\bar{X})$ is their mean, then :

$$\text{Standard deviation } (\sigma) = \sqrt{\frac{(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2 + \ldots\ldots + (X_n - \bar{X})^2}{n}}$$

• For grouped frequency distribution, the standard deviation :

$$\sigma = \sqrt{\frac{f_1(m_1 - \bar{X})^2 + f_2(m_2 - \bar{X})^2 + \ldots\ldots + f_n(m_n - \bar{X})^2}{f_1 + f_2 + \ldots\ldots + f_n}}$$

Where $m_1, m_2, \ldots\ldots m_n$ are the class mark (mid - point of each class interval).

• The square of the standard deviation is variance :

$$\sigma^2 = \sqrt{\frac{(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2 + \ldots\ldots + (X_n - \bar{X})^2}{n}}$$

$$\sigma^2 = \frac{f_1(m_1 - \bar{X})^2 + f(m_2 - \bar{X})^2 + \ldots\ldots\ldots + f_n(m_n - \bar{X})^2}{f_1 + f_2 + \ldots\ldots + f_n}$$, for grouped frequency distribution

Note : A small standard deviation (compared to actual data values) indicates

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that the data are clustered tightly around the mean and it has high degree of uniformity and homogeneity.

Coefficient of Variation

The standard deviation is an absolute measure of dispersion. Coefficient of variation is the relative measure of dispersion given by :

$$\text{Coefficient of variation: C.V} = \frac{\text{standard deviation}}{\text{Mean}} = \frac{\sigma}{\bar{X}} \times 100$$

If we do not take percentage, we have the coefficient of standard deviation

$$\therefore \text{C.V} = \frac{\sigma}{\bar{X}}$$

- Coefficient of variation is a unit less relative measure that is used to measure the degree of consistency.

Illustrative Example

19. Find the variance, standard and coefficient of variation of the data :

4, 7, 3, 2, 8, 6

Solution :

STEP 1 : Find the mean of the data :

$$\bar{X} = \frac{4+7+3+2+8+6}{6} = 5$$

STEP 2 : Find the square of the sum of all the deviation from the mean.

$$= \frac{(4-5)^2 + (7-5)^2 + (3-5)^2 + (2-5)^2 + (8-5)^2 + (6-5)^2}{6}$$

$$= \frac{1+4+4+9+9+1}{6} = 4.7$$

$\therefore$ The variance: $\sigma^2 = 4.7$

Standard deviation is the square root of variance

$$\text{S.D} : = \sqrt{\text{Variance}} = \sqrt{4.7} = 2.17$$

$$\therefore \text{Coeffiient of variation: C.V} = \frac{\sigma}{\bar{X}} \times 100 = \frac{2.17}{5} \times 100 = 43.4\%$$

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20. Find the variance, standard deviation and coefficient of variation :
4, 7, 9, 8

Solution :

$$\text{Mean} : \bar{X} = \frac{4+7+9+8}{4} = \frac{28}{4} = 7$$

$$\therefore \text{Variance} : = \sigma^2 = \frac{(4-7)^2+(7-7)^2+(9-7)^2+(8-7)^2}{4} = \frac{14}{4} = 3.5$$

$$\therefore \text{Standard deviation} : \sigma = \sqrt{\frac{14}{4}} = \sqrt{3.5} = 1.87$$

$$\therefore \text{Coefficient of varation} : \text{C.V} = \frac{\sigma}{\bar{X}} = \frac{1.87}{7} = 0.27 = 27\%$$

21. The following is a simple frequency distribution of a data with variable X

X	3	5	6	7
frequency	2	5	2	1

Find the variance, standard deviation and coefficient of variation.

Solution :

$$\text{Mean} : \bar{X} = \frac{2\times 3 + 5\times 5 + 2\times 6 + 1\times 7}{2+5+2+1} = \frac{6+25+12+7}{10} = \frac{50}{10} = 5$$

$$\therefore \text{Variance} : \sigma^2 = \frac{2(3-5)^2+5(5-5)^2+2(6-5)^2+1(7-5)^2}{4} = \frac{14}{10} = 1.4$$

$$\therefore \text{Standard deviation} : \sigma = \sqrt{\text{Variance}} = \sqrt{1.4} = 1.18$$

$$\therefore \text{Coefficient of varation} : \text{C.V} = \frac{\sigma}{\bar{X}} \times 100 = \frac{1.18}{5} \times 100\%$$

$$= 0.24 \times 100\% = 24\%$$

22. The following is a frequency distribution table of a grouped data with variable X

Variable : X	3 - 7	8 - 12	13 - 17	18 - 22
frequency : f	4	6	8	2

Calculate the variance, standard variation and coefficient of variation of the data.

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Solution :

First find the class mark m = 5, 10, 15, 20.

Mean of class mark: $\bar{X} = \frac{4\times5+6\times10+8\times15+2\times20}{4+6+8+2} = \frac{240}{20} = 12$

Variance: $\frac{\sum_f (m_i - \bar{X})^2}{\sum f} = \frac{4(5-12)^2+6(10-12)^2+8(15-12)^2+2(20-12)^2}{20}$

$\therefore$ Variance : $\sigma^2 = \frac{196+24+72+128}{20} = \frac{420}{20} = 21$

Standard deviation : $\sigma = \sqrt{21} = 4.58$

Coefficient of variation : $C.V = \frac{\sigma}{\bar{X}} \times 100\% = \frac{4.58}{12} \times 100 = 0.382 \times 100$

$= 38.2$

23. The time needed to type a sample of 8 business letters in an office is 7, 8, 6, 8, 9, 7, 5, 6 minutes. What are the mean $(\bar{X})$ and the standard deviation (σ) of the data in minute ?

A. $\bar{X} = 7, \sigma = \sqrt{1.5}$

B. $\bar{X} = 7, \sigma = \sqrt{2}$

C. $\bar{X} = 8, \sigma = \sqrt{2}$

D. $\bar{X} = 8, \sigma = \sqrt{1.5}$

Solution :

Mean : $\bar{X} = \frac{2\times7+2\times8+2\times6+1\times9+1\times5}{8} = 7$

Variance : $\sigma = \frac{2(7-7)^2+2(8-7)^2+2(6-7)^2+1(9-7)^2+1(5-7)^2}{8} = 1.5$

Standard deviation : $\sigma = \sqrt{1.5} = 1.22$

Answer: A

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8.1.5 Comparison of two or more Data Series

We normally use the coefficient of standarddeviation $\left(\frac{S.D}{\bar{X}}\right)$ or coefficient of variation $\left(\frac{S.D}{\bar{X}} \times 100\right)$

- If the series is more uniform (or more consistence), then it has lower value of coefficient of variation.
- Series with greatest coefficient of variation has maximum variability and minimum consistence

Illustrative Example

24. The average score on mathematics final examination was 85, with a standard deviation of 5; the average score on an English final exam was 110, with standard deviation of 8. Which class was more variable ?

Solution :

The coefficient of variation:

$$\text{For Maths} = \frac{\sigma}{\bar{X}} \times 100\% = \frac{5}{85} \times 100 = 5.9\%$$

$$\text{For English} = \frac{\sigma}{\bar{X}} \times 100\% = \frac{8}{110} \times 100 = 7.3\%$$

English final exam is more variable than math final exam since the coefficient of variation is larger for English.

25. The mean monthely income of company A is Birr 40, 000, with standard deviation of Birr 4000. The mean monthly income of company B is Birr 20, 000, with standard deviation of Birr 2400.

Which company has more variable income ?

Solution :

Coefficient of variation :

$$\text{For company A} = \frac{\sigma}{\bar{X}} \times 100\% = \frac{4000}{40,000} \times 100\% = 10\%$$

$$\text{For company B} = \frac{\sigma}{\bar{X}} \times 100\% = \frac{2400}{20,000} \times 100\% = 12\%$$

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Since the coefficient of variation is larger for company B, company B has more variable income.

26. Ten students took two tests, test I and test II. The mean mark of test is 5 with standard deviation 2. Which one of the following is true ?

A. The coefficient of variation of test I is 25 %

B. The coefficient of variation of test II is 30 %

C. The students mark in test II is more consistent than test I.

D. The students mark in test I and test II have the same degree of variability.

Solution :

(i) Coefficient of variation of Test I $= \frac{\sigma}{\bar{X}} = \frac{2}{5} = 0.4 = 40\%$

(ii) Coefficient of variation of Test II $= \frac{\sigma}{\bar{X}} = \frac{2}{6} = 0.333 = 33.3\%$

Interpretation :

- Smaller coefficient of variation is more consistent and high degree of uniformity.
- Students mark in test I show more variable results.

27. There are three children in a room with ages four, five, and six. If a five year old child enters the room, then which of the following statement is True ?

A. Mean age and standard deviation will stay the same

B. Mean age will stay the same but the standard deviation will increase

C. Mean age and standard deviation will increase.

D. Mean age will stay the same but the standard deviation will decrease.

Solution :

Age	4	5	6
f_1	1	1	1
f_2	1	2	1

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$\bar{X}_1 = \frac{4+5+6}{1+1+1} = \frac{15}{3} = 5$

$\bar{X}_2 = \frac{4+2\times5+6}{1+2+1} = \frac{20}{4} = 5$ $\therefore$ Mean age will stay the same.

$\sigma_1 = \sqrt{\frac{(4-5)^2+(5-5)^2+(6-5)^2}{3}} = \sqrt{\frac{2}{3}}$

$\sigma_2 = \sqrt{\frac{(4-5)^2+2(5-5)^2+(6-5)^2}{1+2+1}} = \sqrt{\frac{2}{4}} = \sqrt{\frac{1}{2}}$ since $\sigma_1 > \sigma_2$

Answer : D

28. Fana Telivision (Fana Lamrote Music) competition, five judge s a ward the following rating to the top three competitors.

Competitor : X	58	71	83	98	40
Competitor : Y	74	72	68	70	66
Competitor : Z	60	82	68	53	67

Who will be the winner to be a warded one prize ?

A. Competitor Z B. Competitor X C. Competior Y

Solution :

Sum of score of each is 350.

Mean of $X = \frac{350}{5} = 70$, mean $Y = \frac{350}{5} = 70$, and mean of $Z = \frac{350}{5} = 70$

Since the mean of each competitor are equal, and inadequate to measure hence it must be supported by measure of dispersion. The deviation from the mean for three candidat :

Competitor	Standard deviation
X	$S.D = \sqrt{\frac{(58-70)^2+(71-70)^2+(83-70)^2+(98-70)^2+(40-70)^2}{5}}$ $S.D = \sqrt{\frac{12^2+1^2+13^2+28^2+30^2}{5}} = 20$

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Y	$S.D = \sqrt{\frac{(74-70)^2+(72-70)^2+(68-70)^2+(70-70)^2+(66-70)^2}{5}}$ $S.D = \sqrt{\frac{16+4+4+0+16}{5}} = 2.83$
Z	$S.D = \sqrt{\frac{(60-70)^2+(82-70)^2+(88-70)^2+(53-80)^2+(67-70)^2}{5}}$ $S.D = \sqrt{\frac{100+144+324+289+9}{5}} = \sqrt{173} = 13.2$

The competitor with the least coefficient of variation is the winner of the award.

Answer : C

29. The mean and standard deviation of marks obtained by 50 students of a class in three subjects, mathematics, physics and chemistry are given below :

Subject	Maths	Chemistry	Physics
Mean	45	35	40
Standard deviation	12	15	20

Which of the three subjects show the highest variability in marks and which shows the least ?

Solution :

Coefficient of variation :

$$\text{For Maths} = \frac{S.D}{Mean} \times 100 = \frac{12}{45} \times 100 = 26.7\%$$

$$\text{For Chemistry} = \frac{S.D}{Mean} \times 100 = \frac{15}{35} \times 100 = 42.9\%$$

$$\text{For Physics} = \frac{S.D}{Mean} \times 100 = \frac{20}{40} \times 100 = 50\ \%$$

Physics marks have highest variability and maths marks have lowest variability.

30. Consider the following two data series A and B

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Data A : 3, 5, 4, 8 and Data B : 7, 9, 10, 2

a. Which data series is more uniform

b. Which data series is more variable

Solution :

For data A Mean of A: $\bar{X} = \frac{3+5+4+8}{4} = 5$

Variance: $A = \frac{(3-5)^2+(4-5)^2+(8-5)^2}{4} = \frac{14}{4} = 3.5$

S.D of $A = \sqrt{3.5} = 1.87$

Coefficient of variaion of $A = \frac{S.D}{mean} \times 100 = \frac{1.87}{5} \times 100 = 37.4$

For data B Mean : $\bar{X} = \frac{7+9+10+2}{4} = \frac{28}{4} = 7$

Variance: $\sigma^2 = \frac{(7-7)^2+(9-7)^2+(10-7)^2+(2-7)^2}{4} = \frac{38}{4} = 9.5$

S.D: $\sigma = 9.5 = 3.08$

Coefficient of variaion of B : $= \frac{S.D}{mean} \times 100 = \frac{3.08}{7} \times 100 = 44$

a. Data series A is more uniform and consistent.

b. Data series B has more varability

31. Coefficient of variation of two distributions are 60 and 70 and their standard deviations are 21 and 16 respectively. What are their means

Solution :

$CV = \frac{S.D}{mean} \times 100$, this implies $\frac{C.V}{100} = \frac{S.D}{mean}$

$\therefore$ Mean : $\bar{X}_1 = \frac{21}{0.6} = 35$ and mean : $\bar{X}_2 = \frac{16}{0.7} = 22.86$

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Theorem :

For the data : $X_1, X_2, X_3, \ldots\ldots, X_n$ the standard deviation (σ) is given by :

$$\sigma = \sqrt{\frac{\sum X_i}{n} - (\bar{X})^2}$$

To proof: $S.D = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n}}$

$$= \sqrt{\frac{\sum X_i^2}{n} - \frac{\sum 2X_i\bar{X}}{n} + \frac{\sum \bar{X}^2}{n}}$$

$$\sqrt{\frac{\sum X_i^2}{n} - 2\bar{X}^2 + \bar{X}^2} \text{ where } \sum \bar{X}^2 = n\bar{X}^2$$

$$\sqrt{\frac{\sum X_i^2}{n} - \bar{X}^2}$$

Illustrative Example

32. The mean of 8 observations is 6 and their variance is 9.25. If six of the observations are 3, 4, 7, 9, 9, and 10.

a. Find the remaining two observation.

b. Find the coefficient of variation.

Solution :

a. Let the other two observations be x_7 and x_8.

$$\text{Mean} = \frac{\sum X_i}{8} = 6 \text{ implies } \sum X_i = 48$$

$$\sum X_i = 3+4+7+9+9+10+X_7+X_8 = 48$$

$$X_7 + X_8 = 48 - 42 = 6 \ldots\ldots\ldots (*)$$

Using the theorem: variance : $\sigma^2 = \frac{\sum X_i^2}{n} - \bar{X}^2 = 9.25$

$$\frac{\sum X_i^2}{8} - 6^2 = 9.2 \text{ implis that } \sum X_i^2 = 8(9.25+36) = 362$$

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$\sum X_i^2 = (3)^2 + (4)^2 + (7)^2 + (9)^2 + (10)^2 + X_7^2 + X_8^2 = 362$

$\therefore X_7^2 + X_8^2 = 362 - 336 = 26$ .....(**)

Solving (*) and (**), we get $X_7 = 1$, $X_8 = 5$

b. Coefficient of variation $= \frac{S.D}{\bar{X}} \times 100 = \frac{\sqrt{9.25}}{6} \times 100\% = 50.69\%$

11. The mean and coefficient of variation of 8 observations are 8 and 6.25 respectively. If six of the observations are 2, 2, 8, 8, 14, and 14. Find the other two observation

Solution :

First, we find the variance recall that, coefficient of variation :

$C.V = \frac{S.D}{mean} \times 100 = \frac{\sqrt{Variance}}{8} \times 100 = 56.25$

$\therefore \sqrt{Variance} = \frac{56.25 \times 8}{100} = 4.5$

$Variance = (4.5)^2 = 20.25$

Let x and y are the other two observations.

$Mean = \frac{2+2+8+8+14+14+x+y}{8}$ 8

$\therefore x + y = (8 \times 8) - 48 = 16$ ..... (*)

From, the theorem : variance $= \frac{\sum X_i^2}{n} - (\bar{X})^2 = 20.25$

$\frac{\sum X_i^2}{8} - 8^2 = 20.25$

$\sum X_i^2 = 8(20.2 + 8) = 674$

$\sum X_i^2 = 2^2 + 2^2 + 8^2 + 8^2 + 14^2 + 14^2 + x^2 + y^2 = 674$

$x^2 + y^2 = 674 - 528 = 146$..... (**)

Solving (*) and ((* *) together :

$(x+y)^2 = x^2 + y^2 + 2xy$

$(16)^2 = 146 + 2xy$

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$2xy = 256 - 146 = 110$

$xy = 55 \quad \therefore \quad x = 11$ and $y = 5$

34. Calculate the mean and variance of the first n natural numbers

Solution :

$$\text{Mean} : \bar{X} = \frac{1+2+3+.....+n}{n} = \frac{\frac{n}{2}(n+1)}{n} = \frac{n+1}{2}$$

$$\text{Variance} = \frac{\sum X_i^2}{n} - \bar{X}$$

$$= \frac{1^2+2^2+3^2+.....+n^2}{n} - \left(\frac{n+1}{2}\right)^2$$

$$= \frac{\frac{n}{6}(n+1)(2n+1)}{n} - \left(\frac{n+1}{2}\right)^2$$

$$= \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}$$

$$= \frac{n+1}{2}\left[\frac{2n+1}{3} - \frac{n+1}{2}\right] = \frac{n+1}{2} \cdot \frac{n-1}{6} = \frac{n^2-1}{12}$$

35. The mean marks and standard deviation of 100 students were four be 40 and 6 respectively. Later on it was observed that a score o and 20 was misread as 70 and 90. Find the correct mean and stan deviation.

Solution :

$$\text{Correct mean} = \frac{(\text{Incorrect } \Sigma X) - (\text{incorrect Item}) + (\text{Correct Item Su}}{n}$$

$$\text{Incorrect mean} = \frac{\text{Incorrect sum } (\Sigma X)}{n} = 40$$

$$\frac{\text{Incorrect sum } (\Sigma X)}{100} = 40$$

$$\text{Incorrect sum } (\Sigma X) = 40 \times 100 = 4000$$

$$\text{Correct sum } (\Sigma X) = (\text{Incorrect sum}) - (\text{Incorrect item sum}) + (\text{correct item sum})$$

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$= 4000 - (70 + 90) + (40 + 20) = 3900.$

$\therefore$ Correct mean $= \frac{\text{correct sum}}{n} = \frac{3900}{100} = 39$

Standard deviation : $\sigma = \sqrt{\frac{\sum X_i^2}{n} - (\bar{X})^2}$

$6 = \sqrt{\frac{\text{Incorrect}\,(\sum X_i^2)}{100} - (40)^2}$

$36 = \frac{\text{Incorrect}\,(\sum X_i^2)}{100} - 1600$

Incorrect $(\sum X_i^2) = 100 \times (36 + 1600) = 163,600$

Correct $(\sum X_i^2) = \text{Incorrect}\,(\sum X_i^2) - (70^2 + 90^2) + (40^2 + 20^2)$

$= 163,600 - (4900 + 8100) + (1600 + 400) = 152,600$

$\therefore$ correct : S.D $= \sqrt{\frac{152,600}{100} - (39)^2}$

$= \sqrt{1526 - 1521} = \sqrt{5} = 2.24$

Note :

(i) If the data : $X_1, X_2, X_3, \ldots\ldots X_n$ is increased by an amount K, then the new variance remains unchanged.

(ii) If the data : $X_1, X_2, X_3, \ldots\ldots X_n$ is multiplied by constant K, then the new variance : $\sigma^2 = K^2$ (old variance),

new S. D = K odd standard deviation.

36. Consider the data given below 5, 8, 4, 3, 9, 7.

a. Find the standard deviation

b. Add 3 to each value, and then find the standard deviation

c. Subtract 3 to each value, and then find the standard deviation.

d. Multiply each value by 3 and then find the standard deviation

Solution :

a. Mean : $\bar{X} = \frac{5 + 8 + 4 + 3 + 9 + 7}{6} = \frac{36}{6} = 6$

Variance : $\sigma^2 = \frac{(5-6)^2 + (8-6)^2 + (4-6)^2 + (3-6)^2 + (9-6)^2 + (7-6)^2}{6}$

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$$= \frac{1+4+4+9+9+1}{6} = \frac{28}{6} = 4.67$$

Standard deviation : $\sigma = \sqrt{4.67} = 2.16$

b. After adding 3 to each value we obtain the set 8, 11, 7, 6, 12, 10

The new mean : $\bar{X} = \frac{8+11+7+6+12+10}{6} = \frac{54}{6} = 9$

new mean = old mean + 3 = 6 + 3 = 9

New variance : $\sigma^2 = \frac{(8-9)^2+(11-9)^2+(7-9)^2+(6-9)^2+(12-9)^2+(10-9)^2}{6}$

$$= \frac{1+4+4+9+9+1}{6} = \frac{28}{6} = 4.67$$

$\therefore$ new standared deviation : $\sigma = \sqrt{4.67} = 2.16$

Conclusion : old standard deviation = new standard deviation.

Adding the same constant number to each value the deviation is stay the same.

c. Subtracting the same constant number to each data value the deviation doesn't change.

Old standard deviation = New standard deviation

d. Multiplying by 3, the data : 15, 24, 12, 9, 27, 21.

new mean : $\bar{X} = \frac{15+24+12+9+27+21}{6} = \frac{108}{6} = 18$

new variance : $\sigma^2 = \frac{(15-18)^2+(24-18)^2+(12-18)^2+(9-18)^2+(27-18)^2+(21-18)^2}{6}$

$$= \frac{9+36+36+81+81+9}{6} = \frac{252}{6} = 42$$

New standared variation : $\sigma^2 = \sqrt{42} = 6.48$

Conclusion : New standard deviation = 3 × (old deviation)

37. The variance of 20 observation is 3. If each observation is multiplied by 5, find ;

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a. the new variance b. the new standard deviation

Solution :

Let the data be $X_1, X_2, \ldots\ldots X_{20}$.

$$\text{Variance: } \sigma^2 = \frac{\sum_{i=1}^{20}(X_i - \bar{X})^2}{20} = 3.0$$

$$\sum_{i=1}^{20}(X_i - \bar{X})^2 = 20 \times 3 = 60$$

New data will be $5X_1, 5X_2, \ldots\ldots, 5X_n$.

We known that the new mean is $5\bar{X}$

$$\text{New varianvce : } \sigma^2 = \frac{\sum_{i=1}^{20}(5X_i - 5\bar{X})^2}{20} = \frac{\sum_{i=1}^{20} 25(X_i - \bar{X})^2}{20}$$

$$= 25 \times \left(\frac{\sum_{i=1}^{20}(X_i - \bar{X})^2}{20} \right) = 25 \times 3 = 7$$

8.2 Use of Frequency Curves

A frequency distribution can be described with the help of the following measures.

1. **Averages** These are measures of central tendency (mean, median, mode) of the data, determines the skewness of a distribution
2. **Measure of dispersions :** These are measures of variation, scatter or spread) of the data, determine the degree of dispersion about the central tendency.
3. **Measures of skewness** - These measures determine the degree of a symmetry of the distribution on the two sides of the point of greatest concentration in the data.

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8.2.1 Skewness

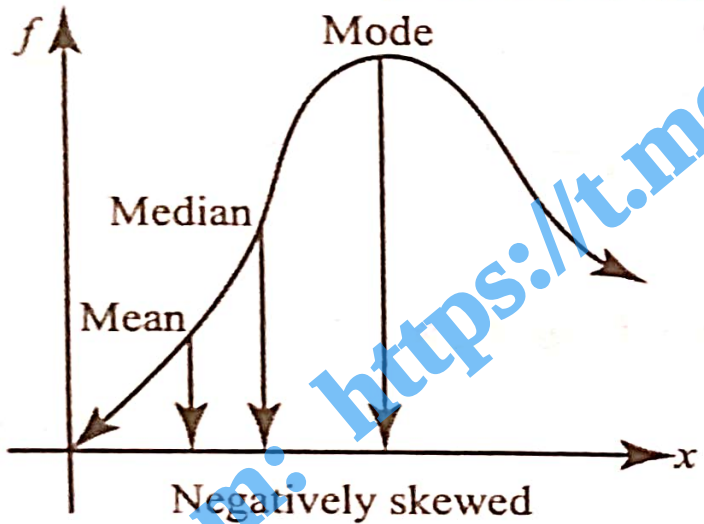
Skewness is the degree of a symmetry, or departure from symmetry, of a distribution.

For a symmetrical distribution, the frequencies are symmetrically distributed about the mean, that is variates equidistant from the mean have equal frequencies. In the case of such distribution, the **mean**, **mode**, and **median** coincide and the median lies half way between the two quartiles.

Thus **mean = mode = median** and $Q_3 - M_d = M_d - Q_1$.

Skewness means lack of symmetry in a frequency distribution. Skewness indicates whether the curve is turned more to one side than to other, i.e whether the curve has longer tail on one side.

Skewness can be positive, as well as negative



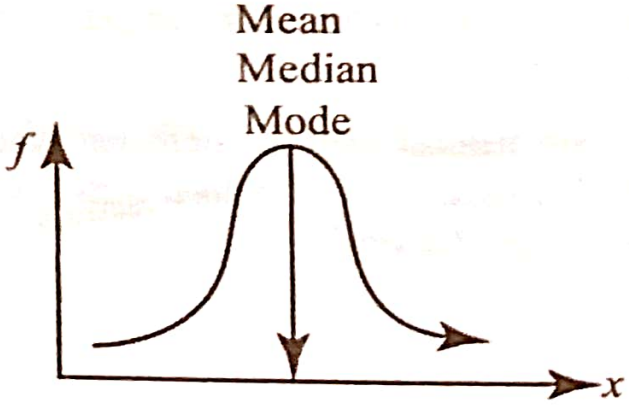
Negatively skewed
(left skewed)

(i) Skewness is negative if the longer tail is on the left side and the mean is on the left of the median.

Negatively skewed (left skewed)

- Mean < median < mode

(ii) No skewness if the distribution is symmetrical in symmetrical distribution the data values equally distributed on both side of the mean.



- The quartiles are equidistant from the mean. In addition when the distribution is unimodal, the mean, median, and mode are the same and are at the center of distribution. That is **mean = median = mode**.

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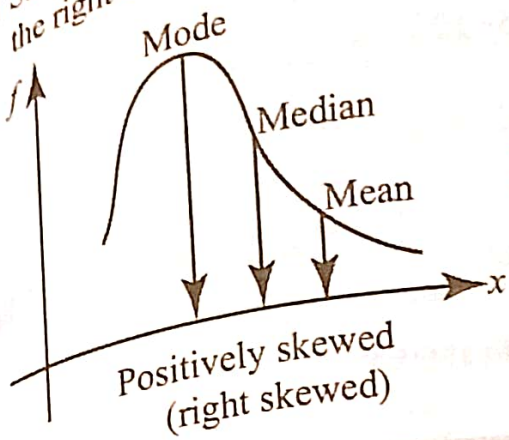
(iii) **Skewness is positive**, if the longer tail of the distribution lies towards the right and the mean is on the right of the median.



Positively skewed (right skewed)

- Mean > median > mode.

Illustrative Example

38. Consider the following frequency distribution.

X	5 - 15	15 - 25	25 - 35	35 - 45
f	10	20	15	5

a. Calculate the mean, median and mode.

b. Describe the skewness of the distribution.

Solution :

a. To find the mean :

- The mean mark of the class is 10, 20, 30 and 40.

$$\text{Mean} : \bar{X} = \frac{10\times10+20\times20+15\times30+5\times40}{10+20+15+5}$$

$$= \frac{100+400+450+200}{50} = \frac{1150}{50} = 23$$

To find the Median Class : $\left(\frac{n}{2}\right)^{th}$ item $\left(\frac{50}{2}\right)^{th}$ item = 25^{th} item

$\therefore$ median class is 15 - 25, $B_L = 15$, $i = 10$, $f_C = 20$, $Cf_B = 10$

$$M_d = B_L + \frac{i}{f_C}\left(\frac{n}{2} - Cf_B\right) = 15 + \frac{10}{20}(25-10) = 22.5$$

To find the mode : The modal class is 15 - 25

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$B_L = 15, i = 10 f_m = 20, f_1 = 10, f_2 = 15$

Hence, $d_1 = |f_m - f_1| = |20-10| = 10$ $d_2 = |f_m - f_2| = |20-15| = 5$

Mode : $M_o = B_L + i\left(\frac{d_1}{d_1 + d_2}\right) = 15+10\left(\frac{10}{10+5}\right) = 21.7$

$\therefore$ Mean = 23, median = 22.5, and mode = 21.7

b. Since, mean > median > mode, then the skewness is positive and the distribution falls toward the right.

8.2.2 Measures of Skewness

A measure of skewness gives a numerical expression and the direction of asymmetry in a distribution. It gives information about the shape of the distribution and the degree of variation on either side of the central value.

Relative measures of skewness are called the **coefficient of skewness.**

Skewness can be measured by calculating :

(i) **Pearson's coefficient of skewness** is defined as :

$$\alpha = \frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$$

(ii) **Bowley's coefficient of skewness** based on quartiles is defined as

$$\beta = \frac{Q_3 + Q_1 - 2(\text{median})}{Q_3 - Q_1}$$

Interpretation :

If the coefficient of skewness is

(i) Zero then the distribution is symmetrical

(ii) Positive value, then the distribution is skewed positively (skewed to right)

(iii) Negative value, then the distribution is skewed negatively (skewed left)

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Illustrative Example

39. Find the coefficient of skewness for each distribution, and describe the shape of the distribution.

a. Mean = 12, median = 10, standard deviation = 3

b. Mean = 24, median = 28, standard deviation = 5

c. Mean = 16.7, median = 16.7, standard deviation = 7

d. $Q_1 = 2$, $Q_3 = 8$ and median = 6

e. $Q_1 = 6$, $Q_3 = 16$ and median = 11

Solution :

a. Pearson coefficient of skewness :

$$\alpha = \frac{3(\text{mean} - \text{median})}{\text{S.D}} = \frac{3(12-10)}{5} = \frac{6}{5} = 1.2$$

Since $\alpha = 1.2 > 0$, then the shape of the distribution is positively skewed (skewed to right).

b. Pearson's coefficient of skewness :

$$\alpha = \frac{3(\bar{X} - M_d)}{\text{S.D}} = \frac{3(24-28)}{7} = -1.71$$

Since $\alpha = -1.71 < 0$, then the shape of the distribution is negatively skewed and skewed toward left.

c. Pearson's coefficient of skewness :

$$\alpha = \frac{3(\text{mean} - \text{median})}{\text{S.D}} = \frac{3(16.7-16.7)}{7} = 0$$

Because $\alpha = 0$, then the distribution is symmetrical.

d. Bowley's coefficient of skewness :

$$\beta = \frac{Q_3 + Q_1 - 2(M_d)}{Q_3 - Q_1} = \frac{8+2-2(6)}{8-2} = \frac{-2}{6} = -0.33$$

Because $\beta = -0.33 < 0$, then the shape of the distribution is skewed negatively, (skewed to ward left)

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e. Bowley's coefficient of skewness :

$$\beta = \frac{Q_3 + Q_1 - 2(\text{median})}{Q_3 - Q_1} = \frac{16 + 6 - 2(11)}{16 - 6} = 0$$

Since $\beta = 0$, then the shape of the distribution is symmetrical.

40. Consider the following data distribution :

X	5	15	25	35	45
f	5	10	20	5	10

Calculate pearson's coefficient of skewness and describe the shape of distribution.

Solution :

$$\therefore \text{Mean} = \frac{5\times 5+10\times 15+20\times 25+5\times 35+10\times 45}{5+10+20+5+10} = \frac{1350}{50} = 27$$

$\therefore$ Median $= 25$ and stand deviation : $\sigma = 12.25$

Pearson's coeffiient of skewness is $\alpha = \frac{3(\text{mean} - \text{median})}{\text{standard - deviation}}$

$$= \frac{3(27-25)}{12.25} = 0.98$$

$\therefore$ The shape of distribution is skewed positively (skewed toward right)

41. Consider the following data distribution table :

Class :	0 - 6	6 - 12	12 -18	18 - 24	24 - 30
Frequency	8	10	12	9	5

Calculate the quartile coefficient of skewness

Solution :

The K^{th} quartile is: $Q_k = \left(\frac{K\times n}{4}\right)^{th}$ item

(i) First quartile class is $= \left(\frac{44}{4}\right)^{th}$ item $= 11^{th}$ item, class of Q_1 is 6 -12

$$\therefore Q_1 = B_L + \frac{i}{f_C}\left(\frac{n}{4} - Cf_B\right) = 6 + \frac{6}{10}(11-8) = 7.8$$

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(ii) 2^{nd} quartile class is $= \left(\frac{2\times 44}{4}\right)^{th}$ item $= 22^{th}$ item, class of Q_2 is 12 - 18

$$\therefore Q_2 = B_L + \frac{i}{f_C}\left(\frac{Kn}{4} - Cf_B\right) = 12 + \frac{6}{12}(22-18) = 14$$

(iii) $Q_3 = \left(\frac{3\times 44}{4}\right)^{th}$ item $= 33^{rd}$ item, class of Q_3 is 18 - 24

$$\therefore Q_3 = B_L + \frac{i}{f_C}\left(\frac{Kn}{4} - Cf_B\right) = 18 + \frac{6}{9}(33-30) = 20$$

Bowley's coefficient: $\beta = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$

$$\beta = \frac{20+7.8-2(14)}{20-7.8} = \frac{-0.2}{12.2} = -0.02$$

Since $\beta = -0.02 < 0$, then the distribution of the data shape is skewed negatively (skewed to left).

2. Find Bowley's coefficient of skewness for the following data and describe the shape of skewness of the distribution : 13, 17, 16, 14, 11, 19, 21, 10, 24, 18, 12

Solution :

First write in order 10, 11, 12, 13, 14, 16, 17, 18, 19, 21, 24

$$Q_2 = \text{median} = \left(\frac{n+1}{2}\right)^{th} \text{item} = \left(\frac{11+1}{2}\right)^{th} \text{item} = 6^{th}\text{item} = 19$$

$$Q_1 = \left(\frac{1(11+1)}{4}\right)^{th} \text{item} = 3^{rd} \text{item} = 16$$

$$Q_3 = \left(\frac{3(11+1)}{4}\right)^{th} \text{item} = 9^{th} \text{item} = 24$$

Bowley's coefficient: $\beta = \frac{Q_3 + Q_1 - 2(Q_2)}{Q_3 - Q_1}$

$$= \frac{24+16-2(19)}{24-16} = \frac{2}{8} = 0.25$$

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Since $\beta = 0.25 > 0$, then the distribution is skewed positively (to toward right).

Conceptual Example

43. If the mode is to the left of the median and the mean is to the right median, then the distribution is ___

A. no skew
B. positively skewed
C. negatively skewed
D. can't determined

Solution :

Mode < median < mean, positively skewed

44. When data are catagorized as, for example, places of residence as rural, suburban, urban, the most appropriate measure of ce tendency is the

A. Mean
B. Mode
C. Median
D. Midra

45. Classify each distribution as symmetrical, negatively skewed, positively skewed.

a. Lower quartile : $Q_1 = 7$, median = 10 and $Q_3 = 13$
b. Mean : $\bar{X} = 7$, median = 6, mode = 5
c. Mean : $\bar{X} = 4$, median = 6, and standard deviation = 2

Solution :

a. Bowley's coefficient: $\beta = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1} = \frac{13+7-2(10)}{13-7} = 0$

$\therefore$ No skew, which is symmetrical

b. Positively skewed, because mode < median < mean

c. Negatively skewed, because by pearson's coefficient.

$$\alpha = \frac{3(\text{mean} - \text{median})}{S.D} = \frac{3(4-6)}{2} = -3$$

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8.3 Sampling Techniques

In your previous grade, the two major areas of statistics are descriptive and **inferential.**

(i) **Descriptive statistics** consists of the collection, organization, summarization and presentation of data.

(ii) **Inferential statistics** consists of generalizing from sample to populations, performing estimation, determining relationships among variables, and making predication inferential statistics is based on probability theory. Here, it is important to distingush between a **sample** and a **population.**

A population consists of all subjects (human or otherwise) that are being studied.

In most cases the population under study are large, due to the expense, time, size of population, etc, it is not possible to use the entire population for a statistical study, therefore, researchers use population into group, group into subgroups called samples to get the necessary data for their study.

A sample is a group of subjects selected from a population

8.3.1 Types of Sampling

There are various types of sampling techniques that are in application. They are broadly catagorized into two.

(i) **Non Probability Sampling Techniques**

Non random sampling techniques, do not follow the scientific rules of sample selection. It is bias or does not have equal chance.

The most known non - probability sampling techniques are :

- Purposive or judgement sampling
- Quota sampling
- Sequential sampling, etc

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(ii) Probability Sampling Techniques (Random Sampling techniques)

- It is scientific
- It allows equal chance to every unit

Under probability sampling techniques there are four basic method used to obtain samples :

1. Simple random sampling method
2. Systematic sampling method
3. Stratified sampling method
4. Cluster sampling method

1. Simple Random Sampling

Random samples are selected by using chance methods or random numbers. Every individuals has equal chance of being selected.

Methods of drawing random sample are :

- Using lottery method
- Using random number table

2. Systematic Sampling

In a systematic sampling subjects are selected by using every K^{th} number after the first subject is randomly selected from 1 through K.

3. Stratified Sampling

Subjects are selected by dividing up the population into group (strata) and subjects within groups are randomly selected

4. Cluster Sampling

Researchers select cluster samples by using intact groups that is representative of the population.

- Cluster sampling divides the population into subgroups, but each subgroup has similar characteristics to the entire group. In this case the sampling unit we select is randomly entire subgroups

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For example

If researcher wanted to do study involving the patients in the hospitals in Addis Ababa City, it would be very costy and time consuming to try to obtain a random sample of patients of over a large area. Instead a few hospitals could be selected at random and the patient in these hospitals would be interviewed in a cluster.

5. Multi Stage Sampling (Phase Sampling)

Multi - stage sampling is more complex form of cluster sampling, in which smaller groups successively selected from large population

- The population is 1st divided into large group, known as 1st stage units. Every first stage unit is divided into smaller groups known as second stage units. These 2nd stage units are further divided into third stage etc.

Illustrative Example

46. Name the four basic sampling methods:

Solution :

Random, systematic, cluster and stratified

47. Why are samples used in statistics

Solution :

Samples are used to save money and time when the population is large

48. The group of all subjects under a study is called ____

Answer : a population

49. A group of subjects selected from the group of all subjects under study is called ____

A. Population
C. Statistics
B. Sample
D. Absolute dispersion

Answer : B

50. A researcher divided subjects into two groups according to gender and then selected members from each group for her sample. What sampling method was the researcher using ?

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A. Cluster
B. Random
C. Stratified
D. Systematic

Answer : C

51. In research study, participants should be assigned to group using ___ methods, if possible

A. Cluster
B. Random
C. Stratified
D. Systematic

Answer : B

52. Classify each sample as random systematic, stratified, or cluster.

a. In a large school district, all teachers from two buildings are interviewed to determine whether they belive the students have less homework to do now than in previous years

b. Every 10^{th} customer entering a shopping mall is asked to select her or his favorite store.

c. Teaching super visors are selected using random numbers in order to determine annual salaries increament

d. Every 20^{th} bottled oil manufactured is checked to determine its fat content.

e. University graduate of a large city are divided into four groups according to gender (male or female) and according to whether they graduate in natural science or social science. Then 5 are selected from each group and interviewed to determine whether they have been working by their own private company ?

Answer

a. Cluster samples : because we selected by using an intact group that is representative of the population.
b. Systematic sample
c. Random - sample
d. Systematic
e. Stratified

SUMMARY OF THE UNIT

1. Range = L - S
2. Mean deviation is the mean of the absolute deviations of the observation

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3. If mean deviation is calculated about the mean, then

$$MD(\bar{X}) = \frac{\sum |X_i - \bar{X}|}{n}, \text{ For individual series}$$

$$MD(\bar{X}) = \frac{\sum f_i |X_i - \bar{X}|}{\sum f_i}, \text{ For frequency distribution}$$

4. If mean deviation is about median

$$MD(M_d) = \frac{\sum |X_i - \text{median}|}{n}, \text{For individual series}$$

$$MD(M_d) = \frac{\sum f_i |X_i - \text{median}|}{\sum f_i}, \text{ For frequency destribution}$$

5. Standard deviation is

$$S.D: \sigma = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n}}, \text{ For individual series}$$

$$S.D: \sigma = \sqrt{\frac{\sum f_i (X_i - \bar{X})^2}{\sum f_i}}, \text{ For frequency distribution}$$

$$\text{Variance}: \sigma^2 = (S.D)^2$$

$$\text{Coefficient of Variation}: CV = \frac{S.D}{\text{Mean}} \times 100$$

Summary of Sampling Method

(i) **Random samples** are selected by random numbers

(ii) **Systematic samples** are selected by using every K^{th} number after the first subject is randomly selected from 1 through K.

(iii) **Stratified samples** are selected by dividing up the population into group, and subjects within groups are randomly selected.

(iv) **Cluster samples** are selected by using an intact group that is representative of the population.

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Test Your Understanding

1. Find the mean deviation about mean

   $MD(\bar{X})$ and coefficient of mean deviation about mean for the following data

   a. 4, 7, 8, 10, 12, 13, 17, and 9

   b.

x	5	7	9	10	12	15
f	8	6	2	2	2	6

   c.

x	0 - 6	6 - 12	12 - 18	18 - 24	24 - 30
f	8	10	12	9	5

2. Find the mean deviation about the median, $MD(M_d)$ and coefficient of mean deviation about the median for the following data distribution :

   a. 8, 7, 6, 9, 4, 11, 12, 15, 17

   b.

x	15	21	27	30	32
f	3	5	6	7	4

3. Find the range, inter - quartile range coefficient of range, coefficient of quartile deviation.

class	10 - 15	15 - 20	20 - 25	25 - 30	30 - 35
frequency	4	6	5	3	2

4. Find mean deviation about the mode and coefficient of mean from mode of the following data distribution.

   a. 6, 6, 7, 77, 8, 8, 10

   b.

x	10	11	12	13	15	16
f	2	1	3	1	2	1

5. Find the quartile deviation and coefficient of quartile deviation

   a. 16, 22, 26, 30, 40, 35, 22, 31, 38

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b.

x	10	15	20	25	30
f	2	4	6	8	5

c.

class	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
frequency	6	8	14	16	4	2

6. Find mean, variance, standard deviation an coefficient of variation

a. 3, 5, 6, 7, 9, 18

b.

x	7	8	9	10	11	12
f	2	3	1	2	1	1

c.

class	1 - 5	6 - 10	11 - 15	16 - 20
frequency	4	6	8	2

Answer Check

a. Mean : $\bar{X} = \frac{4+7+8+10+12+13+17+9}{8} = \frac{80}{4} = 10$

(i) $MD(\bar{X}) = \frac{\sum |X_i - \bar{X}|}{n} = \frac{24}{8} = 3$

(ii) Coefficient of mean deviation : $CMD = \frac{MD}{\bar{X}} = \frac{3}{10} = 0.3$

b. Mean : $\bar{X} = \frac{\sum f_i X_i}{\sum f_i} = \frac{234}{26} = 9$

(i) Mean deviation : $MD(\bar{X}) = \frac{\sum f_i |X_i - \bar{X}|}{\sum f_i} = \frac{88}{26} = 3.38$

(ii) Coefficient of mean deviation : $CMD = \frac{MD(\bar{X})}{\bar{X}} = \frac{3.38}{9} = 0.38$

c. Mean: $\bar{X} = \frac{\sum f_i m_i}{\sum f_i} = \frac{618}{44} = 14$

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(i) Mean deviation : $MD(\bar{X}) = \frac{\sum f_i|m_i - \bar{X}|}{\sum f_i} = \frac{394}{44} = 8.95$

(ii) Coefficient of mean deviation : $CMD = \frac{MD(\bar{X})}{\bar{X}} = \frac{8.95}{44} = 0.64$

2a. Write the data in order : 4, 6, 7, 8, 9, 11, 12, 15, 17, then :

Median : $M_d = \left(\frac{9+1}{2}\right)^{th}$ iem $= 5^{th}$ item $= 9$

(i) Mean deviation : $(M_d) = \frac{\sum|X_i - M_d|}{n} = \frac{30}{9} = 3.33$

(ii) Coefficient of mean deviation : $CMD = \frac{MD(M_d)}{M_d} = \frac{3.33}{9}$

3. Range : 35 -10 =25, coefficient of range $= \frac{35-10}{35+10} = 0.56$

$Q_1 = \left(\frac{20}{4}\right)^{th}$ item $= 5^{th}$ item, class of Q_1 is 15 - 20

$Q_1 = 15.83$

$Q_3 = \left(\frac{3\times 20}{4}\right) = 15^{th}$ item, class of Q_3 is 20-25

$\therefore Q_3 = 20 + \frac{5}{5}(15 - 10) = 25$

inter -quartile range $= Q_3 - Q_1 = 25 - 15.83 = 9.17$

Coefficient quartile deviation : $\frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{9.17}{40.83} = 0.225$

4. a. Mode : $M_o = 7$

(i) Mean deviation about mode: $MD(M_o) = \frac{\sum|X_i - Mode|}{n}$

$MD(M_o) = \frac{2|6-7| + 3|7-7| + 2|8-7| + |10-7|}{8} = \frac{7}{8} = 0.875$

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(ii) Coefficient of mean deviation from mode

$$CMD = \frac{MD(M_o)}{M_o} = \frac{0.875}{7} = 0.125$$

b. Mode : Mo = 12

(i) Mean deviation about mode: $MD(M_o) = \frac{\sum f_i |X_i - M_o|}{\sum f_i}$

$$MD(M_o) = \frac{2|10-12|+1|11-12|+3|12-12|+1|13-12|+2|15-12|+|16-12|}{2+1+3+1+2+1}$$

$$MD(M_o) = \frac{4+1+0+1+6+4}{10} = \frac{16}{10} = 1.6$$

(ii) Coefficient of mean deviation about

$$\text{Mode : } CMD = \frac{MD(M_o)}{M_o} = \frac{1.6}{12} = 0.133$$

a. Arrange the data in order 16, 22, 22, 26, 30, 31, 35, 38, 40.

$$Q_1 = \left(\frac{1(n+1)}{4}\right)^{th} \text{item} = \left(\frac{10}{4}\right)^{th} \text{item} = (2.5)^{th} \text{item}$$

$$= 2^{nd} \text{ item} + 0.5\left(3^{rd} - 2^{nd}\right)\text{item} = 22$$

$$Q_3 = \left(\frac{3(n+1)}{4}\right)^{th} \text{item} = (7.5)^{th} \text{ item}$$

$$Q_3 = 7^{th} \text{item} + 0.5\left(8^{th} - 7^{th}\right)\text{item}$$

$$Q_3 = 35 + 0.5(38-35) = 36.5$$

(i) Quartile deviation : $QD = \frac{1}{2}(Q_3 - Q_1) = 36.5 - 22 = \frac{14.5}{2}$

(ii) Coefficient of quartile deviation :

$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{36.5 - 22}{36.5+22} = \frac{14.5}{58.5} = 0.25$$

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b. $Q_1 = \left(\frac{1(25+1)}{4}\right)^{th}$ item $= (6.5)^{th}$ item

$= 6^{th}$ item $+ 0.5\left(7^{th} - 6^{th}\right)$ item

$= 15 + 0.5(20-15) = 17.5$

$Q_3 = \left(\frac{3(25+1)}{4}\right)^{th}$ item $= (19.5)^{th}$ item $= 25$

(i) Quartile deviation : $Q.D = \frac{1}{2}\left(Q_3 - Q_1\right) = 3.75$

(ii) Coefficient of quartile deviatipon:

$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{25 - 17.5}{25 + 17.5} = \frac{7.5}{42.5} = 0.18$$

c. The K^{th} quartile item is $\left(\frac{K \times n}{4}\right)^{th}$ item

$Q_1 = \left(\frac{1 \times 50}{4}\right)^{th}$ item $= (12.5)^{th}$ item,

The class of Q_1 is 10 - 20 $B_L = 10$, $i = 10$, $f_C = 8$, $Cf_B = 6$

$$Q_1 = B_L + \frac{i}{f_C}\left(\frac{Kn}{4} - Cf_B\right) = 10 + \frac{10}{8}(12.5 - 6) = 18.13$$

$Q_3 = \left(\frac{3 \times 50}{4}\right)^{th}$ item $= (37.5)^{th}$ item, Q_3 lies on 30 - 40.

$B_L = 30$, $i = 10$, $f_C = 16$, and $Cf_B = 28$

$$Q_3 = 30 + \frac{10}{16}(37.5 - 28) = 35.94$$

(i) Quartile deviation : $QD = \frac{1}{2}\left(Q_3 - Q_1\right) = \frac{1}{2}(35.94 - 18.13)$

$= 8.905$

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(ii) Coefficient of quartile deviation:

$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{35.94 - 18.13}{35.94 + 18.13} = \frac{17.81}{54.07} = 0.33$$

6a. Mean : $\bar{X} = \frac{3+5+6+7+9+18}{6} = \frac{48}{6} = 8$

(i) Variance : $\sigma^2 = \frac{(3-8)^2+(5-8)^2+(7-8)^2+(9-8)^2+(18-8)^2}{6} = 22.67$

(ii) $S.D = \sqrt{\text{Variance}} = \sqrt{22.67} = 4.76$

(iii) Coefficient of variance : $\frac{S.D}{\text{mean}} \times 100 = \frac{4.76}{8} \times 100 = 59.5$

b. Mean : $\bar{X} = \frac{14+24+9+20+11+12}{10} = \frac{90}{10} = 9$

(i) Variance: $\sigma^2 = \frac{2(7-9)^2+3(8-9)^2+1(9-9)^2+2(10-9)^2+(11-9)^2+(12-9)^2}{2+3+1+2+1+1}$

$= \frac{24}{10} = 2.4$

(ii) S.D : $\sigma = \sqrt{2.4} = 1.55$

(iii) Coefficient of variation $= \frac{S.D}{\text{mean}} \times 100 = \frac{1.55}{9} \times 100 = 17.21$

c. The class mark of each interval : 3, 8, 13, 18

Mean of class mark : $\bar{X} = \frac{4\times 3 + 6\times 8 + 8\times 13 + 2\times 18}{20} = \frac{200}{20} = 10$

(i) Variance : $\sigma^2 = \frac{4(3-10)^2+6(8-10)^2+8(13-10)^2+2(18-10)^2}{4+6+8+2}$

$= \frac{420}{20} = 21$

(ii) Standard deviation : $\sigma^2 = \sqrt{21} = 4.58$

(iii) Coefficient of variation : $\frac{S.D}{\text{mean}} \times 100 = \frac{4.58}{10} \times 100 = 45.8$

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UNIT 9

9. Probability and Counting Principle

9.1 Fundamental Counting Principle

There are two fundamental counting principles used in this uni

(i) Multiplication principle

(ii) Addition principle

9.1.1 Multiplication Principle

If an event can occur in m ways and if when this has occurred another events can occur n ways then both event can occur in order in m × n different ways to make the entire sequence.

Illustrative Example

1. In how many different ways can 6 people be seated in a row

Solution

We have 6 empty seats: ____, ____, ____, ____,

1st person has six choice, 2nd person has 5 choice, 3rd person can 4 choices .....

$$6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720 \text{ way}$$

2. In a class room 5 seats are vacant on a bench. In how many ways can 3 students sit on them.

Solution

Number of ways for 1st student

to occupy set = 5 ways ___ ___ ___

Number of ways for 2nd student to occupy seat = 4 ways

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Number of ways for 3rd student to occupy seat = 3 way

∴ By multiplication principle, the total number of ways for all the three to occupy seats: $5 \times 4 \times 3 = 60$

3. Addis Ababa car plate number code consists of one letters from the English alphabet followed by five digits

a. How many different ways of can plate are possible, if the digit many be repeated

b. How many different ways of car plate are possible, if digit may not be repeated.

Solution

a. Each letter can be printed in 26 different ways, the first digit in 10 way, and each of the other 4 digit in 10 ways.

∴ $26 \times 10 \times 10 \times 10 \times 10 \times 10 = 2600000$.

b. $26 \times 10 \times 9 \times 8 \times 7 \times 6 = 786240$.

4. An employee identification code for an Ethiopian teacher association consists of 2 letters from the set {A, B, C, D} followed by 4 digits.

a. How many ID are possible if both letter and digits may be repeated.

b. How many ID are possible if letter and digits may not be repeated.

Solution

a. $4 \times 4 \times 10 \times 10 \times 10 \times 10 = 160{,}000$

b. $4 \times 3 \times 10 \times 9 \times 8 \times 7 = 60{,}480$

5. Abebe has 4 pairs of trousers, 5 shirts and 2 pairs of shoes. In how many different ways can he dress.

Solution: A person can choose one trousers and one shirt and one shoes. Since there are 4 trousers choices, 5 shirt choice and 2 shoes choices, the total number of possible different dress is

∴ $4 \times 5 \times 2 = 40$.

6. A coin is tossed and a die is rolled. Find the number of out comes for the sequence of event.

Solution

Coin can land either head (H) or tail (T) up and die can land with any

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one of six numbers showing face up,

H	T
1, 2, 3, 4, 5, 6	1, 2, 3, 4, 5, 6

$H_1, H_2, H_3, H_4, H_5, H_6, T_1, T_2, T_3, T_4, T_5, T_6$

$\therefore$ By fundamental counting principle: $2 \times 6 = 12$ ways

7. A paint manufacture wishes t manufacture several different pain The categories include:

Color: Red, blue, white, black, green, brown, yellow.

Type: Latex, oil.

Texture: Flat, semi gloss, high gloss

Use: Out door, indoor

How many different kinds of paints can be made if a person can select one color, one type, one texture and one use?

Solution

A person can choose one color, and one type and one texture and one use. Since there are 7 color choice, 2 type choice, 3 texture choice and 2 use choices, the total number of possible different paints is:

Color	Type	Texture	Use	
7	2	3	2	$= 7 \times 2 \times 3 \times 2 = 84$

8. In a class there are 20 boys and 15 girls. The teacher wants to select 1 boy and 1 girl to represent the class in function monitor. In how many ways can the teachers make this selection?

Solution

No. of ways of selecting on boy out of 20 boys = 20 ways

No. of ways of selecting one girl out of 15 girls = 15 ways

$\therefore$ By Fundamental principle of counting total number of ways of selecting one boy and one girl $= 20 \times 15 = 300$

9. a. In how many ways can 3 boys and two girls sit in a row

b. In how many ways can 3 boys and two girls sit in a row if the boys

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and girls are each to sit together.

Solution

a. $5 \times 4 \times 3 \times 2 \times 1 = 120$ b. $4 \times 3 \times 2 \times 1 = 24$

9.1.2 Addition Principle

Suppose event A can occur in **m** - ways and a second event B can occur in **n** - ways, and suppose both events can not occur at the same time. Then A and B can occur in **(m + n)** ways

- If A and B are disjoint (mutually exclusive) then $n(A \cup B) = n(A) + n(B)$

Illustrative Example

10. Suppose a book shelf has 5 maths, 3 physics, 6 biology and 4 English reference.
    a. How many ways a student can choose one of the reference book
    b. How many ways a student can choose one of each type of reference book.

Solution

a. Here, the addition rule applies: $n = 5 + 3 + 6 + 4 = 16$

b. Here, the product rule applies: $n = 5 \times 3 \times 6 \times 4 = 360$

11. Suppose a box has 4 red balls, 3 white balls, and 2 green balls. How many ways can draw
    a. Just one of the ball in a box b. One of each kind of ball

Solution

a. Here, the addition rule applies $4 + 3 + 2 = 9$

b. Here, the multiplication rule applies $4 \times 3 \times 2 = 24$

12. A teacher has 5 different books that he wishes to arrange side by side. How many different arrangements are possible.

Solution

Five choice will be made, one for each space that will hold a book. Any of the five books could be chosen for the firsts place. There are 4 choice for the 2nd place. There are 3 choice for the 3rd space, 2 choice for the 2nd space:

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$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

$\therefore$ By fundamental counting principle the number of different possible arrangements is

The Use of the multiplication principle often leads to products such as 5 × 4 × 3 × 2 × 1, the product of all natural numbers from 5 down to 1. it is called 5 factorial, denoted by !5!

n - factorial

The symbol **n!** read "n factorial" denotes the product of all natural numbers from **n down to 1.**

- $n! = n \times (n-1) \times (n-2) \times (n-3) \ldots\ldots \times 3 \times 2 \times 1$

$3! = 3 \times 2 \times 1$ • 3 factorial

$2! = 2 \times 1$ • 2 - factorial

$0! = 1$

9.2 Permutation

What is a permutation?

A permutation is an arrangement of objects in a specific order.

Definition: Permutation Rule

The arrangement of n - objects in a specific order using r - objects at a time is called a permutation of n - objects taking r - objects at a time. It is denoted by $P(n, r) = {}^nP_r$, and is defined as:

$$ {}^nP_r = \frac{n!}{(n-r)!} $$

For example, for the set of letters: A, B, C

- Permutation of the three letters taken all at a time are

  ABC, ACB, BAC, BCA, CAB, CBA. $P(3, 3) = 3! = 3 \times 2 \times 1 = 6$.

- Permutation of the three letters taken two at a time are

  AB, AC, BA, BC, CA, CB

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$$P(3, 2) = \frac{3!}{(3-2)!} = \frac{3 \times 2 \times 1}{1} = 6$$

Illustrative Example

13. Suppose a business owner has a choice of five locations in which to establisher business. She decides to rank each location according to certain criteria, such as price of the store and parking facilities. How many different ways can she rank the five locations

Solution

There are $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

14. In how many ways 6 differently colored marbles be arranged in a row?

Solution

Number of arrangements of 6 marbles in a row is

$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720.$

15. In how may ways can 8 people be seated on a bench if only 3 seats are available?

Solution

Number of arrangements is 8 people taken 3 at a time:

$$P(8, 3) = \frac{8!}{(8-3)!} = \frac{8!}{5!} = 8 \times 7 \times 6 = 336$$

16. If 10 students compete in a race, in how many ways can the first three places be taken.

Solution: $P(10, 3) = \frac{10!}{7!} = 10 \times 9 \times 8 = 720$

17. How many different ways can the leader and vice - leader be selected for a research project if there are seven scientists available?

Solution: $P(7, 2) = \frac{7!}{(7-2)!} = \frac{7!}{5!} = 7 \times 6 = 42$

18. Four difference mathematics book, three different physics books, two different chemistry books are to be arranged on a shelf. How many

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different arrangements are possible:

a. If the books in each particular subject must all stand together.

b. If only the physics books must stand together?

Solution

a. The mathematics books can be arranged among themselves in P (4, 4) = 4! ways, the physics books in P (3, 3) = 3! the chemistry books in P (2, 2) = 2! and the three groups in P (3, 3) = 3! ways. Thus the required number of arrangement is 4! 3! 2! 3! = 864.

b. Consider the three physics books as one big book. Then we have 7 books that can be arranged in P (7, 7) = 7! ways. In all of those ways the physics books are together. But the physics books can be arranged among themselves in P (3, 3) = 3!. Therefore the required number of arrangements is 7! 3! = 241, 920

19. How many different ways can five people - A, B, C, D and E - sit in a row at a move theatre if:

a. A and B must sit together

b. C and E must sit at the end of each row.

Solution

a. Since the two person sit together, and consider them as one object. Now we have 4 objects: AB, C, D, E.

These can be arranged in P(4, 4) = 4!

The two persons can be arranged among themselves in P(2, 2) = 2!

∴ By **F.P.C**, total number of ways

$$= 4!\ 2! = 4 \times 3 \times 2 \times 1 \times 2 \times 1 = 48$$

b. Arrangements in which C and E sit at the end of each row in P(2, 2) = 2! ways (namely by C, E and E, C) and the remaining 3 person can be arranged in P (3, 3) = 3!.

∴ By **F. P.C** , total number of arrangements = 2! . 3! = 12 ways.

20. In how many ways can 4 boys and 3 girls be arranged so that no two girls sit together.

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Solution

In order to alternate boys and girls,

$B_1\ G_1\ B_2\ G_2\ B_3\ G_3\ B_4$: $4 \cdot 3 \cdot 3\ 2 \cdot 2 \cdot 1 \cdot 1 = 144$

Permutation of Identical Objects

The number of arrangements of n - objects alike, and so on for different type then the number of **permutation** is

$$\frac{n!}{n_1! \times n_2! \times n_3 \times .... \times n_r!} \text{ where } n = n_1 + n_2 + ... + n_r$$

21. How many different ways of five digits can be made using the digit

4, 4, 2, 2, 2

Solution

The are 4 two, 2 three: $n = 5, n_1 = 2, n_2 = 3$

$\therefore$ Number of arrangements $= \frac{5!}{2!\,3!} = \frac{5 \times 4}{2} = 10$

22. Find the number of (arrangement permutation) can be made from the letters of the world:

a. LAMP b. FOOD c. APPLE

Solution

a. Total number of lattes in "LAMP" are 4 and all are different

$P(4, 4) = 4! = 4 \times 3 \times 2 \times 1 = 24$

b. The world "FOOD" contains 4 letters F - occurs once, O - occurs twice and D - occurs once, that is $n = 4, n_1 = 1, n_2 = 2, n_3 = 1$

$\therefore$ Total number of arrangement is $\frac{4!}{2!\,1!\,1!\,1!} = 4 \times 3 = 12$

c. In the world "APPLE" there 5 letters. "P" occurs twice, and the rest are all different, that is $n = 5, n_1 = 1, n_2 = 2, n_3 = 1, n_4 = 1$

$\therefore$ Total number arrangements is $\frac{5!}{2!\,1!\,1!\,1!} = 5 \times 4 \times 3 = 60$

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23. How many different arrangement can be made from the letters in the worlds "STATISTICS".

Solution

The world contains: three S, three T, one A, two I, one C, that is:
$n = 10, n_1 = 3, n_2 = 3, n_3 = 1, n_4 = 2, n_5 = 1$

$\therefore$ Total number arrangements is

$$\frac{10!}{3!\,3!\,1!\,2!\,1!} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4}{3 \times 2} = 100,800$$

24. I how many ways can 4 red, 3 yellow, and 2 green bells be arranged in a row

Solution

There are $4 + 3 + 2 = 9$ balls: $n = 9, n_1 = 4\ n_2 = 3, n_3 = 2.$

$\therefore$ Total number arrangements is $\frac{9!}{4!\ 3!\ 2!} = 126.$

25. How many worlds can be formed with the letters of the "EFFECT" if:

a. there is no restriction b. these start with F

Solution

a. In the world "EFFECT" total number of letters = n = 6
"E" occurs twice, F occurs twice the rest are different.

$\therefore$ Total number a arrangement is

$$\frac{6!}{2!\,2!} = \frac{6 \times 5 \times 4 \times 3}{2} = 180$$

a. The restriction is starting letter is F:

F is fixed, then two F, as one object, E occurs twice

$\therefore$ Total number a arrangement is $\frac{5!}{2!} = 5 \times 4 \times 3 = 60$

26. How many worlds can be formed with the letters of the world "MATHEMATICS" if

a. there is no restriction

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b. these start with "H".

c. these begin with "M"

Solution

a. In the word "MATHEMATICS" total number of letters = n = 11
Here, all the letters are nor different:

M - occurs twice, T - occurs twice: A - occurs twice

Using the formula, required number = $\frac{11!}{2!\,2!\,2!}$

b. The restriction is starting letter H.

H - is fixed = 1 way.

Hence, required number of words is = $\frac{10!}{2!\,2!\,2!}$

c. The restriction is stating letter M is fixed - 1 - way .

T - occurs twice, A - occurs twice all the rest are different

$\therefore$ No. of arrangement is = $\frac{10!}{2!\,2!}$

Circular Permutation

The circular permutations are permutations of certain things in the form of a circle. Number of circular permutations of n different things is (n - 1)!

Illustrative Example

27. Find the number of ways in which 6 persons sitting.

a. in row b. in a circle

Solution

a. $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

b. $(6 - 1)! = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

28. There are 6 men and 3 femals to dine at a round table. In how many ways can they seat so that no two femals sit together?

Solution

In order to have two femals together, we shall first arrange all the 6

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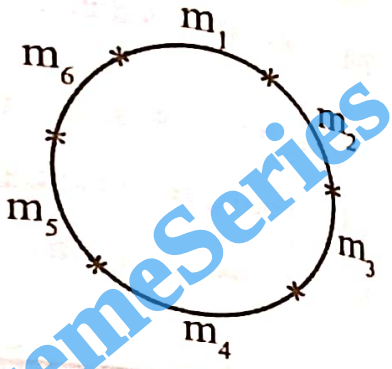
men and then we shall arrange femal in between men.

- No. of ways of arranging men

  $= (6 - 1)! = 5! = 120$

- Females can occupy seats marks "*"
- No. of ways arranging famals is =

$$P(6,3) = \frac{6!}{(6-3)!} = 6 \times 5 \times 4 = 120$$



$\therefore$ By F.P.C, the reacquired number of arrangements = $120 \times 120 = 1400$

9.3 Combination

What is a Combination?

A selection of distinct objects with out regard to order is called a combination

The difference between a permutation and combination is that in a combination, the order or arrangement of the object is not important but the order is important in a permutation.

As an example: Given letters, A, B, C and D list the permutations and combination for selecting two letters. These are the listings:

Permutations			Combination		
AB	AC	AD	AB	AC	AD
BA	BC	BD	BC	BD	CD
CA	CB	CD			
DA	DB	DC			

Note that in permutations, AB different from BA. But in combination, AB is the same as BA, So only AB is listed.

- The number of permutation of the letters A, B, C and D taken two at a times is $P(4, 2) = \frac{4!}{2!} = 4 \times 3 = 12$

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The number of combination of the letters A, B, C, and D taken two at a time is $C(4, 2) = \frac{4!}{2!\,(4-2)!} = \frac{4 \times 3}{2} = 6$

Combination Rule

The number of combinations of r objects selected from n - object is denoted by ${}_nC_r = C(n, r) = \binom{n}{r}$ and is given by the formula:

$$C(n, r) = \frac{n!}{(n-r)!\,r!}, \quad 0 < r \leq n$$

Since any combination of n - objects taken r at a time determine r! permutation of the objects in the combination, we can conclude that

$$P(n, r) = r!\,C(n, r)$$

$$\therefore \quad C(n, r) = \frac{P(n, r)}{r!} = \frac{n!}{(n-r)!\,r!}$$

Note that : $C(n, n) = 1$ and $\binom{n}{r} = \binom{n}{n-r}$

$$\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$$

Illustrative Example

29. Compute each of the following

a. $C(8, 5)$ b. $C(7, 3)$ c. $C(20, 18)$

Solution

a. $c(8, 5) = \frac{8!}{3!\,5!} = \frac{8 \times 7 \times 6 \times 5!}{3 \times 2 \times 5!} = 56$

b. $C(7, 3) = \frac{7!}{4!\,3!} = \frac{7 \times 6 \times 5 \times 4!}{3 \times 2 \times 4!} = 35$

c. $C(20, 18) = \frac{20!}{2!\,18!} = \frac{20 \times 19 \times 18!}{2 \times 18!} = 190$

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30. If C (n, 3) = C (n, 5), find the value of n

Solution

Since $\binom{n}{r} = \binom{n}{n-r}$, then

$\binom{n}{3} = \binom{n}{n-3}$, so that $\binom{n}{n-3} = \binom{n}{5}$

$\therefore \quad n - 3 = 5, n = 8$

31. $\binom{7}{3} + \binom{7}{2} = \binom{8}{r}$, find the value of r.

Solution

We have the telation $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$

$\therefore \quad r = 3$

32. If C (18, r) = C (18, r + 2), then find the value of r

Solution

We use $\binom{18}{r} = \binom{18}{18-r}$

$\binom{18}{18-r} = \binom{18}{r+2}$, then $18 - r = r + 2$

$16 = 2r$, imples $r = \frac{16}{2} = 8$

33. In how many ways can a committee of 4 students to be chosen out
7 students

Solution

$$\binom{7}{4} = \frac{7!}{(7-4)!\,4!} = \frac{7 \times 6 \times 5 \times 4!}{3 \times 2 \times 1 \times 4!} = 35$$

34. In a club there are 7 women and 5men. A committee of 3 women a
2 men is to be formed. In how many way can such a committee b

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formed?

Solution

• No. of ways selecting 3 women from 7 women

$$= C(7, 3) = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \times 3 \times 2} = 35 \text{ way}$$

• No. of ways of selecting 2 men from 5 men is

$$\binom{5}{2} = \frac{5!}{3!\ 2!} = \frac{5 \times 4}{2} = 10 \text{ way}$$

∴ By F.P.C, number of ways of committees $= 35 \times 10 = 350$

35. A class contains 9 boys and 3girls. A committee of 4 student formed. How many ways a teacher can select a committee consists of :

a. 4 student

b. 2 boys and 2girls

c. exactly 1 girls

d. at least 1 girls

e. at most 4 boys

Solution

a. $\binom{12}{4} = \frac{12!}{8!\ 4!} = \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1} = 495$

b. No of ways to select 2 boys from 9 boys is

$$\binom{9}{2} = \frac{9!}{7!\ 2!} = \frac{9 \times 8}{2} = 36$$

No of ways to select 2 girls from 3girls is

$$\binom{3}{2} = \frac{3!}{2!} = 3$$

∴ By F.PC, number of committee is given by:

$$\binom{9}{2}\binom{3}{2} = 36 \times 3 = 108$$

c. Exactly 1 girls means the rest 3 are boys.

No. of selecting 1 girl from 3girl is $C(3, 1) = 3$

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No. of selecting 3 boys from 9 boys is:

$$\binom{9}{3} = \frac{9!}{6!\,3!} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$$

$\therefore$ By F.P.C, number of ways of committee formed is

$$\binom{3}{1}\binom{9}{3} = 3 \times 84 = 252$$

d. At least one girl: $\binom{9}{3}\binom{3}{1} + \binom{9}{2}\binom{3}{2} + \binom{9}{1}\binom{3}{3}$

$= (84)(3) + (36)(3) + 9(1) = 369.$

e. At most 4 boys mean: 4 boys and 0 girls (4 boys, 0 girls), 3 boys and 1 girls, or 2 boys and 2 girls, or 1 boy and 3 girls.

$\therefore$ $\binom{9}{4}\binom{3}{0} + \binom{9}{3}\binom{3}{1} + \binom{9}{2}\binom{3}{2} + \binom{9}{1}\binom{3}{3}$

$= (126)(1) + (84)(3) + (36)(3) + (9)(1)$

36. In how many way can select at least 5 books out of 8 books

Solution

At least 5 means, greater than or equal so 5:

$$\binom{8}{5} + \binom{8}{6} + \binom{8}{7} + \binom{8}{8} = 56 + 28 + 8 + 1 = 93$$

37. In a party every person shakes hands with every other person. If there was a total of 105 hand shakes in the party, find the number of person who were present in the party.

Solution: Let n = the number of person. One handshake take place between two persons. Therefore the number of hand shakes i

$$\binom{n}{2} = 105 \text{ this implies } \frac{n(n-1)}{2} = 105$$

$\therefore$ $n(n-1) = 210 = 15 \times 14, n = 15$

$\therefore$ Number of persons who was present in the party are 15.

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9.4 Binomial Theorem

A baronial is an algebraic expression of two terms which is connected by the operation (+) or (-)

Example: $x + y$, $x - y$, $3a + b^2$ or $a^4 - b$, e.t.c

Recall that : $(a + b)^2 = a^2 + 2ab + b^2$

The expression $a^2 + 2ab + b^2$ is called the **binomial expansion** of $(a + b)^2$ Here , we expand several power of $(a + b)$.

$n = 0 : (a + b)^0 = 1$

$n = 1 : (a + b)^1 = 1a + 1b$

$n = 2 : (a + b)^2 = a^2 + 2ab + b^2$

$n = 3 : (a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$n = 4 : (a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$

$n = 5 : (a + b)^5$
$= a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5a^3b^4 + b^5$

			1			
		1		1		
	1		2		1	
1		3		3		1
1	4		6		4	1
1	5	10		10	5	1

The coefficients for the expansion of $(a + b)^n$ follow a triangular array of numbers called **pascal's triangle.**

Expand: $(a - 2b)^4$ by using pascal's theorem:

$n = 4: (a - 2b)^4 = a^4 + 4a^3(-2b) + 6a^2(-2b)^2 + 4a(-2b)^3 + (-2b)^4$
$= a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4.$

Determining the coefficients of a binomial expansion using pascal's triangle would be cumbersome for expansions of higher degree. Instead we can compute the coefficients of a binomial expansion using the following formula.

Binomial Theorem

Let n be a positive integer. The expansion of $(a + b)^n$ is given by:

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{n}b^n$$

$$= \sum_{r=0}^{n}\binom{n}{r}a^{n-r}b^r$$

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Note: $\binom{n}{r} = \frac{n!}{(n-r)!\,r!}$

Illustrative Example

38. Expand: $(2x - y)^3$

Solution

$$(2x - y)^3 = \binom{3}{0}(2x)^3 + \binom{3}{1}(2x)^{3-1}(-y)^1 + \binom{3}{2}(2x)^{3-2}(-y)^2 + \binom{3}{3}(-y)^3$$

$$= 1.8x^3 + 3.2^2(-y) + 3(2x)^1(-y)^2 + 1.(-1)y^3$$

$$= 8x^3 - 12$$

39. Find the fourth term in the expansion of $(x - 2y)^7$

Solution

The coefficients of each terms are:

$$\binom{7}{0}, \binom{7}{1}, \binom{7}{2}, \binom{7}{3}, \binom{7}{4}, \binom{7}{5}, \binom{7}{6}, \binom{7}{7}$$

The 4^{th} term is $\binom{7}{3}x^4(-2y)^3 = -35.(8y^3) = -280x^4y^3$

40. Find the 3^{rd} terms in the expansion of $\left(x + \frac{1}{x^2}\right)^6$.

Solution

The coefficients of each terms are

$$\binom{6}{0}, \binom{6}{1}, \binom{6}{2}, \binom{6}{3}, \binom{6}{4}, \binom{6}{5}, \binom{6}{6}$$

The 3^{rd} term is $\binom{6}{2}x^4\left(\frac{1}{x^2}\right)^2 = \binom{6}{2}x^4 \cdot \frac{1}{x^4} = 15$

41. Find the coefficient of x^3y^5 in the expansion of $(x+y)^8$.

Solution

The coefficient of x^3y^5 is given by $\binom{8}{r}x^{8-r}y^r$ compare three

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exponents: Here, $x^{8-r} = x^3$ and $y^r = y^5$, $r = 5$

$$\therefore \quad \binom{8}{5} x^3 y^5 = 56x^3 y^5$$

42. Find the coefficient of $x^5 y^3$ in the expansion of $\left(\frac{5}{2}x - \frac{2}{5}y\right)^8$.

Solution: The coefficient of expansion: $\binom{8}{r}\left(\frac{5}{2}x\right)^{8-r}\left(-\frac{2}{5}y\right)^r$

Comparing the exponents with $x^5 y^3$.

Then $x^{8-r} = x^5$ this implies $8 - r = 5$, $r = 3$

$$\therefore \quad \binom{8}{3}\left(\frac{5}{2}x\right)^5\left(-\frac{2}{5}y\right)^3 = 56.\frac{5^5}{2^5}.\frac{(-2)^3}{5^3}x^5 y^3$$

$$= -350x^5 y^3$$

43. Find the coefficient of $x^6 y^3$ in the expansion of $(x + 2y)^9$

Solution

The coefficient of $x^6 y^3$ is given by $\binom{9}{r}x^{9-r}(2y)^r = \binom{9}{r}2^r x^{9-r} y^r$

Here $x^{9-r} = x^6 \Rightarrow 9 - r = 6$, $r = 3$

$$\therefore \quad \binom{9}{3}2^3 x^6 y^3 = 672x^6 y^3$$

44. Find the term independent of in the expansion of $\left(x^2 - \frac{1}{x}\right)^9$

Solution

$\binom{9}{r}\left(x^2\right)^{9-r}\left(-x^{-1}\right)^r$, we make the exponent of x is zero:

$18 - 2r - r = 0$ and $r = 6$

$$\therefore \quad \binom{9}{6}\left(x^2\right)^3\left(-x^{-1}\right)^6 = 84.$$

45. What is the sixth term in the expanded form of $(x + 2y)^8$?

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A. $1024\,xy^7$ B. $448x^5y^3$ C. $1792x^3y^5$ D. $1120x^4y^4$

Solution

The coefficients of the terms are

$$\binom{8}{0}, \binom{8}{1}, \binom{8}{2}, \binom{8}{3}, \binom{8}{4}, \binom{8}{5}, \binom{8}{6}, \binom{8}{7}, \binom{8}{8},$$

Then $\binom{8}{5} x^3 (2y)^5 = (56)(32)\, x^3\, y^5 = 1792\, x^3\, y^5$

Answer: C

9.5 Theory of Probability

What is probability?

Probability as general concept can defined as the chance of an event occurring.

The basic concepts of probability include random experiment probability, sample space, probability of simple event, probability of compound event.

9.5.1 Random Experiment and Their out comes

An **Experiment** is a test with an uncertain out comes.

Random experiment is an experiment whose all possible results (out comes) are known and which can be repeated under identical condition, but not possible to predict the outcomes in advance.

The possible result of a random experiment is called its out come

- **Sample space** is the set of all possible out comes of an experiment

Some example of random experiment and their out come.

Random experiment	Sample Space
• Tossing of a coin	S = {H, T}
• Tossing of two coin	S = {HH, HT, TH, TT}
• Rolling a single die	S = {1, 2, 3, 4, 5, 6}
• A family of two children	S = {BB, BG, GB, GG}

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Illustrative Example

46. List the sample space for the following random experiment:

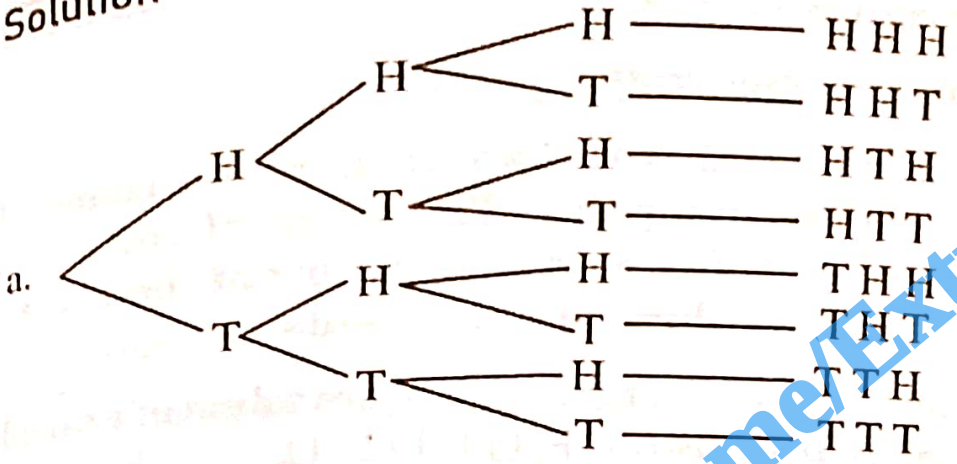
a. A coin is tossed three times

b. A coin is tossed and a die is thrown

c. A family of three children

Solution

a.



∴ Sample space: S = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}.

b. Tossing coin = {H, T} and throwing die = {1, 2, 3, 4, 5, 6} then sample space: S = {H_1, H_2, H_3, H_4, H_5, H_6, T_1, T_2, T_3, T_4, T_5, T_6}.

c. Each child could be either male or femal.
Hence there are eight possibilities:

$(B + G)^3 = B^3 + 3B^2G + 3BG^2 + G^3$

= BBB + BBG + BGB + GBB + BGG + GBG + GGB + GGG

∴ S = {BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG}

Events and Types of Events

What is an event?

The possible collection of outcomes of random experiment are called **events**

- An event is a subset of the sample space denoted by E.

In tossing of single coin, the sample Space S = {H, T}.

The outcomes {H}, {T} is an event.

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An events is called an **impassible event** if it can never occur. On the other hand, an event which is sure to occur is called a **certain (sure event). Equally likely Events** are events that have equal chance of occuring

Example: If a single die is rolled, each outcome: 1, 2, 3, 4, 5 and 6 has the same chance of occuring.

Simple event is an event with one out come. A single event, like Head (H) on tossing coin is simple event

simple event contains only one sample point.

Compound event is an event with two or more out comes. It contains more than one sample point. The event of getting an odd number when a die is rolled is compound event, because it consists of three out comes or three simple events.

Exhaustive Event: A set of events $E_1, E_2, E_3, ..., E_n$ are subset of a sample space S are said to be *exhaustive events* if $E_1 \cup E_2 \cup E_3 \cup, ... \cup E_n = S$

Mutually Exclusive Event

Two events E_1 and E_2 are **mutually exclusive events** if they can not occur at the same time and $E_1 \cap E_2 = \{\}$

- If one events occurs, other cannot occur at the same time.
- In tossing a coin, the events "Head" and "Tail" are mutually exclusive.

Illustrative Example

47. Determine which events are mutually exclusive and which are not when single die is rolled.
    a. Getting an odd number and getting an event number.
    b. Getting a 4 and getting an even number.
    c. Getting an odd number and getting number less than 4.
    d. Getting a number greater than 4 and getting a number less than 4.

Solution

a. The event are mutually exclusive, since the first event can be: $E_1 = \{1, 3, 5\}$ and the second event can be $E_2 = \{2, 4, 6\}$

$E_1 \cap E_2 = \phi$.

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b. Note mutually exclusive, since the first event $E_1 = \{3\}$ and the second event $E_2 = \{1, 3, 5\}$. Hence, 3 is contained in both events. $E_1 \cap E_2 = \{3\}$

c. Not mutually exclusive since the first event: $E_1 = \{1, 3, 5\}$ and the second event: $E_2 = \{1, 2, 3\}$. Hence, 1, and 3 are contained in both events. $E_1 \cap E_2 = \{1, 3\}$

d. Mutually exclusive since event: $E_1 = \{5, 6\}$ and event: $E_2 = \{1, 2, 3\}$. and $E_1 \cap E_2 = \phi$

48. Which one of the following forms represents mutually exclusive and exhaustive set of events if a die is thrown once?

A. $\{1, 2\}, \{2, 4, 6\}, \{2, 3, 5\}$

B. $\{1, 3\}, \{2, 4\}, \{5, 6\}$

C. $\{2, 4, 6\}, \{3, 5\}$

D. $\{2, 4, 6\}, \{1, 2, 3, 5\}$

Solution

The sample space: $S = \{1, 2, 3, 4, 5, 6\}$

A. Exhaustive but not mutually exclusive, since {2} contained in 1st, 2nd and 3rd event.

B. Mutually exclusive and exhaustive since the events are disjoint and the union of the events is equal to $S = \{1, 2, 3, 4, 5, 6\}$.

C. Mutually exclusive but not exhaustive

D. Exhaustive but not mutually exclusive

Answer: B

49. Two coins are tossed. Describe

a. Two events which are mutually exclusive

b. Two events which are not mutually exclusive.

c. Two events which are mutually exclusive and exhaustive?

Solution

$(H + T)^2 = H^2 + 2HT + T^2 = \{HH, HT, TH, TT\}$

The sample space: $S = \{HH, HT, TH, TT\}$.

a. Let event: $E_1 = \{HH\}$ and $E_2 = \{HT\}$ are mutually exclusive, because $E_1 \cap E_2 = \phi$.

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b. Let event : E_1 = {HH, HT} and E_2 = {TT, HT} are not mutually exclusive, because $E_1 \cap E_2$ = {HT}.

c. Let event E_1 = {HH, TT} and E_2 = {HT, TH} are mutually exclusive and exhaustive, since $E_1 \cap E_2 = \varnothing$ and $E_1 \cup E_2 = S$

50. Which one of the following forms represents an exhaustive set of events if a die is thrown once?

A. {1, 2}, {2}, {4, 5}, {5, 6}

B. {1, 2, 3}, {3, 4}, {6}

C. {1, 2}, {4}, {4, 5, 6}

D. {1, 2, 3}, {4, 5}, {6}

Solution

In throwing die, the sample space: S = {1, 2, 3, 4, 5, 6}

$E_1 \cup E_2 \cup E_3$ = {1, 2, 3} ∪ {4, 5} ∪ {6} = {1, 2, 3, 4, 5, 6} = S

Answer: D

51. When three coins are tossed once. Suppose that

Event E_1 = getting three heads.

Event E_2 = getting two heads and one tail.

Event E_3 = getting three tail.

Event E_4 = getting a head on the first coin.

Event E_5 = getting at least one head.

Then identify:

a. Which events simple event

b. Which events are compound event

c. Which events are mutually exclusive event

d. Which events are exhaustive event

e. Which events are exhaustive and mutually exclusive event.

Solution

Sample space S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

E_1 = {HHH}, E_2 = {HHT, HTH, THH}, E_3 = {TTT}

E_4 = {HHH, HHT, HTH, HTT}.

E_5 = {HHH, HHT, HTH, THH, HTT, THT, TTH}.

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a. Events E_1 and E_3 are simple events.

b. Events E_2, E_4 and E_5 are compound event.

c. E_1 and E_2, E_1 and E_3, E_2 and E_3, E_3 and E_4, E_3 and E_5 are mutually exclusive event, since

$E_1 \cap E_2 = \phi$ $\quad E_2 \cap E_3 = \phi$ $\quad E_3 \cap E_5 = \phi$

$E_1 \cap E_3 = \phi$ $\quad E_3 \cap E_4 = \phi$

d. E_3 and E_5 are exhaustive events because $E_3 \cup E_5 = S$

e. E_3 and E_5 are mutually exclusive and exhaustive events because $E_3 \cap E_5 = \phi$ and $E_3 \cup E_5 = S$

ndependent and Dependent Events

Then the out come of one event affect the out come of a second vents, the two events are **dependent events.** When the utcome of one events does not affect the out come of a second event, e two events are **independent events.**

Illustrative Example

2. Classify each pair of events as *dependent* or *independent*

a. Rolling a die, then toss a coin

b. Drawing a ball from a box not replacing it, and then drawing a second ball

c. Drawing a queen from deck of cards, then drawing a king from the remaining cards in the same deck of cards.

d. Having excellent grade and getting scholarship.

e. A month is selected at random a number from 1 to 30 is selected at random.

f. A month is selected at random, a day of that month is selected at random

Solution

a. Independent, since the two events do not affect each other.

b. Dependent, since drawing the first ball affects the possible out comes of drawing the 2nd ball

c. Dependent

d. Dependent

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e. Independent f. Dependent

Complement Event

Events E_1 and E_2 are complementary if when event E_1 occurs E_2 does not occurs, when event E_2 occurs, E_1 does not occurs.

- $P(E) = 1 - P(\text{not } E)$

Illustrative Example

53. Find the complement of each event

a. Rolling a die and getting 3.

b. Selecting a day of the week and getting a week day

Solution

a. Getting a 1, 2, 4, 5 or 6

b. Getting: Saturday or Sunday

54. ETV meteorologists announces that your area has 60% chance of rain tomorrow and a 15% chance of snow. What is the probability that it will not rain or snow tomorrow? (Assume that it will not rain and snow)

Solution

Since they are mutually exclusive

$P(\text{rain or snow}) = P(\text{rain}) + P(\text{snow}) = 0.6 + 0.15 = 0.75$

Thus the probability that it will not rain or snow is $1 - 0.75 = 0.25$

55. If the probability that a person lives in an industrialized country of the world is $\frac{1}{4}$, find the probability that person does not live in an industrialized country.

Solution

Using the probability of complement:

Let A = Person living an industrialized country.

$$P(\text{not } A) = 1 - P(A) = 1 - \frac{1}{4} = \frac{3}{4}$$

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9.6 Probability of an Event

There are two basic approach of probability of an event:

1. Classical probability
2. Empirical or relative frequency probability

Classical Probability

Classical probability use sample space to determine the numerical probability that an event will happen.

Classical probability of an event assumes that all outcomes in the sample space are equally likely to occur, Then the probability of an event is the quotient of the number of outcomes in the event divided by the total number of outcomes.

DEFINITION: Classical probability Event E

The probability of any event E is

$$P(E) = \frac{\text{Number of out come in E}}{\text{Total number of outcomes in the sample space S}}$$

Symbol: This probability denoted by $P(E) = \frac{n(E)}{n(S)}$

Illustrative Example

6. Suppose that a box contains two red, three blue, and five green marbles. If one marble is selected at random, find the probability of the event:

a. Red marble is selected
b. White marble is selected
c. Blue marble is selected
d. Green marble is selected

Solution

Each of the 10 marbles in the box is equally likely to be selected.

The sample space as $S = \{R_1, R_2, B_1, B_2, B_3, G_1, G_2, G_3, G_4, G_5,\}$

a. Event : $E = \{R_1, R_2\}$, $P(E) = \frac{n(E)}{n(S)} = \frac{2}{10} = \frac{1}{5}$.

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b. Event : $E = \{\}$, $P(E) = \frac{n(E)}{n(S)} = \frac{0}{10} = 0$

c. Event : $E = \{B_1, B_2, B_3\}$, $P(E) = \frac{n(E)}{n(S)} = \frac{3}{10}$

d. Event : $E = \{G_1, G_2, G_3, G_4, G_5\}$, $P(E) = \frac{n(E)}{n(S)} = \frac{5}{10} = \frac{1}{2}$

57. There are 7 people on a committee 4 girls and 3 boys. One person will be chosen at random to be chair person.
What is the probability that the chair person will be a girls.

Solution : $P(E) = \frac{n(E)}{n(S)} = \frac{4}{10} = \frac{2}{5}$

58. If a die is rolled twice, then find probability of getting

a. a sum of 5 or 6

b. a sum of 9

c. a sum less than 4 or greater than 9

Solution

Each die has 6 equally likely outcome. **By F.P.C** the number of ways the pair of die can fall is $6 \times 6 = 36$

$S = \{(1, 2), (1, 2), (1, 3) \ldots\ldots (6, 5), (6, 6)\}$.

a. Let E_1 represent the event with sum 6:

$E_1 = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$ and

E_2 event with sum 5: $E_2 = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$.
The two event E_1 and E_2 are mutually exclusive then:

$$P(E_1 \text{or } E_2) = P(E_1) + P(E_2) = \frac{n(E_1)}{n(S)} + \frac{n(E_2)}{n(S)} = \frac{5}{36} + \frac{4}{36} = \frac{9}{36} = \frac{1}{4}$$

b. Event with sum 9 : $E = \{(3, 6), (4, 5), (5, 4), (6, 3)\}$

$$P(E) = \frac{n(E)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

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c. $E_1 = \{(1,1),(1,2),(2,1)\}$ and $E_2 = \{(4,6),(5,6),(6,4),(5,5),(6,5),(6,6)\}$

$$P(E_1 \text{ or } E_2) = \frac{n(E_1)}{n(S)} + \frac{n(E_2)}{n(S)} = \frac{3}{36} + \frac{6}{36} = \frac{9}{36} = \frac{1}{4}$$

59. Two coins are tossed at the time. Find the probability of getting:

a. One head

b. One tail

c. at least one tail

d. More than 2 head

Solution

$S = (H + T)^2\ H^2 + HT + TH + T^2 = \{HH, HT, TH, TT\}$.

Note: HT and TH are distance sample point because HT represent head on first coin and tail on the second coin where as TH represent tail on the first coin and head on the second coin.

a. Event getting one head: E = {HT,TH}

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{2}{4} = \frac{1}{2}.$$

b. Event getting one tail: E = {HT, TH}

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{2}{4} = \frac{1}{2}.$$

c. Event getting at least one tail: E = {HT, TH, TT}

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{3}{4}.$$

d. Event getting more than two head: E = {}.

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{0}{4} = 0. \text{ (Imposible event)}$$

60. There coins are tossed. Find the probability of getting:

a. exactly one tail b. at most one head c. at least one head

Solution: The sample space:

$S = (H + T)^3 = \{H^3 + 3H^2T + 3HT^2 + T^3\}$

$\therefore \quad S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

a. Event getting one tail: E = {HHT, HTH, THH}

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$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{3}{8}$$

b. Event at most one head: $E = \{HTT, THT, TTH, TTT\}$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{4}{8} = \frac{1}{2}$$

c. Event at least one head:

$E = \{HHH, HHT, HTH, THH, HTT, THT, TTH\}$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{7}{8}$$

61. A coin and a die are thrown. What is the probability of getting

a. head

b. an even number

c. head and an event number

d. a tail and an odd number

Solution

Here, $S = \{H_1, H_2, H_3, H_4, H_5, H_6, T_1, T_2, T_3, T_4, T_5, T_6\}$.

The sample point "H1" means head on coin and "1" on the die.

a. Event getting a head: $E = \{H_1, H_2, H_3, H_4, H_5, H_6\}$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{6}{12} = \frac{1}{2}$$

b. Event getting an even number: $E = \{H_2, H_4, H_6, T_2, T_4, T_6\}$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{6}{12} = \frac{1}{2}$$

c. Event head and even number: $E = \{H_2, H_4, H_6\}$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{3}{12} = \frac{1}{4}$$

d. $E = \{T_1, T_3, T_5\}$, and $P(E) = \frac{3}{12} = \frac{1}{4}$

62. If a die is rolled three times. Find the probability of getting

a. Sum 4

b. Sum 16

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Solution

Here, $S = \{(1, 1, 1), (1, 1, 2), ..., (6, 6, 5), (6, 6, 6)\}$

Number of possible out come: $n(S) = 6.6.6 = 216$

a. Event sum 4 : $E = \{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{3}{216} = \frac{1}{7^2}$$

b. Event sum 16: $E = \{(4, 6, 6), (6, 4, 6), (6, 6, 4), (5, 5, 6), (5, 6, 5), (6, 5, 5)\}$.

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{6}{216} = \frac{1}{36}$$

63. A number is selected at random from the set of integers from 1 to 20. Find the probability that the number selected is:

a. Multiple of 2 or 3 b. Multiple of 2 and 3.

Solution: $S = \{1, 2, 3, ... 19, 20\}$

a. Event which multiple of 2 or 3:

$E = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 3, 6, 9, 12, 15, 18\}$

$= \{2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20\}$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{13}{20}$$

b. Event, multiple of 2 and 3: $E = \{6, 12, 18\}$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{3}{20}$$

3. Empirical (relative frequency probability)

The difference between classical and **empirical probability** is that classical probability assumes that certain outcomes are equally likely, while empirical probability relies on estimated number of observation and actual experience to determine the likelihood of outcomes.

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DEFINITION: Empirical Probability

Given a frequency distribution, the probability of an event (E_i) being in a given class is

$$P(E) = \frac{\text{frequency of E}}{\text{total number of observation}} = \frac{f_E}{n}$$

Illustrative Example

64. A 100 bulb were found to be defective in lot of 800 bulb. Find the probability that a bulb selected from the lot is defective

Solution

$$P(\text{defective}) = \frac{f_c}{800} = \frac{100}{800} = \frac{1}{8}$$

65. In a sample of 50 people, 21 had type O blood, 22 has type A blood, 5 had type B blood, and 2 had type AB blood. Find the probabilities of the following person blood type:

a. A person has type O blood.

b. A person has type A or type B blood.

c. A person has neither type A nor type O blood.

d. A person does not have type AB blood.

Solution

Type	A	B	AB	O
frequency	22	5	2	21

Sum: $22 + 5 + 2 + 21 = 50$

a. $P(\text{O-blood}) = \frac{f}{n} = \frac{21}{50}$

b. $P(\text{A or B}) = \frac{22}{50} + \frac{5}{50} = \frac{27}{50}$

c. $P(\text{neither A nor O}) = \frac{5}{50} + \frac{2}{50} = \frac{7}{50}$

d. $P(\text{not AB}) = 1 - P(\text{AB}) = 1 - \frac{2}{50} = \frac{48}{50}$

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66. Hospital recods indicated that maternity patients stayed in the hospital for the umber of days shown in the distribution.

Number of days stayed	3	4	5	6	7
frequency	15	32	56	19	5

Find these probability:

a. A patient stayed exactly 5 days.

b. A patient stayed less than 6 days.

c. A patient stayed at most 4 days.

d. A patient stayed at least 5 days.

Solution: sum of frequency: $15 + 32 + 56 + 19 + 5 = 127$

a. $P(5) = \frac{56}{127}$

b. Let than 6 days means 3, 4, or 5 days:

$\therefore$ P (less than 6 days) $= \frac{56}{127} + \frac{32}{127} + \frac{15}{127} = \frac{103}{127}$.

c. At most 4 days mean 3 or 4 days:

$\therefore$ P (at most 4 days) $= \frac{15}{127} + \frac{32}{127} = \frac{47}{127}$.

d. At least 5 days means 5, 6, or 7 days.

$\therefore$ P (at least 5 days) $= \frac{56}{127} + \frac{19}{127} + \frac{5}{127} = \frac{80}{127}$.

iii. Axiomatic Approach of probability

For any event E in a sample space S, we define a function P on S and call it probability function on S, if:

For any event in S

1. $P(E) \geq 0$
2. $P(S) = 1$
3. $0 \leq P(E) \leq 1$
4. if $E = \phi$, then $P(E) = 0$ then the event E is impassible event

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5. $P(E) + P(\text{not } E) = 1$

$P(\text{not } E) = 1 - P(E)$

Illustrative Example

67. A sample space S consists of four element, that is $E = \{e_1, e_2, e_3, e_4\}$. Determine whether which of the following function does S become valid probability space.

a. $P(e_1) = \frac{1}{3}$, $P(e_2) = \frac{1}{4}$ $P(e_3) = \frac{1}{5}$ and $P(e_4) = \frac{1}{2}$

b. $P(e_1) = \frac{1}{3}$, $P(e_2) = \frac{1}{5}$ $P(e_3) = -\frac{1}{6}$ and $P(e_4) = \frac{1}{3}$

c. $P(e_1) = 0.2$, $P(e_2) = 0.3$ $P(e_3) = 0.4$ and $P(e_4) = 0.1$

d. $P(e_1) = \frac{1}{8}$, $P(e_2) = \frac{1}{8}$ $P(e_3) = \frac{1}{4}$ and $P(e_4) = \frac{1}{2}$

e. $P(e_1) = 0.4$, $P(e_2) = 0.3$ $P(e_3) = 0.29$ and $P(e_4) = 0.1$

Solution

a. Not valid, since the sum of the value $\left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{2} = \frac{67}{60}\right)$ is greater than one.

b. Not valid since $P(e_3)$ is negative, the function does not define S as a probability space

c. Valid, since the sum of the value $0.2 + 0.3 + 0.4 + 0.1 = 1.0$

d. Valid since the sum of the value

$$P(e_1) + P(e_2) + P(e_3) + P(e_4) = \frac{1}{8} + \frac{1}{8} + \frac{1}{4} + \frac{1}{2} = 1$$

e. Not valid, since the sum of the value is grater than 1:

$0.4 + 0.3 + 0.29 + 0.1 = 1.09 > 1$.

68. A box contains 4 white, 6 black and 5 red balls. A ball is drawn at random. Find the probability of getting:

a. red balls

b. not red ball

c. non - white ball

d. What ball or black balls

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Solution

a. P(red ball) $= \frac{5}{15} = \frac{1}{3}$

b. P (not red ball) $= 1 - \frac{1}{3} = \frac{2}{3}$

c. P (non - white ball) $= \frac{6+5}{15} = \frac{11}{15}$

d. P (white or black) $= \frac{4}{15} + \frac{6}{15} = \frac{10}{15} = \frac{2}{3}$

Calculating probability using permutation and combination

By using combination analysis and fundamental principle of counting we can determine the total number of sample space and the required number of outcomes in a given event.

Illustrative Example

69. In a class, there are 8 boys and 5 girls. A committee of 3 students is selected in class. What is the probability that the committee has:

a. all boys
b. all girls
c. 2 boys and 1 girl
d. at least one boy
e. at least one girl

Solution

Total number of student $= 5 + 8 = 13$

a. P (all boys) $= \frac{\text{No.of selections of 3 boys from 8 boys}}{\text{No.of selections of 3 student from 13 student}}$

$$= \frac{C(8,3)}{C(13,3)} = \frac{8\times7\times6}{3\times2\times1} \times \frac{3\times2\times1}{13\times12\times11} = \frac{28}{143}$$

Method II: P(all boys) $= \frac{8}{13}\cdot\frac{7}{12}\cdot\frac{6}{11} = \frac{28}{143}$

b. P (all girls) $= \frac{\text{No of selections of 3 girls from 5 girls}}{\text{No of selections of 3 student 13 student}}$

$$= \frac{C(5,3)}{C(13,3)} = \frac{5\times4\times3}{3\times2\times1} \times \frac{3\times2\times1}{13\times12\times11} = \frac{5}{143}$$

Method II: P(all girls) $= \frac{5}{13}\cdot\frac{4}{12}\cdot\frac{3}{11} = \frac{5}{143}$

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c. $P\ (2\text{boys and }1\text{girls}) = \dfrac{\begin{pmatrix}\text{No.of selection}\\ \text{of 2 boys out of 8}\end{pmatrix}\begin{pmatrix}\text{No.of selection of}\\ \text{1 girl out of 5 girls}\end{pmatrix}}{\text{No of selections of 3 student from 13}}$

$$= \frac{C(8,2)\cdot C(5,1)}{C(13,3)} = \frac{28\times 5}{286} = \frac{70}{143}$$

d. P (at least one boy)= 1 - P(no - boys) = 1 - P (all girls)

$$= 1 - \frac{5}{143} = \frac{138}{143}$$

e. P (at least 1 girl) = 1 - P(no girls) = 1 - P (all boys)

$$= 1 - \frac{28}{143} = \frac{115}{143}$$

70. A box conains 9 items of which 3 are defective. If a man 4 items are drawn at random from the box. What is the probability that

a. 3 items are defective

b. 1 items are defective

c. all items are non - defective

d. 2 items are defective

e. at least 1 item defective

Solution

From set of 9 item 3 are defective and 6 are non - defective

a. $P\begin{pmatrix}3\text{ - defective,}\\ 1\text{ - non def}\end{pmatrix} = \dfrac{\begin{pmatrix}\text{No.of selecting}\\ \text{3 def out of 3}\end{pmatrix}\begin{pmatrix}\text{No of selecting 1}\\ \text{non defecting out of 6}\end{pmatrix}}{\text{No of selecting 4 items out of 9}}$

$$= \frac{C(3,3)\cdot C(6,1)}{C(9,4)} = \frac{6}{126} = \frac{1}{21}$$

b. For 1 defective, 3 - non defective

$$\therefore \quad P\begin{pmatrix}1\text{ - defective,}\\ 3\text{non def}\end{pmatrix} = \frac{C(3,1)\cdot C(6,3)}{C(9,4)} = \frac{60}{126} = \frac{10}{21}$$

c. $P(\text{all items are non - defective}) = \dfrac{C(6,4)\cdot C(3,0)}{C(9,4)} = \dfrac{5}{42}$

d. if 2 item is defective, 2 non - defective

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$$P(2\text{ - def and 2 is non def}) = \frac{\begin{pmatrix}\text{No.of selecting}\\ \text{2 def. out of 3}\end{pmatrix}\begin{pmatrix}\text{No. of selecting}\\ \text{2 non - def.}\\ \text{out of 6}\end{pmatrix}}{\text{No. of 4 item out of 9 item}}$$

$$= \frac{C(3,2)\cdot C(6,2)}{C(9,4)} = \frac{45}{126} = \frac{5}{14}$$

c. $P(\text{at least 1 defective}) = 1 - P(\text{all - non defective})$

$$= 1 - \frac{C(6,4)}{C(9,4)} = 1 - \frac{15}{126} = \frac{37}{42}$$

11. box contains 5 red, 3 white and 4 blue balls. If 3 balls are drawn at random. Find the probability that

a. all are red

b. all are white

c. two red and 1 blue

d. all the balls are different color

Solution

Total no. of balls $= 5 + 3 + 4 = 12$

a. $P(\text{all 3 red}) = \frac{\text{No.of selection of 3 red out of 5 red}}{\text{No.of selection of 3 out of 12 ball}}$

$$= \frac{C(5,3)}{C(12,3)} = \frac{10}{220} = = \frac{1}{22}$$

Method II: Let R_1 = 1st red draw, R_2 = 2nd red draw

$$P(3\text{ red}) = P(R_1\ R_2\ R_3) = \frac{5}{12}\cdot\frac{4}{11}\cdot\frac{3}{10} = \frac{1}{22}$$

b. $P(3\text{ white}) = \frac{\text{No.of selection of 3 white out of white}}{\text{No of selection of 3 ball out of 12 ball}}$

$$= \frac{C(3,3)}{C(12,3)} = \frac{1}{220}$$

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c. P (two red and 1 blue) $= \frac{C(5,3)\,C(4,1)}{C(12,3)} = \frac{40}{220} = \frac{2}{11}$

d. P (all balls are different)

$$= \frac{C(5,1) \cdot C(3,1) \cdot C(4,1)}{C(12,3)} = \frac{5 \times 3 \times 4}{220} = \frac{3}{11}$$

72. A box contains 3 red and 5 green balls identical in all aspects except color. If two balls are drawn at random, then the probability that the two balls are not green is:

A. $\frac{9}{14}$ B. $\frac{15}{28}$ C. $\frac{3}{28}$ D. $\frac{5}{14}$

Solution

$$P(\text{not green}) = 1 - P(\text{green}) = 1 - \frac{\binom{5}{2}\binom{3}{0}}{\binom{8}{2}} = 1 - \frac{10}{28} = \frac{18}{28} = \frac{9}{14}$$

Answer: A

9. 6. 1. Odds in favor of and odds Against an Event

Usually probability statements are given in terms of odds, a comparison of P (E) with P(E').

For example: Suppose $P(E) = \frac{3}{7}$. Then $P(E') = 1 - \frac{3}{7} = \frac{4}{7}$

These probability predicts that event E will occur 3 out of 7 times and E will occur 4 out of 7 times. Then we say the odds in favour E are 3 to 4.

DEFINITION: The odds in favor of an event E are defined as the ratio of P(E) to P (E')

(i) If the odds favouring Event E are m to n, then:

$$P(E) = \frac{m}{m+n} \text{ and } P(E') = \frac{n}{m+n}$$

- The odds against the event E are defined as the ratio of P(E') to P(E).

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- Odds in favor $= \frac{P(E)}{P(E')} = \frac{P(E)}{1-P(E)}$
- Odds against $= \frac{P(E')}{P(E)} = \frac{1-P(E)}{P(E)}$

Illustrative Example

A single fair die is rolled. Find the odds in favor of and against.

a. getting 4 b. getting 2, 3, 4, or 5

Solution: sample space: $S = \{1, 2, 3, 4, 5, 6\}$

a. Let the event $E = \{4\}$, $P(E) = \frac{n(E)}{n(S)} = \frac{1}{6}$ and $P(E') = 1 - \frac{1}{6} = \frac{5}{6}$

∴ The odds in favor of getting 4 $= \frac{P(E)}{P(E')} = \frac{\frac{1}{6}}{\frac{5}{6}} = \frac{1}{5}$

∴ The odds against getting 4 are $\frac{P(E')}{P(E)} = \frac{\frac{5}{6}}{\frac{1}{6}} = \frac{5}{1}$

b. Event $E = \{2, 3, 4, 5\}$, $P(E) = \frac{4}{6} = \frac{2}{3}$ and $P(E') = 1 - \frac{2}{3} = \frac{1}{3}$

∴ The odds in favour of an event $E = \{2, 3, 4, 5\}$

are $P(E)$ to $P(E') = \frac{2}{3} \div \frac{1}{3} = \frac{2}{1}$ or 2 to 1.

∴ The odds against an event E are

$P(E')$ to $P(E) = \frac{1}{3} \div \frac{2}{3} = \frac{1}{2}$ or, 1 to 2

74. A marble is drawn from a box containing 5 yellow, 4 white and 2 blue marble. Find the odds in favor of and odd against each of the following event

a. Drawing a yellow marble
b. Drawing a white marble
c. Drawing a blue marble
d. Not drawing a white marble

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Solution

a. Event E = Darwing yellow marble

$$P(E) = \frac{5}{11} = \text{and } P(E') = 1 - \frac{5}{11} = \frac{6}{11}$$

$\therefore$ odd in favor of E are P(E) to P (E'):

$$\frac{5}{11} \div \frac{6}{11} = \frac{5}{6} \text{ or } 5 \text{ to } 6$$

$\therefore$ odd against an event E are the ratio of P(E') to P (E) are

$$\frac{6}{5} \div \frac{5}{11} = \frac{6}{5} \text{ or } 6 \text{ to } 5.$$

b and c exercise left for you

d. Event: E = not drawing white marble

$$P(\text{white}) = \frac{4}{11} \text{ and } P(\text{not white}) = 1 - \frac{4}{11} = \frac{7}{11}$$

$\therefore$ $P(E) = \frac{7}{11}$ and $P(E') = 1 - \frac{7}{11} = \frac{4}{7}$.

$\therefore$ The odds in favor of no drawing white marble is P(E') to P(E): $\frac{7}{11} \div \frac{4}{11} = \frac{7}{4}$ or 7 to 4

$\therefore$ The odds against is P(E') to P(E) $\frac{4}{11} \div \frac{7}{11} = \frac{4}{7}$ or 4 to 7.

75. Suppose the weather forecaster says that the probability of rain tomorrow is $\frac{1}{4}$. Find the odds in favor of rain tomorrow.

Solution

Let E = the event rain tomorrow.

E' = the event no rain tomorrow.

Since $P(E) = \frac{1}{4}$, then $P(E') = 1 - \frac{1}{4} = \frac{3}{4}$.

The odds in favor of rain are:

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$$\frac{P(E)}{P(E')} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{4} \cdot \frac{3}{4} = \frac{1}{3}$$ write as 1 to 3

On the other had, the odds that will not rain or the odds against rain are $\frac{3}{4} \div \frac{1}{4} = \frac{3}{1}$, wirtten 1 to 3.

76. If the odds in favor of a particular horse's winning a race are 2 to 5, what is the probability that the horse will win the race?

Solution

The odds indicates the chance of 2 out of 7 (2 + 5 = 7) that horse will win.

$$\therefore \quad P(\text{winning}) = \frac{2}{2+5} = \frac{2}{7}$$

77. The odds that particular package will be delivered on time are 18 to 5

a. Find the probability that the package will be deliver on time.

b. Find the odds against the package being delivered on time

Solution

a. odds of 18 to 5 show 18 favourable changes out of 18 + 5 = 23 chances

$$\therefore \quad P(\text{deliver in time}) = \frac{18}{18+5} = \frac{18}{23}$$

b. $P(E') = 1 - P(E) = 1 - \frac{18}{23} = \frac{5}{23}$ chances the package will not be delivered on time. So the odds against the package being delivered on time are P (E ') to P (E): $\frac{5}{23} \div \frac{18}{23} = \frac{5}{18}$ or 5 : 18

78. The odd in favor of an events are 3:8. Find the probability of its occurrence and non - occurrence.

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Solution

The odd favouring of event E are m to n, then: probability of its occurrence :

$$P(E) = \frac{m}{m+n} = \frac{3}{3+8} = \frac{3}{11} \text{ and of non - occurrence is}$$

$$P(E') = \frac{n}{m+n} = \frac{8}{3+8} = \frac{8}{11}$$

79. The odd against certain event are 4:9

a. Find the probability of its occurrence

b. Find the probability of non - occurrence

Solution

odd against: $\frac{P(E')}{P(E)} = \frac{4}{9}$, then $\frac{1 - P(E)}{P(E)} = \frac{4}{9}$

$9 - 9PE = 4P(E)$ implies $13P(E = 9$

a. $P(E) = \frac{9}{13}$ b. $P(E') = 1 - \frac{9}{13} = \frac{4}{13}$

9.7 The Rules of Addition of Probability

The probability of two or more events can be determined by the addition rules. The first addition rule is used when the events are mutually excursive.

Addition Rule 1:

When two event A and B are mutually exclusive, the probability that A or B will occurs is

$$P(A \text{ or } B) = P(A) + P(B)$$

$$P(A \cup B) = P(A) + P(B)$$

Addition Rule 2:

If two events, A and B, are not mutually exclusive, then the probability that either A or B occurs is:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

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$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

Illustrative Example

80. Two dice are rolled. Find the probability of getting doubles or a sum of 8.

Solution

Let A = {(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)}

B = {(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)}, A ∩ B = {(4, 4)}

since, A and B are not mutually exclusive, then by addition rule 2:

$P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$

$= \frac{6}{36} + \frac{5}{36} - \frac{1}{36} = \frac{10}{36} = \frac{5}{18}$

81. Two dice are rolled. Find the probability of getting doubles or a sum of 9.

Solution

Event double: A = {(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)}

Sum 9: B = {(3, 6), (4, 5), (5, 4), (6, 3)} and A ∩ B = {}.

Since A and B are mutually exclusive, then by addition rule 1:

$P(A \text{ or } B) = P(A) + P(B) = \frac{6}{36} + \frac{4}{36} = \frac{10}{36}$

82. Two dice are rolled. Find the probability of getting the first die shows 3 or, the sum 7.

Solution

Let 1st die 3: A = {(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)}

Sum 7: B = {(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)} and A ∩ B = {(3,4)}.

Since A and B are not mutually exclusive, then by addition rule 2:

$P(A \cup B) = P(A) + P(B) - (A \cap B)$

$= \frac{6}{36} + \frac{6}{36} - \frac{1}{36} = \frac{11}{36}$

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83. A box contains 5 red pens and 4 blue pens. Two pens are drawn at random. What is the probability that both are of the same color? (Assume, pens differ only in color)

A. $\frac{5}{18}$ B. $\frac{1}{6}$ C. $\frac{4}{9}$ D. $\frac{5}{9}$

Solution

Let A = 2 pens are red and B = 2 pens are blue

then P (selecting either 2 red pens or 2 blue pens)

Since A and B are mutually exclusive events, then by the Rule of Addition.

$$P(A \text{ or } B) = P(A) + P(B)$$

$$= \frac{C(5, 2) + C(4, 2)}{c(9, 2)} = \frac{10 + 6}{36} = \frac{4}{9}$$

Method II : P(RP) + P(BB)

$$\frac{5}{9}\cdot\frac{4}{8} + \frac{4}{9}\cdot\frac{3}{8} = \frac{32}{72} = \frac{4}{9}$$

Answer: C

Note: The sample space 52 Playing card standard deck

The sample space S of 52 card is divide into four 13 - elements suits named: spades, clubs, Heart and Diamonds. Each cars is numbered from 2, to 10, or jack (J), queen (Q), King (K), or ace (A).

- The 26 black cards (Spade and club) and 26 red cards (Diamonds and Hearts)
- There are 12 face cards (jack, queen, and kings).

Clubs:	A	2	3	4	5	6	7	8	9	10	K	Q	J
Spades:	A	2	3	4	5	6	7	8	9	10	K	Q	J
Hearts:	A	2	3	4	5	6	7	8	9	10	K	Q	J
Diamonds:	A	2	3	4	5	6	7	8	9	10	K	Q	J

84. If a single card is drawn from a deck. Find the probability of getting

a. An ace or a king
b. an ace or a spade
c. a 4 or a 7
d. a 4 or a red card

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c. a face card or hearts

Solution: n (S) = 52

a. n (ace) = 4 and n (king) = 4 an ace and a king re mutually exclusive

$$\therefore \quad P(\text{ace or king}) = P(\text{ace}) + P(\text{king}) = \frac{4}{52} + \frac{4}{52} = \frac{2}{13}$$

b. n (spade) = 13 and n (ace) = 4, spade and ace are not mutually exclusive, they overlap.

$$P(\text{ace or sped}) = P(\text{ace}) + P(\text{spade}) - P(A \cap S)$$

$$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52}$$

c. Let A = event 4 and B = event 7, then

$$P(A \text{ or } B) = P(A) + P(B) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}$$

d. Let A = event 4 and B = event red card and n (A ∩ B) = 2, not mutually exclusive

$$P(A \text{ or } B) = P(A) + P(B) - P(A \cap B) = \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{7}{13}$$

e. n (F) = 12 and n (H) = 13, n (F ∩ H) = 3

$$P(F \text{ or } H) = P(F) + P(H) - P(F \cap H) = \frac{12}{52} + \frac{13}{52} - \frac{3}{52} = \frac{11}{26}$$

85. In a hospital unit there are 10 nurses and 7 - medical doctors; 8 nurse and 3 medical doctors are females. If a staff person is selected, find the probability that the subject is a nurse or a male

Solution

The sample space is shown here:

Staff	Females	Males	Total
Nurses	8	2	10
Medical doctor	3	4	7
Total	11	6	17

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The probability is

$$P(\text{nurse or male}) = P(\text{nurse}) + P(\text{male}) - P(\text{male nurse})$$

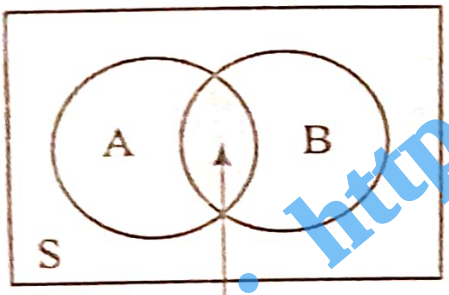
$$= \frac{10}{17} + \frac{6}{17} - \frac{2}{17} = \frac{15}{17}$$

9.8 The Rule of Multiplication of Probability

The multiplication rule can be used to find the probability of two or more events that occur in sequence. Multiplication rule of probability defines interms of **Independent and Dependent events**

- **Multiplication Rule - 1: Independent Events.** If the outcome of one events A does not affect the outcome of the second events B then the two event A and B are called **Independent event.**

DEFINITION: Probability of Independent Event



P(A and B)

If two events, A and B Are independent, then the probability of both events occurring is the product of the probability of A and the probability of B.

Symbols: P (A and B) = P (A)· (P(B)

$P(A \cap B) = P(A) \cdot P(B)$

Illustrative Example

86. A single die is rolled and a coin is tossed. Find the probability of getting

a. 3 on the die and head on the coin

b. Prime number on the die and tail in the coin.

Solution

a. Sample space: S = {1, 2, 3, 4, 5, 6}, $P(3) = \frac{1}{6}$

and {H, T}, $P(H) = \frac{1}{2}$ By multiplication rule 1:

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P (3 and head) = P (3). P (head) = $\frac{1}{6}.\frac{1}{2} = \frac{1}{12}$

$\therefore$

b. P (Prime) = $\frac{3}{6} = \frac{1}{2}$ and P (T) = $\frac{1}{2}$

P (Prime and tail) = P (Prime) P (tail) = $\frac{1}{2}.\frac{1}{2} = \frac{1}{4}$

87. A box contains 5 black balls and 3 white balls. A balls is drawn out and replaced in the box. Then a ball is drawn again. What is the probability that

a. both of them are white

b. both of them are black

c. the first ball was black and the second white.

Solution

a. P (both white) = P (white) · P (white) = $\frac{3}{8}.\frac{3}{8} = \frac{9}{64}$

b. P (both black) = P (black) · P (black) = $\frac{5}{8}.\frac{5}{8} = \frac{25}{64}$

c. P (black, then white) = P (black) · P (white) = $\frac{5}{8}.\frac{3}{8} = \frac{15}{64}$

88. A card is drawn from deck and replaced, then second card is drawn. Find the probability of getting a queen and then an ace.

Solution

P (queen) = $\frac{4}{52}$, and since the card is replaced, then P (ace) = $\frac{4}{52}$

$\therefore$ P (queen and ace) = P (queen) · P(ace)

$$= \frac{4}{52}.\frac{4}{52} = \frac{1}{169}$$

89. Two card are drawn from a ordinary deck of 52 cards. Find the probability that they are both heart if the first card is:

a. replaced

b. not replaced

Solution

Let H = event "heart" on the first draw and let h = event "heart on the

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2nd draw

a. If the first card is replaced, H and h are independent event.

$\therefore$ P (both drew heart) = P (Hh) =

$$P(H) \cdot P(h) = \frac{13}{52} \cdot \frac{13}{52} = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$$

b. If the first card is not replaced H and h are dependent.

$$P(Hh) = \frac{13}{52} \cdot \frac{12}{52} = \frac{1}{17}$$

Multiplication Rule 2 - Dependent Event

If the outcome of one event A affects the outcome of another event B, then events A and B are **dependent events.**

For example, drawing a card from a deck, not returning it, then drawing second card are dependent events because the drawing of the second card is dependent on the drawing of the first card.

DEFINITION: Probability of Dependent Events

If two events, A and B, are dependent then the probability of both events occurring is the product of the probability of A and the probability of B after A occurs.

Symbols: P (A and B) = P (A) P (B following A) = $P(A) \cdot P(B\backslash A)$

$$P(A \cap B) = P(A) \cdot P(B|A)$$

90. A box contains 4 red, 5 white and 6 blue balls. Three balls are drawn at random and not replaced. What is the probability that

a. all of them are white

b. all of them are red

c. all of them are blue

Solution

a. P (3 white) drawing of the first white balls affects drawing of next white ball from which to choose. So the events are dependent

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1st ball: P (white) $= \frac{5}{15} \longleftarrow \frac{\text{No of whit ball}}{\text{total No of ball}}$

2nd ball: P (white) $= \frac{4}{14} \longleftarrow \frac{\text{No of white remaining}}{\text{no of ball remain}}$

3rd ball: P (white) $= \frac{3}{13}$

$\therefore$ P (white, white, white) = P (white) P (white)

$$= \frac{5}{15} \cdot \frac{4}{14} \cdot \frac{3}{13} = \frac{2}{91}$$

Another method : P (three white) $= \frac{C(5, 3)}{C(15, 3)} = \frac{2}{91}$

b. P (red, red, red) = P (red) P (red) P (red)

$$= \frac{4}{15} \cdot \frac{3}{14} \cdot \frac{2}{13} = \frac{4}{455}$$

Anther method: P (3 red) $= \frac{C(4, 3)}{C(15, 3)} = \frac{4}{55}$

c. P (blue, blue, bule) = P (blue) P (blue) P (blue)

$$= \frac{6}{15} \cdot \frac{5}{14} \cdot \frac{4}{13} = \frac{4}{91}$$

Anther method : P (3 blue) $= \frac{C(6, 3)}{C(15, 3)} = \frac{4}{91}$

91. Two cards are drawn at random from a pack of 52 cards with out replacement. What is the probability of

a. both are Ace b. both are king c. both are heart

Solution

a. P (both Ace) = P (Ace) · P (Ace) $= \frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$

b. P (both king) = P (K) · P (K) $= \frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$

c. P (both heart) = P (H) · P (H) $= \frac{13}{52} \cdot \frac{12}{51} = \frac{1}{17}$

92. Three cards are drawn from an ordinary deck and not replaced.

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Find the probability of these:

a. Getting 3 king

b. Getting 3 spade

c. Getting a club, heart, and a diamond in order.

d. Getting an ace, a king, and a queen in order

Solution

a. $P(3 \text{ king}) = \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{2}{50} = \frac{1}{13} \cdot \frac{1}{17} \cdot \frac{1}{25} = \frac{1}{5,525}$

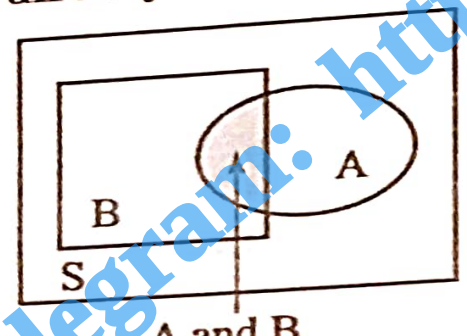
b. $P(3 \text{ spade}) = \frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} = \frac{11}{850}$

c. $P(\text{club, heart and diamod}) = \frac{13}{52} \cdot \frac{13}{51} \cdot \frac{13}{50} = \frac{169}{10200}$

d. $P(\text{ace, king, queen}) = \frac{4}{52} \cdot \frac{4}{51} \cdot \frac{4}{50} = \frac{8}{16,575}$

9. 8. 1. Conditional Probability:

The **conditional probability** of an event B in relationship of an event A is the probability that event B occurs after event A has already occurred, denoted by P (B/A) defined as:



$$P(B/A) = \frac{P(A \text{ and } B)}{P(A)} = \frac{P(A \cap B)}{p(A)}$$

- $P(A \cap B) = P(A) \cdot P(B|A)$
  $= P(B) \cdot P(A|B)$
- P (B|A) measures, the relative probability of B with respect to the reduced space A.

- P (A|B) denotes the conditional probability of event A, given that B has occurred.
- P (B|A) denotes the conditional probability of event B, given that A has occurred.

Illustrative Example

93. Two balls are selected from a box of containing 6 red and 4 blue one

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after the other. Find the probability that the second ball is selected is:

a. blue ball given that the first was red ball.

b. red ball given that the first was blue ball.

c. blue ball given that the first was blue ball.

d. red knowing that the first ball is red.

Solution

a. $P(\text{Red then blue}) = P(R)\,P(B/R) = \frac{6}{10} \cdot \frac{4}{9}$

$\therefore \quad P(B|R) = \frac{4}{9}$

b. $P(\text{Blue then red} = P(B)\,P(R/B) = \frac{4}{10} \cdot \frac{6}{9}$

$\therefore \quad P(R|B) = \frac{6}{9} = \frac{2}{3}$

c. $P(\text{Blue then blue}) = P(B_1)\,P(B_2|B_1) = \frac{4}{10} \cdot \frac{3}{9}$

$\therefore \quad P(B_2|B_1) = \frac{3}{9} = \frac{1}{3}$

d. $P(\text{Red, then red}) = P(R_1)\,P(R_2/R_2) = \frac{6}{10} \cdot \frac{5}{9}$

$\therefore \quad P(R_2|R_1) = \frac{5}{9}$

94. Two coins were tossed, and it is known that at least one was a head. Find the probability that both were head.

A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{1}{3}$ D. $\frac{2}{3}$

Solution

S = {HH, HT, TH, TT}. Because of the condition that at least one coin was a head, the sample space is reduced to : E_1 = {HH, HT, TH}. since only one out come in this reduced sample space is 2 heads.

E_1 = at least one head = {HH, HT, TH} and E_2 = {HH}

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$$\therefore \quad P(E_2|E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)} = \frac{1}{3}$$

95. If two fair dice are rolled. Find the probability of the sum 9, given that the sum is greater than 8.

Answer: C

Solution: S {(1, 1), (1, 2)..... ,(6, 5), (6,6)}. Because of the condition that the sum is greater than 8 the sample space is reduced to

E_1 = {(3, 6), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)}

E_2 = the sum 9 = {(3, 6), (4, 5), (5, 4), (6, 3)}

$$\therefore \quad P(E_2|E_1) = \frac{4}{10} = \frac{2}{5} = 0.4$$

96. A box contains 5 black and 3 white balls, two balls are drawn at random with out replacement. What is the probability that the second ball is black, knowing that the first ball is black?

A. $\frac{5}{8}$ B. $\frac{5}{14}$ C. $\frac{25}{64}$ D. $\frac{4}{7}$

Solution

$$P(\text{black, then back}) = P(B_1) \cdot P(B_2|B_1) = \frac{5}{8} \cdot \frac{4}{7}$$

$$\therefore \quad P(B_2|B_1) = \frac{4}{7}$$

Answer: D

97. Two machines work A and B work independently. The probability that both machines A and B work is 0.4. If the conditional probability that machines B works given that machine A works is 0.5, then the conditional probability that machine A works given that machine B works is _____ .

A. 0.8 B. 0.3 C. 0.5 D. 0.7

Solution

$P(A \cap B) = 0.4$ and since A and B are independent $P(B|A) = 0.5 = P(B)$.

$P(A \cap B) = P(B)\, P(A|B)$, this implies $0.4 = 0.5\, P(A|B)$

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$$\therefore \quad P(A|B) = \frac{0.4}{0.5} = 0.8$$

Answer: A

98. A box contain black balls and white balls. A person selects two balls with out replacement. If the probability of selecting a black ball and a white ball is $\frac{15}{56}$ and the probability of selecting a black ball on the first draw is $\frac{3}{8}$. Find the probability of selecting the white ball on the second draw, given that the first ball is selected was a black ball.

Solution

Let B = Selecting a black ball and W = selecting a white ball

$$\text{when } P(W|B) = \frac{P(B \cap W)}{P(B)} = \frac{\frac{15}{56}}{\frac{3}{8}} = \frac{15}{56} \cdot \frac{8}{3} = \frac{5}{7}$$

Sequential Events of Probability

A tree diagram can be used to show the probabilities of certain out comes occurring when two or more trials take place in succession. The out come is written at the end of the branch and the fraction on the branch and the fraction on the branch given the probability of the out come occurring.

Illustrative Example

99. A box contains 3 red, and 5 blue balls. Two ball are drawn one after the other at random and not replaced.

Find the probability that the balls will be:

a. Red following by blue

b. Red and blue in any order

c. the same color

Solution: Draw a tree diagram showing the probabilities of each out come of the two events.

1st draw 2nd draw

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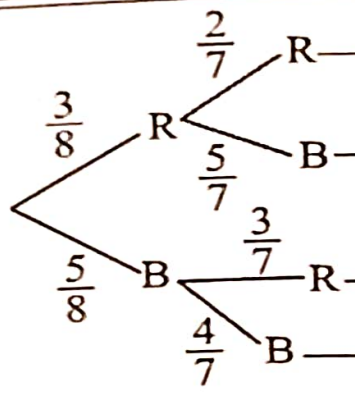

$P(R, R) = \frac{3}{8} \cdot \frac{2}{7} = \frac{3}{28}$

$P(R, B) = \frac{3}{8} \cdot \frac{5}{7} = \frac{15}{56}$

$P(B, R) = \frac{5}{8} \cdot \frac{3}{7} = \frac{15}{56}$

$P(B, B) = \frac{5}{8} \cdot \frac{4}{7} = \frac{5}{14}$

a. P (red followed by red) = P (R, R) = $\frac{3}{28}$

b. P (different color) = P (R, B) + P (B, R) = $\frac{15}{56} + \frac{15}{56} = \frac{15}{28}$

c. P (same color) = P (R, R) + P (B, B) = $\frac{3}{28} + \frac{5}{14} = \frac{13}{28}$

100. A box contains 9 white balls and 7 red balls. Two balls are drawn at random and not replaced. What is the probability that one of them is white and the other red?

Solution

Total number of balls = 9 + 7 = 16

P (W and R) = P (W) P (R/W) + P (R) P (W/R)

$$= \frac{9}{16} \cdot \frac{7}{15} + \frac{7}{16} \cdot \frac{9}{15} = \frac{21}{40}$$

Another method:

1 white ball from 9 white ball C (9, 1).

1 red ball from 7 red ball : C (7, 1)

$$\therefore \quad P(W \text{ and } R) = \frac{C(9,1) \times C(7,1)}{C(16, 2)} = \frac{21}{40}$$

101. Suppose a box contains 10 items of which 4 are defective. Three items are drawn random one after the other. Find probability that all three are non- defective.

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Solution

4 is defective 6 items are non -defective.

Let N_1, N_2 and N_3 denote the events "non defect item on 1st draw, 2nd draw and 3rd draw respectively. Then N_1 N_2 N_3 denotes the event that all 3 items drawn re non - defective

$$P(N_1\ N_2\ N_3) = P(N_1)\cdot P(N_2|N_1)\cdot P(N_3|N_1N_2)$$

$$= \left(\frac{6}{10}\right)\left(\frac{5}{9}\right)\left(\frac{4}{8}\right) = \frac{1}{6}$$

Second method : $\dfrac{C(6,3)}{C(10,3)} = \dfrac{1}{6}$

102. A class has 8 boys and four girls. Suppose three students are selected at random from the class. Find the probability that they are:

a. all boys

b. all girls

Solution

a. $P(B_1\ B_2\ B_3) = P(B_1)\cdot P(B_2|B_1)\cdot P(B_3|B_1B_2) = \dfrac{8}{12}\cdot\dfrac{7}{11}\cdot\dfrac{6}{10} = \dfrac{14}{55}$

Second method: There are C (12, 3) = 220 ways to select three students out of the 12students, and C (8, 3) = 56 ways to select three boys out of 8 boys.

$$\therefore \quad P(\text{all boys}) = \frac{C(8,3)}{C(12,3)} = \frac{56}{220} = \frac{14}{55}$$

b. $P(G_1\ G_2\ G_3) = P(G_1)\cdot P(G_2/G_1)\cdot P(G_3/G_1G_2)$

$$= \frac{4}{12}\cdot\frac{3}{11}\cdot\frac{2}{10} = \frac{1}{55}$$

Second method : $P(\text{all girls}) = \dfrac{C(4,3)}{C(12,3)} = \dfrac{4}{220} = \dfrac{1}{55}$

103. A bag contains 2 red, 3 white and 1 black balls. We draw two balls one after the other without replacement. What is the probability that the first ball is red and the second ball is black?

A. $\frac{1}{15}$ B. $\frac{1}{18}$ C. $\frac{1}{10}$ D. $\frac{1}{12}$

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Solution

$$P(\text{red, then black}) = \frac{2}{6} \cdot \frac{1}{5} = \frac{2}{30} = \frac{1}{15}$$

104. A box contains 4 red, 3 white and 5 blue balls. If 3 balls are drawn at random and not replaced. What is the probability that the balls are drawn in order red, white and blue

Answer: A

A. $\frac{1}{15}$ B. $\frac{1}{22}$ C. $\frac{3}{22}$ D. $\frac{6}{55}$

Solution

$$P(P\ W\ B) = P(R) \cdot P(W/B) \cdot P(B/RB)$$

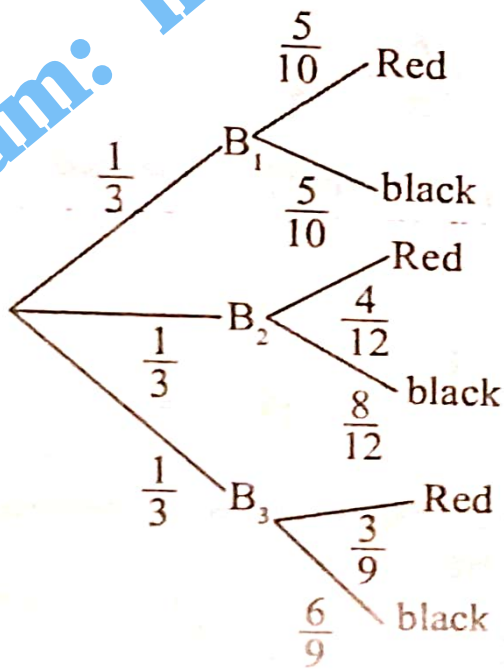
$$= \frac{4}{12} \cdot \frac{3}{11} \cdot \frac{5}{10} = \frac{1}{22}$$

105. Box - I contains 5 red and 5 black balls, Box - II contains 4 red and 8 black balls and Box - III contains 3 red and 6 black balls. One of the box is selected at random and a ball is drawn. If the ball drawn is red, find the probability of that it is drawn from Box - III.

Answer: B

Solution: Let B_1 = box - I, B_2 = box - II and B_3 = box - III.

Using tree diagram:



$\therefore P(B_3 \text{ is selected / red})$

$$= \frac{\frac{1}{3} \cdot \frac{3}{9}}{\frac{1}{3} \cdot \frac{5}{10} + \frac{1}{3} \cdot \frac{4}{12} + \frac{1}{3} \cdot \frac{3}{9}} = \frac{6}{7}$$

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Summary:

Addition and multiplication laws of probability are:

(i) Mutually exclusive events: $P(A \cup B) = P(A) + P(B)$.

(ii) Not mutually exclusive events:

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

(iii) $P(A \cap B) = P(A) \cdot P(B)$ for independent events.

$P(A \cap B) = P(A) \cdot P(B|A)$ for dependent events.

$= P(B) \cdot P(A|B)$ for dependent events.

Conceptual Example

106. Given that $P(A) = \frac{2}{3}$, $P(B) = \frac{1}{2}$, $P(A \cap B) = \frac{1}{3}$ and $P(A \cup B) = \frac{5}{6}$, then determine whether each of the following are true of false.

a. A and B are mutually exclusive events.

b. A and B are independent events.

Solution

a. False, since $P(A \cup B) \neq P(A) + P(B)$

b. True, since $P(A \cap B) = P(A) \cdot P(B) = \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3}$

107. For events A and B, $P(A) = \frac{3}{4}$ $P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{3}{8}$ determine whether the event A and B are :

a. Mutually excursive events

b. Independent events.

Solution

a. Not mutually exclusive events since $P(A \cup B) \neq P(A) + P(B)$.

b. Independent event, since $P(A \cap B) = P(A) \cdot P(B)$

108. If A and B are independent event such that

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$P(A) = \frac{1}{3}$ and $P(B) = \frac{3}{4}$, then find $P(A \cup B)$

Solution

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= \frac{1}{3} + \frac{3}{4} - \left(\frac{1}{3} \cdot \frac{3}{4}\right) = \frac{10}{12} = \frac{5}{6}$$

109. If A and B are dependent events such that

$P(A) = \frac{5}{8}$ and $P(B/A) = \frac{3}{7}$ find $P(A \cap B)$.

Solution

$$P(A \cap B) = P(A)\, P(B|A) = \frac{5}{8} \cdot \frac{3}{7} = \frac{15}{56}$$

110. If A and B are dependent events such that

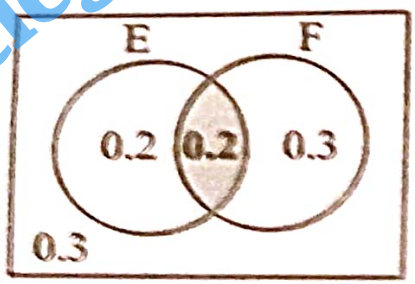
$P(A) = \frac{2}{5}$ and $P(A \cap B) = \frac{3}{10}$. Find $P(B|A)$.

Solution

$$P(A \cap B) = P(A) \cdot P(B/A)$$
$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{3}{10} \cdot \frac{5}{2} = \frac{3}{4}$$

111. Given $P(E) = 0.4$, $P(F) = 0.5$ and $P(E \cup F) = 0.7$ Find $P(E/F)$.

Solution



$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$
$$P(E \cap F) = 0.4 + 0.5 - 0.7 = 0.2$$
$$\therefore \quad P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{0.2}{0.5} = 0.4$$

112. Determine whether each of the following statements is true of false, and explain why.

a. The probability of the union of two events can be found by adding

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the probability of each event.

b. The probability of drawing the king of red from a deck of cards is an example of empirical probability

c. If two events are mutually exclusive, then they are independent.

d. In classical probability, all outcomes in the sample space are equally likely.

e. An event and its complement can occur at the same time.

f. When objects are arranged in a specific order, the arrangement is called a combination.

g. When a meteorologist says that there is 25% chance if rain is an example of classical type of probability

h. Complementary events are mutually exclusive

i. Mutually exclusive events they can occur at the same time.

Solution

a. False, since $P(A \cup B) = P(A) + P(B) - P(A \cup B)$

b. False, classical type of probability

c. False

d. True

e. False (mutually exclusive)

f. false, since order is not important

g. False, it is emprical type

h. True

i. false

Concept Complete

113. The set of all possible outcomes of a probability experiment is called __

Answer: Sample space

114. A/an _____ is a subset of the sample space of an experiment.

Answer: Event

115. If $P(E) = 0$, then E is called

Answer: impossible event

116. If $P(E) = 1$, then E is called _____

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Answer: Certain event

117. Fatuma can solve 90% of the problem given in book and Mesfin can solve 70%. What is the probability that at least one of them will solve the probability?

A. 0.77 B. 0.87 C. 0.67 D. 0.97

Solution:- $P(F) = 0.9$, and $P(F) = 0.7$.

P (problem not solved by Fatuma) $= 1 - 0.9 = 0.1$

P (problem not solved by Mesfin) $= 1 - 0.7 = 0.3$

$\therefore$ P (problem solved) = 1 - P(problem not solved by both)

$1 - (0.1)(0.3) = 1 - 0.03 = 0.97$

Second method: $P(F \cup M) = P(F) + P(M) - P(F \cap M)$

$= 0.9 + 0.7 - (0.9)(0.7) = 0.97$

118. A probability that A hits target is $\frac{1}{5}$ and the probability that B hits target is $\frac{2}{3}$. Both shoot at the target. Find the probability that at least one of them hits the target.

A. $\frac{13}{15}$ B. $\frac{2}{5}$ C. $\frac{11}{15}$ D. $\frac{2}{15}$

Solution :- $P(A) = \frac{1}{5}$ and $P(B) = \frac{2}{3}$

We assume that A hitting target is independent of B hitting the target, the Addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{5} + \frac{2}{3} - \left(\frac{1}{5} \cdot \frac{2}{3}\right) = \frac{11}{15}$$

A. Second method: P (at least one of them) = 1 - P (not hitting)

$$1 - \left(1 - \frac{1}{5}\right) \cdot \left(1 - \frac{2}{3}\right) = 1 - \frac{4}{15} = \frac{11}{15}$$

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19. A problem of Mathematics is given to three student A, B and C whose chance of solving it are $\frac{1}{3}, \frac{1}{4}, \frac{1}{5}$ respectively.

What is the probability that the probable me will be solved?

A. $\frac{47}{60}$ B. $\frac{4}{5}$ C. $\frac{3}{5}$ D. $\frac{3}{4}$

Solution: $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$ and $P(C) = \frac{1}{5}$

P (problem solved) + P (Problem not solved) = 1

P (Problem solved) = 1 - P (Problem not solved)

$$= 1 - \left[\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{5}\right)\right]$$

$$= 1 - \left[\frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5}\right] = 1 - \frac{2}{5} = \frac{3}{5}$$

∴ The probability of problem will is $\frac{3}{5}$

Second method: by the Addition rule:

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

$$= \frac{1}{3} + \frac{1}{4} + \frac{1}{5} - \left(\frac{1}{3} \cdot \frac{1}{4}\right) - \left(\frac{1}{3} \cdot \frac{1}{5}\right) - \left(\frac{1}{4} \cdot \frac{1}{5}\right) + \frac{1}{3} \cdot \frac{1}{4} \cdot \frac{1}{5}$$

$$= \frac{20 + 15 + 12 - 5 - 4 - 3 + 1}{60} = \frac{36}{60} = \frac{3}{5}$$

Review Exercise 9

1. A computer password must have two letters, followed by three digits. How many passwords car be made if
   a. no - restriction.
   b. No - letter or digit may be used more than once.
2. Abebe has 4 pairs of trousers, 5 shirts and 2 pairs shoes. In many different ways can he dress.

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3. A class contains 15 male students and 10 femal students. How many ways the class can elect:
   a. Class representative.
   b. Two class representative, one male and one femal.
   c. Class president and vice president
4. The desert menu of a restaurant has 5 kinds of cake, 3 kinds of cookies and 2 kinds of ice cream. Find the number of ways a person can select
   a. One of the deserts
   b. One of each of the three kinds of deserts.
5. Given {C, D, E}.
   a. List all the permutations of three elements from the set.
   b. List all the combinations of three elements from the set.
6. There are 12 seeds in a package. Five seeds are defective. If four seeds are selected at random, determine the number of ways in which:
   a. 4 defective seeds can be selected.
   b. 4 non - defective seeds can be selected.
   c. 2 defective and 2 non - defective seeds.
7. Find the number of arrangement can be made from
   a. 4 black, 3 red, 2 white
   b. 2 black, 2 red, 2 white
8. A committee of 3 persons has to be formed from 5 men and 6 women. In how many ways can this be done when the committee consists of:
   a. 1 men and 2 women
   c. at least 2 men
   b. all men
9. A box containing 6 red, 4 white and 5 blue bass. One ball is drawn at random. Find the probability that is:
   a. red or white
   c. not blue
   b. white or blue
   d. odd in favor of red
10. From a deck of 52 playing cards, one card is drawn. Find the probability that it is:
   a. Either face card or heart
   b. Either queen or jack

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c. Either king or black

11. A box contains 5 white and 4 red balls If two balls are drawn one after the other with out replacement. Find the probability that

a. both are red

b. both are white

c. the first ball is red and the second is white.

d. the first ball is white and the second is red.

e. both ball different color

f. both the same color.

12. A box contains 6 red balls, and 4 bule balls. Two balls are drawn one after the other with out replacement. Find the probability that:

a. the 2nd ball is blue given that the first ball is red

b. the second ball is red given that the first ball is blue.

c. the second ball is red given that first ball is red.

d. the second ball is blue given that the first ball is blue.

13. What is the probability that a family with two children has two boys, given that they at least one boys.

A. $\frac{1}{4}$ B. $\frac{3}{4}$ C. $\frac{1}{3}$ D. $\frac{2}{3}$

Answer to Review Ex. 9

1. a. $26 \times 26 \times 10 \times 10 \times 10 = 676{,}000$ ways

   b. $26 \times 25 \times 10 \times 9 \times 8 = 468{,}000$ ways

2. $4 \times 5 \times 2 = 40$ ways

3. a. Addition principle: $15 + 10 = 25$

   b. multiplication principle : $15 \times 10 = 150$

   c. $25 \times 24 = 600$

4. a. Addition principle : $5 + 2 = 10$

   b. multiplication principle: $5 \times 3 \times 2 = 30$

5. a. CD, DC, CE, EC, DE and ED. b. CD, CE, DE

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6. a. $C(5, 4) = 5$ b. $C(7, 4) = 35$

c. $(5, 2) \times C(7, 2) = 10 \times 35 = 350$

7. a. $\frac{9!}{4!\,3!\,2!} = 630$ b. $\frac{6!}{2!\,2!\,2!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 2 \times 2} = 90$

8. a. $\binom{5}{1}\binom{6}{2} = (5)(15) = 75$

b. $\binom{5}{3} = \frac{5!}{3!\,2!} = \frac{5 \times 4}{2} = 10$ ways

c. 2 men, 1 women or 3 men, no women $\binom{5}{2}\binom{6}{1} + \binom{5}{3}\binom{6}{0} = 10.6 + 10.(1) = 70$

9. a. $P(\text{red or white}) = \frac{6}{15} + \frac{4}{15} = \frac{10}{15} = \frac{2}{3}$

b. $P(\text{white or blue}) = \frac{4}{15} + \frac{5}{15} = \frac{9}{15} = \frac{3}{5}$

c. $P(\text{not blue}) = 1 - P(\text{blue}) = 1 - \frac{5}{15} = \frac{10}{15} = \frac{2}{3}$

d. $P(\text{odd in favor or red}) = \frac{P(\text{red})}{P(\text{not red})} = \frac{6}{15} \cdot \frac{15}{9} = \frac{2}{3}$

10. a $P(F \text{ or } H) = P(F) + P(H) - P(F \cap H) = \frac{12}{52} + \frac{13}{52} - \frac{3}{52} = \frac{11}{26}$

b. $P(Q \text{ or } J) = P(Q) + P(J) = P(Q \cap J) = \frac{4}{52} + \frac{4}{52} - \frac{0}{52} = \frac{2}{13}$

c. $P(K \text{ or } B) = P(K) + P(B) - P(K \cap B) = \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{7}{13}$

11. a. $P(R_1 \text{ then } R_2) = P(R_1) \cdot P(R_2|R_1) = \frac{4}{9} \cdot \frac{3}{8} = \frac{1}{6}$

b. $P(W_1, \text{ then } W_2) = P(W_1) \cdot P(W_2|W_1) = \frac{5}{9} \cdot \frac{4}{8} = \frac{5}{18}$

c. $P(R_1 \text{ then } W_2) = P(R_1) \cdot P(W_2|R_1) = \frac{4}{9} \cdot \frac{5}{8} = \frac{5}{18}$

d. $P(W_1 \text{ then } R_2) = P(R_2) \cdot P(R_2|W_1) = \frac{5}{9} \cdot \frac{4}{8} = \frac{5}{18}$

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e. $P(RW \text{ or } WR) = P(R) \cdot P(W|R) + P(W) \cdot P(R|W)$

$$= \frac{4}{9} \cdot \frac{5}{8} + \frac{5}{9} \cdot \frac{4}{8} = \frac{5}{9}$$

f. $P(R_1 R_2 \text{ or } W_1 W_2) = P(R_1) \cdot P(R_2|R_1) + P(W_1) \cdot P(W_2|W_1)$

$$= \frac{4}{9} \cdot \frac{3}{8} + \frac{5}{9} \cdot \frac{4}{8} = \frac{4}{9}$$

12. a. $P(B|R) = \frac{4}{9}$ ⟵ since $P(R \text{ then } B) = P(R) \cdot P(B|R)$

b. $P(R|B) = \frac{6}{9}$, since $P(B \text{ then } R) = P(B) \cdot P(R|B)$

c. $P(R_1|R_2) = \frac{5}{9}$, since $P(R_1, \text{ then } R_2) = P(R_1) P(R_2|R_1)$

d. $P(B_1|B_2) = \frac{1}{3}$

13. Sample space: S = {BB, BG, GB, GG} and reduced sample event E_1 = {BB, BG, GB}, event E_2 = {BB}

$$\therefore \quad P(E_2|E_1) = \frac{1}{3}$$

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UNIT 10

10. Sequences and Series

What is a Sequence

Intuitively, **a sequence** is:

- ✓ List of things (number) in rule
- ✓ Pairing between positive integer and real number
- ✓ Special type of function whose domain is the set of positive integer.

Number sequence

DEFINITION:

i) A sequence is a function with domain natural number

ii) A sequence that has a last term is called **finite sequence.**

A sequence that does not have a last term is called **infinite sequence.**

- ✓ 1, 3, 5, 7, 9 ⟵ finite sequence
- ✓ 1, 3, 5, 7, 9, … ⟵ infinite sequence
- ✓ Domain of finite sequence is {1,2,3,…,n}
- ✓ Domain of infinite sequence is N

Notation of sequence

Let function **a(n)** = 2n – 1, domain of **a** is the set of natural number.

Then a(1) = 1, a(2) = 3, a(3) = 5, a(4) = 7

- ✓ Rather than use function notation for sequence use a subscript notation $\{a_n\}$, n = 1, 2, 3, ….
  $\Rightarrow a_1 = 1, a_2 = 3, a_3 = 5, a_4 = 7$ and $a_n = 2n - 1$
- ✓ Elements in the range of sequence $\{a_n\}$ are called **the term of a sequence.**
- ✓ The terms are 1, 3, 5, 7, …., 2n – 1 ….
- ✓ The first term is denoted by $a_1 = 1$

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✓ The 2^{nd} term is denoted by $a_2 = 3$
✓ The 3^{rd} term is denoted by $a_3 = 5$
✓ The n^{th} term is denoted by $a_n = 2n - 1$
✓ a_n is called the general term or the n^{th} term of the sequence.
✓ a_n generate all the terms of sequence
✓ The first three terms denoted by: a_1, a_2, and a_3
✓ The first five terms denoted by : a_1, a_2, a_3, a_4 and a_5
✓ The first n^{th} terms denoted by: a_1, a_2, a_3, ... a_n
✓ The sequence $\{a_1, a_2, a_3, \ldots, a_n \ldots\}$ is denoted by $\{a_n\}$ or $\{a_n\}_{n=1}^{\infty}$

Sequence can be described by:

i) listing the term
ii) writing the general term.
iii) drawing graphs
iv) using recurrence relation.

Illustrative Example

1. Find the first four terms, 8^{th} term and 10^{th} term of sequence.
   a) $a_n = \{1 + (-1)^{n+1}\}$ b) $a_n = \{6\}$

Solution: a) $a_n = \{1 + (-1)^{n+1}\}$

$\Rightarrow a_1 = 1 + (-1)^2 = 2, a_2 = 1 + (-1)^3 = 0, a_3 = 1 + (-1)^4 = 2,$
$a_4 = 1 + (-1) = 0$

$\therefore$ The first four terms are $\{2, 0, 2, 0\}$

$\Rightarrow a_8 = 1 + (-1)^9 = 1 - 1 = 0 \Rightarrow a_{10} = 1 + (-1)^{11} = 1 - 1 = 0$

b) $a_n = \{6\} \Rightarrow a_1 = a_2 = a_3 = a_4 = 6 \Rightarrow a_8 = a_{10} = 6$

✓ A sequence can be described by **recurrence relation.**
✓ **A recursively defined sequence** is one in which each succeeding term of the sequence is defined by using some of the preceding terms and together with several terms.

The Fibonic sequence the fibonic sequence is defined recursively by: $a_1 = 1, a_2 = 1, a_k = a_{k-1} + a_{k-2}$ for $k \geq 3$

$\Rightarrow a_3 = a_2 + a_1 = 1 + 1 = 2$
$\Rightarrow a_4 = a_3 + a_2 = 2 + 1 = 3$ and $a_5 = a_4 + a_3 = 3 + 2 = 5$

This recursive sequence 1, 1, 2, 3, 5, 8, ... is called **the Fabonic sequence.** (The Fibonic sequence is extended by adding two consecutive terms to get the next term)

2. Find the first five terms of sequence defined recursively
   a) $a_1 = 1$ and $a_n = na_{n-1}, n \geq 2.$

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b) $a_1 = 1, a_2 = 3$, and $a_n = \frac{1}{2}(a_{n-1} + a_{n-2}), n \geq 3$

Solution: a) $a_1 = 1 \Rightarrow a_2 = 2a_1 = 2(1) = 2 = 2!$

$\Rightarrow a_3 = 3a_2 = 3(2) = 6 = 3!$

$\Rightarrow a_4 = 4a_3 = 4(6) = 24 = 4!$

$\Rightarrow a_5 = 5a_4 = 5(24) = 120 = 5!$

$\Rightarrow a_n = na_{n-1} = n!$

b) $a_3 = \frac{1}{2}(a_2 + a_1) = \frac{1}{2}(3+1) = 2$

$a_4 = \frac{1}{2}(a_3 + a_2) = \frac{1}{2}(2+3) = \frac{5}{2}$

$a_5 = \frac{1}{2}(a_4 + a_3) = \frac{1}{2}\left(\frac{5}{2} + 2\right) = \frac{9}{4}$

3. Which the following expression define sequence or not

a) $a_n = n^2 + 1$ b) $a_n = a_{n-1} + 1$ for $n \geq 1, a_1 = 1$

c) $a_n^2 = 3n + 1$ d) $a_1 = \pm 1$, $a_2 = 3$, and $a_n = (a_{n-1})(a_{n-2}), n \geq 3$

Solution: Sequence is special type of function with domain natural number

a) $a_n = n^2 + 1$ is function $\Rightarrow$ sequence

b) $a_n = a_{n-1} + 1$ is function $\Rightarrow$ recursively define sequence

c) $a_n^2 = 3n+1 \Rightarrow a_n = \pm\sqrt{3n+1} \Rightarrow a_1 = \pm 2 \Rightarrow (1,2), (1,-2)$

Not function $\Rightarrow$ not sequence

d) $a_3 = (a_2)(a_1) = (3)(\pm 1) = \pm 3 \Rightarrow (3,3), (3,-3), \rightarrow$ not

There are two standard sequence **Arithmetic and geometric sequence**

10.1 Arithmetic sequence (Arithmetic progression)

✓ It is characterized by a common difference between successive term. A sequence with a constant difference between consecutive term is called an **arithmetic sequence (arithmetic progression)**

i.e. $A_2 - A_1 = A_3 - A_2 = \ldots A_n - A_{n-1} = d$, for all $n \geq 1$

✓ The difference between consecutive term is called **the common difference (d).**

$\Rightarrow A_2 = A_1 + d, A_3 = A_2 + d, \ldots A_{n+1} = A_n + d$

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$\Rightarrow A_2 = A_1+d$, $A_3=A_1+2d$, and $A_4=A_1+3d$,...

$\Rightarrow A_n = A_1 + (n-1)d$

Which of the following is an arithmetic sequence

a) 3, 7, 11, 15, ...

b) $\frac{1}{3}, 1, \frac{5}{3}, \frac{7}{3}, \ldots$

c) $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots$

d) 1, 4 +2p, 7+4p, 10+6p,...

e) 1, 4, 9, 16, ...

Solution:

a) $7-3 = 11-7 = 15-11 = 4 \Rightarrow$ A.P, with d = 4

b) $1-\frac{1}{3} = \frac{5}{3}-1 = \frac{7}{3}-\frac{5}{3} = \frac{2}{3} \Rightarrow$ A.P, with $d = \frac{2}{3}$

c) $\frac{1}{3}-\frac{1}{2} \neq \frac{1}{4}-\frac{1}{3} \Rightarrow$ not A.P

d) $4+2p-1 = 7+4p-(4+2p) = 3+2p \Rightarrow$ A.P with $d = 3+2p$

5. State whether each of the following sequence is an arithmetic sequence. If it is state the common difference (d).

a) $a_n = 5-2n$

b) $a_n = 4n+1$

c) $a_n = n^2$

d) $a_n = \frac{5+3n}{2}$

e) $a_n = 7$

f) $a_n = (-1)^n$

Solution: a) $a_{n+1} = 5-2(n+1) = 3-2n$

$\Rightarrow a_{n+1}-a_n = 3-2n-(5-2n) = -2 \Rightarrow d = -2 \Rightarrow$ A.P

b) $a_{n+1}-a_n = 4(n+1)+1-(4n+1) = 4 \Rightarrow d = 4 \Rightarrow$ A.P

c) $a_{n+1}-a_n = (n+1)^2-n^2 = n^2+2n+1-n^2 = 2n+1$

$\Rightarrow$ not constant and not arithmetic sequence

d) $a_{n+1}-a_n = \frac{5+3(n+1)}{2}-\left(\frac{5+3n}{2}\right) = \frac{5+3n+3-5-3n}{2} = \frac{3}{2}$

$d = \frac{3}{2} \Rightarrow A.P$

e) $a_{n+1}-a_n = 7-7 = 0 \Rightarrow d = 0 \Rightarrow$ A.P

f) -1, 1, -1, 1, ...

$\Rightarrow 1-(-1) \neq -1-1 \Rightarrow 2 \neq -2 \Rightarrow$ not A.P.

If $\{A_n\}$ is n^{th} terms of arithmetic sequence with first term A_1 and common difference d is given by $A_n = A_1 + (n-1)d$, for all $n \geq 1$

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6. Find 16^{th}, 25^{th} and n^{th} terms of sequence -4, 1, 6,...

Solution: $A_1 = -4$, $d = 1 - (-4) = 5$, $A_n = A_1 + (n-1)d$

$\Rightarrow A_{16} = -4 + (16-1)5 = -4 + 75 = 71$

$\Rightarrow A_{25} = -4 + (25-1)5 = -4 + 120 = 116$

$\Rightarrow A_n = -4 + (n-1)5 = -4 + 5n - 5 = 5n - 9$

7) The 5^{th} term of an A.P is 19 and the 11^{th} term is 43

Find a) d b) A_1 c) A_{20} d) n^{th} term

Solution: $A_5 = A_1 + 4d = 19$

$A_{11} = A_1 + 10d = 43$

a) $A_5 - A_{11} = -6d = -24 \Rightarrow d = 4$

b) To find A_1 from $A_5 = A_1 + 4d = 9 \Rightarrow A_1 + 4d = 19$

$\Rightarrow A_1 = 19 - 4(4) = 19 - 16 = 3$

c) $A_{20} = A_1 + 19d = 3 + 19(4) = 79$

d) n^{th} term, $A_n = A_1 + (n-1)d \Rightarrow A_n = 3 + (n-1)4 = 4n - 1$

8. If $A_7 = 27$, $d = 5$, then find a) A_{20} b) n^{th} term

Solution: $A_7 = A_1 + 6d \Rightarrow A_1 + 6(5) = 27 \Rightarrow A_1 = -3$

a) $A_{20} = -3 + 19(5) = 92$

b) $A_n = A_1 + (n-1)d \Rightarrow A_n = -3 + (n-1)5 = 5n - 8$

9. Which term of the sequence 6, 14, 22,... is

a) 246 b) 258

Solution: a) $A_1 = 6$, $d = 14 - 6 = 8 \Rightarrow A_n = A_1 + (n-1)d$

$\Rightarrow 246 = 6 + (n-1)8 \Rightarrow 8n - 2 = 246 \Rightarrow 8n = 248 \Rightarrow n = 31$

$\Rightarrow a_{31} = 246$ $(31)^{th}$ term is 246

b) $A_n = A_1 + (n-1)d \Rightarrow 6 + (n-1)8 = 258$

$\Rightarrow 8n - 2 = 258$ and $8n = 260 \Rightarrow n = \dfrac{260}{8} = \dfrac{65}{2} \notin N$

$\therefore$ 258 is not part of sequence

10. The p^{th} term of an A.P is q and the q^{th} term is P. Find $(p+q)^{th}$ term.

Solution: $A_n = A_1 + (n-1)d$

$$\Rightarrow \begin{cases} A_p = A_1 + (p-1)d = q \\ A_q = A_1 + (q-1)d = p \end{cases}$$

$\Rightarrow A_p - A_q = d(p-1) - d(q-1) = d(p-1-q+1) = d(p-q)$

$\Rightarrow q - p = d(p-q)$ and $d = \dfrac{p-q}{q-p} = -1$

$\Rightarrow A_q = A_1 + (q-1)(-1) = p \Rightarrow A_1 = p + q - 1$

$\Rightarrow A_{(p+q)} = A_1 + (p+q-1)d = p + q - 1 + (p+q-1)(-1)$

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$\Rightarrow A_{(p+q)} = p + q - 1 - p - q + 1 = 0$

11. Find the total number of integer between 100 and 1000 divisible by 7

Solution: $A_1 = 105, \quad d = 7$

Also integer less than 1000 divisible by 7 is 994.

$\Rightarrow \{105, 112, 119, 126, 133, \ldots, 994\}$

$\Rightarrow A_n = 994 \Rightarrow A_n = A_1 + (n-1)d$

$\Rightarrow 994 = 105 + (n-1)7 \Rightarrow 7n = 994 - 105 + 7 = 896$

$\Rightarrow 7n = 896 \Rightarrow n = \dfrac{896}{7} = 128 \Rightarrow$ the total number of integer is 128

12. In each part, find the common difference of a_n:

a) $a_n = 4n + 6$ b) $a_n = \dfrac{3-5n}{2}$ c) $a_n = 7 - 3n$

Solution: a) $d = a_{n+1} - a_n \Rightarrow d = 4(n+1)+6-(4n+6)$

$= 4n + 4 + 6 - 4n - 6 = 4$

b) $d = a_{n+1} - a_n \Rightarrow d = \dfrac{3-5(n+1)-(3-5n)}{2}$

$= \dfrac{3-5n-5-3+5n}{2} = \dfrac{-5}{2}$

c) $d = a_{n+1} - a_n \Rightarrow d = 7 - 3(n+1) = 7 - 3n - 3 - 7 + 3n = -3$

13. In each part, find i) common difference ii) n^{th} term of an A.P

a) $A_{20} = -1$ and $A_{30} = -6$ c) $A_1 = 8$ and $A_6 = 23$

b) $A_7 = 32$ and $A_{12} = 57$ d) $A_5 = -13$ and $a_9 = -25$

Solution: a) $A_{20} = A_1 + (20-1)d = -1 \Rightarrow A_1 + 19d = -1$

$\Rightarrow A_{30} = A_1 + 29d = -6$

$\Rightarrow A_{20} - A_{30} = (A_1 + 19d) - (A_1 + 29d)$

$\Rightarrow -1 - (-6) = -10d \Rightarrow d = \dfrac{-1}{2}$

$\therefore$ common difference $d = \dfrac{-1}{2}$

To find A_1, put $d = \dfrac{-1}{2}$ in $A_{20} = A_1 + 19d = -1$

$\Rightarrow A_1 + 19\left(\dfrac{-1}{2}\right) = -1$ and $A_1 = -1 + \dfrac{19}{2} = \dfrac{17}{2}$

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The n^{th} term, $A_n = \frac{17}{2} + (n-1)\left(\frac{-1}{2}\right) = 9 - \frac{1}{2}n$

b) $A_7 = A_1 + 6d = 32$ and $A_{12} = A_1 + 11d = 57$

$\Rightarrow A_{12} - A_7 = A_1 + 11d - (A_1 + 6d)$

$\Rightarrow 57 - 32 = 11d - 6d$ and $5d = 25 \Rightarrow d = 5$

$\therefore$ common difference, $d = 5$

To find A_1, put $d = 5$ in $A_7 = A_1 + 6d = 32$

$\Rightarrow A_1 + 6(5) = 32$ and $A_1 = 32 - 30 = 2$

$\therefore$ The n^{th} term $A_n = 2 + (n-1)5 = 5n - 3$

c) and (d) exercise left for you (Tip)

14. A man accepts a position with an initial salary of Birr 14,400 per year. If it is known that his salary will increase at the end of every year by birr 1,440. what will be his annual salary:

a) at the beginning of the 11^{th} year b) at the beginning of 5^{th} year c) at the end of 5^{th} year

Solution: $A_1 = 14{,}400$, $d = 1{,}440$

The beginning of the 11^{th} year is the same as the end of the 10^{th} year.

a) $A_{10} = A_1 + 9d = 14{,}400 + 9(1{,}440) = 27{,}360$

$\Rightarrow$ His annual salary at the beginning of the 11^{th} year is 27,360 birr

b) Beginning of 5^{th} year is the same as end of 4^{th} year

$\Rightarrow A_4 = A_1 + 3d = 14{,}400 + 3(1{,}440) = 18{,}720$

$\Rightarrow$ His annual salary at the beginning of 5^{th} year is 18,720

c) $A_5 = A_1 + 4d \Rightarrow 14{,}400 + 4(1{,}440) = 20{,}160$

$\Rightarrow$ His annual salary at the end of 5^{th} year is birr 20,160

15. A person is scheduled to get a rise of Birr 50 every 6 months, during his/her 5 years on the job. If his/her starting salary is Birr 1600 per month what will his/her annual salary be at the end of 3^{rd} year.

Solution: Let A_1 = his annual salary

$\Rightarrow A_1 = 1600 \times 12 = 19{,}200$ birr per year

$\Rightarrow d = 50$ birr

The number (period) of increment, in 3 year is $n = 2 \times 3 = 6$

At the end of 3^{rd} year (A_7)

$\Rightarrow A_7 = A_1 + 6d = 19{,}200 + 6(50) =$ Birr 19500

16. Find the 25^{th} term of an arithmetic sequence

a) 5, 9, 13, 17 b) 6, 10, 14, 18

Solution: a) $A_n = A_1 + (n-1)d \Rightarrow A_{25} = A_1 + 24d$

$\Rightarrow A_{25} = 5 + 24(4) = 5 + 96 = 101$

b) $A_{25} = 6 + 24(4) = 6 + 96 = 102$

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10.1.1 Arithmetic Mean between a and b

The terms of an arithmetic sequence that are between two given terms are called **arithmetic means** of the given terms.

- ✓ If $\{a, C_1, C_2, C_3, ..., C_k, b\}$ is an arithmetic sequence then $C_1, C_2, C_3, ..., C_k$ are k-arithmetic mean between a and b.
- ✓ $b = (k + 2)^{th}$ term of A.P

Illustrative Example

17. a) Insert 3 arithmetic mean between 11 and 19
b) Insert 4 arithmetic mean between 2 and 3
c) Insert n arithmetic mean between 2 and 3

Solution; a) Let, m_1, m_2, m_3 be the 3 arithmetic means

$11, m_1, , m_3, 19$ are A.P

$\Rightarrow A_5 = 19 \Rightarrow A_1 + 4d = 19$

$\Rightarrow 11 + 4d = 19 \Rightarrow 4d = 19 - 11 = 8$

$\Rightarrow d = 2$

$\Rightarrow m_1 = A_2 = A_1 + d = 11 + 2 = 13$

$\Rightarrow m_1 = 13, m_2 = 15, m_3 = 17$

$\therefore \{13, 15, 17\}$ are 3 arithmetic mean between 11 and 19

Explanation

$d = \frac{A_5 - A_1}{5 - 1} = \frac{19 - 11}{4} = 2$

b) $2, m_1, m_2, m_3, m_4, 3$ are A.P

$\Rightarrow A_6 = A_1 + 5d = 3$ and $2 + 5d = 3 \Rightarrow d = \frac{3-2}{5} = \frac{1}{5}$

$\Rightarrow m_1 = A_2 = A_1 + d = 2\frac{1}{5} = \frac{11}{5}$

$\Rightarrow m_2 = A_3 = A_1 + 2d = 2 + 2\left(\frac{1}{5}\right) = \frac{12}{5}$

$\Rightarrow m_3 = A_4 = A_1 + 3d = 2 + 3\left(\frac{1}{5}\right) = \frac{13}{5}$

$\Rightarrow m_4 = A_5 = A_1 + 4d = 2 + 4\left(\frac{1}{5}\right) = \frac{14}{5}$

$\therefore \left\{\frac{11}{5}, \frac{12}{5}, \frac{13}{5}, \frac{14}{5}\right\}$ are 4 arithmetic mean between 2 and 3

Explanation

$d = \frac{A6 - A_1}{6 - 1} = \frac{3-2}{5} = \frac{1}{5}$

c) $2, m_1, m_2, m_3, \ldots, m_n, 3$

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$\Rightarrow A_{n+2} = 3$ and $A_{n+2} = A_1 + (n+2-1)d = 2 + (n+1)d = 3$

$\Rightarrow (n+1)d = 3 - 2 \Rightarrow d = \dfrac{1}{n+1}$ ← common difference

$\Rightarrow m_1 = 2 + d = 2 + \dfrac{1}{n+1} = \dfrac{2n+3}{n+1}$

$\Rightarrow m_2 = 2 + 2d = 2 + \dfrac{2}{n+1} = \dfrac{2n+4}{n+1}$

$\Rightarrow m_n = 2 + nd = 2 + n\left(\dfrac{1}{n+1}\right) = 2 + \dfrac{n}{n+1}$

$= \dfrac{2n+2+n}{n+1} = \dfrac{3n+2}{n+1}$

$\Rightarrow m_1 = a + d = a + \dfrac{b-a}{n+1}$

$$\{m_1, m_2, m_3, \ldots, m_n\} = \left\{a + \frac{(b-a)}{n+1}, a + 2\frac{(b-a)}{n+1}, \ldots, a + \frac{n(b-a)}{n+1}\right\}$$

Sigma notation and partial sum of A.P

✓ A **series** is the indicated sum of the terms of a sequence. (series is the sum of terms of given sequence.

If $a_1, a_2, a_3, a_4, \ldots a_n$ are terms of sequence)

Thus $S_1 = a_1$ ← sum of first term.

$S_2 = a_1 + a_2$ ← sum of the first two terms of **sequence**

$S_3 = a_1 + a_2 + a_3$ ← sum of the first three terms of **sequence**

$S_4 = a_1 + a_2 + a_3 + a_4$ ← sum of the first four terms of a_n **sequence**

$S_n = a_1 + a_2 + a_3 + \ldots + a_n$ ← sum of the first n terms of **sequence**

$S = a_1 + a_2 + a_3 + \ldots + a_n \ldots$ ← sum of all the terms of **sequence**

The sigma (Σ) notation

The Greek letter (Σ) is used to write all the term of infinite and finite series

i) $S_2 = \sum_{k=1}^{2} a_k = a_1 + a_2$

ii) $S_3 = \sum_{k=1}^{3} a_k = a_1 + a_2 + a_3$

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iii) $S_4 = \sum_{k=1}^{4} a_k = a_1 + a_2 + a_3 + a_4$

iv) $S_n = \sum_{k=1}^{n} a_k = a_1 + a_2 + a_3 + \ldots + a_n$ ⟵ sum of n^{th} partial sum

v) $S = \sum_{k=1}^{\infty} a_k = a_1 + a_2 + a_3 + \ldots + a_n + \ldots$ ← sum of all not partial sum. (Infinite series)

In this notation, the letter k is the **index of summation**, 1 is the lower limit,

✓ $\sum_{k=1}^{n} Ca_k = C\sum_{k=1}^{n} a_k$

✓ $\sum_{k=1}^{n} (a_k \pm b_k) = \sum_{k=1}^{n} a_k \pm \sum_{k=1}^{n} b_k$

✓ $\sum_{k=1}^{n} C = C + C + C + \ldots + C = nC$

✓ $\sum_{k=m}^{n} C = (n - m + 1)C$

Example:a) $\sum_{k=1}^{5} 7 = 7 + 7 + 7 + 7 + 7 = 5 \times 7 = 35$

b) $\sum_{k=4}^{9} 7 = (9 - 4 + 1)7 = 42$

c) $\sum_{k=15}^{28} 8 = (28 - 15 + 1)8 = 14 \times 8 = 112$

Illustrative Example

18. Find the sum (k^{th} partial sum)

a) $\sum_{k=1}^{5} k$ b) $\sum_{k=3}^{9} (5k - 4)$ c) $\sum_{k=2}^{6} 3k$

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d) $\sum_{k=4}^{11} 12$ f) $\sum_{n=1}^{12}\left(\frac{3}{n}-\frac{3}{n+1}\right)$

Solution: a) $\sum_{k=1}^{5} k = 1+2+3+4+5 = 15$

b) $\sum_{k=3}^{9}(5k-4) = 5\sum_{k=3}^{9} k - \sum_{k=3}^{9} 4$

$= 5(3+4+5+6+7+8+9) - (9-3+1)4 = 210 - 20 = 190$

c) $\sum_{k=2}^{6} 3k = 3(2+3+4+5+6) = 60$

d) $\sum_{k=4}^{11} 12 = (11-4+1)12 = 96$

e) $\sum_{n=1}^{n}\left(\frac{3}{n}-\frac{3}{n+1}\right) = 3\sum_{n-1}^{n}\left(\frac{1}{n}-\frac{1}{n+1}\right)$

$= 3\left[\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+\ldots+\left(\frac{1}{n}-\frac{1}{n+1}\right)\right]$

$S_n = 3\left(1-\frac{1}{n+1}\right) = 3-\frac{3}{n+1}$

$\Rightarrow S_{12} = 3-\frac{3}{12+1} = \frac{39-3}{13} = \frac{36}{13}$

$\Rightarrow S_{99} = 3-\frac{3}{100} = \frac{297}{100} = 2.97$

$\Rightarrow S_{200} = 3-\frac{3}{201} = 3\left(1-\frac{1}{201}\right) = 3\left(\frac{200}{201}\right) = \frac{600}{201}$

Note: $(a_1 - a_2) + (a_2 - a_3) + (a_3 - a_4) + \ldots$ is called **telescoping series** a_2 is canceled by 2nd term, a_3 is canceled by 3rd term and so on so that, the nth partial sum of this series is $S_n = a_1 - a_{n+1}$

19. Find the indicated partial sum (S_n) of the following sequence.

a) $a_n = \frac{1}{n}-\frac{1}{n+1}$, S_{10}, S_{50}, S_{100}, S_n

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b) $a_n = \ln\left(\frac{n+1}{n}\right)$, S_n, S_{100}, S_{200},...

c) $a_n = \frac{1}{n+2} - \frac{1}{n+3}$, S_n, S_{20}, S_{50}, S_{100}, S_{200}

Solution: a) $\sum_{n=1}^{n} a_n = \sum_{n=1}^{n}\left(\frac{1}{n} - \frac{1}{n+1}\right)$

$$= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \ldots + \left(\frac{1}{n} - \frac{1}{n+1}\right)$$

$$\Rightarrow S_n = 1 - \frac{1}{n+1} = \frac{n}{n+1} \Rightarrow S_{10} = 1 - \frac{1}{11} = \frac{11-1}{11} = \frac{10}{11}$$

$$\Rightarrow S_{50} = \frac{50}{51} \text{ and } S_{100} = \frac{100}{101}$$

b) $\sum_{n=1}^{n} a_n = \sum_{n=1}^{n} \ln\left(\frac{n+1}{n}\right) = \ln 2 + \ln\frac{3}{2} + \ln\frac{4}{3} + \ldots + \ln\frac{n+1}{n}$

$$= \ln\left(2 \times \frac{3}{2} \times \frac{4}{3} \times \ldots \times \frac{n}{n-1} \times \frac{n+1}{n}\right) = \ln(n+1)$$

$$\Rightarrow S_n = \ln(n+1) \Rightarrow S_{100} = \ln(101) \Rightarrow S_{200} = \ln(201)$$

c) $\sum_{n=1}^{n} a_n = \sum_{n=1}^{n}\left(\frac{1}{n+2} - \frac{1}{n+3}\right) = \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \ldots + \left(\frac{1}{n+2} - \frac{1}{n+3}\right)$

$$\Rightarrow S_n = \frac{1}{3} - \frac{1}{n+3} = \frac{n+3-3}{3(n+3)} = \frac{n}{3(n+3)}$$

$$\Rightarrow S_{20} = \frac{20}{3(23)} = \frac{20}{69} \Rightarrow S_{50} = \frac{50}{3(50+3)} = \frac{50}{159}$$

$$\Rightarrow S_{100} = \frac{100}{3(100+3)} = \frac{100}{309}$$

10.2 Sum of Arithmetic Progressions

Consider the following arithmetic sum written in **increasing order** and then in decreasing order. Paired vertically and added.

The sums form a new series in which all the addends are the same

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$$+\begin{cases} S_8 = 6+9+12+15+18+21+24+27 \\ S_8 = 27+24+21+18+15+12+9+6 \end{cases}$$

$$2S_8 = 33+33+33+33+33+33+33+33$$

$\Rightarrow 2S_8 = 8(33) = 264 \Rightarrow S_8 = 132$

This methods generates n^{th} partial sum of A.P.

Let $\{A_n\}_{n=1}^{\infty}$ be an A.P, and S_n be the sum of the first n terms of an A.P. That is $S_n = A_1 + A_2 + A_3 + \ldots + A_n$

$$+\begin{cases} S_n = A_1 + (A_1 + d) + (A_1 + 2d) + \ldots + (A_n - d) + A_n \\ S_n = A_n + (A_n - d) + (A_n - 2d) + \ldots + A_1 + d + A_1 \end{cases}$$

$$2S_n = (A_1 + A_n) + (A_1 + A_n) + (A_1 + A_n) + \ldots + (A_1 + A_n)$$

$\Rightarrow 2S_n = n(A_1 + A_n)$, but $A_n = A_1 + (n-1)d$

$$\Rightarrow S_n = \frac{n}{2}(A_1 + A_n) = \frac{n}{2}(2A_1 + (n-1)d)$$

The **n^{th} partial sum of an arithmetic** sequence $\{A_n\}_{n=1}^{\infty}$ with common difference **d** is

i) $S_n = \dfrac{n(A_1 + A_n)}{2}$, ii) $S_n = \dfrac{n}{2}[2A_1 + (n-1)d]$

Illustrative Example

20. Find the sum of the first:
   a) 100 natural number
   b) 100 odd natural number
   c) 50 even natural number
   d) 100 natural number exactly divisible by 7
   e) 100 terms of A.P between 3 and 4
   f) n-natural number
   g) n-odd natural number
   h) n-even natural number

Solution: we use $S_n = \dfrac{n}{2}(A_1 + A_n)$

a) $1 + 2 + 3 + \ldots + 100 \Rightarrow S_{100} = \dfrac{100}{2}(1+100) = 5{,}050$

b) The n^{th} odd natural number is $A_n = 2n - 1$

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$\Rightarrow$ the 100^{th} odd natural number is

$A_{100} = 2(100) - 1 = 199$ and $A_1 = 1, A_{100} = 199$

$$\Rightarrow S_{100} = \frac{100}{2}(1+199) = (50)(200) = 10{,}000$$

c) We can use $S_n = \frac{n}{2}[2A_1 + (n-1)d] \Rightarrow A_1 = 2, d = 2$

$$\Rightarrow S_{50} = \frac{50}{2}(2(2) + (50-1)(2)] = 25(4+98) = 2{,}500 = 2550$$

d) The sequence is $\{7, 14, 21, \ldots\}$, $S_n = \frac{n}{2}[2A_1 + (n-1)d]$

$$\Rightarrow A_1 = 7, d = 7 \Rightarrow S_{100} = \frac{100}{2}(2(7) + (99)(7)) = 35{,}350$$

e) $A_1 = 3, A_{100} = 4, \Rightarrow S_{100} = \frac{100}{2}[3+4] = 50 \times 7 = 350$

f) n^{th} partial sum is denoted by $S_n = \frac{n}{2}[2A_1 + (n-1)d]$

$$\Rightarrow S_n = \frac{n}{2}(2(1) + (n-1)1) = \frac{n^2+n}{2}$$

g) $A_1 = 1, d = 2 \Rightarrow S_n = \frac{n}{2}(2(1) + (n-1)2) = \frac{n}{2}(2n) = n^2$

h) $A_1 = 2, d = 2 \Rightarrow S_n = \frac{n}{2}(2(2) + (n-1)2) = \frac{n}{2}(2n+2) = n^2 + n$

21. Find the sum of two digit numbers which leave the remainder 2 when divided by 3.

Solution: by division algorithm, $A_n = 3n + 2$

$\Rightarrow$ 11, 14, 17, …, 98

$\Rightarrow$ $98 = 11 + (n-1)3 \Rightarrow 3n = 90 \Rightarrow n = 30$

Use, $S_n = \frac{n}{2}(A_1 + A_n) \Rightarrow S_{30} = \frac{30}{2}(11+98) = 15(109) = 1635$

22. Find the sum of the sequence

a) $18 + 21 + 24 + \ldots + 90$ b) $12 + 19 + 26 + \ldots + 89$

Solution: First determine the number of terms

a) $A_n = A_1 + (n-1)d = 18 + (n-1)3 = 90$ and $3n = 75$

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$\Rightarrow n = 25$

$\Rightarrow S_{25} = \frac{25}{2}(18+89) \Rightarrow 1350$

b) $A_n = A_1 + (n-1)d \Rightarrow 89 = 12 + (n-1)7 \Rightarrow 7n = 84 \Rightarrow n = 12$

$\Rightarrow S_{12} = \frac{12}{2}(12+89) \Rightarrow 606$

23. Find the sum pf the first 20 terms of $2 + 5 + 8 + \ldots$

Solution: $S_{20} = \frac{20}{2}(2(2)+(20-1)3) = 10 \times 61 = 610$

24. Find the sum of all 3 digit numbers between 200 and 400 which are divisible by 13.

Solution: The least 3 digit numbers greater than 200 divisible by 13 is 208, $A_1 = 208$, and $A_n = 390$

To find n, $A_n = A_1 + (n-1)d \Rightarrow 390 = 208 + (n-1)13$

$\Rightarrow 13n = 390 - 208 + 13 = 195$

$\Rightarrow n = \frac{195}{13} = 15$

$S_{15} = \frac{15}{2}(208+390) = 4485$

25. If $A_5 = 15$ and $A_{24} = 91$, then find

a) S_{20} b) S_{30} c) S_n

Solution: a) $S_n = \frac{n}{2}(2A_1 + (n-1)d) \Rightarrow S_{20} = \frac{20}{2}(2A_1 + 19d)$

To find d, $A_{24} - A_5 = (A_1 + 23d) - (A_1 + 4d)$

$\Rightarrow 91 - 15 = 19d \Rightarrow d = \frac{76}{19} = 4$

To find A_1, $A_5 = A_1 + 4d \Rightarrow 15 = A_1 + 4(4) \Rightarrow A_1 = 15 - 16 = -1$

$\Rightarrow S_{20} = \frac{20}{2}(2(-1)+(20-1)4) = \frac{20}{2}(74) = 740$

b) $S_{30} = \frac{30}{2}(2(-1)+(30-1)4) = 15(-2+116) = 1710$

c) $S_n = \frac{n}{2}(2(-1)+(n-1)4) = 2n^2 - 3n$

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10.2.1 Finding the maximum partial sum of A.P

Example: Find the maximum sum

a) $9 + 6 + 3 + 0 + -3 + -6 \ldots$

b) $13 + 10 + 7 + 4 + 1 + -2 + -5 + \ldots$

a) $9 + 6 + 3 + 0 = 18$ ⟵ maximum sum

b) $13 + 10 + 7 + 4 + 1 = 35$ ⟵ maximum sum

Note: Maximum sum is obtained when the n^{th} term $(A_n \geq 0)$ (the sum is maximum if we add only all positive term)

Illustrative Example

1. Find the maximum sum:

a) $A_1 = 42, d = -5$

b) $A_1 = 19, d = -2$

Solution: a) A.P $= 42, 37, 32, \ldots$

$\Rightarrow A_n \geq 0$ and $A_1 + (n-1)d \geq 0$

$\Rightarrow (42 + (n-1)(-5)) \geq 0 \Rightarrow 47 \leq 5n \Rightarrow n \leq 9\frac{2}{5}$

Since n is natural number $\Rightarrow n = 9$

$\Rightarrow S_9 = \frac{9}{2}(2(42) + (9-1)(-5)) = 198$ ⟵ maximum sum

b) A.P. $= 19, 17, 15, \ldots \Rightarrow n^{th}$ term $\geq 0 \Rightarrow A_n \geq 0$

$\Rightarrow (19 + (n-1)(-2)) \geq 0 \Rightarrow -2n + 21 \leq 0 \Rightarrow n \leq 10\frac{1}{2}$

$\Rightarrow n = 10$

$\Rightarrow S_9 = \frac{10}{2}(2(19) + (9)(-2)) = 100$ ⟵ maximum sum

27. Find S_6 for the given arithmetic sequence

a) $A_1 = 6, d = 3$

b) $A_1 = 5, d = 4$

c) $A_{10} = 11, d = 2$

Solution: use $S_n = \frac{n}{2}(2A_1 + (n-1)d)$

a) $S_6 = \frac{6}{2}(2(6) + (5)(3)) = 3(27) = 81$

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b) $S_6 = \frac{6}{2}(2(5)+5(4)) = 90$

c) **First, find A_1,** $A_{10}=A_1+9d \Rightarrow 11 = A_1+18 \Rightarrow A_1 = -7$

$\Rightarrow S_6 = \frac{6}{2}(2(-7)+5(2)) = -12$

28. Find the indicated term of an arithmetic sequence

a) $S_{10} = 210, A_1 = 3,$ Find A_{10}

b) $S_{20} = 600, A_{16} = 41,$ find A_1, A_{20}, A_n.

c) $S_{10} = 145, A_9 = 25,$ find S_9, d, A_n.

d) $S_n = 300, A_1 = 3, d = 4,$ find n, A_n

Solution: a) $S_{10} = \frac{10}{2}(A_1 + A_{10})$

$\Rightarrow 210 = 5(3 + A_{10})$ and $A_{10} + 3 = 42 \Rightarrow A_{10} = 39$

b) $A_{16} = A_1 + 15d$ and $41 = A_1 + 15d \Rightarrow A_1 = 41 - 15d$

$S_{20} = \frac{20}{2}(2A_1 + 19d) \Rightarrow 600 = 10[2(41-15d)+19d]$

$\Rightarrow 60 = 82 - 30d + 19d$

$\Rightarrow 60 - 82 = -11d \Rightarrow d = \frac{-22}{-11} = 2$

So that $A_1 = 41 - 15d = 41 - 15(2) = 11$

$\Rightarrow A_{20} = A_1 + 19d = 11 + 19(2) = 49$

$\Rightarrow A_n = A_1 + (n-1)d = 11 + (n-1)(2) = 2n + 9$

c) $A_9 = A_1 + 8d \Leftrightarrow 25 = A_1 + 8d \Rightarrow A_1 = 25 - 8d$

$\Rightarrow S_{10} = \frac{10}{2}[2(25-8d)+9d] \Leftrightarrow 145 = 5[50 - 16d + 9d]$

$\Rightarrow 29 = 50 - 7d$ and $-7d = -21 \Rightarrow d = \frac{21}{7} = 3$

$\therefore A_1 = 25 - 8(3) = 25 - 24 = 1$

- $S_9 = \frac{9}{2}[A_1 + A_9] = \frac{9}{2}[1+25] = 117$
- $A_n = A_1 + (n-1)d \Rightarrow = 1 + (n-1)3 = 3n - 2$

d) $S_n = \frac{n}{2}(2A_1 + (n-1)d) = n(2n+1) = 300 \Rightarrow 2n^2 + n - 300 = 0$

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$$\therefore n = \frac{-1 \pm \sqrt{1-4(2)(-300)}}{2(2)} = \frac{-1+\sqrt{2401}}{4} = \frac{-1+49}{4} = 12$$

$\Rightarrow A_n = A_1 + (n-1)d = 3 + (n-1)4 = 4n - 1$

If $S_8 = 12$ and $S_{16} = 56$, then find

a) Common difference (d) b) the n^{th} term

29. *Solution:*

$$S_8 = \frac{8}{2}[2A_1 + 7d] \Rightarrow \begin{cases} 2A_1 + 7d = 3 \\ 2A_1 + 15d = 7 \end{cases} \Rightarrow d = \frac{4}{8} = \frac{1}{2}$$

$$S_{16} = \frac{16}{2}[2A_1 + 15d] \Rightarrow \quad 0 - 8d = -4$$

b) From S_8, $2A_1 + 7\left(\frac{1}{2}\right) = 3 \Rightarrow A_1 = \frac{-1}{4}$

$\Rightarrow A_n = A_1 + (n-1)d$ and $A_n = \frac{-1}{4} + (n-1)\left(\frac{1}{2}\right) = \frac{2n-3}{4}$

30. Find the sum of 12 terms of the A.P, -7, -3, -1, … starting from the 8^{th} term

Solution: $a_1 = A_8 = A_1 + 7d = -7 + 7(4) = 21$

$a_{12} = A_{19} = A_1 + 18d = -7 + 18(4) = 65$

$$\Rightarrow S_{12} = \frac{12}{2}[a_1 + a_{12}] = 6[21+65] = 6(86) = 516$$

31. Find S_n (n^{th} partial sum) of arithmetic sequence.

a) $a_n = 3n + 2$ b) $a_n = 4n - 3$ c) $a_n = 1 - 2n$

Solution: a) $a_1 = 3(1) + 2 = 5$, $a_n = 3n + 2$

$$\Rightarrow S_n = \frac{n}{2}[a_1 + a_n] = \frac{n}{2}[5+3n+2] = \frac{3n^2+7}{2}$$

b) $$S_n = \frac{n}{2}[a_1 + a_n] = \frac{n}{2}[1+4n-3] = 2n^2 - n$$

c) $$S_n = \frac{n}{2}[a_1 + a_n] = \frac{n}{2}[-1+1-2n] = -n^2$$

32. Find the n^{th} term (a_n), common difference (d) for the given n^{th} partial sum of arithmetic sequence.

a) $S_n = 2n^2 + 3n$ b) $S_n = 2n^2 + n$ c) $S_n = 3n^2 - 5n$

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Solution: $S_n = \underbrace{a_1 + a_2 + a_3 + \ldots + a_{n-1}}_{S_{n-1}} + a_n \Rightarrow a_n = S_n - S_{n-1}$

a) $a_n = S_n - S_{n-1} \Rightarrow a_n = (2n^2 + 3n) - (2(n-1)^2 + 3(n-1))$
$\Rightarrow a_n = (2n^2 + 3n) - (2(n-1)^2 + 3(n-1))$
$\Rightarrow a_n = 2n^2 + 3n - 2n^2 + 4n + 1 - 3n = 4n + 1$
$\Rightarrow a_1 = 5, a_2 = 9 \Rightarrow d = a_2 - a_1 = 9 - 5 = 4$

b) $a_n = s_n - s_{n-1} \Rightarrow a_n = (2n^2 + n) - (2(n-1)^2 + (n-1))$
$\Rightarrow a_n = 2n^2 + n - 2n^2 + 4n - 2 + -n + 1 = 4n - 1 \Rightarrow d = 4$

c) exercise left for you (Tip!)

33. If $S_n = n^2 + 3n - 2$, find

a) 5^{th} term b) 12^{th} term

Solution: $a_n = s_n - s_{n-1}$

a) $a_5 = s_5 - s_4 = (5^2 + 3(5) - 2) - (4^2 + 3(4) - 2)$
$\Rightarrow a_5 = 25 + 15 - 2 - 16 - 12 + 2 = 12$

b) $a_{12} = s_{12} - s_{11} \Rightarrow a_{12} = (144 + 36 - 2) - (121 + 33 - 2) = 26$

34. Find

a) $\sum_{n=1}^{30}(3n-1)$ b) $\sum_{n=1}^{50}(2n+5)$ c) $\sum_{n=1}^{60}(n+2)$

Solution: $S_n = \frac{n}{2}[A_1 + A_n]$

a) $S_{30} = \frac{30}{2}[A_1 + A_{30}] = 15[1 + 89] = 1350$

b) $S_{50} = \frac{50}{2}[A_1 + A_{50}] = 25[7 + 105] = 2800$

c) $S_{60} = \frac{60}{2}[A_1 + A_{60}] = 30[3 + 62] = 1950$

Finding Sum of arithmetic mean between a and b

Let $m_1, m_2, m_3, \ldots, m_n$ be n-arithmetic means between a and b

$\Rightarrow a, m_1, m_2, m_3, \ldots, m_n, b$

So that, $m_1 + m_2 + m_3 \ldots, + m_n = \frac{n}{2}[m_1 + m_n]$

- $m_1 = a + d$ and $b - m_n = d \Rightarrow m_n = b - d$

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$\therefore m_1 + m_2 + m_3 + \ldots + m_n = \frac{n}{2}[a+d+b-d] = \frac{n}{2}[a+b]$

Illustrative Example

35. Find the sum of:
a) 3 arithmetic mean between 3 and 4
b) 10 arithmetic mean between 7 and 13
c) n arithmetic mean between 3 and 4
d) n arithmetic mean between 4 and 6
e) n arithmetic mean between a and b

Solution:

a) $S_3 = \frac{3}{2}[3+4] = \frac{21}{2} = 10.5$

b) $S_{10} = \frac{10}{2}[7+13] = \frac{(10)(20)}{2} = 100$

c) $S_n = \frac{n}{2}[3+4] = \frac{7n}{2}$

d) $S_n = \frac{n}{2}[4+6] = 5n$

e) $S_n = \frac{n}{2}[a+b]$

36. When n-arithmetic means are inserted between 11 and 19, the sum of the resulting terms is 150. Find
a) the value of n b) common difference

Solution: $11 + m_1 + m_2 + m_3 + \ldots + m_n + 19 = 150$

$S_k = \frac{k}{2}[A_1 + A_k] = \frac{k}{2}[11+19] = 150 \Rightarrow k = 10$

$\Rightarrow$ n =(total number of term) – (two end point term)

$\Rightarrow n = k - 2 \Rightarrow n = 10 - 2 = 8$

b) $A_{10} = A_1 + 9d \Rightarrow 19 = 11 + 9d \Rightarrow 9d = 8 \Rightarrow d = \frac{8}{9}$

$\therefore$ common difference, $d = \frac{8}{9}$

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37. How many terms of the series may be taken their sum is $65 - 7 + -4 + -1 + 2 \ldots\ldots$

Solution: Let $S_n = 65$, $n = ?$

$$\Rightarrow S_n = \frac{n}{2}[2(-7)+(n-1)(3)] \Rightarrow 65 = \frac{n}{2}[3n-3-14]$$

$$\Rightarrow 3n^2 - 17n = 130 \Rightarrow 3n^2 - 17n - 130 = 0$$

$$\Rightarrow 3n^2 - 30n + 13n - 130 \Rightarrow 3n(n-10) + 13(n-10) = 0$$

$$\Rightarrow (3n+13)(n-10) = 0 \Rightarrow n = 10$$

38. A man saved birr 16,500 in ten years. In each year after the first he saved birr 100 more than he did in the preceding year. How much did he saved in the first year.

Solution: $S_{10} = 16{,}500$, $d = 100$

$$\Rightarrow S_{10} = \frac{10}{2}[2A_1 + 9d] \Rightarrow 16500 = 5[2A_1 + 900]$$

$$\Rightarrow 2A_1 + 900 = 3{,}300 \Rightarrow 2A_1 = 2{,}400$$

$\therefore$ $A_1 = 1200$ birr $\longleftarrow$ saving in the first year

39. A man saves birr 500 of his salary in certain year and in each subsequent year he will save birr 150 more than the previous year. How much will he save during the ninth year

Solution:

$A_1 = 500, d = 150 \Rightarrow A_9 = A_1 + 8d = 500 + 8(150) = 1700$

40. **Theater seating:** The seating section in a theater hall has total of 10 rows. The first row has 27 seats, the second row has 29 seats, and the third row has 31 seats, and so on.

a) How many seats are there in the tenth row.

b) Howmanyseats are there in the section(seating capacityofhall)

Solution: $A_1 = 27$, $d = 2$

a) $A_{10} = A_1 + (9d) = 27 + 9(2) = 45$ seats

b) $S_{10} = \frac{n}{2}[A_1 + A_{10}] = \frac{10}{2}(27+45) = 360$ total seats

41. **Contest prize:** A contest offers 15 prizes. The first prize is worth birr 5,000 and each successive prize is worth birr 250 less than the preceding prize. Find the value of the fifteenth prize and the total value of the prize

Solution: $A_1 = 5{,}000$ $d = -250$

$\Rightarrow A_{15} = A_1 + 14d \Rightarrow 5{,}000 + 14(-250) = 5000 - 3500 = 1500$

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$$\Rightarrow S_{15} = \frac{15}{2}[5000+1500] = 48{,}750$$

42. **Prize money:** A contest will have 5 prizes totaling birr 5000 and there will be a birr 100 difference between successive prizes. Find the first prize

Solution: $S_5 = 5000, \quad d = -100$

$$A_5 = A_1 + 4d = A_1 + 4(-100) = A_1 - 400$$

$$\Rightarrow S_5 = \frac{5}{2}[A_1 + A_5] = \frac{5}{2}[A_1 + A_1 - 400]$$

$$\Rightarrow 5000 = \frac{5}{2}[2A_1 - 400] \Rightarrow 2A_1 - 400 = 2000$$

$$\Rightarrow 2A_1 = 2400 \text{ and } A_1 = \frac{2400}{2} = 1200$$

43. Elon Academy offers a job applicants with choice of two salary increase plans.
With firm A, starting salary of birr 10,000 with increase of birr 1000 per year.
With firm B, offers starting salary of birr 12,000 with increase of birr 600 per year.
a) What would the annual salary be in the 9th year at firm A, at firm B?
b) Over the first 9 year, how much would be earned at firm A, at firm B?
c) Which firm should you choose?

Solution: a) The annual salary in both firm (plan) forms arithmetic sequence

With firm A(plan A): 10,000, 11,000, 12,000....

$\Rightarrow A_1 = 10{,}000, \quad d = 1000$

$\Rightarrow A_9 = A_1 + 8d \Rightarrow A_9 = 10{,}000 + 8(1000) = 18{,}000.$

With firm B, 12,000, 12,600, 13,200...

$\Rightarrow A_1 = 12{,}000, \quad d = 600$

$\Rightarrow A_9 = a_1 + 8d = 12{,}000 + 8(600) = 16{,}800$

b) over the first 9 year

with firm A: $S_9 = \frac{9}{2}(10{,}000+18{,}000) = 126{,}000$

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with firm B: $S_9 = \frac{9}{2}(12{,}000 + 16{,}800) = 129{,}600$

d) Firm B offers more than firm A by 3600.

10.3 Geometric sequence

✓ A geometric sequence is characterized by a common ratio (r) between successive term.

✓ Let $\{G_n\}$ be a geometric sequence if and only if.

$$\Rightarrow \frac{G_2}{G_1} = \frac{G_3}{G_2} = \frac{G_4}{G_3} = \ldots = \frac{G_{n+1}}{G_n} = r,\ r \in R\setminus\{0\}$$

$$\Rightarrow G_2 = G_1 r,\ G_3 = G_2 r = G_1 r^2,\ \boxed{G_n = G_1 r^{n-1}}$$

Illustrative Example

44. If $G_1 = 9$, $r = \frac{1}{3}$, then find

a) G_6, b) n^{th} term

Solution:

a) $G_6 = G_1 r^5 = (9)\left(\frac{1}{3}\right)^5 = \frac{1}{27}$

b) $G_n = G_1 r^{n-1} = (9)\left(\frac{1}{3}\right)^{n-1} = 9(3^{1-n})$

45. If $G_3 = 5$, and $G_6 = -40$

Find a) common ratio (r) b) G_1 c) G_8

Solution: $G_6 = G_1 r^5$ and $G_3 = G_1 r^2$

$$\Rightarrow \frac{G_6}{G_3} = \frac{G_1 r^5}{G_1 r^2} = \frac{-40}{5} \Rightarrow r^3 = -8 \Rightarrow r = -2 \leftarrow \text{common ratio}$$

b) $G_8 = G_1 r^7 = \left(\frac{5}{4}\right)(-2)^7 = -160$

46. Find the indicted term of geometric sequence

a) $G_3 = \frac{16}{3}$ *and* $G_5 = \frac{64}{27}$, Find G_1 and r

b) $G_4 = 3$ and $G_7 = 24$, find G_1, r and G_n

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c) $G_2 = 2+\sqrt{2}$ and $G_5 = 4\sqrt{2}+4$, Find r, G_1 and G_n

d) $G_4 = 3$ and $G_6 = 9$, Find r and G_1

e) $G_4 = 4$ and $G_7 = 12$, Find r and G_1 and G_{10}

Solution: a) $G_3 = G_1r^2$ and $G_5 = G_1r^4$

$$\Rightarrow \frac{G_5}{G_3} = \frac{G_1r^4}{G_1r^2} = \frac{64}{27}\cdot\frac{3}{16} \Rightarrow r^2 = \frac{4}{9} \Rightarrow r = \pm\frac{2}{3}$$

$$\Rightarrow G_3 = G_1r^2 \Rightarrow \frac{16}{3} = G_1\left(\frac{2}{3}\right)^2 \Rightarrow G_1 = \frac{16}{3}\cdot\frac{9}{4} = 12$$

b) $G_4 = G_1r^3$ and $G_7 = G_1r^6$

$$\Rightarrow \frac{G_7}{G_4} = \frac{G_1r^6}{G_1r^3} = \frac{24}{3} \Rightarrow r^3 = 8 \Rightarrow r = 2$$

$$\Rightarrow G_4 = G_1r^3 \Rightarrow 3 = G_1(2)^3 \Rightarrow G_1 = \frac{3}{8}$$

c) $G_2 = G_1r$ and $G_5 = G_1r^4$

$$\Rightarrow \frac{G_5}{G_2} = \frac{G_1r^4}{G_1r} \Rightarrow r^3 = \frac{4\sqrt{2}+4}{2+\sqrt{2}} = 2\sqrt{2} = \sqrt{8} \Rightarrow r = \sqrt{2}$$

$$\Rightarrow G_2 = G_1r \Rightarrow 2+\sqrt{2} = G_1(\sqrt{2}) \Rightarrow G_1 = \frac{2+\sqrt{2}}{\sqrt{2}} = \sqrt{2}+1$$

$$\Rightarrow G_{11} = G_1r^{10} = (\sqrt{2}+1)(\sqrt{2})^{10} = 32\sqrt{2}+32$$

d) $G_4 = G_1r^3$ and $G_6 = G_1r^5$

$$\Rightarrow \frac{G_6}{G_4} = \frac{G_1r^5}{G_1r^3} \Rightarrow r^2 = \frac{9}{3} = 3 \Rightarrow r = \pm\sqrt{3}$$

$$\Rightarrow G_4 = G_1r^3 \Rightarrow 3 = G_1\left(\pm\sqrt{3}\right)^3 \Rightarrow G_1 = \frac{3}{\pm\sqrt{27}} = \frac{3}{\pm3\sqrt{3}} = \frac{3}{\sqrt{3}}$$

e) $G_4 = G_1r^3$ and $G_7 = G_1r^6$

$$\Rightarrow \frac{G_7}{G_4} = \frac{G_1r^6}{G_1r^3} \Rightarrow r^3 = \frac{12}{4} = 3 \Rightarrow r = \sqrt[3]{3}$$

$$\Rightarrow G_4 = G_1r^3 \Rightarrow 4 = G_1(3) \Rightarrow G_1 = \frac{4}{3}$$

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$$\Rightarrow G_{10} = G_1 r^9 \Rightarrow G_{10} = \left(\frac{4}{3}\right)\left(\sqrt[3]{3}\right)^9 = 36$$

47. If $x+1, x+3$ and $x+7$ are consecutive terms of a G.P. Find the value of x.

Solution: $\frac{x+3}{x+1} = \frac{x+7}{x+3} \Rightarrow (x+3)^2 = (x+1)(x+7)$

$\Rightarrow x^2 + 6x + 9 = x^2 + 8x + 7 \Rightarrow 8x - 6x = 9 - 7$

$\Rightarrow 2x = 2 \Rightarrow x = 1$, G.P: 2, 4, 8.

48. Find the term of the sequence:

a) $4, 2, 1, \frac{1}{2}, \frac{1}{4}, \ldots$ which is less than $\frac{1}{1,024}$

b) 5, 50, 500, ... which exceeds 500,000.

Solution: a) $G_n = G_1 r^{n-1} \Rightarrow G_n = (4)\left(\frac{1}{2}\right)^{n-1} < 2^{-10}$

$\Rightarrow 2^{3-n} < 2^{-10} \Leftrightarrow 3 - n < -10 \Rightarrow -n < -13 \Rightarrow n > 13$

$\therefore$ The smallest value of $n = 14$

49. $G_n = G_1 r^{n-1} \geq 500,000 \Rightarrow (5)(10)^{n-1} \geq 500,000$

$\Rightarrow 10^{n-1} \geq 100,000 \Rightarrow 10^{n-1} \geq 10^5 \Leftrightarrow n - 1 \geq 5$

$\Rightarrow n \geq 6$ $\therefore$ The smallest value of $n = 6$.

50. **Growth of bacteria:** A certain culture initially contains 10,000 bacteria and increase by 20% every hour. Find the number of bacteria after

a) t hours b) 10 hours

Solution: The population growth at the end of n-hour is given by

$G_n = [1 + \text{rate}]^n G_1$

a) $G_t = G_1[1+0.2]^t = 10,000[1.2]^t$

b) $G_{10} = (10,000)[1.2]^{10} \approx 61,917$

51. **Population growth:** The population of a certain town is increasing at the rate of 5% per year. If the present number of population is 48,000. Find the population after

a) t year b) 4 year c) 10 year

Solution: $G_t = G_1(1 + \text{rate})^t$

a) $G_t = 48,000\,(1 + 0.05)^t = 48,000\,(1.05)^t$

b) $G_4 = (48,000)(1.05)^4 \approx 58,344.$

c) $G_{10} = (48,000)(1.05)^{10} \approx 78,186.$

52. Hana deposits birr 6000 in a bank account that earns an annual interest rate of 8%. Find the amount she has at end of

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a) the first year c) the 4th year
b) the second year d) the nth year

Solution: The amount at the end of n year is given by
$G_n = G_1[1+\text{rate}]^n$

a) At the first year $= 6{,}000(1 + 0.08) = 6480$
b) $G_2 = 6000(1+0.08)^2 = 6{,}998$
c) $G_4 = 6000(1+0.08)^4 = 6{,}000(1.08)^4 \approx 8{,}162$
d) $G_n = 6{,}000(1.08)^n$

53. If the value of **Lifan car** depreciated by 25% per year. If its present value is birr 240,000. What will be the value at the end of
a) 2 year b) 8 year c) nth year

Solution: The depreciation value at the end of t year is given by
$G_t = G_1(1 - \text{rate})^t$

a) $G_2 = 240{,}000\ (1 - 0.25)^2 = 240{,}000\ (0.75)^2 = 135{,}000$
b) $G_8 = 240{,}000\ (1 - 0.25)^8 = 240{,}000\ (0.75)^8 = 24{,}027$
c) $G_n = 240{,}000\ (0.75)^n$

54. A car that is bought for birr 120,000 depreciate in value by birr 8000 per year. How long will it take for the car to make lose of 20% of its value?

Solution: Let the cost of the car at the end of n year (A_n and G_n)
✓ Arithmetic depreciation is $A_n = 120{,}000 - 8000n$
✓ Geometric depreciation is $G_n = 120{,}000\ (1 - 0.2)$
$\Rightarrow A_n = G_n \Rightarrow 120{,}000 - 8000n = 96{,}000$
$\Rightarrow 120{,}000 - 96{,}000 = 8{,}000n \Rightarrow 24{,}000 = 8{,}000n \Rightarrow n = 3$
$\therefore$ It takes 3 year to make loss of 20% of its value.

Geometric Mean

If a and b are positive real number then a positive number c is called the **geometric mean** of a and b if a, c, b is geometric sequence.

$$\Rightarrow r = \frac{c}{a} = \frac{b}{c} \Rightarrow c^2 = ab \Rightarrow c = \sqrt{ab}$$

Product of n geometric mean between a and b is $\left(\sqrt{ab}\right)^n = (ab)^{\frac{n}{2}}$

55. Find the geometric mean for each pair:
a) 3 and 12 b) -3 and -27 c) $\sqrt{5}+\sqrt{2}$ *and* $\sqrt{5}-\sqrt{2}$

Solution: a) $G = \sqrt{(3)(12)} = 6$

b) $G = -\sqrt{(-3)(-27)} = -\sqrt{81} = -9$

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c) $G = \sqrt{(\sqrt{5}+\sqrt{2})(\sqrt{5}-\sqrt{2})} = \sqrt{5-2} = \sqrt{3}$

56. For two positive numbers **a** and **b** show that the arithmetic mean between **a** and **b** is greater than or equal to their geometric mean.

Solution: Arithmetic mean, $A = \frac{1}{2}(a+b)$

Geometric mean, $G = \sqrt{ab}$

To show $A \geq G \Rightarrow A - G \geq 0$

$$\Rightarrow \frac{1}{2}(a+b) - \sqrt{ab} \geq 0 \Rightarrow a+b-2\sqrt{ab} \geq 0$$

$$\Rightarrow \left(\sqrt{a}-\sqrt{b}\right)^2 \geq 0 \Rightarrow a+b-2\sqrt{ab} \geq 0$$

$$\Rightarrow \frac{a+b}{2} \geq \sqrt{ab} \Rightarrow A \geq G$$

57. Insert four arithmetic mean and 5 geometric mean between 3 and 9.

Solution: Let m_1, m_2, m_3, m_4 are four A.M

$\Rightarrow 3, m_1, m_2, m_3, m_4, 9$ are 6 A.P

Common diff. $d = \frac{A_6 - A_1}{5} = \frac{9-3}{5} = \frac{6}{5}$

$$\Rightarrow A.M = 3+\frac{6}{5}, 3+\frac{12}{5}, 3+\frac{18}{5}, 3+\frac{24}{5}$$

When 5 geometric means are inserted between 3 and 9, then

$G_7 = G_1 r^6$

$$\Rightarrow 9 = 3r^6 \Rightarrow r^6 = \frac{9}{3} = 3 \text{ and } r = \sqrt[6]{3}$$

$$\Rightarrow G.M = 3\sqrt[6]{3},\ 3\sqrt[3]{3},\ 3\sqrt{3}, 3\sqrt[3]{9},\ 3\sqrt[6]{3^5}$$

Note :

If A_1, A_2, A_3,... and A_n are n arithmetic mean between any two real numbers a and b and G_1, G_2, G_3... and G_n are 'n' geometric mean between a and b then:

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i) $\dfrac{A_1+A_2}{G_1G_2}=\dfrac{a+b}{ab}$ ⟶

Explanation
$A_1 - a = b - A_2$
$\Rightarrow A_1 + A_2 = a+b$
$\dfrac{G_1}{a}=\dfrac{b}{G_2}$
$\Rightarrow G_1G_2 = ab$

ii) $\dfrac{A_1+A_2+A_3}{G_1\cdot G_2\cdot G_3}=\dfrac{\frac{3}{2}(a+b)}{\sqrt{(ab)^3}}$

iii) $\dfrac{A_1+A_2+A_3+\ldots+A_n}{G_1\cdot G_2\cdot G_3\ldots G_n}=\dfrac{\frac{n}{2}(a+b)}{\sqrt{(ab)^n}}$

10.3.1 Sum of Geometric progression

The n^{th} partial sum S_n of a geometric sequence with first term G_1 and common ratio r is given by

$$S_n=\begin{cases} nG_1, & \text{if } r=1 \\ \dfrac{G_1(1-r^n)}{1-r}=\dfrac{G_1(r^n-1)}{r-1}, & \text{if } r\neq 1 \end{cases}$$

To proof: $S_n = G_1 + G_1r + G_1r^2 + \ldots + G_1r^{n-2} + G_1r^{n-1}$

$rs_n = rG_1 + G_1r^2 + G_1r^3 + \ldots + G_1r^{n-1} + G_1r^n$

Subtracting rS_n from S_n

$\Rightarrow S_n - rS_n = G_1 - G_1r^n \Rightarrow S_n(1-r) = G_1(1-r^n)$

$\Rightarrow S_n = \dfrac{G_1(1-r^n)}{1-r}=\dfrac{G_1(r^n-1)}{r-1}$

Illustrative Example

58. Find the indicated sum:

a) $G_1 = -2$, $r = 3$, Find S_6

b) $G_3 = 9$, and $G_5 = \dfrac{81}{4}$, Find r_1G_1 and S_4

c) $G_4 = 24$, and $G_7 = 192$, Find r_1G_1 and S_{10}

Solution: Here

a) $S_6=\dfrac{G_1(r^6-1)}{r-1}=\dfrac{-2(3^6-1)}{3-1}=-728$

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b) $G_3 = G_1r^2$ and $G_5 = G_1r^4 \Rightarrow \dfrac{G_5}{G_3} = \dfrac{G_1r^4}{G_1r^2} = \dfrac{81}{4(9)}$

$$\Rightarrow r^2 = \frac{9}{4} \Rightarrow r = \pm\frac{3}{2},\ Let\ r = \frac{3}{2}$$

$$G_3 = G_1r^2 \Rightarrow 9 = G_1\left(\frac{9}{4}\right) \Rightarrow G_1 = 4$$

$$\Rightarrow S_4 = \frac{G_1(r^4 - 1)}{r - 1} = 4\left[\frac{\left(\frac{3}{2}\right)^4 - 1)}{\frac{3}{2} - 1}\right] = \frac{65}{2}$$

c) $$\frac{G_7}{G_4} = \frac{G_1r^6}{G_1r^3} = \frac{192}{24} \Rightarrow r^3 = 8 \Rightarrow r = 2$$

$$G_4 = G_1r^3 \Rightarrow 24 = G_1(8) \Rightarrow G_1 = 3$$

$$S_{10} = \frac{G_1[r^{10} - 1]}{r - 1} = \frac{3(2^{10} - 1)}{2 - 1} = 3{,}069$$

59. Find the partial sum of each geometric sequence

a) $\sum_{n=1}^{4} 5\left(3^{n-1}\right)$ b) $\sum_{n=1}^{7} 64\left(\frac{-1}{2}\right)^{n-1}$ c) $\sum_{n=1}^{4} (2)^n 3$

Solution: a) $G_1 = 5, r = 3 \Rightarrow S_4 = \sum_{n=1}^{4} 5\left(3^{n-1}\right) = \dfrac{5[3^4 - 1]}{3 - 1} = 200$

b) $G_1 = 64, r = \dfrac{-1}{2} \Rightarrow S_7 = \dfrac{G_1[r^7 - 1]}{r - 1}$

$$= \frac{64\left(\frac{-1}{128} - 1\right)}{\frac{-1}{2} - 1} = 64\left(\frac{129}{128}\right)\frac{2}{3} = 43$$

c) $G_1 = 6, r = 2 \Rightarrow S_4 = \dfrac{6[2^4 - 1]}{2 - 1} = 6(15) = 90$

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10.4 Infinite Geometric Series

✓ An infinite series is the sum of an infinite sequence of numbers $\{a_n\}_{n=1}^{\infty}$ is denoted by $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \ldots + a_n + \ldots$

✓ If the sum of all terms of the sequence is unique finite number, then the series $\sum_{n=1}^{\infty} a_n$ is called **convergent**, other wise divergent.

✓ If the sequence is geometric sequence, we call **an infinite geometric series.**

✓ If $\{G_n\}_{n=1}^{\infty}$ is a geometric series with common ration r, then the n^{th} partial sum:

$$S_n = \frac{G_1(1-r^n)}{1-r} = \frac{G_1}{1-r} - \frac{G_1 r^n}{1-r}, r \neq 1$$

Convergence of Geometric series

* If $|r| < 1$, the geometric series:

$G_1 + G_1 r + G_1 r^2 + \ldots + G_1 r^n + \ldots$ converges to the sum $S = \frac{G_1}{1-r}$

$$\sum_{n=1}^{\infty} G_1 r^n = \frac{G_1}{1-r}, |r| < 1$$

* If $|r| \geq 1$, the series **diverges.**

Illustrative Example

60. Find the sum of the infinite geometric series

a) $2 + 1 + \frac{1}{2} + \ldots$

b) $2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \ldots$

c) $\frac{1}{3} - \frac{2}{9} + \frac{4}{27} - \frac{8}{81} + \ldots$

d) $3 + \sqrt{3} + 1 + \frac{1}{\sqrt{3}} + \ldots$

e) $2 + \frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \ldots$

f) $3 + 0.3 + 0.03 + \ldots$

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Solution: If $|r| < 1$ then sum $S_\infty = \dfrac{G_1}{1-r}$

a) $G_1=2, r=\dfrac{1}{2} \Rightarrow S_n = \dfrac{2\left[1-\left(\dfrac{1}{2}\right)^n\right]}{1-\dfrac{1}{2}} = 4\left[1-\left(\dfrac{1}{2}\right)^n\right] \leftarrow n^{th}$

partial sum AS $n \to \infty$, $S_\infty = \dfrac{G_1}{1-r} = \dfrac{2}{1-\dfrac{1}{2}} = 4$

a) $G_1=2, r=\dfrac{1}{3}, S_n = \dfrac{2\left[1-\left(\dfrac{1}{3}\right)^n\right]}{1-\dfrac{1}{3}} = 3\left[1-\left(\dfrac{1}{3}\right)^n\right] \leftarrow n^{th}$ partial sum

$\Rightarrow S_\infty = \dfrac{G_1}{1-r} = \dfrac{2}{1-\dfrac{1}{3}} = 3 \leftarrow$ (the sum of all terms of the sequence)

c) $G_1 = \dfrac{1}{3}, r = \dfrac{-2}{3} \Rightarrow S_\infty = \dfrac{G_1}{1-r} = \dfrac{\dfrac{1}{3}}{1+\dfrac{2}{3}} = \dfrac{1}{3}\cdot\dfrac{3}{5} = \dfrac{1}{5}$

d) $G_1 = 3, r = \dfrac{1}{\sqrt{3}} \Rightarrow S_\infty = \dfrac{G_1}{1-r} = \dfrac{3}{1-\dfrac{1}{\sqrt{3}}} = \dfrac{3\sqrt{3}}{\sqrt{3}-1}$

e) $2+\dfrac{1}{5}+\dfrac{1}{5^2}+\dfrac{1}{5^2}....\infty = 2+\sum_{n=1}^{\infty}\left(\dfrac{1}{5}\right)^n, G_1 = \dfrac{1}{5}, r = \dfrac{1}{5}$

$\Rightarrow S_\infty = 2+\dfrac{G_1}{1-r} = 2+\dfrac{\dfrac{1}{5}}{1-\dfrac{1}{5}} = 2+\dfrac{1}{4} = \dfrac{9}{4}$

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f) $G_1 = 3, r = \frac{1}{10} \Rightarrow S_\infty = \frac{G_1}{1-r} = \frac{3}{1-\frac{1}{10}} = \frac{3}{1-\frac{1}{10}} = \frac{10}{3}$

61. In each part, determine whether the series converges, and if so find its sum.

a) $\sum_{k=0}^{\infty}\left(\frac{5}{4^k}\right)$

b) $\sum_{n=1}^{\infty} 3\left(\frac{1}{2}\right)^{n-1}$

c) $\sum_{k=0}^{\infty}\left(\frac{3}{5}\right)^k$

d) $\sum_{n=1}^{\infty}\left(\frac{1}{9}\right)\left(\frac{1}{3}\right)^{n-1}$

e) $\sum_{k=1}^{\infty} 3^{k-2}$

f) $\sum_{n=1}^{\infty}(-1)^{n+1}\left(\frac{2}{3}\right)^n$

g) $\sum_{k=1}^{\infty}\frac{3^{k+1}}{4^{k-1}}$

h) $\sum_{k=1}^{\infty}\left(\frac{2}{3}\right)^{k-1}\left(\frac{3}{4}\right)^{k+1}$

Solution: a) $\sum_{k=0}^{\infty} 5\left(\frac{1}{4^k}\right)$ has $G_1 = 5, r = \frac{1}{4}$

$r < 1 \Rightarrow$ converges to $S_\infty = \frac{G_1}{1-r} = \frac{5}{1-\frac{1}{4}} = \frac{20}{3}$

b) $G_1 = 3, r = \frac{1}{2} \Rightarrow r < 1 \Rightarrow$ convergent.

The series converges to $S_\infty = \frac{G_1}{1-r} = \frac{3}{1-\frac{1}{2}} = 6$

c) $G_1 = 1, r = \frac{3}{5} \Rightarrow r < 1$, Converges to $S_\infty = \frac{1}{1-\frac{3}{5}} = \frac{5}{2}$

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d) $G_1 = \frac{1}{9}, r = \frac{1}{3} < 1$, the series, Converges to $S_\infty = \frac{\frac{1}{9}}{1-\frac{1}{3}} = \frac{1}{6}$

e) $\sum_{k=1}^{\infty} 3^{k-2} = \sum_{k=1}^{\infty} 3^k \cdot 3^{-2} = \sum_{k=1}^{\infty} 3^k \cdot \frac{1}{9} \Rightarrow G_1 = \frac{1}{9}, r = 3$

$\Rightarrow r > 1 \Rightarrow$ diverge

$\therefore$ the sum doesn't exist.

f) $\sum_{n=1}^{\infty} (-1)^{n+1} \cdot \left(\frac{2}{3}\right)^n = \frac{2}{3} - \frac{4}{9} + \frac{8}{27} - \frac{16}{81} + \cdots$

$G_1 = \frac{2}{3}, r = \frac{-2}{3} \Rightarrow \left|\frac{-2}{3}\right| < 1$

$\Rightarrow$ Converges to the sum, $S_\infty = \frac{G_1}{1-r} = \frac{\frac{2}{3}}{1+\frac{2}{3}} = \frac{2}{3} \cdot \frac{3}{5} = \frac{2}{5}$

g) $\sum_{k=1}^{\infty} \frac{3^{k+1}}{4^{k-1}} = \sum_{k=1}^{\infty} \frac{3^1 \cdot 3^k}{4^{-1} \cdot 4^k} = \sum_{k=1}^{\infty} 12\left(\frac{3}{4}\right)^k$

$G_1 = 9, r = \frac{3}{4} \Rightarrow \frac{3}{4} < 1$ $\therefore$ Convergent

The series converges to the sum, $\frac{G_1}{1-r} = \frac{9}{1-\frac{3}{4}} = 36$

h) $\sum_{k=1}^{\infty} \left(\frac{2}{3}\right)^{k-1} \cdot \left(\frac{3}{4}\right)^{k+1} = \sum_{k=1}^{\infty} \left(\frac{2}{3}\right)^k \cdot \left(\frac{3}{4}\right)^k \cdot \left(\frac{3}{2} \cdot \frac{3}{4}\right)$

$= \sum_{k=1}^{\infty} \frac{9}{8}\left(\frac{2}{3} \cdot \frac{3}{4}\right)^k$

$\Rightarrow \sum_{k=1}^{\infty} \frac{9}{8}\left(\frac{1}{2}\right)^k \Rightarrow G_1 = \frac{9}{16}, r = \frac{1}{2} < 1$ $\therefore$ Convergent

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$\Rightarrow$ The series converges to the sum, $\frac{G_1}{1-r}=\frac{\frac{9}{16}}{1-\frac{1}{2}}=\frac{9}{8}$

Find the sum

a) $\sum_{n=0}^{\infty}\left(\frac{2^n+5}{3^n}\right)$

$\sum_{n=0}^{\infty}\left(\frac{5}{2^n}-\frac{1}{3^n}\right)$

Solution: a) $\sum_{n=0}^{\infty}\left(\frac{2^n+5}{3^n}\right)=\sum_{n=0}^{\infty}\left(\frac{2^n}{3^n}\right)+\sum_{n=0}^{\infty}\left(\frac{5}{3^n}\right)$

$=\frac{1}{1-\frac{2}{3}}+\frac{5}{1-\frac{1}{3}}=3+\frac{15}{2}=\frac{21}{2}$

b) $\sum_{n=0}^{\infty}\left(\frac{5}{2^n}-\frac{1}{3^n}\right)=\sum_{n=0}^{\infty}\frac{5}{2^n}-\sum_{n=0}^{\infty}\left(\frac{1}{3^n}\right)=\frac{5}{1-\frac{1}{2}}-\frac{1}{1-\frac{1}{3}}=\frac{17}{2}$

63. Find the sum if it exist

a) $\sum_{k=0}^{\infty}\cos(\pi k)$ b) $\sum_{k=0}^{\infty}\frac{\cos(\pi k)}{3^k}$

Solution: $\sum_{k=0}^{\infty}\cos(\pi k)=1+(-1)+(1)+(-1)\ldots=\begin{cases}0\\1\end{cases}$

$$S_n=\begin{cases}0 \text{ if } n \text{ is even}\\1 \text{ if } n \text{ is odd}\end{cases}$$

Since it is not unique number, It is divergent

$\therefore$ The sum does not exist

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b) $\sum_{k=0}^{\infty}\frac{\cos(\pi k)}{3^k}=1-\frac{1}{3}+\frac{1}{9}-\frac{1}{27}+\ldots, G_1=1, r=-\frac{1}{3}$

$$\therefore S_\infty=\frac{G_1}{1-r}=\frac{1}{1-\left(\frac{-1}{3}\right)}=\frac{1}{1+\frac{1}{3}}=\frac{3}{4}$$

64. Find the sum of the series: $\frac{1}{2}+\frac{1}{3^2}+\frac{1}{2^3}+\frac{1}{3^4}+\frac{1}{2^5}+\ldots$

Solution: $\left(\frac{1}{2}+\frac{1}{2^3}+\frac{1}{2^5}+\ldots\right)+\left(\frac{1}{3^2}+\frac{1}{3^4}+\frac{1}{3^6}+\ldots\right)$

$$=\left[\left[\frac{\frac{1}{2}}{1-\frac{1}{4}}\right]\right]+\left[\frac{\frac{1}{9}}{1-\frac{1}{9}}\right]=\frac{2}{3}+\frac{1}{8}=\frac{19}{24}$$

A geometric series for a repeating decimal

65. Express the repeating decimal as a fraction using the infinite sum

a) $\frac{5}{10}+\frac{5}{100}+\frac{5}{1000}+\ldots$

b) $0.\dot{8}$

c) $0.2\dot{5}$

d) $24.\dot{3}$

Solution:

a) $G_1=\frac{5}{10}, r=\frac{1}{10}\Rightarrow S_\infty=\frac{G_1}{1-r}=\frac{\frac{5}{10}}{1-\frac{1}{10}}=\frac{5}{10}\cdot\frac{10}{9}=\frac{5}{9}$

b) $0.\dot{8}=0.\dot{8}+0.0\dot{8}+0.00\dot{8}+\ldots=\frac{8}{10}+\frac{8}{100}+\frac{8}{1000}+\ldots$

$$\Rightarrow G_1=\frac{8}{10}, r=\frac{1}{10},+\Rightarrow S_\infty=\frac{G_1}{1-r}=\frac{\frac{8}{10}}{1-\frac{1}{10}}=\frac{8}{10}\cdot\frac{10}{9}=\frac{8}{9}$$

c) $0.2\dot{5}=0.2+0.05+0.005+\ldots=0.2+\frac{5}{100}+\frac{5}{1000}+\ldots$

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$$\Rightarrow S_\infty = 0.2 + \left[\frac{\frac{5}{100}}{1-\frac{1}{10}}\right] = 0.2 + \frac{5}{100}\cdot\frac{10}{9} = \frac{2}{10} + \frac{5}{90} = \frac{23}{90}$$

d) $24.\dot{3} = 24 + 0.3 + 0.03 + 0.003 + \ldots \Rightarrow G_1 = 0.3, r = \frac{1}{10}$

$$\Rightarrow S_\infty = 24 + \left[\frac{\frac{3}{10}}{1-\frac{1}{10}}\right] = 24 + \frac{3}{10}\cdot\frac{10}{9} = 24 + \frac{1}{3} = 24\frac{1}{3}$$

66. Evaluate in each part of the infinite series
a) $0.2 + 0.22 + 0.222 + 0.2222 + \ldots$
b) $0.3 + 0.33 + 0.333 + 0.3333 + \ldots$

Solution:

a) $0.2 + 0.22 + 0.222 + 0.2222 + \ldots = \frac{2}{10} + \frac{22}{100} + \frac{222}{1000} + \ldots$

$$\Rightarrow 2\left[\frac{1}{10} + \frac{11}{100} + \frac{111}{1000} + \ldots\right]$$

$$\Rightarrow \frac{2}{9}\left[\frac{9}{10} + \frac{99}{100} + \frac{999}{1000} + \ldots\right]$$

$$\Rightarrow \frac{2}{9}\left[\frac{10-1}{10} + \frac{10^2-1}{100} + \frac{10^3-1}{1000} + \ldots \frac{10^n-1}{10^n}\right]$$

$$\Rightarrow \frac{2}{9}\left[1-\frac{1}{10} + 1-\frac{1}{10^2} + 1-1\frac{1}{10^3} + \ldots 1-\frac{1}{10^n}\right]$$

$$\Rightarrow \frac{2}{9}\left[(1+1+1+\ldots+1) - \left(\frac{1}{10} + \frac{1}{10^2} + \frac{1}{10^3} + \frac{1}{10^n}\right)\right]$$

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$$\Rightarrow \frac{2}{9}\left[n-\left(\frac{\frac{1}{10}\left(1-\frac{1}{10}\right)^n}{1-\frac{1}{10}}\right)\right]=\frac{2}{9}\left[n-\frac{1}{9}\left(1-\left(\frac{1}{10}\right)^n\right)\right]$$

$$\Rightarrow S_n=\frac{2}{9}n-\frac{2}{81}\left(1-\frac{1}{10^n}\right)$$

b) Exercise left for you.

$$S_n=\frac{3}{9}n-\frac{3}{81}\left(1-\frac{1}{10^n}\right)=\frac{1}{3}n-\frac{1}{27}\left(1-\frac{1}{10^n}\right)$$

Geometric series for product of exponential form

67. Find the product

a) $2^1\cdot 2^{\frac{1}{3}}\cdot 2^{\frac{1}{9}}\cdot 2^{\frac{1}{27}}\ldots$ b) $3^x\cdot 3^{x^2}\cdot 3^{x^3}\ldots 3^{x^n}$ *for* $|x|<1$

Solution: a) $2^1\cdot 2^{\frac{1}{3}}\cdot 2^{\frac{1}{9}}\cdot 2^{\frac{1}{27}}\ldots=2^{1+\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\frac{1}{3n}}$

$$=2^{\frac{G_1}{1-r}}=2^{\frac{1}{1-\frac{1}{3}}}=2^{\frac{3}{2}}=\sqrt{2^3}=2\sqrt{2}$$

b) $3^x\cdot 3^{x^2}\cdot 3^{x^3}\ldots=3^{x+x^2+x^3\ldots+x^n}=3^{\frac{G_1}{1-r}}=3^{\frac{x}{1-x}}$

68. Find the value of k;

a) $4^k\cdot 4^{k^2}\cdot 4^{k^3}\ldots.=16$ b) $e^1\cdot e^k\cdot e^{k^2}\cdot e^{k^3}\ldots.=e^3$

Solution: a) $4^k\cdot 4^{k^2}\cdot 4^{k^3}\ldots.=4^{k+k^2+k^{3+\ldots}}=4^2$

$\Leftrightarrow k+k^2+k^3+\cdots=2\Rightarrow G_1=k, r=k$

Since series convergent

$$\Rightarrow \frac{G_1}{1-r}\Rightarrow\frac{k}{1-k}=2\Rightarrow k=2-2k\Rightarrow 3k=2\Rightarrow k=\frac{2}{3}$$

b) $e^1\cdot e^k\cdot e^{k^2}\cdot e^{k^3}\ldots.=e^3\Rightarrow e^{1+k+k^2+k^{3+\ldots}}=e^3$

$\Leftrightarrow 1+k+k^2+k^3+\cdots=3\Rightarrow G_1=1, r=k$

$$\Rightarrow \frac{1}{1-k}=3\Rightarrow 1=3-3k\Rightarrow -3k=-2\Rightarrow k=\frac{2}{3}$$

69. Find the value of m: if

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a) $\sum_{k=0}^{\infty} e^{mk} = 9$ b) $\sum_{k=1}^{\infty} 6^{mk} = \frac{1}{5}$

Solution: Here

a) $\sum_{k=0}^{\infty} e^{mk} = 9 \Rightarrow 1 + e^{m} + e^{2m} + e^{3m} + \ldots = 9$

the series converges to the sum 9. $G_1 = 1, r = e^{m}$

$$\Rightarrow \frac{G_1}{1-r} \Rightarrow \frac{1}{1-e^{m}} = 9 \Rightarrow 1 = 9 - 9e^{m} \Rightarrow -9e^{m} = -8$$

$$\Rightarrow e^{m} = \frac{8}{9} \Rightarrow m = \log_e^{\frac{8}{9}} = \ln\frac{8}{9} = \ln 8 - \ell n 9$$

$$\therefore \quad m = \ln\frac{8}{9} = \ln\frac{8}{9} = \ln 8 - \ln$$

b) $\sum_{k=1}^{\infty} 6^{mk} = 6^{m} + 6^{2m} + 6^{3m} + \ldots = \frac{1}{5} \Rightarrow G_1 = 6^{m}, r = 6^{m}$

$$\Rightarrow S_{\infty} = \frac{G_1}{1-r}$$

$$\Rightarrow \frac{6^{m}}{1-r} = \frac{1}{5} \Rightarrow (5)6^{m} = 1 - 6^{m} \Leftrightarrow 5(6)^{m} + 6^{m} = 1$$

$$\Rightarrow 6^{m}(5+1) = 1 \Rightarrow 6^{m} = \frac{1}{6} \Leftrightarrow 6^{m} = 6^{-1} \Leftrightarrow m = -1$$

70. Solve a) $\sum_{n=1}^{\infty} x^{n} = \frac{3}{8}$ b) $\sum_{n=0}^{\infty} y^{n} = x$

Solution: a) $\sum_{n=1}^{\infty} x^{2n-1} = x + x^{3} + x^{5} + \ldots = \frac{3}{8}, G_1 = x, r = x^{2}$

$$\Rightarrow \frac{G_1}{1-r} = \frac{x}{1-x^{2}} \Rightarrow \frac{3}{8} \Rightarrow 8x = 3 - 3x^{2} \Rightarrow 3x^{2} + 8x - 3 = 0$$

$$\Rightarrow x = \frac{1}{3}, x = -3 \text{ we take only } x = \frac{1}{3}$$

b) $\sum_{n=1}^{\infty} y^{n} = 1 + y + y^{2} + \ldots = x, G_1 = 1, r = y$

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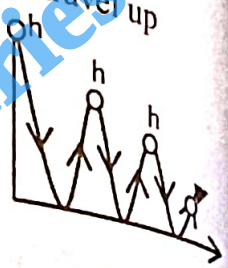
$$\Rightarrow S_\infty = \frac{G_1}{1-r} = x = \frac{1}{1-y} \Leftrightarrow x - xy = 1$$

$$\Rightarrow x - 1 = xy \Leftrightarrow y = \frac{x-1}{x}, \text{ for } x \neq 0$$

A bouncing Ball problem

A ball dropped from a height of **h meter**. In each bounce, it rebounds r of the height from which it falls. Find the total distance the ball travel up and down.



Solution:

- ✓ **Down ward series:** $h+hr+hr^2 + hr^3 + \ldots + hr^{n-1} + \ldots$
- ✓ **Upward series:** $hr + hr^2 + hr^3 + \ldots + hr^n + \ldots$

Total distance is adding down ward and upward series:

$$\Rightarrow S = h + 2hr + 2hr^2 + 2hr^3 + \ldots + 2hr^n + \ldots$$

$$S = h + \frac{2hr}{1-r} = h\left[\frac{1+r}{1-r}\right] \longleftarrow \text{the total vertical distance}$$

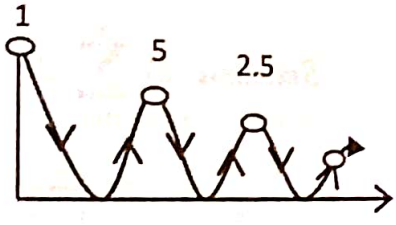
Illustrative Example

71. A ball is dropped from height of 16 meter. Each time it drops h meter, it rebounds **0.81h** meters. Find the total vertical distance traveled by the ball.

Solution: Total distance traveled $= h\left[\frac{1+r}{1-r}\right]$, $h = 16, r = 0.81$

$$\Rightarrow S = 16\left[\frac{1+0.81}{1-0.81}\right] = 16\left(\frac{1.81}{0.19}\right) = 152.42m$$

72. A ball is dropped from height of 10 meter. It rebounds one-half the distance after each fall.



a) How far does it bound up on the seventh rebounds.

b) Find the total distance the ball has traveled when it hits the ground the sixth time.

b) Find the total (maximum) distance the ball travels before comes to rest.

Solution: a) $G_n = Gr^{n-1}$, $= hr^{n-1}$, $h = 10$, $r = \frac{1}{2}$

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$$\Rightarrow G_5 = (10)\left(\frac{1}{2}\right)^{5-1} = (10)\left(\frac{1}{2}\right)^4 = \frac{10}{16} = \frac{5}{8}m$$

a) Total distance the ball traveled when hits the ground the sixth time $\Rightarrow S_6 = h + 2hr + 2hr^2 + 2hr^3 + 2hr^4 + 2hr^5$

$$= 10 + 2(10)\left(\frac{1}{2}\right) + 2(10)\left(\frac{1}{4}\right)2(10)\left(\frac{1}{8}\right) + 2(10)\left(\frac{1}{16}\right) + 2(10)\left(\frac{1}{32}\right)$$

$$= 10 + 10 + 5 + \frac{10}{4} + \frac{10}{8} + \frac{10}{16} + \frac{10}{32} \approx 29m$$

c) Total distance before comes to rest is

$$\Rightarrow S = h\left[\frac{1+r}{1-r}\right] = 10\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right) = 10(3) = 30m$$

73. A ball is dropped from height of 5 meter the ball rebounds 60% of the distance after each fall. Find the total distance

Solution: h = 5m, r = 60% = 0.6

$$\text{Total distance } S = 5\left[\frac{1+0.6}{1-0.6}\right] = \frac{5(1.6}{0.4} = 20m$$

74. Let $a_n = n + \cos n\pi$, n = 1, 2, 3, … be the n^{th} term of sequence. Then which of the following is true…

a) $a_5 = 6$ b) $a_{10} = 9$ c) $a_7 + a_8 = 15$ d) $a_7 - a_6 = 1$

Solution: a) $a_5 = 5 + \cos(5\pi) = 5 + (-1) = 4$

b) $a_{10} = 10 + \cos(10\pi) = 10 + 1 = 11$

c) $a_7 + a_8 = 7 + \cos(7\pi) + (8 + \cos 8\pi) = 7 + (-1) + 8 + (1) = 15$

d) $a_7 - a_6 = 7 + (-1) - (6 + 1) = -1$ **Answer: C**

75. If $\{A_n\}$ is an arithmetic sequence with $A_1 = 5$ and $A_5 = 21$, then the partial sum $\sum_{n=1}^{20} A_n$ is equal to ……

A) 760 B) 780 C) 860 D) 870

Solution: $d = \frac{A_5 - A_1}{4} = \frac{21-5}{4} = 4$

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$$\sum_{n=1}^{n} A_n = S_n = \frac{n}{2}(2A_1 + (n-1)d \Rightarrow S_{20} = \frac{20}{2}(2(5)+19(4)) = 860$$

76. What is the sum of the series $\sum_{n=0}^{\infty} 2^n 3^{2-n}$? .....

A) 6 B) 18 C) 27 D) ∞

Solution:

$$\sum_{n=0}^{\infty} 2^n 3^{2-n} = \sum_{n=0}^{\infty} 2^n \cdot 3^2 \cdot 3^{-n} = \sum_{n=0}^{\infty} 9\left(\frac{2}{3}\right)^n, G_1 = 9,\ r = \frac{2}{3}$$

$$\Rightarrow S_\infty = \frac{G_1}{1-r} = \frac{9}{1-\frac{2}{3}} = 9 \times 3 = 27$$

Answer: C

77. Find the sum of all integers between 30 and 201 which are
a) multiple of 4 b) multiple of 7

Solution: $S_n = \frac{n}{2}[A_1 + A_n]$ and $n = \frac{A_n - A_1}{d} + 1$

a) $n = \frac{200-32}{4} + 1 = 32$ and $S_{43} = \frac{43}{2}[32 + 200] = 4644$

b) $n = \frac{196-35}{7} + 1 = 24$ and $S_{24} = \frac{24}{2}[35 + 196] = 2772$

78. A ball is thrown vertically from the ground up to a height of 20 meter. Each time it drops h meters, it rebounds $\frac{3}{4}$ of the height. What is the total vertical distance travelled by the ball before it comes to rest?

A) 80m B) 160m C) 120m D) 60m

Solution: $G_1 = h = 20m,\ r = \frac{3}{4}$

$$\Rightarrow S_\infty = \frac{2G}{1-r} = \frac{2h}{1-r} = \frac{2(20)}{1-\frac{3}{4}} = 160m$$

79. Suppose $\{a_n\}$ is an A.P such that $a_5 = 21$ and $a_{10} = 66$ then a_2 is equal to:

A) -6 B) -15 C) 3 D) 9

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Solution: $d = \frac{a_{10} - a_5}{10-5} = \frac{66-21}{5} = \frac{45}{5} = 9$

Also $\frac{a_5 - a_2}{5-2} = 9 \Rightarrow \frac{21-a_2}{3} = 9 \Rightarrow 21 - a_1 = 27 \Rightarrow a_2 = -6$

80. $\sum_{n=1}^{\infty} \frac{(-1)^n}{(3^n)^2}$ is equal to

A) $\frac{9}{8}$ B) $\frac{9}{10}$ C) $\frac{-9}{10}$ D) $\frac{-1}{10}$

Solution: $\sum_{n=1}^{\infty} \frac{(-1)^n}{(3^n)^2} = \frac{-1}{3^2} + \frac{1}{3^4} + \frac{-1}{3^6} + \ldots \quad G_1 = \frac{-1}{9},\ r = -\frac{1}{9}$

$$\Rightarrow S_\infty = \frac{G_1}{1-r} = \frac{-\frac{1}{9}}{1+\frac{1}{9}} = \frac{-1}{9} \cdot \frac{9}{10} = -\frac{1}{10}$$

Answer: D

81. Let $S_n = 4n^2$, n = 1, 2, 3, ... be the sum of the first n terms a sequence $\{a_n\}_{n=1}^{\infty}$. Then a_3

A) 8 B) 12 C) 16 D) 20

Solution:

$a_n = S_n - S_{n-1} \Rightarrow a_3 = S_3 - S_2 = 4(3)^2 - 4(2)^2 = 36 - 16 = 20$

82. Let a sequence $\{a_n\}_{n\geq 1}$ with $a_1 = 2$ and $a_n = 3a_{n-1}$ for all $n \geq 2$.

Then $\sum_{i=1}^{20} a_i$ is equal to

A) $3^{20} - 1$ B) $1 - \frac{1}{3^{20}}$ C) $1 - \frac{1}{2^{21}}$ D) $3^{-20} - 1$

Solution: $a_1 = 2;\ a_2 = 3a_1 = 3(2) = 2,\ a_3 = 3a_2 = 3(6) = 18$

$\Rightarrow$ 2, 6, 18, 54 .... $\Rightarrow G_1 = 2,\ r = 3$. It is G.P

$$\Rightarrow \sum_{i=1}^{20} a_i \Rightarrow S_{20} = G_1\left[\frac{r^n - 1}{r-1}\right] = 2\left[\frac{3^{20}-1}{3-1}\right] = 3^{20} - 1$$

Answer: A

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83. Let $\{a_n\}_{n\geq 1}^{\infty}$ be sequence with $a_1 = 3$ and $a_{n+1} = na_n$ then $a_n =$ ___

A) $3n!$ B) $3(n-1)!$ C) $(3n)!$ D) $3(n+1)!$

Solution: $a_{n+1} = na_n \Rightarrow a_{1+1} = 1a_1 = 3,\ a_2 = 3(1)$

$\Rightarrow n = 2 \Rightarrow a_{2+1} = 2a_2 = 2(3) = 6 \Rightarrow a_3 = 6 = 3 \times 2 \times 1 = 3(2 \times 1)$

$\Rightarrow n = 3 \Rightarrow a_{3+1} = 3a_3 = 3(6) = 18$

$\Rightarrow a_4 = 18 = 3(3 \times 2 \times 1) = 3(3 \times 2 \times 1)$

$\Rightarrow a_5 = 72 = 3(4 \times 3 \times 2 \times 1) \Rightarrow a_n = 3(n-1)!\ n \geq 1$

84. Let $\{a_n\}$ be A.P 20, 16, 12, 8, ... Then $a_{200} - a_{203}$ **Answer: B**

A) -12 B) 8 C) 12 D) 16

Solution: $d = \dfrac{a_m - a_k}{m-k} \Rightarrow \dfrac{a_{200} - a_{203}}{200-203} = -4$

$\Rightarrow \dfrac{a_{200} - a_{203}}{-3} = -4 \Rightarrow a_{200} - a_{203} = 12$ **Answer: C**

Review Exercise 10

1. If $A_5 = 8$ and $d = 3$, then find:
   a) A_1 b) A_{20} c) nth term (A_n) d) A_{n+1}
2. If $A_4 = 11$ and $A_8 = 13$ then find
   a) Common difference b) A_1 c) nth term
3. Find the total number of natural number between 100 and 1000 divisible by six.
4. Let $\{a_n\}$ be an arithmetic sequence: 28, 25, 22, . . . Then find
   a) $a_{307} - a_{309}$ b) $a_{200} - a_{204}$
5. If $G_1 = 8$, $r = \dfrac{1}{2}$, then find
   a) G_6 b) G_8 c) G_n
6. If $G_4 = 3$ and $G_7 = 81$ then find:
   a) Common ratio (r) b) G_1 c) G_n
7. If $x + 1$, $-5x + 4$, and $2x + 8$ are consecutive G.P, find the value of x.
8. If $A_6 = 23$ and $A_{15} = 50$, then find:
   a) S_{10} b) S_{20} c) $\sum_{n=1}^{24} A_n$ d) $\sum_{n=1}^{12} A_n$
9. Find the indicated term of an A. P:
   a) If $S_8 = 96$, $A_1 = 5$, then find A_8, A_4, and S_{10}
   b) $S_6 = 60$, $A_8 = 19$, then find A_1, A_{10} and A_n
10. If $S_4 = 40$ and $S_{10} = 160$, then find:

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a) d b) n^{th} term c) $\sum_{n=1}^{n} A_n$ d) $\sum_{n=1}^{20} A_n$

11. Find the nth partial sum (S_n) for the given nth term arithmetic sequences.
a) $A_n = 3n + 1$ b) $A_n = 2n + 3$ c) $a_n = 2 - 5n$

12. Find the nth term (A_n) and common difference for the given nth partial sum (S_n) of arithmetic sequence.
a) $S_n = 3n^2 + 2n$ b) $S_n = \frac{1}{2}(Sn^2 - n)$

13. a) If $A_4 = 11, A_7 = 20$ of an A. P then find $\sum_{n=1}^{20} A_n$
b) If $A_5 = 4, A_{12} = 18$ of an A.P then find $\sum_{n=1}^{30} A_n$

14. Find the sum of all number from 100 to 300.
a) Which are exactly divisible by 8.
b) Which are exactly divisible by 7.

15. a) Find the number of integers between 32 and 395 that are divisible by 6.
b) Find the number of negative integers greater than -500 that are divisible by 33.

16. Find the sum
a) $\sum_{k=1}^{20}(3k-5)$ b) $\sum_{k=10}^{28}(2k+3)$ c) $\sum_{k=4}^{18}(4k-1)$
d) $\sum_{n=1}^{99}\log\frac{n+1}{n}$ e) $\sum_{k=1}^{100}\frac{1}{(k+1)(k+2)}$ f) $\sum_{k=0}^{69}x^3$

17. Find the sum of all 3 digit whole numbers that are divisible by:
a) 11 b) 13 c) 24 d) 31

18. Find the sum of 16 terms of the A.P -10, -8, -16, -4, . . . starting from the 10^{th} term.

19. If $G_4 = 24$, and $G_7 = 192$, then find the sum fo the first indicated terms.
a) S_6 b) S_8 c) S_{10} d) S_n

20. Find the partial sum of each geometric sequence.

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a) $\sum_{k=1}^{7} 3(2^{k-1})$ b) $\sum_{k=2}^{8} 32\left(\frac{1}{3}\right)^{k}$ c) $\sum_{k=1}^{10}\left(\frac{3}{2}\right)^{k}$

21. If $G_4 = 4$, and $G_6 = 16$, then find $(r > 0)$.

a) r b) S_6 c) $\sum_{n=1}^{n} G_n$

22. Find the sum of infinite geometric series.

a) $1-\frac{1}{2}+\frac{1}{4}-\frac{1}{8}+\ldots$ c) $\frac{1}{5}-\frac{2}{S^2}+\frac{2^2}{S^4}-\frac{2^3}{S^6}+\ldots$

b) $2+\frac{2}{3}+\frac{2}{9}+\frac{2}{27}+\ldots$ d) $\frac{1}{2}+\frac{1}{3}+\frac{1}{8}+\frac{1}{9}+\frac{1}{32}+\frac{1}{27}+\ldots$

23. Find the sum if exists

a) $\sum_{n=0}^{\infty}(0.7)^n$ b) $\sum_{n=2}^{\infty}(0.7)^n$ c) $\sum_{n=0}^{\infty}(-0.6)^n$

d) $\sum_{n=1}^{\infty} 20(0.9)^{n-1}$ e) $\sum_{n=1}^{\infty} 10(0.25)^{n-1}$

f) $\sum_{n=1}^{\infty} 5\left(-\frac{1}{3}\right)^{n-1}$ g) $\sum_{k=2}^{\infty}(5^{4-k})$ h) $\sum_{k=3}^{\infty}\left(-\frac{3}{2}\right)^{1-k}$

i) $\sum_{k=1}^{\infty} 6\left(\frac{4}{3}\right)^{2-k}$ j) $\sum_{k=1}^{\infty}\frac{3^{k+2}}{4^{k-1}}$ k) $\sum_{k=1}^{\infty}(5^k)(7^{1-2k})$

l) $\sum_{k=0}^{\infty}\left(\frac{2}{3^{k-1}}-\frac{3}{4^k}\right)$

24. Find the product of each of the following expression.

a) $3.3^{\frac{1}{3}}.3^{\frac{1}{9}}.3^{\frac{1}{27}}\ldots$ b) $4.4^{\frac{1}{5}}.4^{\frac{1}{25}}.4^{\frac{1}{125}}\ldots$

25. Suppose: $6^k.6^{k^2}.6^{k^3}\ldots = 36$, then find the value of k.

26. Find the value (s) of k if

a) $\sum_{m=1}^{\infty} 4^{mk} = \frac{1}{15}$ b) $\sum_{m=0}^{\infty} 3^{mk} = \frac{27}{26}$ c) $\sum_{m=0}^{\infty}\left(\frac{3}{2}\right)^{mk} = \frac{9}{5}$

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Answer to Review Exercise 10

a. $A_1 = -4$ b. $A_{20} = A_1 + 19d = -4 + 57 = 53$

c. $A_n = 3n - 7$ d. $A_{n+1} = 3n - 4$

4. a. $d = \frac{A_8 - A_4}{8-4} = \frac{13-11}{4} = \frac{1}{2}$ b. $A_1 = \frac{19}{2}$ c. $A_n = \frac{1}{2}n + 9$

5. $\{102, 108, \ldots 996\} \Rightarrow A_n = A_1 + (n-1)6$

$\Rightarrow 996 = 102 + 6n - 6 \Rightarrow n = 150$

6. a. $d = \frac{a_{307} - a_{309}}{307 - 309} \Rightarrow a_{307} - a_{309} = 6$

b. $d = \frac{a_{200} - a_{204}}{200 - 204} = -3 \Rightarrow a_{200} - a_{204} = 12$

7. a. $G_6 = \frac{1}{4}$ b. $G_8 = (8)\left(\frac{1}{27}\right) = \frac{1}{16}$

c. $G_n = 8\left(\frac{1}{2}\right)^{n-1} = 16\left(\frac{1}{2}\right)^n$

8. a. $r = 3$ b. $G_1 = \frac{1}{9}$ c. $G_n = G_1 r^{n-1}$

9. $\frac{-5x+4}{x+1} = \frac{2x+8}{-5x+4} \Rightarrow (-5x+4)^2 = (x+1)(2x+8) \Rightarrow x = 2.$

10. a. $S_{10} = \frac{10}{2}[A_1 + A_{10}] = 5\,(8 + 35) = 215$

b. $S_{20} = \frac{20}{2}[A_1 + A_{20}] = 730$

c. $\sum_{n=1}^{24}(5+3n) = 12\,(8 + 77) = 1020$

d. $\sum_{n=1}^{12}(5+3n) = 6\,(8 + 4) = 294$

11. a. $A_8 = 19$, $A_4 = 11$ and $S_{10} = S\,[2A_1 + 9d] = S(28) = 140$

b. $A_1 = 5$, $A_{10} = 23$ and $A_n = 2n + 3$

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10. a. $d = 2$ b. $A_n = 2n + 5$ c. $\sum_{n=1}^{n} 2n + 5 = n^2 + 6n$

d. $S_{20} = 520$

11. a. $S_n = \frac{n}{2}[4 + 3n + 1] = \frac{1}{2}(3n^2 + 5n)$ b. $S_n = n^2 + 4n$

c. $S_n = \frac{1}{2}(5n^2 + n)$

12. a. $A_n = S_n - S_{n-1} = 6n - 1$ b. $A_n = S_n - S_{n-1} = Sn - 3$

13. a. $S_{20} = 10\,[A_1 + A_{20}] = 10\,[2 + 59] = 610$

b. $S_{30} = \frac{30}{2}[A_1 + a_{30}] = 750$

14. a. $S_{25} = \frac{25}{2}[104 + 296] = 5{,}000$ b. $S_{28} = \frac{28}{2}[105 + 294] = 5516$

15. a. $S_{60} = 12{,}780$ b. $S_{15} = \frac{15}{2}[-395 + -33] = -3{,}120$

16. a. $S_{20} = 530$ b. $S_{19} = \frac{19}{2}[23 + 59] = 779$ c. $S_{15} = 645$

d. $S_{99} = \log 100 = 10$ e. $S_{100} = \frac{25}{51}$ f. $S_{70} = 70x^3$

17. a. $S_{79} = 79 \times 979$ b. $S_{69} = \frac{69}{2}[104 + 988]$ c. $S_{37} = 20{,}424$

18. $S_{16} = 8\,(2(8) + 30) = 368$

19. a. $S_6 = 195$ b. $S_8 = 381$ c. $S_{10} = 3069$

d. $3[2^n - 1]$

20. a. $S_7 = 381$ b. $S_7 = \frac{16}{3}[1 - 3^{-7}]$ c. $3\left[\left(\frac{3}{2}\right)^{10} - 1\right]$

21. a. $r = 2$ b. $S_6 = \frac{65}{2}$ c. $S_n = \frac{1}{2}(2^n - 1)$

22. a. $\frac{2}{3}$ b. 3 c. $\frac{1}{7}$ d. $\frac{3}{2}$

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23. a. $\frac{10}{3}$ b. $\frac{49}{30}$ c. $\frac{5}{8}$ d. 200

e. $\frac{40}{3}$ f. $\frac{15}{4}$ g. $\frac{125}{4}$ h. $\frac{4}{3}$

i. 32 j. 108 k. $\frac{35}{44}$ l. 5

24. a. $3^{1+\frac{1}{3}+\frac{1}{9}+\cdots} = 3^{\frac{1}{1-\frac{1}{3}}} = 3\sqrt{3}$ b. $4^{\frac{1}{1-\frac{1}{5}}} = \sqrt[4]{1024}$

25. $k = \frac{2}{3}$

26. a. $k = -2$ b. $k = -3$ c. $k = -2$

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UNIT 11

11. Introduction to Calcaulus

Concepts to Master in this units:

(i) Rate of change:
- Average rate of change
- Instantaneous rate of change

(ii) Slope of the tangent line to the graph of a function.

(iii) Definition of Derivative.
- Interpretations of the derivative.

(iv) Basic derivative Rule.
- The product and Quotient Rules.
- The chain rule.

(v) Derivative Application
- Increasing and Decreasing function.
- Maximizing and problem Minimization problem.

Introduction To Calculus

- Understand that the tangent line problem is basic to calculus.
- Understand that the area problem is also basic to calculus.

What is Calculus?

Calculus is the Mathematics of change. If is used to find the rate of change of variable such as, velocity, acceleration, slope of curve, tangent lines, and areas. There are two branches of calculus.

(i) **Differential calculus** introduced with the tangent line problem.
- Differential calculus is the study of derivative.

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(ii) Integral calculus introduced with the area problem.

Differential calculus and Integral calculus relate the fundamental theorem of calculus.

What is a Derivative ?

There are three equally important answer:

- Derivative as a rate of change.
- Derivative as the instantaneous rate of change.
- Derivative as the slope of tangent line

Now here, we explore all three facts of the derivative, basic rule of differentiation, extreme values of a function and related rate of a real life application.

Derivative is a powerful mathematical tool that is used to calculate rate of change and slopes of curves.

What is a Rate of Change

Slope, velocity, acceleration, and electric potential are some examples of **rate of change**

- Slope: $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$ ⟵ Rate of change of y with respect to x
- Velocity: $V = \frac{\Delta S}{\Delta t} = \frac{S_2 - S_1}{t_2 - t_1}$ ⟵ Rate of change of S with respect to t
- Acceleration : $a = \frac{\Delta V}{\Delta t} = \frac{V_2 - V_1}{t_2 - t_1}$
- Electric potential : $V = \frac{\text{Work done}}{\text{charge moved}}$

A Rate of change is a rate that describes how one quantity changes with respects to another quantity.

The two kinds of rate of change are average rate of change and instantaneous rate of change.

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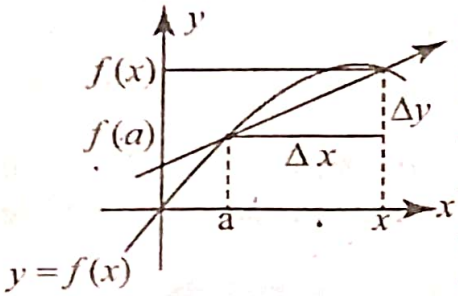
11.1 Average Rate of Change

DEFINITION: The average rate of change of $f(x)$ with respect to x for a function f as x changes from **a** to **b** is given by

$$\frac{f(b) - f(a)}{b - a}$$



is the **average rate of change** of $y = f(x)$ on the interval [a, b].

The average rate of change of $y = f(x)$ on the interval [a, b] is the **slope of secant line** through the point secant line through the point (a, f(a)) and (x, f(b)).

Geometrically:

Average rate of change = slope of secant line = $\frac{f(b) - f(a)}{b - a}$

Illustrative Example

1. Let $f(x) = x^2 + 5x$

   a. Find the average rate of change of f over [2, 4]

   b. Find the slope of secant line over the interval [2, 4]

   c. Find the average rate of change rate of change of f over [a, b]

   Solution

   a. $V_{av} = \frac{f(4) - f(2)}{4 - 2} = \frac{(16 + 20) - (4 + 10)}{2} = \frac{36 - 14}{2} = 11$

   b. $m_{sec} = \frac{f(4) - f(2)}{4 - 2} = \frac{36 - 14}{2} = 11$

   c. $r_{av} = \frac{f(b) - f(a)}{b - a} = \frac{(b^2 + 5b) - (a^2 + 5a)}{b - a}$

   $= \frac{b^2 - a^2}{b - a} + \frac{5(b - a)}{b - a} = b + a + 5$

2. Let $f(x) = x^3 - 3x$. then

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a. find the average rate change f over the interval [2 5].

b. Find the slope of secant line through the point $(-1, f(-1))$ and $(2, f(2))$.

Solution

a. Average rate of change:

$$\frac{f(5) - f(2)}{5 - 2} = \frac{(5^3 - 3(5)) - (2^3 - 3(2))}{3} = 27$$

b. Slope of secant line:

$$\frac{f(2) - f(-1)}{5 - 2} = \frac{(8 - 6) - (-1 + 3)}{3} = \frac{2 - 2}{3} = 0$$

11.2 Instantaneous Rate Change

If we interpret the expression

$$\frac{f(a + h) - f(a)}{h}$$

as the average rate of change in f over smaller and smaller interval from a to a + h, as h closer and closer to zero is the **instantaneous rate of change**

DEFINITION:

The **instantaneous rate of change** for a function f when $y = f(x)$ at $x = a$ is $\frac{f(a + h) - f(a)}{h}$ as h closer to 0.

Illustrative Example

3. Find the instantaneous rate of change of $f(x) = x^2 + 4x$ at $x = 3$

Solution

$$\frac{f(a + h) - f(a)}{h}, \text{ as } h \longrightarrow 0$$

At a = 3: $\frac{f(3 + h) - f(3)}{h}$

1. $f(3 + h) = (3 + h)^2 + 4(3 + h) = h^2 + 10h + 21$

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2. $f(3+h) - f(3) = h^2 + 10h + 21 - 21 = h^2 + 10h$

3. $\dfrac{f(3+h) - f(3)}{h} = \dfrac{h^2 + 10h}{h} = h + 10$, as $h \longrightarrow 0$

$\therefore$ The instantaneous rate of change is 10.

4. A ball dropped from a height of 40meter. The ball's height h at time t is the position function $h(t) = -16t^2 + 40$

   a. Find the average velocity over the intervals form 1 second to 2 second , [1, 2]

   b. Find the instantaneous velocity at t = 1 second.

Solution

a. $V_{av} = \dfrac{h(2) - h(1)}{2 - 1} = \dfrac{-24 - 24}{1} = -48 \text{m/sec}$

b. $r_{inst} = \dfrac{\Delta h}{\Delta t} = \dfrac{h(t) - h(1)}{t - 1} = \dfrac{(-16t^2 + 40) - (24)}{t - 1}$

$= \dfrac{-16t^2 + 16}{t - 1} = \dfrac{16(1 - t^2)}{t - 1} = \dfrac{16(1 - t)(1 + t)}{t - 1}$

$r_{inst} = -16(1 + t)$, as $t \longrightarrow 1$

$\therefore \quad r_{inst} = -16(1 + 1) = -32 \text{m/sec}$

5. Let $f(x) = 6x^2 - 4x + 1$.

   a. find the average rate of change over the interval [1 4]

   b. Find the instantaneous rate of change of $y = f(x)$ at $x = 1$

Solution

a. $\dfrac{f(4) - f(1)}{4 - 1} = \dfrac{81 - 3}{3} = 26$

b. $\dfrac{f(x) - f(1)}{x - 1} = \dfrac{6x^2 - 4x + 1 - 3}{x - 1} = \dfrac{6x^2 - 4x - 2}{x - 1}$

$= \dfrac{(6x + 2)(x - 1)}{x - 1} = 6x + 2, \; x \neq 1$

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$= 6x + 2 = 6 + 2 = 8$, as x close to 1

$\therefore$ The instantaneous rate of change is 8

The slope (Gradient) of tangent line

Gradient (Slope): is a number that describes steepness and diction of a line

(i) Positive slope is increasing line from left right (rise up)

(ii) Negative slope is decreasing line, from left to right (moves down)

(iii) Zero slope is horizontal line.

The slope of tangent line is the instantaneous rate of change of $y = f(x)$ with respect to x when $x = x_0$

DEFINITION: The **slope of tangent line** of the graph $y = f(x)$ at the point $(x_0, f(x_0))$ is $m_{tan} = \dfrac{f(x) - f(x_0)}{x - x_0}$ as x closer and closer to x_0

Equivalently: we can write as

$$m_{tan} = \frac{f(x_0 + h) - f(x_0)}{h}$$ as h closer and closer to zero

Note: The slope of the tangent line to the graph of f at the point $(x_0, f(x_0))$ is also called the slope of the graph of f at $x = x_0$

- The slope of tangent line = The instantaneous rate of change.

6. Let $f(x) = x^2 + 3x$. Then find:
   a. the average rate of change of $y = f(x)$ w.r.t x over $[1, 3]$
   b. the slope of secant line passing through P (-1, -2) and Q (2, 10)
   c. the instantaneous rate of change of y, w.r.t x, at $x = 1$
   d. the slope of graph of $y = f(x)$ at the point, $x = 1$.

Solution

a. $r_{av} = \dfrac{\Delta y}{\Delta x} = \dfrac{f(3) - f(1)}{3 - 1} = \dfrac{18 - 4}{3 - 1} = \dfrac{14}{2} = 7$

b. $m_{sec} = \dfrac{\Delta y}{\Delta x} = \dfrac{f(2) - f(-1)}{2 - (-1)} = \dfrac{10 - (-2)}{3} = \dfrac{12}{3} = 4$

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c. $r_{inst} = as \quad x \to 1, \dfrac{f(x) - f(1)}{x - 1} = as\ x \to 1, \left(\dfrac{x^2 + 3x - 4}{x - 1}\right)$

$= as \quad x \to 1, \dfrac{(x - 1)(x + 4)}{x - 1}$

$= as \quad x \to 1, x + 4 = 1 + 4 = 5$

d. slope of a curve at $x = 1$ is $1 + 4 = 5$

Illustrative Example

7. Let $f(x) = \sqrt{x}$

a. Find the gradient of $y = f(x)$ at $x = x_0$

b. Find the slope of tangent line at $x_0 = 9$

Solution

a. $m_{tan} = \dfrac{f(x_0 + h) - f(x_0)}{h} = \dfrac{\sqrt{x_0 + h} - \sqrt{x_0}}{h}$

Rationalizing the numerator

$m_{tan} = \dfrac{\sqrt{x_0 + h} - \sqrt{x_0}}{h} = \dfrac{\left(\sqrt{x_0 + h} + \sqrt{x_0}\right)}{\left(\sqrt{x_0 + h} + \sqrt{x_0}\right)}$

$= \dfrac{x_0 + h - x_0}{h\left(\sqrt{x_0 + h} + \sqrt{x_0}\right)} = \dfrac{h}{\left(\sqrt{x_0 + h} + \sqrt{x_0}\right)} = \dfrac{1}{\left(\sqrt{x_0 + h} + \sqrt{x_0}\right)}$

$= \dfrac{1}{2\sqrt{x_0}}$, as h close to 0

b. Slope of tangent line at $x_0 = 9$ *is* $m_{tan} = \dfrac{1}{2\sqrt{9}} = \dfrac{1}{6}$

8. Find the slope of tangent line to the curve $f(x) = \dfrac{1}{x - 1}$ at the point where $x = 2$

Solution

we use $\dfrac{f(x + h) - f(x)}{h} = \dfrac{f(2 + h) - f(2)}{h}$

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$f(2+h) - f(2) = \frac{1}{2+h-1} - \frac{1}{2-1} = \frac{1}{h+1} - \frac{1}{1} = \frac{-h}{h+1}$

Now we determine : $\frac{f(2+h) - f(2)}{h} = \frac{-h}{h(h+1)} = \frac{-1}{h+1}$

Thus, as h close to 0, $\frac{-1}{h+1} \longrightarrow \frac{-1}{0+1} = -1$

∴ The slope of tangent line is -1

9. **Critical thinking:** Where does the curve $f(x) = \frac{1}{x}$ have slope - 4?

Solution

The slope of the curve is the slope of tangent line at at $x = a$:

$\frac{f(x) - f(a)}{x - a}$, as x closer and closer to a

$\frac{\frac{1}{x} - \frac{1}{a}}{x - a} = \frac{a - x}{(xa)(x - a)} = \frac{-1}{xa}$

As x close to a, $\frac{-1}{xa} = \frac{-1}{a^2}$,since the slope is - 4,

$\Rightarrow \frac{-1}{a^2} = -4$, this implies a $= \pm\sqrt{\frac{1}{4}} = \pm\frac{1}{2}$

Therefore, $f(x) = \frac{1}{x}$ has slope - 4 at the point $\left(\frac{1}{2}, 2\right)$ and $\left(\frac{-1}{2}, -2\right)$

10. Let $f(x) = 4x - x^2$

a. Find the average rate of change of over the interval [3 5]

b. Find the instantaneous rate of change of f at $x = 3$

c. Find the slope of f at $x = 3$.

Solution

a. $\frac{f(5) - f(3)}{5 - 3} = \frac{(20 - 25) - (12 - 9)}{5 - 3} = \frac{-8}{2} = -4$

∴ The average rate of change is - 4

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b. $\frac{f(x) - f(3)}{x - 3} = \frac{(4x - x^2)}{x - 3} = \frac{-x^2 + 4x - 3}{x - 3}$

$= \frac{-(x - 3)(x - 1)}{x - 3} = -x + 1$

$-x + 1 = -3 + 1 = -2$, as x close to 3.

$\therefore$ The instantaneous rate of change of f at $x = 3$ is -2

c. The slope of graph of f at $x = 3$ is the instantaneous rate change o

f at $x = 3$ will be $-x + 1\big|_{x=3} = -3 + 1 = -2$

11. Find the gradient of the tangent line to the graph of the following

a. $f(x) = x^2 - 3$ at point $(4, f(4))$ b. $f(x) = x^3$, at point $(a, f(a))$

Solution

Gradient meant the slope, so that

a. Gradient of $f = \frac{f(x) - f(4)}{x - 4}$, as x close to 4

$= \frac{(x^2 - 3) - (13)}{x - 4} = \frac{x^2 - 16}{x - 4} = \frac{(x + 4)(x - 4)}{x - 4} = x + 4$

$\therefore$ The gradient of tangent line at $x = 4$ will be $x + 4\big|_{x=4} = 4 + 4 = 8$

b. $\frac{f(x) - f(a)}{x - a} = \frac{x^3 - a^3}{x - a} = \frac{(x - a)(x^2 + ax + a^2)}{x - a} = x^2 + ax + a^2$

$\therefore$ The gradient of $f(x) = x^3$ at $x = a$ is $a^2 + a^2 + a^2 = 3a^2$

11.3 The Derivative

What derivatives measures?

Derivatives measures:

- The rate at which function changes.
- The instantaneous rate of changes.
- The slope of a curve at a point
- Howe fast $f(x)$ is change per unit change of x.

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How to find the derivative of a function?

Derivative of a function can defined as interms of slope of tangent line to the function.

11.3.1 Derivative of Function at a Point

Suppose that a is a number in the dominative of a function f.

The derivative of f at $x = a$ denoted by $f'(a)$ is the **slope of the tangent line to the graph of** $y = f(x)$ at the point $(a, f(a))$.

DEFINITION: Let a is number in the domain of f.

The derivative of $y = f(x)$ at a point a is

$$f'(a) = \frac{f(a+h) - f(a)}{h} \text{ as close to } 0$$

An equivalent way to write this is

$$f'(a) = \frac{f(x) - f(a)}{x - a}, \text{ as } x \text{ closer and closer to a}$$

Illustrative Example

12. Find the derivative of $f(x) = x^2 - 6x$ at $x = 1$

Solution

$$f'(1) = \frac{f(x) - f(1)}{x - 1}, as\ x \longrightarrow 1$$

$$= \frac{x^2 - 6x - (1 - 6)}{x - 1} = \frac{x^2 - 6x + 5}{x - 1} = \frac{(x - 1)(x + 5)}{x - 1}$$

$$= x + 5, \text{ as } x \text{ close to } 1$$

$$= 1 + 5 = 6$$

$$\therefore \quad f'(1) = 6$$

13. Find the derivative of $f(x) = x^3 + 4x$, at $x = -1$

Solution

$$f'(-1) = \frac{f(x) - f(-1)}{x + 1} \text{ as } x \text{ close to } -1$$

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$$= \frac{(x^3 + 4x) - (-1 - 4)}{x + 1} = \frac{x^3 + 4x + 5}{x + 1}$$

$$= \frac{(x + 1)(x^2 - x + 5)}{x + 1} = x^2 - x + 5, \text{ as } x \to -1$$

$$\therefore \quad f'(-1) = (-1)^2 + 1 + 5 = 7$$

14. Let $f(x) = \sqrt{4x + 1}$. Then find

a. $f'(a)$ b. $f'(2)$

Solution

a. $f'(a) = \frac{f(a + h) - f(a)}{h}$ as h close to 0.

$$= \frac{\sqrt{4a + 4h + 1} - \sqrt{4a + 1}}{h}$$ rationalizing the numberator

$$= \frac{\sqrt{4a + 4h + 1} - \sqrt{4a + 1}}{h} \cdot \frac{\sqrt{4a + 4h + 1} + \sqrt{4a + 1}}{\sqrt{4a + 4h + 1} + \sqrt{4a + 1}}$$

$$= \frac{4a + 4h + 1 - (4a + 1)}{h\left(\sqrt{4a + 4h + 1} + \sqrt{4a + 1}\right)} = \frac{4}{\sqrt{4a + 4h + 1} + \sqrt{4a + 1}}$$

As h close to 0

$$f'(a) = \frac{4}{2\sqrt{4a + 1}} = \frac{2}{\sqrt{4a + 1}}$$

b. $f'(2) = \frac{2}{\sqrt{4(2) + 1}} = \frac{2}{\sqrt{9}} = \frac{2}{3}$

15. Let $f(x) = x^3 + 2x + 1$

Find the slope of the tangent line to f at the point a = 2.

Solution

$$f'(2) = \frac{f(x) - f(2)}{x - 2} \text{ as } x \text{ close to } 2.$$

$$= \frac{x^3 + 2x + 1 - (8 + 4 + 1)}{x - 2} = \frac{(x - 2)(x^2 + 2x + 6)}{x - 2}$$

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$= x^2 + 2x + 6$, as x close to 2

$\therefore \quad f'(2) = 4 + 4 + 6 = 14$

Interpretaion :

- $f'(2)$ is the slope of tangent line to the graph of f at $(2, f(2))$.
- $f'(2)$ is the instantaneous rate of change of f at $x = 2$.

11.3.2 Derivative as A Function

We have seen that the derivative of a function f at a fixed number a:

$$f'(a) = \frac{f(a+h) - f(a)}{h}, \text{ as } h \longrightarrow 0$$

Here, we change the flexed number a, replaced by the variable x. This leads us tot eh following definition.

DEFINITION: The **derivative** of $f(x)$ is the function $f'(x)$ given by

$$f'(x) = \frac{f(x+h) - f(x)}{h}, \text{ as } h \text{ close to } 0.$$

Equivalently, $f'(x) = \dfrac{f(x_0) - f(x)}{x_0 - x}$ as x_0 close to x

is called the derivative of f with respect t x.

- The domain of f' is the set of points in the domain of f for which the slope of tangent line exists. If x is a value in the domain of f and if $f'(x)$ exists, we say that f has a derivative (or **differentiable**) as x.

The process of finding a derivative of a function is called **differentiation.**

Illustrative Example

16. Let $f(x) = \sqrt{x - 3}$, then find

a. $\dfrac{f(x+h) - f(x)}{h}$, as h approaches 0. b. $f'(4)$

c. $f'(3)$ d. the domain of f'

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Solution

a. $\frac{f(x+h)-f(x)}{h}$, as h approaches 0.

$$\frac{\sqrt{x+h-3}-\sqrt{x-3}}{h}$$

$$\left(\frac{\sqrt{x+h-3}-\sqrt{x-3}}{h}\right)\cdot\left(\frac{\sqrt{x+h-3}+\sqrt{x-3}}{\sqrt{x+h-3}+\sqrt{x-3}}\right)$$

$$\frac{x+h-3-(x-3)}{h\left(\sqrt{x+h-3}+\sqrt{x-3}\right.} = \frac{h}{h\left(\sqrt{x+h}+\sqrt{x-3}\right.}$$

$$\frac{1}{\sqrt{x+h-3}+\sqrt{x-3}} = \frac{1}{\sqrt{x+0-3}+\sqrt{x-3}}$$

$$= \frac{1}{2\sqrt{x-3}} \text{ as h cose to 0}$$

b. $f'(x) = \frac{1}{2\sqrt{x-3}}$, then $f'(4) = \frac{1}{2\sqrt{4-3}} = \frac{1}{2}$

c. f (3) does not have a derivative why?

Because the graph of f has no tangent line at $x = 3$.

d. The domain of $f' = \{x: x > 3\} = (3, \infty)$

17. Let $f(x) = \frac{2}{x^2}$, then

a. Find $f'(x)$

b. Find $f'(1)$

c. Where does the curve $y = f(x)$ have slope - 8

a. Using the definition of derivative:

$$f'(x) = \frac{f(x+h)-f(x)}{h}, \text{ as h approches 0.}$$

$$\frac{1}{h}\left(\frac{2}{(x+h)^2}-\frac{2}{x^2}\right) = \frac{-2(2x+h)}{x^2(x+h)^2}$$

$$\therefore \quad f'(x) = \frac{-4}{x^3}, \text{ as h closer to 0.}$$

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b. $f'(1) = \frac{-4}{1^3} = -4$ as h closer to 0.

c. we will find a such that $f'(a) = -8$ and $\frac{-4}{a^3} = -8, a^3 = \frac{1}{2}, a = \frac{1}{\sqrt[3]{2}}$

18. **Skill Test :** Determine whether each of the following is True or false

a. $f'(x)$ measures the average rate of change of $y = f(x)$ with respect to x

b. $f'(a) = \frac{f(h) - f(x)}{h}$ as h close to zero.

c. $f'(x) = \frac{f(x + \Delta x) - f(x)}{\Delta x}$ as Δx close to zero.

Answer:

a. False, $f'(x)$ measure the instantaneous rate of change

b. False

c. True

19. The value of $\frac{\sqrt{9 + h} - 3}{h}$ as h approaches zero is the derivative of:

A. $f(x) = \sqrt{9 + x}$ at $x = 3$

B. $f(x) = \sqrt{x}$ at $x = 3$

C. $f(x) = \sqrt{x}$ at $x = 9$

D. $f(x) = \frac{\sqrt{9 + h} - 3}{h}$ at $x = 0$

Solution

$$f'(x) = \frac{f(x + h) - f(x)}{h}, \text{ as h close to 0}$$

$$f'(9) = \frac{f(9 + h) - f(9)}{h} = \frac{\sqrt{9 + h} - 3}{h}$$

Answer: C

11.3.3 Basic Derivative Rule

As you have seen, finding the derivative a function can be time consuming and complex. There are rules that will make the process easier.

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1. Constant Rule: Derivative of a constant.

- The derivative of constant function is zero.

If $f(x) = C$, then $f'(x) = 0$

Example: $f(x) = 6$, then $f'(x) = 0$

$f(x) = \pi$, then $f'(x) = 0$

2. Power Rule:

If $f(x) = x^n$, where n is a real number, then $f'(x) = nx^{n-1}$

Example

$f(x) = x^4$, then $f'(x) = 4x^3$

$f(x) = x^{2/3}$, then $f'(x) = \frac{2}{3}x^{\frac{2}{3}-1} = \frac{2}{3}x^{\frac{-1}{3}}$

$f(x) = \sqrt{x}$, then $f'(x) = \frac{1}{2}x^{\frac{-1}{3}} = \frac{1}{2}x^{\frac{-1}{2}} = \frac{1}{2\sqrt{x}}$

$f(x) = x^{-10}$, then $f'(x) = -10x^{-10-1} = -10x^{-11}$

3. Constant multiple of a power function

If k and ∩ are rational numbers, then

$f(x) = k\,x^n$ then $f'(x) = nkx^{n-1}$

Example

$f(x) = 3x^{-2}$, then $f'(x) = 3(-2)x^{-2-1} = -6x^{-3} = \frac{-6}{x^3}$

$f(x) = 4x^{\frac{1}{3}}$, then $f'(x) = (4)\left(\frac{1}{3}\right)x^{\frac{1}{3}-1} = \frac{4}{3}x^{\frac{-2}{3}} = \frac{4}{3\sqrt[3]{x^2}}$

4. Sum and Difference Rule

If $h(x) = f(x) + g(x)$, then $h'(x) = f'(x) + g'(x)$

If $h(x) = f(x) - g(x)$, then $h'(x) = f'(x) - g'(x)$

Example:

If $f(x) = x^3 - 4x^2 + 6x - 7$, then

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$f'(x) = 3x^2 - 8x + 6$

Illustrative Example

20. Find the derivative of each of the following function with respect to x

a. $f(x) = x^{e-1}$ b. $f(x) = x^{3-\pi}$ c. $f(x) = \frac{2}{x^3}$

Solution

a. $f(x) = x^{e-1}$, then $f'(x) = (e-1)\ x^{e-2}$

b. $f(x) = x^{3-\pi}$, then $f'(x) = (3-\pi)\ x^{2-\pi}$

c. $f(x) = \frac{2}{x^3} = 2x^{-3}$, then $f'(x) = -6x^{-4} = \frac{-6}{x^4}$

21. Let $f(x) = x^{-3} + \frac{1}{x^7}$, the find $f'(x)$

$$f'(x) = -3x^{-4} + -7x^{-8} = \frac{-3}{x^4} + \frac{-7}{x^8} = \frac{-3x^4 - 7}{x^8}$$

22. Let $f(x) = \sqrt{x} + \frac{1}{x}$, then find $f'(x)$

Solution

$$f(x) = \sqrt{x} + \frac{1}{x} = x^{\frac{1}{2}} + x^{-1}$$

$$f'(x) = \frac{1}{2}x^{\frac{1}{2}-1} + (-1)\ x^{-1-1} = \frac{1}{2}x^{\frac{-1}{2}} - x^{-2}$$

$$= \frac{1}{2\sqrt{x}} - \frac{1}{x^2}$$

23. Let $f(x) = \frac{1}{2\sqrt[3]{x^2}}$, then find $f'(x)$

Solution

We use: $x^n = n\,x^{n-1}$ where $n \in Q$

$$f(x) = \frac{1}{2\sqrt[3]{x^2}}, = \frac{1}{2\ x^{\frac{2}{3}}} = \frac{1}{2}x^{\frac{-2}{3}} \text{ then}$$

$$f'(x) = \left(\frac{-2}{3}\right)\left(\frac{1}{2}\right)x^{\frac{-2}{3}-1} = -\frac{1}{3}x^{\frac{-5}{3}} = \frac{-1}{3\sqrt[3]{x^5}}$$

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24. What is the slope of the tangent line through the point $(2, f(2))$ if $f(x) = x^3 + 1$

Solution

$f'(x) = 3x^2$, so that $f'(2) = 3(2)^2 = 12$

The slope of tangent line is = 12

25. Suppose $f'(1) = 3$, and $g'(1) = 5$, then find $(3f + 2g)'(1)$

Solution: $(3f + 2g)'(1) = 3f'(1) + 2g'(1) = 3(3) + 2(5) = 19$

26. If $f(x) = x^3$, then the value of $\frac{f(2+h) - f(2)}{h}$ as h approach 0 is equal to

A. 4 B. -4 C. 12 D. ∞

Solution

By definition and basic rule of derivative:

$$f'(2) = \frac{f(2+h) - f(2)}{h}, \text{ as } h \longrightarrow 0$$

If $f(x) = x^n$, then $f'(x) = nx^{n-1}$

If $f(x) = x^3$, then $f'(x) = 3x^2$

$\therefore \quad f'(2) = 3(2)^2 = 12$

Answer: C

27. Find a if $f(x) = 4x - x^3$ and $f'(a) = -8$

Solution

$f'(x) = 4 - 3x^2 \Rightarrow f'(a) = 4 - 3a^2 = -8$

$-3a^2 = -12, \; a^2 = 4$

$\therefore \quad a = 2$ or $a = -2$

Interpretations of the Derivative

The derivative of f at $x = a$ is $f'(a)$ the same as:

- The slope of the tangent line to the graph of f at $x = a$
- The instantaneous rate of change of $y = f(x)$ with respect to x at $x = a$.

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Note
If $f'(a)$ exists, the tangent line to the graph of f at a is the line through $(a, f(a))$ with slope $f'(a)$. If $f'(a)$ does not exist, then the tangent line might not exist, might be a vertical line or might not be unique.

Illustrative Example

28. Find all points in the domain of f at which $f'(x)$ does not exist.

a. $f(x) = |x|$

b. $f(x) = \sqrt{x}$

c. $f(x) = x^{2/3}$

d. $f(x) = |x + 3|$

e. $f(x) = \sqrt{2x - 1}$

f. $f(x) = (x - 1)^{2/3}$

Solution

a. $f(x) = |x|$, here the domain of f is $(-\infty, \infty)$

$$f'(x) = \frac{f(x) - f(0)}{x - 0} = \frac{|x|}{x} = \pm 1$$

The derivative of f at $x = 0$ s not unique, so that $f'(x)$ does not exist as $x = 0$.

b. $f(x) = \sqrt{x}$, domain of f is $[0, \infty)$

$f'(x) = \frac{1}{2\sqrt{x}}$ and $f'(x)$ does not exist at $x = 0$

c. $f(x) = x^{2/3}$, domain of f is $(-\infty, \infty)$

$f'(x) = \frac{2}{3}x^{\frac{-1}{3}} = \frac{2}{3\sqrt[3]{x}}$ and $f'(x)$ does not exist at $x = 0$

d. $f(x) = |x + 3|$, domiain of is $(-\infty, \infty)$

$$f'(x) = \frac{f(x) - f(-3)}{x + 3} = \frac{|x + 3|}{x + 3} = \pm 1, \text{ not unique}$$

$\therefore$ $f'(x)$ does not exist at $x = -3$

e. $f(x) = \sqrt{2x - 1}$, domain of is $\left[\frac{1}{2}, \infty\right)$. $f'(x) = \frac{1}{\sqrt{2x - 1}}$,

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and $f'(x)$ does not exist at $x = \frac{1}{2}$

f. $f(x) = (x-1)^{-\frac{1}{3}}$, domain of f is $(-\infty, \infty)$ $f'(x) = \frac{2}{3}(x-3)^{\frac{-1}{3}} = \frac{2}{\sqrt[3]{x-1}}$

$\therefore$ $f'(x)$ dose not exist at $x = 1$

11.3.4 Derivative of Product and Quotients of Function

While the derivative of the sum of two functions of the their derivatives, the derivative of the product of two functions is not the product of their derivative.

Product Rule: If f and g are differentiable functions, then f . g is differentiable and $(f\,g)'(x) = f'(x)\,g(x) + f(x)\,g'(x)$

Quotitent Rule : $\left(\frac{f}{g}\right)'(x) = \frac{g(x)\,f'(x) - g'(x)\,f(x)}{[g(x)]^2}$

Note : $\left(\frac{1}{g(x)}\right)' = \frac{g(x)\,(1') - g'(x)\cdot 1}{[g(x)]^2} = \frac{0 - g'(x)}{[g(x)]^2} = \frac{-g'(x)}{[g(x)]^2}$.

Illustrative Example

29. If $f(x) = \sqrt{4x-3}$, $g(x) = x^2 + 6$ and $h(x) = (f\,g)(x)$, then find $h'(x)$.

Solution

$f(x) = \sqrt{4x-3}$, $\quad f'(x) = \frac{4}{2\sqrt{4x-3}} = \frac{2}{\sqrt{4x-3}}$

$g(x) = x^2 + 6$ and $g'(x) = 2x$

Using product property of derivative So that

$h'(x) = (f\,g)'(x) = f'(x)\,g(x) + f(x)\,g'(x)$

$h'(x) = \frac{2}{\sqrt{4x-3}}, (x^2 + 6) + \sqrt{4x-3}\cdot 2x$

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$$= \frac{2x^2 + 12}{\sqrt{4x - 3}} + 2x \cdot \frac{\sqrt{4x - 3} \cdot \sqrt{4x - 3}}{\sqrt{4x - 3}}$$

$$\therefore \quad h'(x) = \frac{2x^2 + 12 + 2x(4x - 3)}{\sqrt{4x - 3}} = \frac{10x^2 - 6x + 12}{\sqrt{4x - 3}}$$

30. Suppose $f(-3) = -2$, $f'(-3) = 4$, $g(-3) = 6$ and $g'(-3) = 7$

a. If $h(x) = (fg)(x)$, then find the value of $h'(-3)$.

b. If $H(x) = \left(\frac{f}{g}\right)(x)$, then find the value of $H'(-3)$.

Solution

a. $h'(x) = (fg)'(x) = f'(x)\, g(x) + f(x)\, g'(x)$

$h'(-3) = (fg)'(-3) = f'(-3)\, g(-3) + f(-3)\, g'(-3)$

$h'(-3) = (4)(6) + (-2)(7) = 10$

b. $H'(x) = \left(\frac{f}{g}\right)'(x) = \frac{f'(x)\, g(x) - f(x)\, g'(x)}{[g(x)]^2}$

$$H'(-3) = \left(\frac{f}{g}\right)'(3) = \frac{f'(-3)\, g(-3) - f(-3)\, g'(-3)}{[g(x)]^2}$$

$$= \frac{(4)(6) - (-2)(7)}{6^2} = \frac{38}{36} = \frac{19}{18}$$

31. Let $h(x) = \frac{2x + 1}{3x - 5}$, then

a. find $h'(x)$

b. Where does the curve $y = h(x)$ have slope -13

Solution:- Use the quotient rule:

$$\text{a. } h'(x) = \left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$$

$$= \frac{(3x - 5)(2x + 1)' - (2x + 1)(3x - 5)'}{(3x - 5)^2}$$

$$= \frac{2(3x - 5) - 3(2x + 1)}{(3x - 5)^2} = \frac{-13}{(3x - 5)^2}$$

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b. $h'(x) = -13$

$$\frac{-13}{(3x-5)^2} = -13 \text{ implies that } (3x-5)^2 = 1$$

$3x - 5 = 1$ or $3x - 5 = -1$

$x = 2$ or $x = \frac{4}{3}$

32. Let $h(x) = \frac{4x}{x-2}$, then

a. find $h'(x)$ b. Where does the tangent line have slope -2.

Solution

Let $f(x) = 4x$ and $g(x) = x - 2$

Let $h(x) = \left(\frac{f}{g}\right)(x)$, then

a. $h'(x) = \frac{f'(x)\,g(x) - f(x)\,g'(x)}{(g(x))^2}$

$$= \frac{(4)(x-2) - 4x(1)}{(x-2)^2}$$

$$\therefore h'(x) = \frac{-8}{(x-2)^2}$$

b. $h'(x) = -2$, then $\frac{-8}{(x-2)^2} = -2$

$(x-2)^2 = 4 \Rightarrow$ or $x - 2 = \pm\sqrt{4}$

$\Rightarrow x - 2 = 2$ or $x - 2 = -2$

$x = 4$ or $x = -2 + 2 = 0$

33. Find $h'(0)$ given that $f(0) = 4$, $f'(0) = -3$, $g(0) = 2$ and $g'(0) = 5$,

a. $h(x) = (f\,g)(x)$ c. $h(x) = \frac{g(x)}{f(x)}$

b. $h(x) = \frac{f(x)}{g(x)}$ d. $h(x) = 3f(x) - 8g(x)$

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Solution

a. $h(x) = (f\,g)(x)$, then $h'(x) = f'(x)g(x) + f(x) + f(x)g'(x)$

$\therefore \quad h'(0) = f'(0)\,g(0) + f(0)\,g(0) + f(0)\cdot g'(0)$

$\therefore \quad h'(0) = (-3)(2) + (4)(5) = 14$

b. $h(x) = \dfrac{f(x)}{g(x)}$, then $h'(x) = \dfrac{g(x)f'(x) - g'(x)f(x)}{(g(x))^2}$

$\therefore \quad h'(0) = \dfrac{g(0)\,f'(0) - g'(0)\,f(0)}{(g(0))^2} = \dfrac{2(-3) - 5(4)}{2^2}$

$\therefore \quad h'(0) = \dfrac{-6 - 20}{4} = \dfrac{-13}{2}$

c. $h(x) = \dfrac{g(x)}{f(x)}$, then $h'(x) = \dfrac{f(x)\,g'(x) - f'(x)g(x)}{[f(x)]^2}$

$\therefore \quad h'(0) = \dfrac{f(0)\,g'(0) - f'(0)g(0)}{[f(0)]^2} = \dfrac{4(5) - (-3)(2)}{4^2}$

$\therefore \quad h'(0) = \dfrac{26}{16} = \dfrac{13}{8}$

4. Find $g'(4)$ given that $f(4) = 8$ and $f'(4) = -6$, if

a. $g(x) = \sqrt{x}\ f(x)$ b. $g(x) = \dfrac{f(x)}{x}$

Solution

a. Use the product rule:

$$g'(x) = \sqrt{x}'\ f(x) + \sqrt{x}\,f'(x) = \frac{f(x)}{2\sqrt{x}} + \sqrt{x}\,f'(x)$$

$$g'(4) = \frac{f(4)}{2\sqrt{4}} + \sqrt{4}\cdot f'(4) = \frac{8}{2\sqrt{4}} + 2(-6)$$

$$\therefore \quad g'(4) = 4 - 12 = -8$$

b. $g(x) = \dfrac{f(x)}{x}$, we use quotient rule

$$g'(x) = \frac{xf'(x) - x'f(x)}{x^2} = \frac{x\,f'(x) - f(x)}{x^2}$$

$$g'(4) = \frac{4\,f'(4) - f(4)}{4^2} = \frac{4(-6) - 8}{16} = \frac{-32}{16} = -2$$

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35. Given that $g(8) = -2$, $g'(8) = -1$, $f(8) = -1$ and $f'(8) = 1$. Then find $F'(8)$

a. $F(x) = \sqrt[3]{x}\ f(x)$ b. $F(x) = \dfrac{3x - 1}{g(x)}$

Solution:- a. Use product property rule

$$F(x) = \sqrt[3]{x}\ f(x) = x^{1/3} f(x), \text{ then}$$

$$F'(x) = \frac{1}{3} x^{-2/3} f(x) + [f'(x)]\ x^{1/3}$$

$$= \frac{1}{3} x^{-2/3} f(x) + \sqrt[3]{x}\ f'(x)$$

$$= \frac{1}{3\sqrt[3]{x^2}} f(x) + \sqrt[3]{x}\ f'(x)$$

$$F'(8) = \frac{1}{3\sqrt[3]{64}} f(8) + \sqrt[3]{8}\ f'(8)$$

$$\therefore \quad F'(8) = \frac{1}{12}(-1) + 2(1) = \frac{-1}{12} + 2\left(\frac{12}{12}\right) = \frac{23}{12}$$

b. Use quatient rule :

$$F'(x) = \frac{(3x - 1)' g(x) - (3x - 1) g'(x)}{[g(x)]^2}$$

$$= 3g(x) - (3x - 1) g'(x)$$

$$F'(8) = 3g(8) - (3(8) - 1) g'(8)$$

$$= \frac{3(-2) - (23)(-1)}{(-2)^2} = -6 + 23 = \frac{17}{4}$$

36. Let $f(x) = \begin{cases} 3x - 1, & x \le 2 \\ x^2 + 1, & x > 2 \end{cases}$ then Find

a. the derivative of f at $x = 1$

b. the derivative of f at $x = 4$

Solution

a. $f'(x) = \begin{cases} 3, \text{ if } & x \le 2 \\ 2x \text{ if } & > 2 \end{cases}$ then $f'(1) = 3$ b. $f'(4) = 2(4) = 8$

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Notation for Derivatives

There are many ways to denote the derivative of a function $y = f(x)$. Besides $f'(x)$, some common alternative notation foe the derivative are as follows:

$$y' = f'(x) = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x).$$

- The symbol: $\frac{dy}{dx}$, which introduced by Leibniz.
- We also read as $\frac{dy}{dx}$ "the derivative of y with respect to x,"
- The value of the derivative at $x = a$ is written

$f'(a)$, or $\left.\frac{dy}{dx}\right|_{x=a}$ or $\left.\frac{df}{dx}\right|_{x=a}$

Illustrative Examples

37. Find the values of the derivative

a. $\left.\frac{dy}{dx}\right|_{x=1}$ if $y = 3x^2 + 4x$

b. $\left.\frac{dy}{dx}\right|_{x=4}$ if $y = x\sqrt{x}$

c. $\left.\frac{dy}{dx}\right|_{x=-1}$ if $y = \frac{2x - 1}{3x + 1}$

Solution

a. $\frac{dy}{dx} = \frac{d}{dx}(3x^2 + 4x) = 6x + 4$

$\left.\frac{dy}{dx}\right|_{x=1} = 6(1) + 4 = 6 + 4 = 10$

b. $y = x\sqrt{x} = x(x^{1/2}) = x^{1+\frac{1}{2}} = x^{3/2}$

$\frac{dy}{dx} = \frac{d}{dx}(x^{3/2}) = \frac{3}{2}x^{\frac{3}{2}-1} = \frac{3}{2}x^{\frac{1}{2}} = \frac{3}{2}\sqrt{x}$

$\left.\frac{dy}{dx}\right|_{x=4} = \frac{3}{2}\sqrt{4} = \frac{3}{2}(2) = 3$

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c. $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\frac{d}{dx}[f(x)] - f(x)\frac{d}{dx}[g(x)]}{[g(x)]^2}$

$$y = \frac{2x - 1}{3x + 1}$$

$$\frac{d}{dx} = \frac{(3x + 1)\frac{d}{dx}(2x - 1) - (2x - 1)\frac{d}{dx}(3x + 1)}{(3x + 1)^2}$$

$$= \frac{(3x + 1)(2) - (2x - 1)(3)}{(3x + 1)^2} = \frac{5}{(3x + 1)^2}$$

$$\left.\frac{dy}{dx}\right|_{x=-1} = \frac{5}{(3(-1) + 1)^2} = \frac{5}{(-2)^2} = \frac{5}{4}$$

38. If h (3) = 4 and h' (3) = -2, then what is the value of $\left.\frac{d}{dx}\left(\frac{h(x)}{x}\right)\right|_{x=3}$

A. $\frac{2}{9}$ B. $\frac{-2}{9}$ C. $\frac{-10}{9}$ D. $\frac{-1}{3}$

Solution

Use the quotient rule

$$\frac{d}{dx}\left(\frac{h(x)}{x}\right) = \frac{x\frac{d}{dx}(h(x)) - h(x)\frac{d}{dx}x}{x^2}$$

$$= \frac{xh'(x) - h(x) \cdot 1}{x^2}$$

Then the value at $x = 3$, is $\frac{h'(3) - h(3)}{3^2} = \frac{3(-2) - 4}{9} = \frac{-10}{9}$

Answer: C

39. Find the rate of change of the area of a square with respect to side length of x

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Solution

Area: $A(x) = x^2$, then $\frac{dA}{dx} = \frac{d}{dx} x^2 = 2x$

40. Find the rate of change of the area of a circle with respect to its radius (r)

Solution : Area: $A \pi r^2$, $\frac{dA}{dx} = \frac{d}{dx} \pi r^2 = 2\pi r$

41. Find the rate of change of the volume of cube with respect to the side length of x.

Solution: Volume of a cube: $V = x^3$

$$\therefore \quad \frac{dv}{dx} = \frac{d}{dx} x^3 = 3x^2$$

11.3.5 The Chain Rule of Derivative

The chain rule is the derivative of composition function. The derivative of $f(g(x))$ is the derivative of the out side function evaluated at the inside function times the derivative of the inside function.

The chain Rule: If g is differentiable at x and f is differentiable at $g(x)$. then the composition fog is differentiable at x.

Symbol: $(fog)'(x) = f'(g(x))\, g'(x)$

Alternatively, *if* $y = f(g(x))$ and $u = g(x)$ then $y = f(u)$ and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Note : $y = (f(x))^n$, then $\frac{dy}{dx} = n\,(f(x))^{n-1}\, f'(x)$ (for any number n).

Illustrative Example

42. Let $F(x) = 10\sqrt{4 + x^2}$, find $F'(x)$

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Solution

Let $F(x) = 4 + x^2$ and $g(x) = 10\sqrt{x}$

Since $f'(x) = 2x$ and $g'(x) = \frac{10}{2\sqrt{x}} = \frac{5}{\sqrt{x}}$

We conclude that : $F'(x) = g'(f(x))\ f'(x)$

$$\therefore \quad F'(x) = 5\,\frac{1}{\sqrt{4 + x^2}}(2x) = \frac{10x}{\sqrt{4 + x^2}}$$

43. Let $f(x) = (x^2 + 4x)^2$, then find $f'(x)$.

Solution: $f(x) = (g(x))^n$, then $f'(x) = n\,(g(x))^{n-1}\,g'(x)$

If $f(x) = (x^2 + 4x)^2$, then $f'(x) = 2(x^2 + 4x)\,(2x + 4)$

44. Let $y = (x^3 - 1)^{50}$, then find $\frac{dy}{dx}$

Solution

$$\frac{dy}{dx} = \frac{d}{dx}(x^3 - 1)^{50} \cdot \frac{d}{dx}\left(x^3 - 1\right)$$

$$= 50\,(x^3 - 1)^{49} \cdot 3x^2 \quad = 150x^2\ (x^3 - 1)$$

45. Let $y = \sqrt[3]{2x + 1}$, then find $\frac{dy}{dx}$

Solution: First rewite y: $y = \sqrt[3]{2x + 1} = (2x + 1)^{1/3}$

$$\therefore \quad \frac{dy}{dx} = \frac{d}{dx}(2x + 1)^{1/3}\ \frac{d}{dx}(2x + 1)$$

$$= \frac{1}{3}(2x + 1)^{\frac{-2}{3}}\ 2 = \frac{2}{3\sqrt[3]{(2x + 1)^2}}$$

46. Let $f(x) = \frac{1}{(4x - 3)^2}$ then find $f'(x)$

Solution: First rewrite f: $f(x) = (4x - 3)^{-2}$, then

$$f'(x) = -2(4x - 3)^{-2-1}\,(4x - 3)' = -2(4x - 3)^{-3}\,(4)$$

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$$\therefore \quad f(x) = \frac{-8}{(4x-3)^3}$$

47. Let $f(x) = \left(\frac{3x+1}{4x-5}\right)^6$, find $f'(x)$

Solution

We begin with $f(x) = (H(x))^n$ then

$$f'(x) = n(H(x))^{n-1} H'(x)$$

Now, let $f(x) = \left(\frac{h(x)}{g(x)}\right)^6$, where

$h(x) = 3x + 1$, $h'(x) = 3$, $g(x) = 4x - 5$, $g'(x) = 4$

$$f'(x) = 6\left(\frac{h(x)}{g(x)}\right)^5 \left(\frac{h(x)}{g(x)}\right)'$$

$$= 6\left(\frac{h(x)}{g(x)}\right)^5 \frac{[g(x)\,h'(x) - h(x)\,g'(x)]}{(g(x))^2}$$

$$= 6\left(\frac{3x+5}{4x-5}\right)^5 \left[\frac{(4x-5)3 - (3x+1)4}{(4x-5)^2}\right]$$

$$= \frac{-114(3x+1)^5}{(4x-5)^7}$$

48. UEE. If $h(x) = \sqrt{1+\sqrt{x}}$

A. $\frac{1}{2\sqrt{1+\sqrt{x}}}$

B. $\frac{x}{2\sqrt{1+\sqrt{x}}}$

C. $\frac{1}{4\sqrt{x+x\sqrt{x}}}$

D. $\frac{x}{4\sqrt{x+x\sqrt{x}}}$

Solution

Let begin with :

Let $f(x) = \sqrt{x}$ and let $g(x) = 1 + \sqrt{x}$

$$f'(x) = \frac{1}{2\sqrt{x}} \text{ and } g'(x) = \frac{1}{2\sqrt{x}}$$

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Now, $h(x) = f(g(x)) = \sqrt{1 + (x)}$

$$h'(x) = f'(g(x))\, g'(x) = \frac{1}{2\sqrt{g(x)}} \cdot \frac{1}{2\sqrt{x}}$$

$$= \frac{1}{2\sqrt{1+\sqrt{x}}} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{4\sqrt{x + x\sqrt{x}}}$$

49. Skill work: Let $f(x) = \sqrt{9x - \sqrt{x}}$, find $f'(x)$. Answer: C

50. If $f(2) = 1, f'(2) = 7, f(8) = 5$ and $f'(8) = -3$ then

a. find $g'(2)$, where $g(x) = [f(x)]^3$

b. find $h'(2)$, where $h(x) = f(x^3)$

Solution

a. $g'(x) = 3(f(x))^2 . f'(x)$

$\therefore \quad g'(2) = 3(f(2))^2 f'(2) = 3(1)^2(7) = 21$

b. $h'(x) = f'(x^3)(x^3)' = f'(x^3)(3x^2)$

$h'(2) = f'(2^3)(3(2)^2) = 12 f'(8) = 12(-3) = -36$

51. Suppose $f'(1) = 2, g(1) = 1$, and $g'(1) = -3$ then the value of $(fog)'(1)$ is equal to

A. -6 B. 6 C. 3 D. -3

Solution

$(fog)(1) = f'(g(1))\, g'(1) = f'(1)\, g'(1) = 2(-3) = -6$

Answer: A

In example 53 - 54, use the table of values to find the indicated derivative of the function.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
2	3	-4	3	4
3	2	-5	7	-2

52. If $h(x) = f(g(x))$, then which of the following is then value of $h'(2)$ is equal to:

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B. -20 C. 20 D. -12

A. 12

Solution

From the above table:

$h'(x) = f'(g(x))\, g'(x)$

$h'(2) = f'(g(2))\, g'(2)$

$= f'(3)(4)$

$= (-5)(4) = -20$

Answer: B

3. If $h(x) = g(f(x))$, then which of the following is the value of $h'(2)$ is equal to:

A. -12 B. -8 C. 20 D. 8

Solution

From the above table:

$h'(x) = g'(f(x)).f'(x)$ and $h'(2) = g'(f(2))f'(2)$

$h'(2) = g'(3)(-4) = (-2)(-4) = 8$

Answer: D

54. Let $f(x) = g\left(\sqrt{4x+1}\right)$. If $g'(3) = \frac{1}{2}$, then $f'(2)$ *is*

A. $\frac{2}{3}$ B. $\frac{1}{3}$ C. $\frac{\sqrt{5}}{2}$ D. $\frac{\sqrt{13}}{2}$

Solution

$$f'(x) = g'\left(\sqrt{4x+1}\right)\frac{d}{dx}\left(\sqrt{4x+1}\,\frac{d}{dx}(4x+1)\right)$$

$$= g'\sqrt{4x+1}\cdot\frac{1}{2\sqrt{4x+1}}(4)$$

$$f'(2) = g'(3)\cdot\frac{1}{6}\cdot 4 = \left(\frac{1}{2}\right)\left(\frac{1}{6}\right)(4) = \frac{1}{3}$$

Answer: B

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11.3.6 Tangent Line and Normal line

REMINDER: The equation of the line through P = (a, b) of m point slope form: $y - b = m(x - a)$.

Based on the definition of **derivative** $f'(a)$ is slope of tangent line at a point $x = s$

$$f'(a) = \frac{f(x) - f(a)}{x - a} \text{ as } x \text{ approach to a.}$$

$$\boxed{f(x) - f(a) = f'(a)(x - a)}$$

The line perpendicular to a tangent line at the point of tangency is called **Normal line.** If the slope of tangent line at $x = a$ is $f'(a)$, then the slope of normal line is $\frac{-1}{f'(a)}$.

The equation of Tangent line

DEFINITION: The tangent line to the graph of $y = f(x)$ at point $(a, f(a))$ is the line through P of slope $f'(a)$. The equation of the tangent line in point - slope form is

$$y - f(a) = f'(a)(x - a)$$

The equation of the normal line in point - slope form is

$$y - f(a) = \frac{-1}{f'(a)}(x - a)$$

Illustrative Example

55. Let $f(x) = x^3 - 3x + 4$

a. Find the equation of tangent line and normal line at P(-2, 2).

b. For what values of x, the tangent line has negative slope, positive slope.

c. For what values of x the tangent line has a horizontal slope

Solution

a. $f'(x) = 3x^2 - 3$ and $f'(-2) = 3(-2)^2 - 3 = 9$

The equation of tangent line is : $y - f(-2) = f'(2)(x - (2))$

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$y - 2 = 9(x + 2)$

$\therefore \quad y = 9x + 20$

The equation of normal to the graph of f at (-2, 2) is

$$y - f(-2) = \frac{-1}{f'(-2)}(x + 2)$$

$$y - 2 = \frac{-1}{9}(x + 2)$$

$$\therefore \quad x + 9y = 16$$

b. $f'(x) = 3x^2 - 3 = 3(x^2 - 1) = 3(x - 1)(x + 1)$

The tangent line has a negative slope if $f'(x) < 0$.

$3(x^2 - 1) < 0 = 3(x - 1)(x + 1) < 0$

Slope of tangent line is negative over the interval (-1, 1)

$\therefore$ Slope of tangent line is positive over the interval $(-\infty\ -1) \cap (1, \infty)$.

c. Slope of tangent line is horizontal if $f'(x) = 0$, then:

$3(x^2 - 1) = 0$, $x = -1$ and $x = 1$

56. Find the point on the graph of $f(x) = 2x^2 + 5x + 4$ at which the slope of the normal line is 3.

Solution

$f'(x) = 4x + 5$

If the slope of normal line is 3, then slope of tangent line is $\frac{-1}{3}$

$\therefore \quad f(a) = 4a + 5 = \frac{-1}{3}$ and $4a = \frac{-1}{3} - 5 = \frac{-16}{3}$

$\therefore \quad a = \frac{-16}{12} = \frac{-4}{3}$

$\therefore \quad \left(\frac{-4}{3}, f\left(\frac{-4}{3}\right)\right) = \left(\frac{-4}{3}, \frac{8}{9}\right)$ is point on the hraph of f.

57. Let $f(x) = \begin{cases} 5x^2 - 3x, & x < 0 \\ x^2 - 3x, & x \geq 0, \text{ then} \end{cases}$

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find a. $f'(0)$ b. $f'(-1)$ c. $f'(1)$

Solution

$$f'(x) = \begin{cases} 10x^2 - 3, & x < 0 \\ 2x - 3, & x \geq 0, \end{cases}$$

a. $f'(0) = -3$,

b. $f'(-1) = -10 - 3 = -13$

c. $f'(1) = 2 - 3 = -1$

58. What is the equation of tangent line having slope - 4 to the graph of $y = 3x^2 + 2x + 1$

A. $y = 4x$ B. $y = 4x + 1$ C. $y = -4x - 2$ D. $y = -4x + 6$

Solution

Let $(a, f(a))$ be point of tangency to $y = 3x^2 + 2x + 1$:

We have, $f'(a) = 6a + 2 = -4$, if $a = -1$

Then $f(a) = f(-1) = 3 - 2 + 1 = 2$

$\therefore$ Equation of tangent line at $x = -1$ is $y - f(-1) = f(-1)(x + 1)$

$y - 2 = -4(x + 1) = -4x - 4$

$y = -4x - 2$

Answer: C

59. Let $f(x) = x^3 - 5x$, for what values of a $f'(a) = 13$

A. $a = \pm 6$ B. $a = \pm\sqrt{6}$ C. $a = \pm 3$ D. $a = \pm\sqrt{3}$

Solution

$$f'(a) = 3a^2 - 5 = 13, \text{ and } 3a^2 = 18$$

$$\therefore \quad a = \pm\sqrt{6}$$

Answer: B

60. Let $f(x) = \frac{1}{\sqrt{x}}$ and $g(x) = 2 - \frac{1}{6}x^3$ be two functions. For what values of x do the graphs of f and g have tangent lines of the same slope?

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B. $\frac{\sqrt{3}}{2}$ C. $\frac{1}{6}$ D. 4

A. 1

Solution: $f(x)=\left(\sqrt{x}\right)^{-11}=x^{\frac{-11}{2}}$

We set: $f'(x)=g'(x)$

$$\Rightarrow \left(\left(\sqrt{x}\right)^{-11}\right)'=\left(2-\frac{1}{6}x^3\right)'$$

$$\Rightarrow \frac{1}{2}x^{\frac{-13}{2}}=\frac{-1}{2}x^2$$

$$\Rightarrow x^{\frac{7}{3}}=1, \text{ then } x=1$$

Answer: A

61. If $g(x)=x f(x)-\sqrt{f(x)}$ and $f'(2)=f(2)=4$, then which of the following is equal to g' (2):

A. 11 B. 8 C. 2 D. 0

Solution

$$g'(x)=x' f(x)+x f'(x)-\frac{f'(x)}{2\sqrt{f(x)}}$$

$$g'(2)=f(2)+2f'(2)-\frac{f'(2)}{2\sqrt{f(2)}}=4+2(4)-\frac{4}{4}=11$$

Answer: A

62. The equation of tangent line to the graph of $f(x)=\frac{3x}{x-1}$ at $(2, f(2))$

A. $x+3y=20$ C. $y-3x=0$

B. $x-3y=-16$ D. $y+3x=12$

Solution: we find both $f'(2)$ and $f(2)$ 0

$$f(x)=\frac{h(x)}{g(x)}, \text{ then } f'(x)=\frac{g(x)h'(x)-h(x)g'(x)}{[g(x)]^2}$$

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$f(x) = \frac{3x}{x-1}$, then $f'(x) = \frac{(x-1)(3) - 3x(1)}{(x-1)^2} = \frac{-3}{(x-1)^2}$

$f'(2) = \frac{-3}{(2-1)^2} = \frac{-3}{1^2} = -3$

$f(2) = \frac{3(2)}{2-1} = \frac{6}{1} = 6$

Since $f'(2)$ is the slope and the tangent line passes through $(2, f(2))$, is $y - f(2) = f'(2)(x-2)$

$y - 6 = -3(x-2)$

$y + 3x = 12$

Solution

Answer: D

Frist rewreite f : $f(x) = \left(1 - \sqrt{x}\right)^{-2}$ then

$f'(x) = -2\left(1-\sqrt{x}\right)^{-3} \cdot \left(1-\sqrt{x}\right)' = \frac{-2\left(1-\sqrt{x}\right)^{-3}}{-2\sqrt{x}}$

$f'(x) = \frac{1}{\left(1-\sqrt{x}\right)^{-3}\sqrt{x}}$

$f'(4) = \frac{1}{\left(1-\sqrt{x}\right)^{-3}\sqrt{4}} = \frac{1}{(-1)(2)} = \frac{-1}{2}$

Answer: D

63. The equation of tangent of tangent line to the graph of $f(x) = \frac{1}{\left(1-\sqrt{x}\right)^2}$ at $x = 4$

Solution

$y - f(4) = f'(4)(x-4)$

$y - 1 = -\frac{1}{2}(x-4)$

$x + 2y = 6$

64. Find the equation of tangent line and normal line to the graph of

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$f(x) = \dfrac{4x^2 - x + 3}{\sqrt{x}}$ at $(1, f(1))$

Solution

First rewrite $f(x) = \dfrac{4x^2}{\sqrt{x}} - \dfrac{x}{\sqrt{x}} + \dfrac{3}{\sqrt{x}}$

$= 4x\sqrt{x} - \sqrt{x} + \dfrac{3}{\sqrt{x}}$

$= 4x^{3/2} - x^{1/2} + 3x^{-1/2}$

$f'(x) = 6x^{\frac{1}{2}} - \dfrac{1}{2\sqrt{x}} + \dfrac{-3}{2}x^{-3/2}$

$f'(x) = 6\sqrt{x} - \dfrac{1}{2\sqrt{x}} - \dfrac{3}{2\sqrt{x^3}}$

$f'(1) = 6 - \dfrac{1}{2} - \dfrac{-3}{2} = \dfrac{12 - 1 - 13}{2} = 4$

$f'(1) = \dfrac{4 - 1 + 3}{\sqrt{1}} = 6$

Equation of tangent line is $y - f(1) = f'(1)(x - 1)$

$y - 6 = 4(x - 1) \Rightarrow y = 4x + 2$

Equation of normal line is $y - f(1) = \dfrac{-1}{f'(1)}(x - 1)$

$y - 6 = \dfrac{-1}{4}(x - 1)$ this implies that $x + 4y = 25$

65. If $h(x) = \sqrt[3]{g(x)}$, $g(2) = 8$ and $g'(2) = 6$ then the equation of tangent line at $(2, h(2))$

A. $y + 2x = 1$ B. $2y + x = -1$ C. $2y - x = 1$ D. $2y - x = 2$

Solution

$h(x) = (g(x))^{1/3}$

$h'(x) = \dfrac{1}{3}(g(x))^{-2/3} \cdot g'(x) = \dfrac{1}{3\sqrt[3]{(g(x))^2}} \cdot g'(x)$

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$$h'(2) = \frac{1}{3\sqrt[3]{(g(2))^2}} \cdot g'(2) = \frac{6}{3\sqrt[3]{(g(x))}} = \frac{2}{4} = \frac{1}{2}$$

$$h(2) = \sqrt[3]{g(2)} = \sqrt[3]{8} = 2$$

Equation of tangent line is: $y - h(2) = h'(2)(x - 2)$

$$y - 2 = \frac{1}{2}(x - 2) \Rightarrow 2y - x = 2$$

66. Find the points on the graph of $f(x) = 12x - x^3$ where the tangent line is horizontal. **Answer: D**

Solution

The tangent line is horizontal at points where the slope $f(x)$ is zero, so, $f'(x) = 12 - 3x^2$, and $f'(x) = 12 - 3x^2 = 0$

$$\therefore \quad x = \pm 2$$

Now $f(x) = 12(2) - (2)^3 = 24 - 8 = 16$ and

$$f(-2) = 12(-2) - (-2)^3 = -24 + 8 = -16$$

Hence, the tangent lines are horizontal at (2, 16) and (-2, - 16)

67. Find the points on the graph of $y = x^2 + 3x - 6$ at which the slope of the tangent line is equal to -7

Solution :- $\frac{dy}{dx} = 2x + 3 = -1$

$2x + 3 - 3 = -7 - 3 = -10$

$\therefore \quad y = (-5)^2 + (5) - 6 = 4$

$\therefore \quad (-5, 4)$ is point on the graph of f

68. Let $f(x) = x^2\sqrt{2x + 12}$, the equation of tangent line to the graph of $y = f(x)$ at $x = 2$

A. $y - 4 = 16(x - 2)$

B. $y = 17(x - 2) + 16$

C. $y = 17(x - 2) - 16$

D. $y = 16(x - 2) + 1$

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Solution

$$f'(x) = \frac{d}{dx}\left(x^2\sqrt{2x+12}\right)$$

$$= x^2\frac{d}{dx}\sqrt{2x+12}\cdot\frac{d}{dx}(2x+12) + \sqrt{2x+12}\cdot\frac{d\,x^2}{dx}$$

$$f'(x) = \frac{2x^2}{2\sqrt{2x+12}} + 2x\sqrt{2x+12}$$

$$f'(2) = \frac{4}{4} + 4\sqrt{16} = 1 + 16 = 17$$

Equation of tangent line is $y - f(2) = f'(2)(x-2)$

$$y - 16 = 17(x-2)$$

Answer: B

69. Find the value of k so that the line $y = -8x + k$ is tangent to the curve $f(x) = 3x^2 + 4x + 1$.

Solution

Let $f(x) = 3x^2 + 4x + 1$, then $f'(x) = 6x + 4$.

The slope of tangent line is -8

Let $(a, f(a))$ be point on tangent line $f'(a) = 6a + 4 = -8$, $a = -2$

Equation of tangent line is $y - f(a) = f'(a)(x - a)$

$y - f(-2) = f'(-2)(x + 2)$, $f(-2) = 5$, $f'(-2) = -8$

$y - 5 = -8(x + 2) = -8x - 16$

$y = -8x - 16 + 5 = -8x - 11$

$\therefore$ The value of k is -11

70. The curve $f(x) = x^2 + ax + b$ and $g(x) = cx^3 - x^2$ have common tangent line at the point (1, 0). Find a, b and c

Solution

Since the curve have common tangent line at (1, 0).

Then $g(1) = f(1) =$ and $g'(1) = f'(1)$.

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$C - 1 = 0 \Rightarrow C = 1$

$\therefore \quad g(x) = x^2 - x^3$ and $g'(x) = 2x - 3x^2$

$g'(1) = 2 - 3 = -1$

Also $f(x) = x^2 + ax + b$, then $f'(x) = 2x + a$

since $f'(1) = g'(1) = 2(1) + a = -1, \; a = -3$

We have $f(1) = 1 + a + b = 0, \quad 1 + (-3) + b = 0 \; b = 2$

$\therefore \quad a = -3, \; b = 2$ and $c = 1$

71. Let $G(x) = f\left(\frac{3x^2 + 1}{x - 1}\right)$. If $f'(0) = 0$ $f'(-1) = 2$, then $G'(0)$

A. -2 B. -1 C. 0 D. 3

Solution

$$G'(x) = f'\left(\frac{3x^2 + 1}{x - 1}\right) \frac{d}{dx}\left(\frac{3x^2 + 1}{x - 1}\right)$$

$$= f'\left(\frac{3x^2 + 1}{x - 1}\right) \frac{(x-1)\frac{d}{dx}(3x^2 + 1) - (3x^2 + 1)\frac{d}{dx}(x - 1)}{(x - 1)^2}$$

$$G'(x) = f'\left(\frac{3x^2 + 1}{x - 1}\right) \frac{6x(x - 1) - (3x^2 + 1)}{(x + 1)^2}$$

$$= f'\left(\frac{3x^2 + 1}{x - 1}\right)\left(\frac{3x^2 - 6x - 1}{x - 1}\right)$$

$\therefore \quad G'(0) = f'(-1)(-1) = 2(-1) = -2$

Answer: A

72. If $f(2) = -3, f'(2) = 4, g(1) = 1, g'(1) = -5$ and $F(x) = f(2x + 2).\, g(1 - x^2)$, then $F'(0)$.

A. 4 B. 8 C. -11 D. -22

Solution

$F'(x) = f'(2x + 2)(2) \cdot g(1 + x^2) + f(2x + 2)\, g'(1 - x^2)(-2x)$.

$F'(0) = f'(2)(2)\, g(1) + f(2)\, g'(1)(0)$

$F'(0) = (4)(2)(1) + (-30(-5)(0) = 8 + 0 = 8$

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Answer: B

73. Let f and g be differentiable functions h $(x) = g(f(3x + 1))$. If $f'(1) = f(1) = 2$ and $g'(2) = 1$, then what is the value of h' (0) ?

A. 12 B. 2 C. 4 D. 6

Solution

$h'(x) = g'(f(3x + 1))\, f'(3x + 1)\,(3x + 1)'$

$= g'(f(3x + 1))\, f'(3x + 1)\,(3)$

Then $h'(0) = g'(f'(1))\, f'(1)\,(3) = g'(2) \cdot (2)\,(3)$

$= (1)\,(2)\,(3) = 6$

Answer: D

Conceptual Example

74. Determine whether each of the following statements is true of false

a. The derivative as the average rate of change of a function .

b. The derivative as the instant aneous rate of change of a function.

c. The slope of the tangent line is the average rate of change

d. The slope of secant line is the slope of tangent line.

e. The slope of secant line to curve is the average rate of change.

f. The derivative of velocity function is accelerative of a function

Solution

a. False c. False e. True

b. True d. False f. True

Vertical Tangent Line and Deriyative

Let a in the domain of f. The vertical tangent line to $y = f(x)$ at $(a, f(a))$ is the vertical line whose derivative of f at 'a' does not exist.

Illustrative Example

75. Find the points on the graph of f where the tangent line is vertical

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a. $f(x) = \sqrt{2x - 1}$

b. $f(x) = \sqrt{x^2 - 4x}$

c. $\sqrt[3]{x - 1}$

d. $\sqrt[3]{x^2}$

Solution

a. $f(x) = \sqrt{2x - 1} = (2x - 1)^{1/2}$

$$f'(x) = \frac{1}{2}(2x - 1)^{\frac{-1}{2}} \cdot 2 = \frac{2}{2\sqrt{2x - 1}} = \frac{1}{\sqrt{2x - 1}}$$

$\therefore$ f has vertical tangent line at $x = \frac{1}{2}$, because $f'\left(\frac{1}{2}\right)$ does not exis

b. $f(x) = \frac{d}{dx}\sqrt{x^2 - 4x} \cdot \frac{d}{dx}(x^2 - 4x) = \frac{2x - 4}{2\sqrt{x^2 - 4x}}$

$f'(x)$ *does not exist if* $x^2 - 4x = 0$

$\therefore$ $x = 0$ and $x = 4$ are point on vertical tangent line.

c. $f'(x) = \frac{d}{dx}(x - 1)^{1/3} = \frac{1}{3}(x - 1)^{-2/3} = \frac{1}{\sqrt[3]{(x - 1)^2}}$

$\therefore$ $x = 1$ is point on vertical tangent line.

d. $f'(x) = \frac{d}{dx}(x^{2/3}) = \frac{2}{3}x^{-1/3} = \frac{2}{3\sqrt[3]{x}}$

$\therefore$ $x = 0$ is point on vertical tangent line.

Conceptual Example

76. Determine whether each of the following statement is true or false. Explain the reason why the statement is false

a. A polynomial function has a tangent line at every point on its graph.

b. The instantaneous rate of change of $y = f(x)$ with respect to x at x_0 is the slope of the tangent line to the graph at $(x_0, f(x_0))$.

c. If $f'(x) = g'(x)$, then $f(x) = g(x)$

d. If $f(x) = x^n$, where n is a real number has a tangent line at every point on its graph

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Answer

1. True b. True c. False d. False

1.4 Increasing and Decreasing Behaviour of function

If x_1 and x_2 are any two numbers between a and b with $x_1 < x_2$, then $f(x_1) < f(x_2)$ or $f(x_1) > f(x_2)$.

DEFINITION A function f is :

(i) **increasing** on (a, b) if $f(x_1) \leq f(x_2)$ when ever $x_1 < x_2$ in (a, b)
[That is, $f(x)$ get larger as x gets larger]

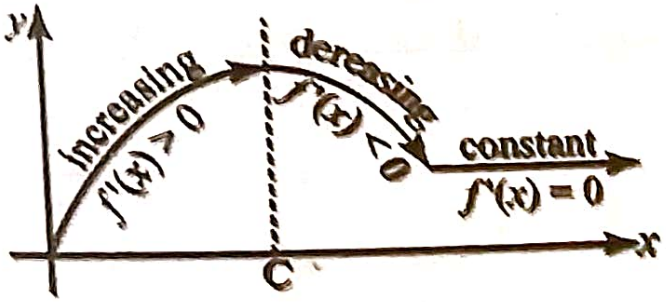
(ii) **decreasing** on (a, b) if $f(x_1) \geq f(x_2)$ whenever $x_1 < x_2$ in (a, b)
[That is, $f(x)$ get smaller as x gets larger]

- We say that f is monotonic on (a, b) if it is either increasing or decreasing on (a, b)
- A function is increasing if, as x moves to the right, its graph move up, and is decreasing if its graph, move down.

How Derivative Tells the Increasing/ Decreasing behaviour of a function

The derivative of f at x, $f'(x)$ represents the slope of the curve $y = f(x)$

- Positive derivative implies that the function is increasing.
- Negative derivative implies that the function is decreasing
- Zero derivative implies that the function is constant on that interval.



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What Does the sign of f'' say about f ?

The sign of derivative tells us if:

- $f'(x) \geq 0 \Rightarrow$ tangent lines have positive slope
  $\Rightarrow f$ is increasing.
- $f'(x) \leq 0 \Rightarrow$ tangent lines have negative slope
  $\Rightarrow f$ is decreasing.

Increasing/Decreasing Test (I/D test)

The sign of derivative (+ , or -)

- If $f'(x) \geq 0$ on an interval, then f is increasing on that interval. That is
- If $f'(x) \leq 0$ is an interval, the f is decreasing on that interval.

Note: If $f' \geq 0$ at each point of (a, b), then f increases on [a, b]

- If $f' \leq 0$ at each point of (a, b), then f decreases on [a, b].

Illustrative Example

77. Find the intervals on which $f(x) = x^4 - \frac{8}{3}x^3 - 6x^2 + 3$ is increasing or decreasing

Solution: (I/D test)

To use increasing/decreasing test we have to known where $f(x) \geq 0$ and where $f'(x) \leq 0$.

$f'(x) = 4x^3 - 8x^2 - 12x = 4x(x^2 - 2x - 3) = 4x(x - 3)(x + 1)$

$\therefore$ The critical numbers are $x = 0$, $x = -1$ and $x = 3$.

We divide the x- axis into intervals

	$-\infty$		-1		0		3	∞
$4x$	- - -		- - -	0	+ + +		+ + +	
$x + 1$	- - -	0	+ + +		+ + +		+ + +	
$x - 3$	- - -		- - -		- - -	0	+ + +	
$f'(x) = 4x(x + 1)(x - 3)$	- - -		+ + +	0	- - -	0	+ + +	

$\therefore$ The behaviour of f decreasing on the interval $(-\infty\ -1] \cup [0\ \ 3]$.

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$\therefore$ f is increasing on the interval $[-1, 0] \cup [3, \infty)$.

78. The largest interval where $f(x) = \dfrac{x}{x^2 + 1}$ is increasing: UEE

A. $(-\infty, 1]$ B. $[1, \infty)$ C. $[0, \infty)$ D. $[-1, 1]$

Solution

$$f'(x) = \frac{(x^2 + 1)^2 \dfrac{d}{dx} x = x \dfrac{d}{dx}(x^2 + 1)}{(x^2 + 1)^2} = \frac{x^2 + 1 - 2x^2}{(x^2 + 1)^2}$$

$$f'(x) = \frac{1 - x^2}{(x^2 + 1)^2}$$

Therefore, $f'(x) \geq 0$, if $1 - x^2 \geq 0$, $-1 \leq x \leq 1$

Answer: D

79. On which of the following intervals is the graph of $f(x) = \dfrac{x}{x^2 + 9}$ decreasing: UEE

A. $(-\infty, \infty)$ B. $[-3, 3]$ C. $(-\infty -3] \cup [3, \infty)$ D. $[0, \infty)$

Solution

$$f'(x) = \frac{(x^2 + 9) \dfrac{d}{dx} x - x \dfrac{d}{dx}(x^2 + 9)}{(x^2 + 9)^2} = \frac{x^2 + 9 - 2x^2}{(x^2 + 9)^2}$$

$$f'(x) = \frac{9 - x^2}{(x^2 + 9)^2}, \quad f'(x) \leq 0, \quad 9 - x^2 \leq 0$$

$x^2 \geq 9$, when $x \leq -3$ and $x \leq 3$

$\therefore$ f is decreasing on $(-\infty, 3] \cup [3, \infty)$

Answer: C

Critical point (number)

DEFINITION: A number C in the domain of f is called a critical number if either $f'(c) = 0$ or $f'(c)$ has no derivative at C.

Illustrative Example

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80. Let $f(x) = (x^2 - 9)^{2/3}$. Then

a. Find the critical number

b. Determine the intervals on which f is increasing or decreasing.

Solution

a. Domain of f is $\mathbb{R}$

$$f'(x) = \frac{2}{3}(x^2 - 9)^{\frac{-1}{3}} \cdot 2x = \frac{4x}{3(x^2 - 9)^{1/3}}$$

$f'(x) = 0$ when $x = 0$

$f'(x)$ does not exist when $x = 3$ and $x = -3$.

$\therefore$ The critical numbers are $x = 0, x = -3$ and $x = 3$.

b. Now we divide the critical number in interval.

	$-\infty$	-3	0	3	∞
$4x$	− − −	− − −	+ + +	+ + +	
$x+3$	− − −	+ + +	+ + +	+ + +	
$x-3$	− − −	− − −	− − −	+ + +	
$f'(x) = \frac{4x}{3(x^2-9)^{1/3}}$	− − −	+ + +	− − −	+ + +	

From the table, we conclude that f is decreasing on the intervals $(-\infty, -3]$ and $[0, 3]$.

f is increasing on the intervals $[-3, 0]$ and $[3, \infty)$.

81. Let $f(x) = \frac{2x^2}{x + 2}$, find all the critical number.

Solution

Domain of $f = \mathbb{R} / \{x = -2\}$

$$f'(x) = \frac{(x+2)\frac{d}{dx}(2x^2) - 2x^2\frac{d}{dx}(x+2)}{(x+2)^2} = \frac{4x(x+2) - 2x^2}{(x+2)^2}$$

$$f'(x) = \frac{2x(x+4)}{(x+2)^2}, \; f'(x) = 0, \text{ when } x = 0 \text{ and } x = -4.$$

$\therefore$ The critical number are $x = 0$ and $x = -4$. However -2 is not in

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the domain of f so that $x = -2$ is not critical number.

82. Which of the following is the set of a critical number of $f(x) = |x^2 - 4|$

A. $\{0\}$ B. $\{-2, 2\}$ C. $\{-2, 0, 2\}$ D. $\{0, 2\}$

Solution: Redefine $f(x) = |x^2 - 4|$

$$f(x) = \begin{cases} x^2 - 4 & \text{if } x < -2 \text{ and } x > 2 \\ 0 & \text{if } x = -2 \text{ and } x = 2 \\ 4 - x^2 & \text{if } -2 < x < 2 \end{cases}$$

$$f'(x) = \begin{cases} 2x & \text{if } x < -2 \text{ and } x > 2 \\ 0 & \text{if } x = -2 \text{ and } x = 2 \\ -2x & \text{if } -2 < x < 2 \end{cases}$$

$f'(x) = 0$, when $x = 0$

When $x < -2, f'(-2) = -4$

When $x > -2\ f'(-2) = (-2)(2) = 4$

$\therefore$ $f'(-2)$ does not exist

when $x > 2, f'(2) = 4$, when $x < 2, f'(2) = -4$

$\therefore$ $f'(2)$ does not exist

$\therefore$ The critical points are theses of $\{-2, 0, 2\}$

Answer: C

83. Let $f(x) = x^{2/3}(4 - x)$. Then

a. Find the critical number

Solution

a. $f'(x) = x^{\frac{2}{3}} \frac{d}{dx}(4 - x) + 4 - x \cdot \frac{d}{dx} x^{2/3}$

$= -x^{2/3} + (4 - x)\left(\frac{2}{3}x^{\frac{2}{3}}\right)$

$= -x^{2/3} + \frac{2(4 - x)}{3\sqrt[3]{x}} = \frac{-3x + 8 - 2x}{3\sqrt[3]{x}} = \frac{8 - 5x}{3\sqrt[3]{x}}$

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Therefore, $f(x) = 0$, if $8 - 5x = 0$, that is $x = \frac{8}{5}$ and $f(x)$ does not exist when $x = 0$. Thus the critical numbers are $\frac{8}{5}$ and 0.

84. Let $f(x) = (x - 4)^{2/3}$ .Then find the critical value of f.

Solution

$$f'(x) = \frac{2}{3}(x - 4)^{\frac{-1}{3}} = \frac{2}{3(x - 4)^{1/3}} = \frac{2}{3\sqrt[3]{x - 4}}$$

We see that $f'(x)$ does not exist when $x = 4$. Since 4 is in the domain of f, we conclude it is a critical value.

11.4.1 Maximum and Minimum Values of a Function

What is Extrema of Function?

Suppose a function f is defined on a interval I.

The **maximum** and **minimum values** of f on an interval I are called **extrema** of the function.

There are two kinds of extrema:

1. **Absolute Extrema**
2. **Relative Extrema (Local extrema)**

1. **An absolute extrema** is referred as an absolute maximum and an absolute minimum.

DEFINITION: Absolute Extrema

Let C be a number in the domain of a function f. We say that:

(i) $f(c)$ is **absolute maximum** of a function f

If $f(c) \geq f(x)$ for every x in the domain of f.

- That is $f(c)$ is an absolute maximum value of f if and only if $(c, f(c))$ is the highest point on the graph of f.
- No greater value of f any where, but $f(c)$

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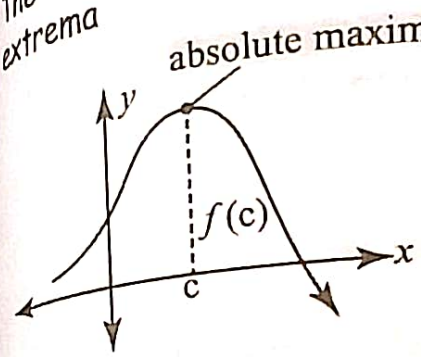
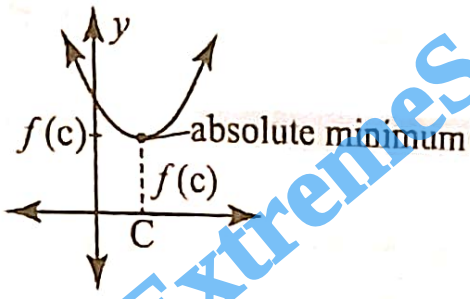
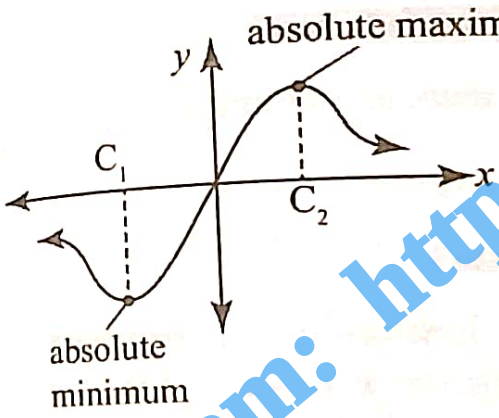
(iii) $f(c)$ is an **absolute minimum** of a function f if $f(c) \leq f(x)$ for every x in the domain of f.

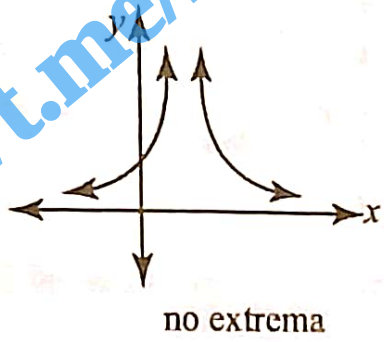
- That is $f(c)$ is an absolute minimum value of f if and only if $(c, f(c))$ is the lowest point on the graph of f.
- No smaller value of f any where but $f(c)$.

The following figure shows several possibilities of absolute extrema









Example 1

a. $f(x) = \sin x$ its absolute maximum value

and $f\left(\frac{3\pi}{2}\right) = -1$ is its absolute minimum on $[-2\pi, 2\pi]$.

b. $f(x) = x^2$ has absolute minimum $f(0) = 0$ but no absolute maximum.

c. $f(x) = \frac{1}{x}$ has neither an absolute maximum nor an absolute minimum

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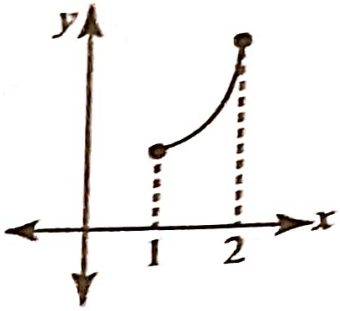
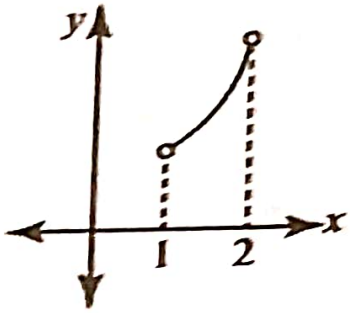
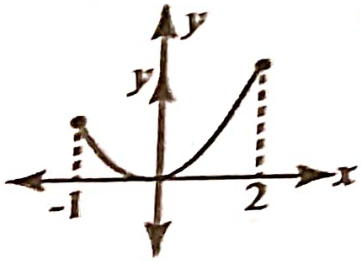
Example 2



a. $f(x) = x^2$ defined on interval $[1, 2]$, has absolute maximum $f(2) = 4$ absolute minimum $f(1) = 1$.



b. $f(x) = x^2$ defined on open interval $(1, 2)$, then f has no extrema. In this case $f(1)$ and $f(2)$ are not defined.



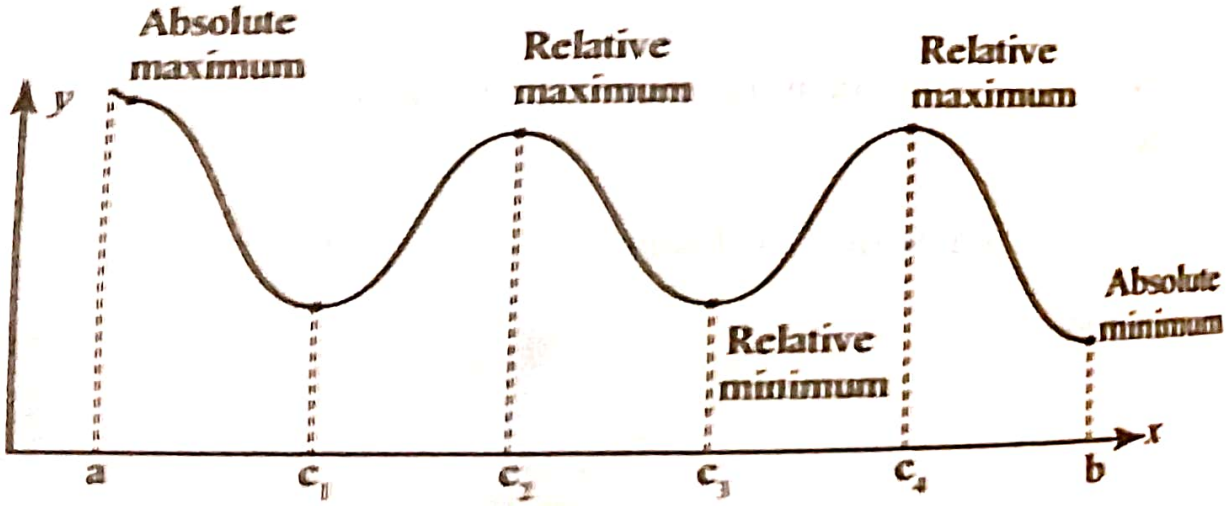
c. $f(x) = x^2$, defined on closed interval $[-1, 2]$

- f has the absolute maximum $f(2) = 4$
- f has the absolute minimum $f(0) = 0$

2. Relative Extrema (Local Extrema)

The term "local" is used because we localize our attention to a sufficiently small open interval containing c where f takes on a largest (or smallest) value at $x = c$. Outside that interval f may take on larger (or smaller) values. Sometimes the word "relative" is used in place of "local".

Consider the following graph of a function $y = f(x)$ on interval $[a, b]$



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(i) $f(c_1)$ are $f(c_3)$ are relative minimum value.

(ii) $f(c_2)$ and $f(c_4)$ are relative maximum value

(iii) $f(a)$ is the absolute maximum value of f on [a, b], although it is a local maximum

(iv) $f(b)$ is the absolute minimum value of f on [a, b], although it is a relative minimum.

DEFINITION: Let C be a number in the domain of a function f. We say that:

(i) $f(c)$ is a **relative maximum** of f if there exists an open interval (a, b) containing c such that $f(x) \leq f(c)$ for all x in (a, b)

- No grater value of f near by c.

(ii) $f(c)$ is a **relative minimum** of f if there exists an open interval (a, b) containing c such that $f(x) \geq f(c)$ for all x in (a, b).

- No smaller value of f near by c.

Finding The Extreme values of a Function f on a closed interval [a, b]

The absolute maximum and minimum values in closed interval [a, b] occur either at the end points a or b of the interval, or

at a critical number of f in (a, b).

To find the absolute extrema of a function on [a, b]

STEP 1: Find all critical number c in (a, b).

STEP 2: Evaluate f at all critical values.

STEP 3: Evaluate f at end point of a and b.

STEP 4: The largest and smallest values in the list:

$f(a), f(b), f(c_1), \ldots f(c_n)$ are the absolute maximum and the absolute minimum of f on the interval.

Illustrative Example

85. Find the absolute extreme value of $f(x) = 2x^3 - 15x^2 + 24x + 7$ on [0 6]

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Solution

The extreme values occur at critical points or endpoints.

STEP 1: Find the critical points

$$f'(x) = 6x^2 - 30x + 24 = 6(x - 1)(x - 4) = 0$$

$\therefore$ The critical points are c = 1 and c = 4

STEP 2: compare values of f (x) at the critical points and endpoints.

At critical point: $x = 1, f(1)\ 2 - 15 + 24 + 7 = 18$

At critical point: $x = 4, f(4) = 128 - 124 + 96 + 7 = -9$

At end point: $x = 0, f(0) = 4$

At end point : $x = 6\ f(6) = 43$

$\therefore$ The absolute maximum value is $f(6) = 43$

The absolute minimum value is $f(4) = -9$

$\therefore$ (6, 43) is maximum point and (4, -9) is minimum point of f.

86. Find the absolute extrema of $f(x) = 2x^3 - 3x^2 - 12x + 5$ on [2, 4].

Solution

STEP 1: Find the critical points.

$$f'(x) = 6x^2 - 6x - 12 - 6(x^2 - x - 2) = 6(x - 2)(x + 1)$$

$\therefore$ The critical points are $x = -1$ and $x = 2$.

STEP 2: compare the values of f (x) at critical points and end points.

At $x = -1 : f(-1) = -2 - 3 + 12 + 5 = 12$

At $x = 2: f(2) = 16 - 12 - 24 + 5 = -15$

At $x = -2: f(-2) = -16 - 12 + 24 + 5 = 1$

At $x = 4: f(4) = 128 - 48 - 48 + 5 = 37$

$\therefore$ The absolute maximum is $f(4) = 37$

The absolute minimum is $f(2) = -15$

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What will be the absolute maximum value of $f(x) = x^4 - 6x^2 + 1$ on $[-2, 3]$?

A. $f(-\sqrt{3})$ B. $f(3)$ C. $f(\sqrt{3})$ D. $f(0)$

Solution

First, we find the critical point:

$f'(x) = 4x^3 - 12x = 4x(x^2 - 3) = 0$

$\therefore \quad x = 0,\ x = -\sqrt{3}$ and $x = \sqrt{3}$

Compute the values of $f(x)$ at critical point and at end points.

$f(-2) = -7,\ f(-\sqrt{3}) = -8,\ f(0) = 1$

$f(\sqrt{3}) = -8$ and $f(3) = 28$

Answer: B

11.4.2 Critical Number and Relative Extreme value

As a consequence of definition of relative extrema, we conclude that every absolute extremum with the exception of an endpoint extremum is also a relative extremum.

Theorem: If a function f has a local extremum at a number C, then $f'(c) = 0$ or $f'(c)$ does not exist. That is $f(c)$ is a local extremum then $f'(c) = 0$ or $f'(c)$ does not exist.

Note: If a function has a local extremum, then it must occur at a critical number; however, not every critical number leads local extremum.

88. What are the value of a and b if $f(x) = ax^3 + bx^4$ has local minimum value at $(1, -1)$

A. $a = 3, b = 4$ C. $a = 4, b = -3$

B. $a = -4, b = 3$ D. $a = 3, b = -4$

Solution

$f(1) = a + b = -1$ ..... (*)

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$f'(x) = 3ax^2 + 4bx^3$. Let $x = 1$ as critical point. Then:

$f'(1) = 3a + 4b = 0$ ..... (**)

Equating (*) and (**)

$$\begin{cases} a + b = -1 \\ 3a + 4b = 0 \end{cases} \Rightarrow + \begin{cases} -3a - 3b = 3 \\ 3a + 4b = 0 \end{cases}$$

$b = 3$ and $a = -4$

89. What are the values of a and b if $f(x) = \frac{ax^2 + b}{x^3}$ has extreme value at (3, 2)?

Answer: B

A. $a = 6, b = 3$

C. $a = -9, b = 27$

B. $a = 9, b = -27$

D. $a = -6, b = 4$

Solution

$$f'(x) = \frac{x^3 \frac{d}{dx}(ax^2 + b) - (ax^2 + b)\frac{d}{dx}x^3}{(x^3)^2}$$

$$\frac{2ax^4 - 3ax^4 - 3x^2b}{x^6} = \frac{x^2(-ax^2 - 3b)}{x^6}$$

$$\therefore \quad f'(x) = \frac{-ax^2 - 3b}{x^4}$$

We assume that $x = 3$ as critical number, so that

(*) $f'(3) = \frac{-9a - 3b}{81} = 0 \Rightarrow b = -3a$ ..... (*)

we have again $f(3) = 2$

(**) $\frac{9a + b}{27} = 2$, this implies $9a + b = 54$ ..... (**)

Now from (*) and (**): $\begin{cases} b = -3a \\ 9a + b = 54 \end{cases}$

$\therefore \quad a = 9$ and $b = -27$

Answer:

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90. For what values of k so that $f(x) = \frac{x}{x^2 + k}$ will have extreme values at $x = 2$

A. $k = 0$ B. $k = 2$ C. $k = 4$ D. $k = 16$

Solution

$$f'(x) = \frac{(x^2 + k)\frac{d}{dk}x - x\frac{d}{dx}(x^2 + k)}{(x^2 + k)^2} = \frac{k - x^2}{(x^2 + k)^2}$$

Since $x = 2$ is extreme velue then

$$f'(2) = \frac{k - 4}{(4 + k)^2} = 0, \text{ this implies that } k = 4$$

Answer: C

91. **Absolute Extreme and Range of f:** Let $f(x) = x^3 - 12x + 1$ be defined on $[0\ 3]$. what is the range of f:

A. $(-\infty, \infty)$ B. $[-15, 17]$ C. $[-8, 1]$ D. $[-15, 1]$

Solution

$f'(x) = 3x^2 - 12$ and critical point of f is : $f'(x) = 0$

$3(x - 2)(x + 2) = 0$ at $x = 2$ and $x = -2$

since $2 \in [0, 3]$, but $-2 \notin [0, 3]$.

Then at critical point: $f(2) = 8 - 24 + 1 = -15$.... minimum value

At the endpoint: $f(0) = 1$. ⟵ maximum value

$f(3) = 27 - 36 - 1 = -8$

∴ The range of f is $[-15, 1]$

Answer: D

92. Let $f(x) = x^4 - \frac{16}{3}x^3 + 6x^2$ for all $x \in (-\infty, \infty)$.

What is the range of f.

A. $(-\infty, \infty)$ B. $[0, \infty)$ C. $[-9, \infty)$ D. $\left[\frac{5}{3}, -9\right]$

Solution

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$f'(x) = 4x^3 - 16x^2 + 12x = 4x(x^2 - 4x + 3) = 0$

$\Rightarrow 4x(x-3)(x-1) = 0,\ x = 0,\ x = 1$ and $x = 3$

$\therefore\ f(0) = 0,\ f(1) = 1 - \frac{16}{3} + 6 = \frac{5}{3}$

$f(3) = 81 - 144 + 54 = -9 \longleftarrow$ absolute minimum

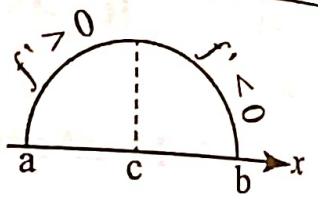
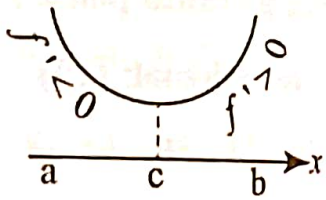
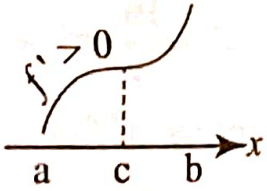
$\therefore$ The range of is $[-9, \infty)$

$\therefore$ Answer: C

11.4.3 First Derivative Test for Local Extreme value

Suppose f has derivative on (a, b) and that C is a critical value in the interval (a, b). We can use for easy reference in the following table

First Derivative Test

Critical Value	Sign of $f(x)$ on the in interval: (a, c) and (c, b)	Conclusion
$x = c$	(a, c) : + (c, b) : -	 $f(c)$ is a relative maximum or local maximum
$x = c$	(a, c): - (c, b): +	 $f(c)$ is a local minimum.
$x = c$	(a, c): + (c, b): +	 no local extremum

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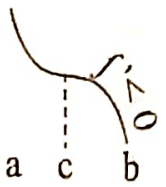
$x = c$	(a, c): - (b, c): -	 no extremum

First Derivative Test for critical points

Let c be a critical point of f. Then

- If $f'(x)$ changes from (+) to (-) at c, then $f(c)$ is a **local maximum** at c
- If $f'(x)$ changes from (-) to (+) at c, then $f(c)$ is a **local minimum** at c.
- If $f'(x)$ does not change the sign at c, then f has no local maximum or minimum at c.
- That is if $f'(x)$ has the same sign on both side of c, then f has no local extreme value at c.

Illustrative Example

93. Let $f(x) = 2x^3 + 3x^2 - 12x + 4$

a. Find the critical points of f.

b. On what intervals f is increasing and deceasing.

c. Determine the local extreme value.

Solution

We begin with by finding derivative of f.

$f'(x) = 6x^2 + 6x - 12 = 6(x - 1)(x + 2) = 0$

a. **Critical points** are c = 1 and C = -2

b. **Find the sign of $f'(x)$ on the intervals between the critical points**

The critical points $x = -2$ and $x = 1$ and divide it into three intervals: On which f' is either positive or negative.

	$-\infty$	-2	1	∞
$6(x - 1)$	- - -		- - -	+ + +
$x + 2$	- - -		+ + +	+ + +
Sign of $f'(x) = 6(x - 1)(x + 2)$	+ + +		- - -	+ + +
Behavior of $f(x)$	Increasing		Decreasing	Increasing

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$\therefore$ f is increasing in interval $(-\infty\ -2]$ and $[1, \infty)$

f is decreasing on interval $[-2, 1]$.

c. Use the first Derivative Test:

At c = -2: sign of $f'(x)$ change from (+) to (-)

$\Rightarrow f(-2) = 24$ is a local maximum value

At c = 1 : sign of $f'(x)$ change from (-) to (+)

$\Rightarrow f(1) = -3$ is a local minimum value.

94. Which of the following is **not true** about $f(x) = x^4 - 4x^3 + 16x + 5$

A. f is strictly decreasing on $(-\infty, -1)$

B. f is strictly increasing on $(-1\ \ 2) \cup (2, \infty)$.

C. f has local maximum value at $x = 2$.

D. f has local minimum value at $x = -1$

Solution: we use first derivative test:

$f'(x) - 4x^3 - 12x^2 + 16 = 4(x + 1)(x - 2)^2 = 0$

The critical points are c = -1 c = 2

	$-\infty$	-1	2	∞
$4(x+1)$	- - -	+ + +	+ + +	
$(x+2)^2$	+ + +	+ + +	+ + +	
Sign of f'	- - -	+ + +	+ + +	

$\therefore$ f is increasing on $[-1, 2] \cup [2, \infty)$.

$\therefore$ f is deceasing on $(-\infty, -1]$

Using the first Derivative Test

At c = -1: sign of $f(x)$ change from (-) to (+)

$\Rightarrow$ $f(-1) = -6$ is local minimum value

At c = 2: sign of $f'(x)$ the same (+) to (-) so that f has neither a local minimum nor a local maximum at c = 2

Answer: C

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Critical Thinking

95. Suppose 5 is a critical number for function f with $f'(x) > 0$ for $x \in (5, 6)$ and $f(x) < 0$ for $x \in (4, 5)$. Then $f(5)$

A. a local maximum

B. a local minimum

C. not a local extremum

D. Can not determine

Solution, Here at,

	4		5		6
Sign of $f'(x)$		- - -		+ + +	

At $C = 5$, $f(x)$ changes (-) to (+).

then $f(5)$ is local minimum.

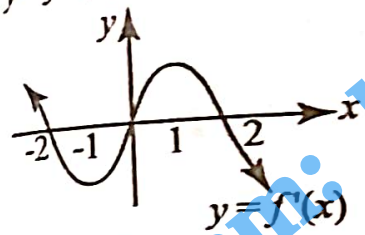
Answer: B

96. Suppose $f'(c) = 0$. Is c a local minimum or maximum?

Answer: Not necessarily: for example, $f(x) = x^3$ with $c = 0$ has $f'(c) = 0$ but $c = 0$ is not local extremum

Illustrative Example

97. Suppose f is differentiable on $(-\infty, \infty)$ and the graph of its derivative $y = f'(x)$ is shown below.



a. Locate the points of f where the local maximum and minimum occur

b. Determine the intervals on which f is increasing and decreasing.

Solution

a. $f'(x) = 0$, at $x = -2$, $x = 0$ and $x = 2$.

From the first derivative test:

At $x = -2$: $f'(x)$ changes from (+) to (-)

$\Rightarrow f(-2)$ is a local maximum

At $x = 0$: $f'(x)$ changes from (-) to (+)

$\Rightarrow f(0)$ is a local minimum.

At $x = 2$: $f'(x)$ changes from (+) to (-)

$\Rightarrow f(2)$ is a local maximum.

b. $f'(x) \geq 0$, on interval $(-\infty, -2]$ and $[0, 2]$

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$\therefore$ f is increasing on $(-\infty, -2]$ and $[0, 2]$

$f'(x) \leq 0$, on interval $[-2, 0]$ and $[2, \infty]$.

$\therefore$ f is decreasing on $[-2, 0]$ and $[2, \infty]$

98. What can you conclude about the local extrema of a differentiable function f from the following table critical numbers for f are c = 1, 3, 4, 6, 8.

Interval	Sign of $f'(x)$
$(-\infty, 1)$	Positive
$(1, 3)$	Negative
$(3, 4)$	Negative
$(4, 6)$	Positive
$(6, 8)$	Positive
$(8, \infty$	Negative

Solution

f has a local maximum at $x = 1$ and $x = 8$

- f has local minimum at $x = 4$
- There are no local extrema at $x = 3$ or $x = 6$.

Theorem: Second Derivative Test for Relative Extrema.

Let f be a continuous on the interval (a, b) and $f'(c) = 0$, for some number $c \in (a, b)$

(i) $f''(c) < 0$, then $f(c)$ is a **relative maximum.**

(ii) $f''(c) > 0$, then $f(c)$ is a **relative minimum.**

Illustrative Example

99. Let $f(x) = \frac{3}{4}x^4 - \frac{27}{2}x^2 + 15$. Find the local extremum of f.

Solution: Here,

$f'(x) = 3x^3 - 27x = 3x(x^2 - 9)$

$\therefore$ The critical numbers are: c = 0, 3 and c = -3

We can use second derivative test

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$f''(x) = 9x^2 - 27$

$f(0) = -27 < 0$, then $f(0)$ is a local maximum.

$f''(-3) = 81 - 27 = 54 > 0$, then $f(-3)$ is a local minimum

$f(3) = 81 - 27 = 54 > 0$, then $f(3)$ is local minimum.

100. Find all local extrema of $f(x) = x^3 + 6x^2 - 36x$

Solution

$f'(x) = 3x^2 + 12x - 36 = 3(x - 2)(x + 6)$

The critical points are at $c = 2, c = -6 = f''(x) = 6x + 12$

At $x = 2, f''(2) = 24 > 0$, then $f(2)$ is a local minimum

At $x = -6, f''(-6) = -24 < 0$, then $f(-6)$ is a local maximum.

101. Which of the following is a local minimum point of the function $f(x) = -3x^4 + 4x^3 + 12x^2 + 1$

A. (-2, -33) B. (-1, 4) C. (0, 1) D. (2, 3)

Solution

$f'(x) = -12x^3 + 12x^2 + 24x = -12x(x^2 - x - 2)$

$= 0 - 12x(x + 1)(x - 2)$

$\therefore$ $x = 0, -1$ and $x = 2$ are critical numbers.

Using 2nd derivative test

$f''(x) = -36x^2 + 24x + 24.$

$f''(0) = 24 > 0$, then $f(0)$ is local minmum.

$f''(-1) = -36 - 24 = -36 < 0$, then $f(-1)$ is a local maximum

$f''(2) = -36 + 48 + 24 = 36 > 0$, then $f(2)$ is a local minimum

So that

$f(0) = 1, f(-1) = -3 + -4 + 12 + 1 = 6$ and $f(2) = -48 + 32 + 24 + 1 = 9$

Answer: C

Motion Along a line: If S(t) is the position function of a particle moving

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on a coordinate line, then:

(i) The instantaneous velocity of the particle at time is defined.

$$V(t) = \frac{dS}{dt}$$

(ii) The instantaneous acceleration of the particle at time t is defined by

$$a(t) = \frac{dV}{dt}$$

Illustrative Example

102. An object moves in a straight line at any time t is given by $S(t) = t^3 - 4t^2 + 9t$, the units being kilometre and hour.

a. Find the velocity at the end of 3hrs.

b. Find the accelerative at the end of 3hrs.

Solution

a. Velosity: $V(t) = \frac{dS}{dt} = 3t^2 - 8t + 9$

$\therefore$ Velocity at the end of 3hrs is

$$\left.\frac{ds}{dt}\right|_{t=3} = 3(3) - 8(3) + 9 = 12\,{}^{km}/_{hr}$$

b. acceleration: $a(t) = \frac{dv}{dt} = 6t - 8$

$\therefore$ acceleration at the end of 3hrs is

$$\left.\frac{dV}{dt}\right|_{t=3} = 6(3) - 8 = 10\,{}^{km}/_{hr^2}$$

11.5 Minimization and Maximization Application

In science engineering, and business one is often interested in the maximum and minimum values of functions. for example, in a global economy, business need to minimization waste and maximum the return on their investment.

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